Moving-Throat PDE Derivation Companion

Reader Ledger

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Reader Build Overview

This build is the collaborator-facing version of the moving-throat PDE derivation ledger. It is meant to be read as a navigation companion to the main derivation, not as a replacement for the full archive.

The authoritative master document remains the archive build, `pde_ledger.tex`. That build preserves the full frontmatter firewalls, the stage ledger, the stage-by-stage appendices, the source-file index, and the internal fill workflow. This reader build keeps only the orientation material needed to move through Parts I–VIII without carrying the full archive wrapper in front of them.

0.1 What this build is for

Use this PDF when the goal is to understand the derivation program at theorem block scale:

- read the main argument from Part I through Part VIII;
- identify which stage range supports a chapter;
- locate the right result-anchor family for future citations;
- decide whether a claim is exact, reduced, numerical, or still open.

0.2 What this build leaves out

The reader build deliberately excludes the heaviest archival layers:

- the full frontmatter firewalls and long governance chapters;
- the stage ledger and Stage 001–253 appendix walk-throughs;
- the full reproducibility registry and source-file index;
- the fill-workflow operating manual.

Those layers still matter. They are simply reserved for the archive build, where they are useful for audit rather than for first-pass reading.

0.3 How to move between builds

The intended workflow is:

1. use this reader build to identify the part, theorem block, and stage range you care about;

2. switch to the archive build when you need stage-by-stage provenance, raw appendix detail, or audit infrastructure;

3. switch to the repo artifacts when you need source notes, SymPy scripts, Mathematica scripts, or saved outputs.

This keeps the readable path and the archival path separate without weakening the provenance chain.

How to Use the Reader Build

The main chapters already carry stage ranges, result-anchor references, and claim-status summaries at the top. That is enough structure for a readable pass if the reader uses the document in the right order.

0.4 Recommended reading paths

Path A: collaborator orientation

Start with the anchor summary in the next chapter, then read only the parts that matter for the current problem. Each part opens with:

- the stage range it covers;
- a source/status table;
- the result-anchor families it mainly serves;
- a claim-status paragraph describing the status ceiling of the block.

Path B: theorem or formula audit

Use the reader build to locate the right chapter and stage range, then move to the archive build for the detailed trail. The normal sequence is:

1. identify the relevant result-anchor family;
2. read the theorem-level statement in the corresponding part;
3. note the stage range and local status language;
4. open the archive build if the claim needs full provenance or appendix detail;
5. use the repo artifacts only when script-level or note-level checking is required.

Path C: future-paper citation support

Future papers should normally cite theorem-block results, not raw stage files. The reader build is the fastest way to find the right result family and the right status language. The archive build remains the place to verify the full audit trail behind that citation.

0.5 Reading discipline

Three rules keep the reader build honest:

1. A compact theorem block is not a status upgrade. If a chapter says a packet is reduced, numerical, or open, later prose must preserve that.
2. A stage range is provenance, not a citation target by itself. Cite the result family first, then use stage ranges when the derivation history matters.
3. A readable build is not a proof compression layer. When the claim is load-bearing, the archive build and repo-level audits are still the decisive references.

Status and Anchor Summary

This chapter keeps only the minimum global navigation state needed for the reader build: the six status tags and the top-level result-anchor families.

0.6 Status tags

Status tag	Reader-build meaning
Exact	Direct consequence of the declared parent theory, exact definitions, or exact algebra with no extra closure step.
Exact Within Closure	Exact once a named closure, branch family, or search class has been declared.
Reduced / Controlled Reduction	Valid only after a controlled reduction, asymptotic regime, or selected branch restriction.
Effective Closure	Organized by a useful model choice or closure that is not yet uniquely forced by the full PDE.
Numerically Located	Defined by exact equations but currently located or checked numerically.
Open	Still depends on actual branch realization or another missing theorem gate.

0.7 Top-level result-anchor families

Anchor	Stages	Main role in the derivation program
MTDC-T1	Parent + 1	Parent-theory import, projection and reduction language, mixed-sector firewall, and first moving-throat target statement.
MTDC-T2	1–2	Moving-surface lift, recovery of (a, L), and distributed wall geometry closure.
MTDC-T3	3	Stable BdG reduction, Schur-complement support kernels, and first grouped P_2 conservative diagnostics.
MTDC-T4	004–022	Maxwell/mixed reduction, outgoing bridge, and the invariant outgoing-normalization product.

Anchor	Stages	Main role in the derivation program
MTDC-T5	023–036	Full grouped bundle, overlap and isotropy machinery, selected-mode normalization language, and support-feasibility front end.
MTDC-T6	037–051	Continuum placement, split-U branch, rank-2 support completion, and coherent tracking front end.
MTDC-T7	052–090	Gain thresholds, Family-1 branch windows, and the first reduced Family-1 verdict chain.
MTDC-T8	091–163	Geometry-lane firewall, retarded closure, outlet/core/mouth compensation, and renormalized normalization residuals.
MTDC-T9	164–200	Weak-axisymmetric transport, quotient coordinates, orbit closure, and the four-scalar verdict packet.
MTDC-T10	201–218	Realization compiler, graph-error form, finite search sieve, and local mixed-ray closure.
MTDC-T11	219–242	Same-charge audit, rigid-mouth chart, selected twin-support placement, and strict actual-branch packet.
MTDC-T12	243–253	Relaxed open-system branch, recovery map, barrier/export companions, and material-threshold layer.

0.8 Status discipline

The reader build uses the same weakest-link rule as the archive build: a result may only be cited at the strongest level supported by its weakest essential input. When in doubt, the archive build and the verification trackers control the stronger interpretation.

0.9 Verification routing

The stage number is the reader-facing audit handle. This build intentionally omits the full verification filename block from each stage page and uses a shared naming convention instead.

All paths below are relative to `research/pde_ledger/`.

- SymPy symbolic audits follow `moving_throat_pde_stageNNN_*_sympy_audit.py`.
- Mathematica symbolic audits follow `moving_throat_pde_stageNNN_*_mathematica_audit.wl`.
- Numerical stress harnesses usually follow `stageNNN*_stress.py` and `stageNNN*_stress.wl`, with occasional shared-range names such as `stage003_021_*`.
- Exact path inventory and trust status live in the repo verification trackers rather than in each printed stage.

Contents

Chapter 1

Parent Theory, Geometry Lift, and Linearized PDE Backbone

Stage range: 001–023

This chapter is the archive entry that later parts treat as shared infrastructure. It imports the parent GNLS/Maxwell equations, promotes the throat geometry from the old (a, L) closure to a distributed shape field, and packages the first reduced wall/support/outgoing operators into the grouped notation reused throughout the ledger.

Part I at a glance

Primary question	How does the derivation program pass from the parent PDE model to a reusable reduced wall/support/outgoing packet?
Starting inputs	The declared $(4 + 1)$-dimensional GNLS/Maxwell parent theory, the old breathing variables (a, L), and the confinement lift $V_{\mathrm{conf}}(\mathbf{X}; a, L) \to V_{\mathrm{conf}}(\mathbf{X}; \Sigma)$.
Delivers	The moving-surface lift; the breathing reduction; the BdG support Schur complement; the projection-first Maxwell law with its matched one-port normal form; and the grouped D, N, P normalization language.
Used by	Part II for overlap and spectral placement, Part III for support completion, and Parts IV–VIII for retarded, quotient, realization, and strict/relaxed branch tests.
Result anchors	**MTDC-T1**, **MTDC-T2**, **MTDC-T3**, and **MTDC-T4**; the closing section hands off directly to **MTDC-T5**.

Stage packets.

Stages	Packet	Status	Carry-forward output
001–002	Parent import and geometry lift	Exact / Exact Within Closure	Shape field $R(\Omega, w, t)$, promoted confinement coupling, and the two-mode breathing reduction recovering $(\delta a, \delta L)$ as collective moments.
003	Conservative BdG support response	Reduced / Controlled Reduction / Exact Within Closure	Stable-mode Schur complements, pole shifts, and the first grouped P_2 conservative support diagnostics.
004–021	Projected EM and parent-action packet	Exact / Reduced / Controlled Reduction / Exact Within Closure	Projection-first Maxwell leakage, matched mixed-sector normal form, parent-action packet, and the first passive/outgoing $l = 2$ bridge.
022–023	Grouped normalization packet	Exact Within Closure / Open	Grouped projectors, normalized D, N, P coefficients, isotropic ratio $P_0 = N_0/D_0$, and anisotropy transport laws reused later.

Claim status. Mixed. The imported parent equations and projection-first Maxwell identities are Exact **inside the declared parent theory and projection kernel. The moving-surface algebra, breathing reduction, Schur complements, grouped projectors, and normalized branch formulas are** Exact Within Closure **once the declared reduced closure is adopted. Stable-mode reduction, matched Maxwell reduction, low-frequency expansion, and outgoing-port attachment are** Reduced / Controlled Reduction**. Actual nonlinear branch realization of the final normalization target is** Open**.**

What this part proves. Part I fixes the parent-theory import, the moving-surface lift, the first reduced wall/BdG response operators, the projection-first Maxwell-to-bundle bridge, and the grouped normalization packet that later parts treat as shared infrastructure.

What this part does not prove. It does not prove that the completed nonlinear moving-throat branch realizes the final outgoing-normalization target; it only derives the reduced object that the branch must realize.

Why later parts need it. Every later part reuses this backbone. Part II builds explicit overlap and selected-mode geometry on top of it, and all later closure, quotient, and realization arguments inherit its operator and normalization language.

1.1 Block contract and theorem-level output

This part has a narrow verification contract. It must show that the later reduced response program is not an unrelated collection of coefficients. The coefficients used later arise from one derivation chain:

$$\begin{aligned} \text{parent GNLS/Maxwell theory} &\longrightarrow \text{moving shape field} \longrightarrow \text{linear wall/BdG/Maxwell system} \\ &\longrightarrow \text{grouped response bundle.} \end{aligned} \tag{1.1}$$

The chain is deliberately status-stratified. The parent action and its Euler–Lagrange identities are exact inside the declared parent model. The shape-field lift is the working effective geometry closure. The projected Maxwell law is exact once the observation kernel is declared, while the old reduction-first Maxwell block survives only as a matched one-port normal form. The stable-mode and Maxwell/mixed Schur complements are exact after their reduced normal-coordinate ansatz is declared. The final outgoing normalization condition is an exact branch test, but its realization by the actual moving-throat branch remains open at this stage.

Theorem 1.1 (Part I backbone theorem)**.** *Under the declared parent* $(4+1)$*-dimensional gauged GNLS plus localized Maxwell theory, and under the moving-surface closure*

$$\Sigma(\mathbf{X},t) = r - R(\Omega, w, t), \qquad r = \sqrt{x^2+y^2+z^2}, \qquad \Omega = \mathbf{x}/r, \tag{1.2}$$

the linearized throat problem admits a reduced wall/support/outgoing response description with the following structure.

1. *The old variables* $a(t)$ *and* $L(t)$ *are recovered as lowest collective moments of* R*, not treated as fundamental moving-throat coordinates.*
2. *The real* $l=2$ *wall/support sector splits into grouped lanes* $20, 21, 22$*, with the* 21 *and* 22 *entries denoting real cosine/sine pairs.*
3. *Stable matter support modes generate conservative Schur-complement corrections to the wall operator.*
4. *Projection-first localized Maxwell theory gives an open-system brane law; its matched mixed* $A_w/F_{\mu w}/J^w$ *normal form generates conservative self-energies and a passive/outgoing transfer factor.*
5. *On the isotropic grouped branch, the leading outgoing quadrupole test reduces to the single invariant ratio*

$$\widehat{m}_0^2 P_0 = \frac{54 G c_s^5}{5 a^5 c^5}, \qquad P_0 = \frac{N_0}{D_0}. \tag{1.3}$$

The derivation of (1.3) *is exact inside the reduced grouped response closure. Whether the actual nonlinear moving-throat branch realizes the required value is not settled in this part.*

Proof. The statement is a compilation of the twenty-three stage-level reductions. The moment recovery and modal decomposition follow from the shape-field definition. The breathing reduction follows by inserting the lowest axisymmetric ansatz into the quadratic wall action. The BdG kernel follows by an exact Schur complement in the declared stable-mode coordinates. The Maxwell sector first follows by exact projection of the localized parent equation, then by matched mouth/bundle transport, parent-action packet bookkeeping, and the reduced Schur complement for the one-port mixed normal form. The final normalization product follows by converting the conservative grouped operator into a normalized response and multiplying by the compact outgoing $l=2$ fingerprint. Each individual step is written below with its own status boundary. □

1.2 Exact parent action and field-equation import

1.2.1 Local goal and import boundary

The moving-throat program does not begin by inventing new field equations. It imports the already-declared parent (4+1)-dimensional matter/gauge theory and then promotes the old geometry closure into a distributed moving-surface variable. The imported parent equations are therefore the fixed background against which the moving-throat extension must be checked.

The exact import contains three pieces:

1. a gauged four-spatial-dimensional GNLS matter sector;
2. a localized $(4+1)$-dimensional Maxwell sector with localization profile $Z(w)$;
3. a confinement coupling that is later promoted from $V_{\text{conf}}(\mathbf{X}; a, L)$ to $V_{\text{conf}}(\mathbf{X}; \Sigma)$.

The geometry lift can modify the confinement argument, the wall response, and the branch data. It does not modify the charge ontology, the minimal-coupling rule, or the exact localized Maxwell equation.

1.2.2 Arena and core fields

The parent arena has coordinates

$$x^M = (t, x, y, z, w), \qquad M, N \in \{0, 1, 2, 3, 4\}. \tag{1.4}$$

Bulk spatial coordinates are

$$\mathbf{X} = (x, y, z, w) \in \mathbb{R}^4, \qquad \mathbf{x} = (x, y, z) \in \mathbb{R}^3, \tag{1.5}$$

where $\mathbf{x}$ is the brane-facing spatial coordinate. The flat bulk metric convention is

$$\eta_{MN} = \text{diag}(-1, +1, +1, +1, +1), \qquad \Box_5 = -\partial_t^2 + \nabla_3^2 + \partial_w^2. \tag{1.6}$$

The core parent fields are

$$\psi(\mathbf{X}, t), \qquad A_M(x), \qquad F_{MN} = \partial_M A_N - \partial_N A_M, \qquad \rho = |\psi|^2. \tag{1.7}$$

The moving-throat extension introduces either a level set $\Sigma(\mathbf{X}, t)$ or the equivalent shape field $R(\Omega, w, t)$, but that variable enters through the promoted confinement coupling.

1.2.3 Charge ontology imported into the PDE program

The electric branch label is

$$\eta_Q = \pm 1, \qquad q_\star = \eta_Q e_\star, \qquad e_\star > 0. \tag{1.8}$$

After zero-mode brane normalization, the observed brane charge is

$$q_{\text{eff}} = \frac{q_\star}{\sqrt{Z_{\text{int}}}}, \qquad e_{\text{eff}} = \frac{e_\star}{\sqrt{Z_{\text{int}}}}, \qquad Z_{\text{int}} = \int_{-\infty}^{+\infty} Z(w)\, dw. \tag{1.9}$$

This firewall matters already in Part I because the mixed channels $A_w, J^w, F_{\mu w}$ are retained microscopically. Keeping those channels alive must not be confused with returning to an older circulation-based definition of electric charge. Circulation remains a magnetic/vortical label; it is not the sign or magnitude of the electric branch.

1.2.4 Matter action and exact GNLS equation

The imported matter Lagrangian density is

$$\mathcal{L}_\psi = \frac{i\hbar}{2}\left(\psi^* D_t\psi - \psi D_t\psi^*\right) - \frac{\hbar^2}{2m}(D_i\psi)^*(D_i\psi) - V_{\rm conf}(\mathbf{X};\Sigma)\rho - U(\rho). \tag{1.10}$$

At the exact parent level before promotion, the confinement potential is usually written as $V_{\rm conf}(\mathbf{X}; a, L)$. In the moving-throat lift the same coupling is read as a surface-dependent potential. The covariant derivatives are

$$D_t\psi = \partial_t\psi + \frac{iq_\star}{\hbar}A_0\psi, \qquad D_i\psi = \partial_i\psi - \frac{iq_\star}{\hbar}A_i\psi. \tag{1.11}$$

The stiff-polytropic equation of state carried into the program is

$$P(\rho) = K\rho^5, \qquad U(\rho) = \frac{K}{4}\rho^5, \qquad h(\rho) = \frac{dU}{d\rho} = \frac{5K}{4}\rho^4, \tag{1.12}$$

with sound-speed ledger

$$c_s^2(\rho) = \frac{1}{m}\frac{dP}{d\rho} = \frac{5K}{m}\rho^4. \tag{1.13}$$

Variation with respect to ψ^* gives the exact GNLS equation

$$\boxed{i\hbar D_t\psi = \left[-\frac{\hbar^2}{2m}D_iD_i + V_{\rm conf}(\mathbf{X};\Sigma) + h(\rho)\right]\psi}. \tag{1.14}$$

The exact number current and continuity equation are

$$\boxed{j^i = \frac{\hbar}{m}\Im(\psi^* D_i\psi), \qquad \partial_t\rho + \partial_i j^i = 0}. \tag{1.15}$$

1.2.5 Localized Maxwell action and exact mixed observables

The localized Maxwell Lagrangian density is

$$\mathcal{L}_{\rm EM} = -\frac{Z(w)}{4\mu_0}F_{MN}F^{MN} - \frac{H(w)}{2\xi\mu_0}(\partial\cdot A)^2 - A_M J^M_{\rm ext}. \tag{1.16}$$

Here $H(w) = 1$ is the old unweighted gauge-fixing convention, while $H(w) = Z(w)$ is the projection-stable localized convention used below. The matter current generated by minimal coupling is not to be double-counted in $J^M_{\rm ext}$. The source entering Maxwell's equation is

$$J^M_{\rm tot} = J^M_\psi + J^M_{\rm ext}. \tag{1.17}$$

Variation with respect to A_N gives

$$\boxed{\partial_M(Z(w)F^{MN}) + \frac{1}{\xi}\partial^N(H(w)\partial\cdot A) = \mu_0 J^N_{\rm tot}}. \tag{1.18}$$

The Bianchi identity is

$$\partial_{[L}F_{MN]} = 0. \tag{1.19}$$

The mixed gauge-invariant fields retained by the PDE program are

$$\boxed{E_w = F_{w0} = -\partial_t A_w - \partial_w A_0, \qquad C_a = F_{aw} = \partial_a A_w - \partial_w A_a}. \tag{1.20}$$

They are suppressed only in the strict far-field brane Maxwell reduction. They remain available in the moving-throat PDE and become essential in the first outgoing bridge.

1.2.6 Madelung identities and the vortex/gauge firewall

Writing

$$\psi = \sqrt{\rho}\, e^{i\theta}, \tag{1.21}$$

the exact gauge-invariant velocity is

$$v_i = \frac{\hbar}{m}\partial_i\theta - \frac{q_\star}{m}A_i. \tag{1.22}$$

The quantum potential is

$$Q(\rho) = -\frac{\hbar^2}{2m}\frac{\nabla_4^2\sqrt{\rho}}{\sqrt{\rho}}. \tag{1.23}$$

The Euler-like form is

$$m(\partial_t + v_j\partial_j)v_i = q_\star(E_i + v_jB_{ij}) - \partial_i\left(V_{\text{conf}} + h(\rho) + Q(\rho)\right), \tag{1.24}$$

and the exact vorticity–gauge identity is

$$\boxed{\Omega_{ij} := \partial_i v_j - \partial_j v_i = -\frac{q_\star}{m}F_{ij}}. \tag{1.25}$$

This equation is frequently useful, but it is not a charge dictionary. It records the relation between gauge field strength and gauge-invariant vorticity away from singular phase defects.

1.2.7 Projection and reduction hooks

For a normalized brane-observation kernel $W(w)$, projection means

$$\langle Q\rangle_W = \int_D W(w)Q(w)\,dw, \qquad \int_D W(w)\,dw = 1. \tag{1.26}$$

Projection commutes with brane derivatives, but not with the transverse derivative:

$$\langle \partial_w Q\rangle_W = [WQ]_{\partial D} - \int_D W'(w)Q(w)\,dw. \tag{1.27}$$

This identity is the reason a projection-first brane theory is generally an open system.

Applied to the matter current, the same projection defines

$$\rho_{\text{brane}}(\mathbf{x},t) = \int W(w)\rho(\mathbf{x},w,t)\,dw, \qquad \mathbf{j}_{\text{brane}} = \int W(w)\mathbf{j}_{xyz}(\mathbf{x},w,t)\,dw. \tag{1.28}$$

Projected continuity is exact once W is declared:

$$\partial_t\rho_{\text{brane}} + \nabla_3\cdot\mathbf{j}_{\text{brane}} = S_{\text{leak}}, \tag{1.29}$$

where

$$S_{\text{leak}} = -[Wj^w]_{-\infty}^{+\infty} + \int W'(w)j^w\,dw. \tag{1.30}$$

Applied to the localized Maxwell equation (1.18), projection gives, for brane index ν,

$$\boxed{\partial_\mu\langle ZF^{\mu\nu}\rangle_W + [WZF^{w\nu}]_{\partial D} - \int_D W'(w)ZF^{w\nu}\,dw + \frac{1}{\xi}\partial^\nu\langle H\,\partial\!\cdot\! A\rangle_W = \mu_0\langle J^\nu\rangle_W}. \tag{1.31}$$

The homogeneous equations project in the measured fields

$$E_{\mathrm{meas},a} = \langle F_{a0} \rangle_W, \qquad B_{\mathrm{meas},ab} = \langle F_{ab} \rangle_W, \tag{1.32}$$

whereas (1.31) uses the source-coupled fluxes

$$D^a_{\mathrm{flux}} = \langle Z F^{a0} \rangle_W, \qquad H^{ab}_{\mathrm{flux}} = \langle Z F^{ab} \rangle_W. \tag{1.33}$$

Identifying measured fields with flux fields is therefore a closure assumption, not a parent identity.

In the zero-mode channel

$$A_\mu(x,w) = a_\mu(x), \qquad A_w = 0, \qquad F^{w\nu} = 0, \qquad J^\nu(x,w) = j^\nu(x) S(w), \tag{1.34}$$

the projected effective parameters are

$$\boxed{\mu^{\mathrm{proj}}_{0,\mathrm{eff}} = \mu_0 \frac{\int W S \, dw}{\int W Z \, dw}, \qquad \xi^{\mathrm{proj}}_{\mathrm{eff}} = \xi \frac{\int W Z \, dw}{\int W H \, dw}}. \tag{1.35}$$

The matched reduction-first brane Maxwell law is recovered by the localized gauge choice and source profile

$$\boxed{H = Z, \qquad S(w) = \frac{Z(w)}{Z_{\mathrm{int}}}, \qquad \mu^{\mathrm{proj}}_{0,\mathrm{eff}} = \frac{\mu_0}{Z_{\mathrm{int}}}}. \tag{1.36}$$

Thus the reduction-first Maxwell sector is retained below as a controlled matched channel and one-port normal form, not as the generic parent projection.

Under the Helmholtz split

$$\mathbf{v}_{\mathrm{brane}} = \nabla_3 \varphi + \mathbf{v}_T, \qquad \nabla_3 \cdot \mathbf{v}_T = 0, \tag{1.37}$$

the exact longitudinal identity is

$$\rho_{\mathrm{brane}} \nabla_3^2 \varphi = S_{\mathrm{leak}} - \partial_t \rho_{\mathrm{brane}} - (\nabla_3 \rho_{\mathrm{brane}}) \cdot (\nabla_3 \varphi + \mathbf{v}_T). \tag{1.38}$$

This is an imported lower-order hook rather than a new Stage 001–023 result. It fixes how the moving-throat response must later interface with the brane-facing scalar sector.

Theorem 1.2 (Parent-theory import)**.** *Within the corrected charge ontology and the declared localized Maxwell/GNLS parent action, the moving-throat derivation inherits the exact equations* (1.14)*,* (1.15)*,* (1.18)*,* (1.20)*, and* (1.25)*. After a brane-observation kernel is declared, it also inherits the exact projected Maxwell law* (1.31)*. Any moving-throat closure must therefore preserve the mixed-channel ontology and may only suppress $A_w, J^w, F_{\mu w}, E_w, C_a$ as part of a controlled matched brane reduction.*

Proof. The equations are direct Euler–Lagrange consequences of the imported parent action and the minimal-coupling definitions. The projected Maxwell law follows by applying (1.27) to the transverse derivative in (1.18). The mixed fields in (1.20) are components of the gauge-invariant field strength. Therefore they cannot be removed by a gauge choice. The zero-mode brane Maxwell reduction may set them to zero as a matched branch assumption, but the moving-throat PDE that is supposed to compute microscopic response data must retain them until such a branch restriction is explicitly imposed. □

1.3 Hybrid level-set / shape-field representation

1.3.1 Purpose of the geometry lift

The old conservative hierarchy used finite-dimensional geometry variables $a(t)$ and $L(t)$. Those variables are adequate as a reduced breathing closure, but they cannot expose the grouped real P_2 support lanes or the full mouth/worldtube response operator. Stage 001 therefore replaces the old finite-dimensional closure by a distributed surface field while preserving a and L as low moments.

The chosen representation is hybrid: the finite throat is represented by a level set, but the level set is parameterized by a brane-spherical mouth shape. Define

$$\boxed{\Sigma(\mathbf{X},t) = r - R(\Omega,w,t), \qquad r = \sqrt{x^2+y^2+z^2}, \qquad \Omega = \frac{\mathbf{x}}{r} \in S^2}. \tag{1.39}$$

The throat surface is $\Sigma(\mathbf{X},t) = 0$. We use the sign convention

$$\Sigma > 0 \quad \text{outside the throat,} \qquad \Sigma < 0 \quad \text{inside the support region.} \tag{1.40}$$

1.3.2 Reference stationary throat

A stationary finite-throat reference branch is written

$$\Sigma_0(\mathbf{X}) = r - R_0(w). \tag{1.41}$$

The reference mouth is at $w = 0$, the throat extends over $0 \le w \le L_0$, and the reference mouth radius is

$$a_0 = R_0(0). \tag{1.42}$$

The bottom may be represented by

$$R_0(L_0) = 0, \tag{1.43}$$

or by an equivalent regular bottom condition. The finite-throat assumption is important because the later support problem uses finite axial spectral data rather than a noncompact Gaussian surrogate.

1.3.3 Promoted confinement potential

The old confinement potential is promoted by replacing

$$V_{\text{conf}}(\mathbf{X}; a, L) \quad \longrightarrow \quad V_{\text{conf}}(\mathbf{X}; \Sigma). \tag{1.44}$$

The minimal smooth-wall choice is

$$V_{\text{conf}}(\mathbf{X}; \Sigma) = V_{\text{wall}}\left(\frac{\Sigma(\mathbf{X},t)}{\ell_c}\right), \tag{1.45}$$

where ℓ_c is a wall-thickness scale. Around Σ_0, the first variation is

$$\delta V_{\text{conf}} = \frac{V'_{\text{wall}}(\Sigma_0/\ell_c)}{\ell_c}\,\delta\Sigma. \tag{1.46}$$

Since $\Sigma = r - R$, one has

$$\delta\Sigma = -\delta R. \tag{1.47}$$

Writing the wall displacement as $\eta = \delta R$, the moving-wall coupling to the matter sector is therefore

$$\boxed{\delta V_{\text{conf}} = -\frac{V'_{\text{wall}}(\Sigma_0/\ell_c)}{\ell_c}\,\eta}. \tag{1.48}$$

This sign is one of the main sign checks for the entire linearized program: a positive outward radial displacement decreases Σ at fixed bulk point and therefore enters the confinement variation with the minus sign in (1.48).

1.4 Collective moment recovery for $a(t)$ and $L(t)$

1.4.1 Mouth-radius and depth moments

The old mouth radius is not discarded. It is recovered as the spherical average of the moving mouth:

$$\boxed{a(t) = \frac{1}{4\pi} \int_{S^2} R(\Omega, 0, t)\, d\Omega}. \tag{1.49}$$

Let $W_b(\Omega, t)$ denote the bottom location, implicitly defined by

$$R(\Omega, W_b(\Omega, t), t) = 0. \tag{1.50}$$

Then the old axial depth is recovered as

$$\boxed{L(t) = \frac{1}{4\pi} \int_{S^2} W_b(\Omega, t)\, d\Omega}. \tag{1.51}$$

Equations (1.49) and (1.51) are the first audit gate for the lift: the new distributed theory must reduce to the old (a, L) closure when restricted to its lowest axisymmetric moments.

1.4.2 Mode decomposition of the throat shape

Write

$$R(\Omega, w, t) = R_0(w) + \eta(\Omega, w, t). \tag{1.52}$$

The wall displacement is expanded in real spherical harmonics:

$$\eta(\Omega, w, t) = q_{00}(w, t) Y_{00}(\Omega) + \sum_{A \in P_2^{\mathrm{real}}} q_A(w, t) Y_A(\Omega) + \eta_{\geq 3}(\Omega, w, t). \tag{1.53}$$

Here

$$P_2^{\mathrm{real}} = \{20, 21c, 21s, 22c, 22s\}. \tag{1.54}$$

The grouped real ledger later compresses the c/s pairs into grouped labels

$$A \in \{20, 21, 22\}, \tag{1.55}$$

with natural weights $(1, 2, 2)$. The grouped labels are lane labels, not spacetime indices.

With the standard normalized harmonic

$$Y_{00} = \frac{1}{2\sqrt{\pi}}, \qquad \int_{S^2} Y_{00}^2\, d\Omega = 1, \qquad \frac{1}{4\pi} \int_{S^2} Y_{00}\, d\Omega = \frac{1}{2\sqrt{\pi}}, \tag{1.56}$$

the physical mouth-average shift obeys

$$\boxed{q_{00}(0, t) = 2\sqrt{\pi}\, \delta a(t)}. \tag{1.57}$$

All grouped real P_2 harmonics have zero average over the sphere. Thus grouped quadrupole deformations do not change the mouth-average radius at linear order.

1.4.3 Axisymmetric residual lane

The axisymmetric wall field is further split as

$$q_{00}(w,t)Y_{00}(\Omega) = 2\sqrt{\pi}\left[\alpha_a(w)\delta a(t) + \alpha_L(w)\delta L(t)\right]Y_{00}(\Omega) + g(w,t)Y_{00}(\Omega), \tag{1.58}$$

where $g(w,t)$ is the residual axisymmetric geometry lane orthogonal to the chosen a and L profiles. In practice, the first truncation drops g; later geometry-completion terms are naturally interpreted as the return of this residual lane.

Proposition 1.3 (Collective-moment recovery)**.** *Inside the shape-field closure $\Sigma = r - R(\Omega, w, t)$, the old collective variables $a(t)$ and $L(t)$ are recovered as the monopole moments* (1.49) *and* (1.51)*. The grouped real P_2 deformations are orthogonal to the mouth-average radius shift at linear order.*

Proof. The formula for $a(t)$ is the spherical average of the mouth radius. The formula for $L(t)$ is the spherical average of the bottom location. The zero-average property of all $l = 2$ spherical harmonics gives the last statement immediately. The normalization bridge (1.57) follows from (1.56). ☐

1.5 Distributed wall action and densitized convention

1.5.1 Quadratic wall action

The first distributed wall action is the minimal passive local quadratic action

$$S_\eta^{(2)} = \frac{1}{2}\int dt\, dw\, d\Omega\, \sqrt{\gamma_0}\left[\mu_\eta(w)(\partial_t\eta)^2 - T_w(w)(\partial_w\eta)^2 - T_\Omega(w)\eta(-\Delta_{S^2})\eta - K_\eta(w)\eta^2\right]. \tag{1.59}$$

The coefficients $\mu_\eta, T_w, T_\Omega, K_\eta$ are branch data of the chosen reference throat. They are not to be refit independently at later stages to force a normalization condition.

After integrating over the reference sphere, the program uses a densitized one-dimensional convention: the background surface weight is absorbed into the effective axial coefficients. With that convention, the modal action is written with measure dw. If an explicit axial weight $g(w) = \sqrt{\gamma_0}$ were retained, the Euler–Lagrange equation would contain the corresponding first-derivative contribution from $g'(w)$. In the ledger formulas below, that bookkeeping has already been absorbed into $\mu_\eta, T_w, T_\Omega, K_\eta$.

1.5.2 Modal wall equation

For a harmonic component $q_{lm}(w,t)Y_{lm}(\Omega)$, using

$$-\Delta_{S^2}Y_{lm} = l(l+1)Y_{lm}, \tag{1.60}$$

the densitized modal wall equation is

$$\boxed{\mu_\eta q_{lm,tt} - \partial_w(T_w\partial_w q_{lm}) + \left[K_\eta + l(l+1)T_\Omega\right]q_{lm} = S_{lm}^{(\psi,A)} + f_{lm}^{\text{ext}}}. \tag{1.61}$$

The source $S_{lm}^{(\psi,A)}$ is generated by variation of the matter and gauge sectors, with the first matter contribution inherited from (1.48).

Equation (1.61) separates the first three relevant sectors:

1. the $l = 0$ collective sector containing δa, δL, and the residual geometry lane $g(w,t)$;

2. the grouped real $l = 2$ sector containing $20, 21c, 21s, 22c, 22s$;

3. higher $l \geq 3$ wall modes.

The present response program keeps higher modes in the full PDE ontology but does not need them for the first grouped-quadrupole bridge.

Proposition 1.4 (Wall modal backbone)**.** *The quadratic wall action* (1.59)*, read in the densitized axial convention, yields the modal operator in* (1.61)*. The scalar and grouped real P_2 sectors therefore split before any matter or gauge coupling is added.*

Proof. Insert $\eta = q_{lm}(w,t)Y_{lm}(\Omega)$ into (1.59), use orthonormality on S^2, and vary the resulting one-dimensional action. The angular stiffness term contributes $l(l+1)T_\Omega q_{lm}$. The remaining source terms are not part of the free wall action and are kept on the right-hand side. □

1.6 Axisymmetric breathing reduction

1.6.1 Two-mode breathing ansatz

Stage 002 checks that the distributed wall lift recovers the old (a, L) closure at the expected conservative linear level. Use the axisymmetric truncation

$$\eta_0(w,t) = 2\sqrt{\pi}\left[\alpha_a(w)\delta a(t) + \alpha_L(w)\delta L(t)\right]. \tag{1.62}$$

Define

$$Q^A = (\delta a, \delta L), \qquad A \in \{a, L\}. \tag{1.63}$$

The truncation is not the final geometry theory. It is the lowest-mode audit that the distributed wall has not lost contact with the older closure.

1.6.2 Reduced matrices

For the axisymmetric branch, the reduced Lagrangian is

$$L^{(0)}_{\text{red}} = \frac{1}{2}M_{AB}\dot{Q}^A\dot{Q}^B - \frac{1}{2}K_{AB}Q^AQ^B, \tag{1.64}$$

where

$$\boxed{M_{AB} = 4\pi \int dw\, \mu_\eta(w)\alpha_A(w)\alpha_B(w)}, \tag{1.65}$$

and

$$\boxed{K_{AB} = 4\pi \int dw\, \left[T_w(w)\alpha'_A(w)\alpha'_B(w) + K_0(w)\alpha_A(w)\alpha_B(w)\right]}. \tag{1.66}$$

The Euler–Lagrange equation is

$$\boxed{M_{AB}\ddot{Q}^B + K_{AB}Q^B = 0}. \tag{1.67}$$

This is the conservative part of the older effective geometry law. Damping and open-system terms can be added only after coupling to exterior, matter, gauge, or leakage channels.

1.6.3 One-mode grouped P_2 comparison

For a single grouped real quadrupole amplitude,

$$\eta_{2m}(\Omega, w, t) = \beta_2(w) q_{2m}(t) Y^{\mathrm{real}}_{2m}(\Omega), \tag{1.68}$$

the reduced one-mode Lagrangian is

$$L_{2m} = \frac{1}{2} M_2 \dot{q}_{2m}^2 - \frac{1}{2} K_2 q_{2m}^2, \tag{1.69}$$

with

$$\boxed{M_2 = \int dw\, \mu_\eta(w) \beta_2(w)^2, \qquad K_2 = \int dw \left[T_w(w) \beta_2'(w)^2 + (K_\eta(w) + 6T_\Omega(w)) \beta_2(w)^2 \right]}. \tag{1.70}$$

Therefore

$$M_2 \ddot{q}_{2m} + K_2 q_{2m} = 0. \tag{1.71}$$

Before anisotropy or support coupling is added, every real P_2 component has the same geometry-only operator. This is the first degeneracy statement behind the later grouped-isotropy gates.

Theorem 1.5 (Breathing reduction as the lowest wall truncation)**.** *Inside the quadratic distributed-wall closure, the two-mode axisymmetric ansatz* (1.62) *reduces exactly to the conservative matrix system* (1.67)*. The old* (a, L) *variables are therefore the lowest collective coordinates of the wall PDE rather than independent microscopic primitives.*

Proof. Substitute (1.62) into the quadratic wall action, use the normalization of Y_{00}, and collect coefficients of $\dot{Q}^A \dot{Q}^B$ and $Q^A Q^B$. The resulting matrices are (1.65) and (1.66). Varying (1.64) gives (1.67). □

1.7 Stable BdG normal-mode reduction and Schur complement

1.7.1 Linearized matter coupling

Let the stationary matter background be

$$\psi(\mathbf{X}, t) = e^{-i\mu_0 t/\hbar} \psi_0(\mathbf{X}), \tag{1.72}$$

and write

$$\psi = e^{-i\mu_0 t/\hbar} (\psi_0 + \delta\psi). \tag{1.73}$$

The linearized density is

$$\delta\rho = \psi_0^* \delta\psi + \psi_0 \delta\psi^*. \tag{1.74}$$

The moving wall enters the linearized GNLS problem through (1.48). In BdG form the schematic system is

$$i\hbar \partial_t \begin{pmatrix} \delta\psi \\ \delta\psi^* \end{pmatrix} = \mathcal{L}_{\mathrm{BdG}} \begin{pmatrix} \delta\psi \\ \delta\psi^* \end{pmatrix} + \mathcal{C}_A[\delta A_M] + \mathcal{C}_\eta[\eta]. \tag{1.75}$$

Stage 003 does not claim to solve the full BdG PDE. It passes to a stable normal-mode reduction in selected $l = 0$ and $l = 2$ sectors. Once that reduction is declared, integrating out the stable modes is exact algebra.

1.7.2 Axisymmetric BdG support kernel

Let $Q^A = (\delta a, \delta L)$ and let X_α be stable scalar BdG normal coordinates with frequencies $\varpi_{0\alpha} > 0$. The reduced quadratic Lagrangian is

$$L^{(0)}_{\rm red} = \frac{1}{2}\dot{Q}^T M_0 \dot{Q} - \frac{1}{2} Q^T K_0 Q + \frac{1}{2}\sum_\alpha \left(\dot{X}_\alpha^2 - \varpi_{0\alpha}^2 X_\alpha^2\right) + \sum_{A,\alpha} C_{A\alpha} Q^A X_\alpha. \tag{1.76}$$

In frequency space, the matter coordinates solve

$$(\varpi_{0\alpha}^2 - \omega^2) X_\alpha = C_{A\alpha} Q^A. \tag{1.77}$$

Substitution gives the exact effective matrix kernel

$$\boxed{D_0^{\rm eff}(\omega) = K_0 - \omega^2 M_0 - C(\Omega_0^2 - \omega^2 I)^{-1} C^T}, \tag{1.78}$$

where

$$\Omega_0^2 = \mathrm{diag}(\varpi_{0\alpha}^2). \tag{1.79}$$

Its low-frequency expansion is

$$D_0^{\rm eff}(\omega) = K_0^{\rm eff} - \omega^2 M_0^{\rm eff} - \omega^4 N_0^{\rm eff} + O(\omega^6), \tag{1.80}$$

with

$$\boxed{K_0^{\rm eff} = K_0 - C\Omega_0^{-2} C^T, \qquad M_0^{\rm eff} = M_0 + C\Omega_0^{-4} C^T, \qquad N_0^{\rm eff} = C\Omega_0^{-6} C^T}. \tag{1.81}$$

Thus stable matter support softens the static stiffness, increases the effective inertia, and generates the next even low-frequency coefficient.

1.7.3 Grouped real P_2 BdG kernels

For each grouped lane $A \in \{20, 21, 22\}$, let q_A be the wall amplitude and $X_{A\alpha}$ be stable quadrupole BdG coordinates with frequencies $\varpi_{A\alpha}$. The reduced Lagrangian is

$$L^{(2)}_{\rm red} = \sum_A \left[\frac{1}{2} M_A \dot{q}_A^2 - \frac{1}{2} K_A q_A^2 + \frac{1}{2}\sum_\alpha (\dot{X}_{A\alpha}^2 - \varpi_{A\alpha}^2 X_{A\alpha}^2) + \sum_\alpha g_{A\alpha} q_A X_{A\alpha}\right]. \tag{1.82}$$

Eliminating the stable coordinates gives

$$\boxed{D_A^{\rm eff}(\omega) = K_A - M_A\omega^2 - \sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^2 - \omega^2}}. \tag{1.83}$$

The low-frequency coefficients are

$$\boxed{\begin{aligned} d_{0A}^{\rm eff} &= K_A - \sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^2}, \\ d_{2A}^{\rm eff} &= -\left(M_A + \sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^4}\right), \\ d_{4A}^{\rm eff} &= -\sum_\alpha \frac{g_{A\alpha}^2}{\varpi_{A\alpha}^6}. \end{aligned}} \tag{1.84}$$

These are the first explicit conservative grouped-lane moments generated by matter support.

1.7.4 First pole-shift formula

For one wall mode q coupled to one stable matter mode X,

$$L = \frac{1}{2}M\dot{q}^2 - \frac{1}{2}Kq^2 + \frac{1}{2}\dot{X}^2 - \frac{1}{2}\varpi^2 X^2 + gqX. \tag{1.85}$$

The exact dispersion relation is

$$(K - M\omega^2)(\varpi^2 - \omega^2) - g^2 = 0. \tag{1.86}$$

With $\Omega_\eta^2 = K/M$, the exact poles are

$$\boxed{\omega_\pm^2 = \frac{\Omega_\eta^2 + \varpi^2 \pm \sqrt{(\Omega_\eta^2 - \varpi^2)^2 + 4g^2/M}}{2}}. \tag{1.87}$$

If $\varpi^2 > \Omega_\eta^2$ and g is weak, the wall-like pole shifts by

$$\boxed{\delta\Omega_\eta^2 = -\frac{g^2/M}{\varpi^2 - \Omega_\eta^2} + O(g^4)}. \tag{1.88}$$

This is the first explicit reduced origin of the pole data that later appear as open throat observables.

1.7.5 Matter-coupled isotropy diagnostic

For grouped triples (x_{20}, x_{21}, x_{22}), define

$$\bar{x} = \frac{x_{20} + 2x_{21} + 2x_{22}}{5}, \qquad a_x = \frac{2x_{20} - x_{21} - x_{22}}{10}, \qquad b_x = \frac{x_{21} - x_{22}}{2}. \tag{1.89}$$

If

$$K_{20} = K_{21} = K_{22}, \quad M_{20} = M_{21} = M_{22}, \quad g_{20,\alpha} = g_{21,\alpha} = g_{22,\alpha}, \quad \varpi_{20,\alpha} = \varpi_{21,\alpha} = \varpi_{22,\alpha}, \tag{1.90}$$

then all grouped coefficients in (1.84) are lane-independent, and therefore the grouped anisotropy defects vanish.

Theorem 1.6 (BdG Schur-complement support kernel)**.** *Inside the stable normal-mode reduction of the linearized matter sector, integrating out the BdG coordinates gives the exact Schur-complement kernels* (1.78) *and* (1.83)*. The low-frequency moments are fixed by the support frequencies and coupling overlaps; grouped P_2 anisotropy can enter only through lane dependence of the wall data, support frequencies, or coupling overlaps.*

Proof. The reduced matter coordinates enter quadratically and linearly. Their frequency-domain equations are algebraic. Solving those equations and substituting back into the wall equations gives the Schur complements. Expanding $(\Omega^2 - \omega^2)^{-1}$ in powers of ω^2 gives the stated low-frequency coefficients. The isotropy statement follows from equality of the three lane data. □

1.8 Projection-first Maxwell/mixed response and outgoing $l = 2$ bridge

1.8.1 Why the mixed sector enters before the outgoing bridge

The BdG support kernel is conservative. It can produce even low-frequency response moments and pole shifts, but by itself it does not produce an honest passive/outgoing odd term. The first available microscopic carrier of an outgoing branch is the localized Maxwell/mixed block, because A_w, $F_{\mu w}$, and J^w are present in the parent ontology and are not gauge artifacts.

Stages 004–021 first project the localized parent Maxwell equation, producing the exact open-system law (1.31). Only after the matched channel (1.36) is declared does the derivation compress the EM sector into a minimal reduced lane containing one wall coordinate Q_l, one brane-like localized Maxwell coordinate A_l, and one mixed-sector coordinate W_l. This is not the full PDE. It is the Stage 021 normal form used to carry both conservative gauge self-energy and passive outgoing transfer into the grouped bundle.

1.8.2 One-lane Maxwell/mixed Lagrangian

The reduced quadratic Lagrangian is

$$\begin{aligned} L_l = {}& \frac{1}{2}M_l\dot{Q}_l^2 - \frac{1}{2}K_lQ_l^2 + \frac{1}{2}\dot{A}_l^2 - \frac{1}{2}\Omega_{A,l}^2A_l^2 + \frac{1}{2}\dot{W}_l^2 - \frac{1}{2}\Omega_{W,l}^2W_l^2 \\ & + R_lA_lW_l + g_{A,l}Q_lA_l + g_{W,l}Q_lW_l. \end{aligned} \tag{1.91}$$

Here A_l is a brane-like localized gauge coordinate, W_l is a mixed $A_w/F_{\mu w}/J^w$-active coordinate, R_l mixes the internal gauge coordinates, and $g_{A,l}, g_{W,l}$ are wall couplings.

1.8.3 Conservative self-energy

With frequency convention $e^{-i\omega t}$, define

$$A_l(\omega) = \Omega_{A,l}^2 - \omega^2, \qquad W_l(\omega) = \Omega_{W,l}^2 - \omega^2, \qquad \Delta_l(\omega) = A_l(\omega)W_l(\omega) - R_l^2. \tag{1.92}$$

Eliminating A_l, W_l gives the conservative self-energy

$$\boxed{\Sigma_l^{\text{EM+mix,cons}}(\omega) = \frac{g_{A,l}^2W_l(\omega) + 2g_{A,l}g_{W,l}R_l + g_{W,l}^2A_l(\omega)}{\Delta_l(\omega)}}. \tag{1.93}$$

The wall operator is

$$D_l^{\text{cons}}(\omega) = K_l - M_l\omega^2 - \Sigma_l^{\text{EM+mix,cons}}(\omega). \tag{1.94}$$

Set

$$\Delta_{0,l} = \Omega_{A,l}^2\Omega_{W,l}^2 - R_l^2, \qquad S_{2,l} = \Omega_{A,l}^2 + \Omega_{W,l}^2, \tag{1.95}$$

$$N_{0,l} = g_{A,l}^2\Omega_{W,l}^2 + 2g_{A,l}g_{W,l}R_l + g_{W,l}^2\Omega_{A,l}^2, \qquad G_{2,l} = g_{A,l}^2 + g_{W,l}^2. \tag{1.96}$$

Then

$$\Sigma_l^{\text{EM+mix,cons}}(\omega) = z_{l,0} + z_{l,2}\omega^2 + z_{l,4}\omega^4 + O(\omega^6), \tag{1.97}$$

with

$$\boxed{\begin{aligned} z_{l,0} &= \frac{N_{0,l}}{\Delta_{0,l}}, \\ z_{l,2} &= \frac{N_{0,l}S_{2,l} - G_{2,l}\Delta_{0,l}}{\Delta_{0,l}^2}, \\ z_{l,4} &= \frac{N_{0,l}(S_{2,l}^2 - \Delta_{0,l}) - S_{2,l}G_{2,l}\Delta_{0,l}}{\Delta_{0,l}^3}. \end{aligned}} \tag{1.98}$$

1.8.4 Outgoing-port dressing and transfer factor

Attach the exterior passive/outgoing channel to the mixed coordinate by replacing

$$W_l(\omega) \longrightarrow W_l(\omega) - \Pi_l^{\mathrm{out}}(\omega). \tag{1.99}$$

To first order in the outgoing port,

$$\Sigma_l^{\mathrm{full}}(\omega) = \Sigma_l^{\mathrm{cons}}(\omega) + \Pi_l^{\mathrm{out}}(\omega)N_l(\omega) + O\Big((\Pi_l^{\mathrm{out}})^2\Big), \tag{1.100}$$

where the transfer factor is

$$\boxed{N_l(\omega) = \frac{\big[A_l(\omega)g_{W,l} + R_l g_{A,l}\big]^2}{\big[A_l(\omega)W_l(\omega) - R_l^2\big]^2}}. \tag{1.101}$$

At zero frequency,

$$\boxed{N_l(0) = \frac{\big[\Omega_{A,l}^2 g_{W,l} + R_l g_{A,l}\big]^2}{\big[\Omega_{A,l}^2\Omega_{W,l}^2 - R_l^2\big]^2} \geq 0}. \tag{1.102}$$

The sign of the outgoing odd term is therefore supplied by the outgoing port; the internal conservative block transfers it with a nonnegative static factor.

1.8.5 Compact outgoing $l = 2$ fingerprint

For the compact outgoing $l = 2$ branch, the normalized outgoing response has the low-frequency form

$$\widehat{Y}_2^{\mathrm{out}}(\omega) = 1 + \frac{a^2\omega^2}{9c_s^2} + \frac{4a^4\omega^4}{81c_s^4} + i\frac{a^5\omega^5}{27c_s^5} + O(\omega^6). \tag{1.103}$$

Thus the outgoing port coefficient is

$$\Gamma_2^{\mathrm{port}} = \frac{a^5}{27c_s^5}. \tag{1.104}$$

Combining (1.101) with (1.104), the wall-level odd quadrupole contribution is

$$\boxed{\delta D_2^{\mathrm{odd}}(\omega) = -iN_2(0)\frac{a^5}{27c_s^5}\omega^5 + O(\omega^7)}. \tag{1.105}$$

In the wall-operator convention $D = K - M\omega^2 - \Sigma$, this is the passive sign. In normalized response language the same branch is read with the positive $+i\omega^5$ fingerprint.

1.8.6 Scalar compatibility check

A naive scalar outlet could introduce a dangerous $i\omega$ monopole term. The reduced mixed-sector model shows how such a term can be delayed. If the scalar/breathing wall coordinate couples to the port-active mixed combination only through a derivative coupling, for example

$$g_{W,0}(\omega) = \eta\omega, \tag{1.106}$$

and if there is no direct non-derivative coupling into the same port-active combination, then

$$N_0(\omega) = O(\omega^2). \tag{1.107}$$

Therefore an outgoing scalar port with $\Pi_0^{\mathrm{out}}(\omega) = i\gamma_1\omega + \cdots$ contributes at wall level only as $O(i\omega^3)$. This is a compatibility mechanism, not yet a full scalar no-go theorem.

Theorem 1.7 (Projection-first Maxwell/mixed outgoing bridge)**.** *The exact projected Maxwell law* (1.31) *retains the transverse mixed channel before any brane reduction. In the matched one-lane Maxwell/mixed normal form, the wall inherits an outgoing odd contribution through the exact transfer factor* (1.101)*. For $l = 2$, the leading wall-level odd coefficient is* (1.105)*; the remaining normalization problem is the computation of the static transfer factor on the actual moving-throat branch.*

Proof. Projection supplies the mixed transverse source term in (1.31); the matched channel (1.36) is the reduction in which it can be represented by a finite one-port packet. Eliminating the two internal gauge coordinates from (1.91) gives the conservative self-energy (1.93). Replacing W_l by $W_l - \Pi_l^{\mathrm{out}}$ and expanding to first order in Π_l^{out} gives (1.101). Multiplying by the compact outgoing $l = 2$ fingerprint coefficient (1.104) gives (1.105). ☐

1.9 Full grouped real P_2 bundle transition

1.9.1 Conservative operator moments and normalized response moments

Stages 022 and 023 lift the projection-first Maxwell packet, through its matched one-port normal form, into the full grouped real bundle. For each grouped lane $A \in \{20, 21, 22\}$, write the conservative operator as

$$D_A^{\mathrm{cons}}(\omega) = D_{A0} + D_{A2}\omega^2 + D_{A4}\omega^4 + O(\omega^6). \tag{1.108}$$

The normalized response is

$$Y_A(\omega) = \frac{D_{A0}}{D_A^{\mathrm{cons}}(\omega)} = 1 + u_2^{(A)}\omega^2 + u_4^{(A)}\omega^4 + O(\omega^6). \tag{1.109}$$

The exact conversion formulas are

$$\boxed{u_2^{(A)} = -\frac{D_{A2}}{D_{A0}}, \qquad u_4^{(A)} = \frac{D_{A2}^2 - D_{A0}D_{A4}}{D_{A0}^2}}. \tag{1.110}$$

This is the first bookkeeping bridge between the wall/BdG/Maxwell operator language and the grouped-response language used by the later retarded quadrupole tests.

1.9.2 Grouped inverse map and weighted projectors

For a grouped triple $x = (x_{20}, x_{21}, x_{22})^T$, define

$$\bar{x} = \frac{x_{20} + 2x_{21} + 2x_{22}}{5}, \qquad a_x = \frac{2x_{20} - x_{21} - x_{22}}{10}, \qquad b_x = \frac{x_{21} - x_{22}}{2}. \tag{1.111}$$

The inverse map is

$$\boxed{x_{20} = \bar{x} + 4a_x, \qquad x_{21} = \bar{x} - a_x + b_x, \qquad x_{22} = \bar{x} - a_x - b_x}. \tag{1.112}$$

The natural grouped metric is

$$G_{\mathrm{grp}} = \mathrm{diag}(1, 2, 2). \tag{1.113}$$

A G_{grp}-orthogonal basis is

$$e_{\bar{x}} = (1, 1, 1), \qquad e_a = (4, -1, -1), \qquad e_b = (0, 1, -1), \tag{1.114}$$

with norms

$$e_{\bar{x}}^T G_{\mathrm{grp}} e_{\bar{x}} = 5, \qquad e_a^T G_{\mathrm{grp}} e_a = 20, \qquad e_b^T G_{\mathrm{grp}} e_b = 4. \tag{1.115}$$

The exact projectors are

$$\boxed{P_{\bar{x}} = \frac{1}{5}\begin{pmatrix} 1 & 2 & 2 \\ 1 & 2 & 2 \\ 1 & 2 & 2 \end{pmatrix}, \quad P_a = \frac{1}{20}\begin{pmatrix} 16 & -8 & -8 \\ -4 & 2 & 2 \\ -4 & 2 & 2 \end{pmatrix}, \quad P_b = \frac{1}{4}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 2 & -2 \\ 0 & -2 & 2 \end{pmatrix}}. \tag{1.116}$$

They satisfy

$$P_{\bar{x}} + P_a + P_b = I_3, \qquad P_i P_j = \delta_{ij} P_i. \tag{1.117}$$

The grouped isotropy condition is simply

$$P_a x = P_b x = 0. \tag{1.118}$$

1.9.3 Outgoing prefactor and normalized branch coefficients

Let the outgoing transfer factor in lane A be

$$N_A(\omega) = N_{A0} + N_{A2}\omega^2 + N_{A4}\omega^4 + O(\omega^6). \tag{1.119}$$

The internal prefactor multiplying the outgoing branch is

$$\mathrm{Pref}_A(\omega) = \frac{D_{A0} N_A(\omega)}{D_A^{\mathrm{cons}}(\omega)^2} = P_{A0} + P_{A2}\omega^2 + P_{A4}\omega^4 + O(\omega^6). \tag{1.120}$$

The exact coefficients are

$$\boxed{\begin{aligned} P_{A0} &= \frac{N_{A0}}{D_{A0}}, \\ P_{A2} &= \frac{D_{A0} N_{A2} - 2D_{A2} N_{A0}}{D_{A0}^2}, \\ P_{A4} &= \frac{D_{A0}^2 N_{A4} - 2D_{A0}(D_{A2} N_{A2} + D_{A4} N_{A0}) + 3D_{A2}^2 N_{A0}}{D_{A0}^3}. \end{aligned}} \tag{1.121}$$

Multiplying by the compact outgoing $l = 2$ fingerprint (1.103), with

$$A_2^{\text{out}} = \frac{a^2}{9c_s^2}, \qquad B_2^{\text{out}} = \frac{4a^4}{81c_s^4}, \qquad G_5^{\text{out}} = \frac{a^5}{27c_s^5}, \tag{1.122}$$

gives

$$\boxed{K_{A0} = P_{A0}, \quad K_{A2} = P_{A2} + A_2^{\text{out}} P_{A0}, \quad K_{A4} = P_{A4} + A_2^{\text{out}} P_{A2} + B_2^{\text{out}} P_{A0}, \quad \Gamma_5^{(A)} = G_5^{\text{out}} P_{A0}}. \tag{1.123}$$

Only the static prefactor P_{A0} enters the leading $i\omega^5$ coefficient.

1.9.4 Full reduced grouped bundle

Stage 023 writes the full coupled lane data explicitly. For each lane A, define BdG support moments

$$B_{A0} = \sum_\alpha \frac{c_{A\alpha}^2}{\varpi_{A\alpha}^2}, \qquad B_{A2} = \sum_\alpha \frac{c_{A\alpha}^2}{\varpi_{A\alpha}^4}, \qquad B_{A4} = \sum_\alpha \frac{c_{A\alpha}^2}{\varpi_{A\alpha}^6}. \tag{1.124}$$

For each conservative Maxwell/mixed internal pair r, set

$$\Delta_{A,r} = \Omega_{U,A,r}^2 \Omega_{W,A,r}^2 - R_{A,r}^2, \qquad S_{A,r} = \Omega_{U,A,r}^2 + \Omega_{W,A,r}^2, \tag{1.125}$$

$$Q_{A,r} = g_{U,A,r}^2 \Omega_{W,A,r}^2 + 2g_{U,A,r} g_{W,A,r} R_{A,r} + g_{W,A,r}^2 \Omega_{U,A,r}^2, \qquad G_{A,r} = g_{U,A,r}^2 + g_{W,A,r}^2. \tag{1.126}$$

Then

$$\boxed{\begin{aligned} Z_{A0}^{(r)} &= \frac{Q_{A,r}}{\Delta_{A,r}}, \\ Z_{A2}^{(r)} &= \frac{Q_{A,r} S_{A,r} - G_{A,r} \Delta_{A,r}}{\Delta_{A,r}^2}, \\ Z_{A4}^{(r)} &= \frac{Q_{A,r}(S_{A,r}^2 - \Delta_{A,r}) - S_{A,r} G_{A,r} \Delta_{A,r}}{\Delta_{A,r}^3}. \end{aligned}} \tag{1.127}$$

The outgoing-transfer moments are

$$\boxed{\begin{aligned} N_{A0}^{(r)} &= \frac{P_{A,r}^2}{\Delta_{A,r}^2}, \\ N_{A2}^{(r)} &= \frac{2P_{A,r}(P_{A,r} S_{A,r} - \Delta_{A,r} g_{W,A,r})}{\Delta_{A,r}^3}, \\ N_{A4}^{(r)} &= \frac{\Delta_{A,r}^2 g_{W,A,r}^2 - 2\Delta_{A,r} P_{A,r}^2 - 4\Delta_{A,r} P_{A,r} S_{A,r} g_{W,A,r} + 3P_{A,r}^2 S_{A,r}^2}{\Delta_{A,r}^4}, \end{aligned}} \tag{1.128}$$

where

$$P_{A,r} = \Omega_{U,A,r}^2 g_{W,A,r} + R_{A,r} g_{U,A,r}. \tag{1.129}$$

Summing over r gives Z_{An} and N_{An}. The total conservative wall coefficients are

$$\boxed{D_{A0} = K_A - B_{A0} - Z_{A0}, \qquad D_{A2} = -(M_A + B_{A2} + Z_{A2}), \qquad D_{A4} = -(B_{A4} + Z_{A4})}. \tag{1.130}$$

1.9.5 Projected Maxwell slippage into grouped slots

The projection-first correction is transported into the same grouped variables by the slot shifts

$$Z_n \mapsto Z_n + \varepsilon z_n, \qquad N_n \mapsto N_n + \varepsilon n_n, \qquad n \in \{0, 2, 4\}. \tag{1.131}$$

At a symmetric interior slice $\varepsilon = O(\sigma^2)$, while a one-sided mouth projection gives $\varepsilon = O(\ell)$. On an isotropic lane, the induced response shifts are

$$\delta u_2 = \frac{D_0 z_2 - D_2 z_0}{D_0^2}, \qquad \delta u_4 = \frac{D_0^2 z_4 - D_0(2D_2 z_2 + D_4 z_0) + 2D_2^2 z_0}{D_0^3}, \tag{1.132}$$

and the static outgoing-prefactor shift is

$$\delta P_0 = \frac{D_0 n_0 + N_0 z_0}{D_0^2}. \tag{1.133}$$

For the eliminated isotropic one-pole compatibility surface

$$\mathcal{C} = \frac{N_0}{P_{0,\text{target}}} - \frac{3S^2}{T}, \qquad S = M + B_2 + Z_2, \quad T = B_4 + Z_4, \tag{1.134}$$

the projected variation is

$$\boxed{\delta\mathcal{C} = \frac{n_0}{P_{0,\text{target}}} - \frac{6Sz_2}{T} + \frac{3S^2 z_4}{T^2}}. \tag{1.135}$$

The static conservative shift z_0 cancels from this eliminated surface. It still changes P_0 through (1.133), so the projection-first EM correction remains live in the normalization branch even when it does not move the one-pole compatibility equation.

1.9.6 Isotropic branch and final Part I normalization test

On the isotropic branch,

$$D_{20,n} = D_{21,n} = D_{22,n} = D_n, \qquad N_{20,n} = N_{21,n} = N_{22,n} = N_n. \tag{1.136}$$

Then

$$\boxed{\begin{aligned} u_2 &= -\frac{D_2}{D_0}, \\ u_4 &= \frac{D_2^2 - D_0 D_4}{D_0^2}, \\ P_0 &= \frac{N_0}{D_0}, \\ P_2 &= \frac{D_0 N_2 - 2D_2 N_0}{D_0^2}, \\ P_4 &= \frac{D_0^2 N_4 - 2D_0(D_2 N_2 + D_4 N_0) + 3D_2^2 N_0}{D_0^3}. \end{aligned}} \tag{1.137}$$

The universal quadrupole-normalization condition is

$$\boxed{\widehat{m}_0^{\,2} P_0 = \widehat{m}_0^{\,2} \frac{N_0}{D_0} = \frac{54 G c_s^5}{5 a^5 c^5}}. \tag{1.138}$$

Using (1.130), the same test is

$$\boxed{\widehat{m}_0^2 \frac{N_0}{K - B_0 - Z_0} = \frac{54 G c_s^5}{5 a^5 c^5}}. \tag{1.139}$$

This is the central output of Part I. The algebraic machinery has compressed the outgoing-normalization issue to a single isotropic ratio. The actual branch still has to compute the numerator, denominator, and source-map factor.

1.9.7 Constant-prefactor branch

The constant-prefactor branch used in the later retarded program is defined by

$$P_2 = 0, \qquad P_4 = 0. \tag{1.140}$$

Using (1.137), this is equivalent to the exact correlation conditions

$$\boxed{N_2 = \frac{2D_2N_0}{D_0}, \qquad N_4 = \frac{2D_0(D_2N_2 + D_4N_0) - 3D_2^2N_0}{D_0^2}}. \tag{1.141}$$

Thus the constant-prefactor branch does not require N_2 and N_4 to vanish individually. It requires them to be locked to the conservative moments D_0, D_2, D_4.

1.9.8 First-order anisotropy transport

Linearize around an isotropic branch:

$$D_{A,n} = D_n + \delta D_{A,n}, \qquad N_{A,n} = N_n + \delta N_{A,n}. \tag{1.142}$$

The response-moment variation is

$$\delta u_2^{(A)} = -\frac{\delta D_{A,2} + u_2 \delta D_{A,0}}{D_0}. \tag{1.143}$$

Therefore

$$\boxed{a_{u,2} = -\frac{a_{D,2} + u_2 a_{D,0}}{D_0}, \qquad b_{u,2} = -\frac{b_{D,2} + u_2 b_{D,0}}{D_0}}. \tag{1.144}$$

Similarly, since $P_0^{(A)} = N_{A0}/D_{A0}$,

$$\delta P_0^{(A)} = \frac{\delta N_{A,0} - P_0 \delta D_{A,0}}{D_0}, \tag{1.145}$$

and hence

$$\boxed{a_{P,0} = \frac{a_{N,0} - P_0 a_{D,0}}{D_0}, \qquad b_{P,0} = \frac{b_{N,0} - P_0 b_{D,0}}{D_0}}. \tag{1.146}$$

These formulas will be used later when weak-axisymmetric anisotropy is transported through the grouped P_2 bundle.

Theorem 1.8 (Full grouped P_2 bundle and isotropic normalization test)**.** *Inside the full reduced wall/BdG/projected-Maxwell grouped-bundle closure, the coupled conservative moments and outgoing-transfer moments are given by* (1.124)–(1.130)*, with projection-first EM slippage transported by* (1.131)–(1.135)*. On the isotropic branch the leading outgoing quadrupole normalization reduces to the single scalar condition* (1.139)*. First-order anisotropy around that branch is transported by* (1.144) *and* (1.146)*.*

Proof. The full grouped bundle is the direct sum of the lane-wise reduced mechanisms from Stage 003 and Stages 004–021. The BdG support moments, conservative Maxwell/mixed moments, and outgoing-transfer moments add to the wall operator as shown in (1.130); projection-first corrections enter the same Z_n, N_n slots by (1.131). The normalized response and prefactor expansions are algebraic inversions of $D_A^{\rm cons}$ and $D_A^{\rm cons2}$, giving (1.137). Equality of the three grouped lanes kills the projector components P_a and P_b, leaving the scalar ratio N_0/D_0. Linearizing the definitions of $u_2^{(A)}$ and $P_0^{(A)}$ gives the anisotropy transport laws. □

1.10 Part I audit summary and handoff to Part II

Part I establishes the following stable outputs.

1. **Exact parent imports.** The gauged GNLS equation, localized Maxwell equation, current conservation, projection language, charge firewall, and mixed-sector gauge invariants are fixed by the declared parent theory.
2. **Moving geometry closure.** The working moving-throat field is $\Sigma = r - R(\Omega, w, t)$, with $a(t)$ and $L(t)$ recovered as collective moments.
3. **Wall modal structure.** The distributed wall action separates the scalar lane, residual geometry lane, grouped real P_2 lane, and higher multipoles.
4. **Breathing reduction.** The old (a, L) closure is the lowest axisymmetric truncation of the distributed wall field.
5. **Matter support Schur complement.** Stable BdG modes generate conservative low-frequency support moments and pole shifts.
6. **Projected Maxwell/mixed outgoing bridge.** Projection-first Maxwell supplies the open-system mixed-channel provenance; the matched one-port normal form supplies a nonnegative outgoing-transfer factor and carries the compact outgoing $l = 2$ $i\omega^5$ fingerprint.
7. **Grouped bundle.** The final Part I object is the grouped coefficient packet (D_A, N_A, P_A) and the isotropic normalization gate (1.139).

This chapter does not prove that the actual nonlinear moving-throat PDE realizes the target value in (1.139). It also does not prove uniqueness of the shape-field closure, solve the full BdG spectrum, solve the full Maxwell/mixed spectrum, or complete the passive/outgoing boundary-value problem. Those remain branch-realization tasks.

- ☐ The parent GNLS, Maxwell, continuity, and mixed-sector equations are imported with corrected charge ontology.
- ☐ The sign in $\delta V_{\rm conf} = -V'_{\rm wall}\eta/\ell_c$ is preserved.
- ☐ The monopole normalization $q_{00} = 2\sqrt{\pi}\,\delta a$ is preserved.
- ☐ The old (a, L) closure is recovered as a two-mode axisymmetric truncation.
- ☐ The geometry-only grouped real P_2 sector is degenerate on an isotropic reference throat.
- ☐ Stable BdG modes are integrated out by Schur complement, not by a fitted response coefficient.

- ☐ The projection-first localized Maxwell block retains $A_w, F_{\mu w}, J^w$ until a controlled matched brane reduction is declared.
- ☐ The outgoing $l = 2$ odd coefficient depends on the static prefactor $P_0 = N_0/D_0$.
- ☐ The grouped metric $G_{\mathrm{grp}} = \mathrm{diag}(1, 2, 2)$ and projectors $P_{\bar{x}}, P_a, P_b$ are used consistently.
- ☐ The final Part I theorem gate is the branch-realization test (1.139), not a free normalization choice.

Part II uses the operator-to-response bridge (1.110) and the grouped projector calculus to build the conservative response and overlap-isotropy tests. Part III uses the same full-bundle data to discuss support completion and continuum placement. Part IV uses the outgoing bridge and scalar compatibility mechanism when the retarded closure, outlet models, core/mouth compensation, and boundary-layer laws are introduced. Part V uses the anisotropy transport formulas (1.144) and (1.146) as the starting point for the weak-axisymmetric quotient and orbit-lock structure.

The citation-safe summary is:

> Stages 001–023 define the moving-throat linearized response backbone. They project the parent Maxwell sector before applying the matched one-port reduction, and reduce the parent wall/matter/gauge system to a grouped real P_2 response bundle with exact conservative moments, exact outgoing-transfer moments, exact grouped projectors, and the isotropic normalization test
>
> $$\widehat{m}_0^2 \frac{N_0}{K - B_0 - Z_0} = \frac{54Gc_s^5}{5a^5c^5}.$$
>
> They do not by themselves prove that the actual nonlinear branch realizes that test.

Chapter 2

Conservative Response Kernels and Grouped P_2 Normalization Language

Stage range: 024–036

Part II converts the grouped packet from Part I into explicit finite-throat placement data. Part I reduced the isotropic outgoing-normalization gate to the ratio

$$\widehat{m}_0^2 \frac{N_0}{K - B_0 - Z_0} = \frac{54Gc_s^5}{5a^5c^5}. \tag{2.1}$$

This chapter asks what that ratio means once the grouped real P_2 bundle is no longer treated as a formal coefficient packet. The job here is to tie the invariant target to actual angular overlaps, finite-throat axial modes, selected wall eigenmodes, and an explicit source-map normalization.

The endpoint is a reduced admissibility geometry rather than a realized nonlinear solution. On the isotropic branch, angular matching is exact and the source-map angular factor is one. On the first weak-axisymmetric perturbation, the grouped defects lie on the diagnostic line $b = 3a$. On the first selected finite-throat branch, the outgoing-normalization problem becomes a scalar softening-depth problem, and then a dimensionless placement test governed by

$$F(\xi, \delta) = R_{\mathrm{target}}, \qquad M_{\mathrm{mix}} \leq G(\xi, \delta). \tag{2.2}$$

This is still not a proof that the actual nonlinear moving-throat branch realizes the target. It is the exact reduced geometry that the actual branch must land on.

Part II at a glance

Primary question	Which concrete overlaps, eigenmodes, and source-map choices turn the Part I target into a branch-placement test?
Starting inputs	The Part I grouped D, N, P packet, the isotropic outgoing-normalization target, and the first finite-throat selected-branch closure.
Delivers	Explicit angular overlaps, finite-throat axial constants, the selected wall eigenmode, the natural source-map rule, and the dimensionless admissibility pair (F, G).
Used by	Part III for support completion and continuum placement, Part IV for retarded and mouth/core compensation formulas, and Part V for the weak-axisymmetric signature and outgoing-prefactor slope language.

Result anchors **MTDC-T5**.

Stage packets.

Stages	Packet	Status	Carry-forward output
024–027	Angular and finite-throat data	Exact Within Closure / Reduced / Controlled Reduction	Real-STF overlaps, isotropy collapse, weak-axisymmetric signature $b = 3a$, explicit D/N constants, and the first profile-dressed normalization gates.
028–031	Selected spectral placement	Exact Within Closure / Reduced / Controlled Reduction	Loaded wall eigenproblem, Schur-complement loading law, selected prefactor $P_{0,-}$, and stable-side unique-crossing control.
032–034	Source-map and microscopic normalization	Exact Within Closure / Open	Natural D/N source map, coupling-level normalization equation, stability gate, and the scalar softening-depth normal form $x = A - \lambda_-$.
035–036	Dimensionless admissibility frontier	Exact Within Closure / Open	Monotone shape function $F(\xi, \delta)$, support frontier $G(\xi, \delta)$, and the final two-condition selected-branch test.

Claim status. Mixed. The angular algebra, finite-throat mode integrals, Schur complements, eigenvalue formulas, monotonicity statements, and dimensionless branch functions are Exact Within Closure **once the declared reduced closure is adopted. The use of stable normal modes, first finite-throat axial profiles, local bilinear kernels, and the selected D/N source map is** Reduced / Controlled Reduction**. Actual realization of the required physical point** $(R_{\text{target}}, M_{\text{mix}})$ **by the nonlinear moving-throat branch remains** Open**.**

What this part proves. Part II turns the grouped conservative packet into explicit finite-throat overlap data, selected-mode spectral placement, and the two-function admissibility geometry (F, G) for the outgoing quadrupole test.

What this part does not prove. It does not show that the actual branch lands on the required point of that geometry. It only derives the reduced placement problem and the exact branch tests attached to it.

Why later parts need it. Part III uses the (F, G) geometry to build support completion, Part IV reuses the selected prefactor in the retarded and mouth/core chains, and Part V inherits the weak-axisymmetric signature and slope language introduced here.

2.1 Block contract and theorem-level output

Part II has a different verification contract from Part I. Part I showed that the coupled wall, support, and outgoing system compresses into grouped conservative coefficients and an outgoing transfer ratio.

Part II must show that those coefficients are not arbitrary labels. It must tie them to actual finite-throat overlaps, selected axial profiles, and stable-branch spectral placement.

The stage chain is

$$\begin{aligned}\text{grouped bundle} &\longrightarrow \text{angular isotropy} \longrightarrow \text{finite-throat axial overlaps} \\ &\longrightarrow \text{selected wall eigenmode} \longrightarrow \text{dimensionless admissibility frontier.}\end{aligned} \tag{2.3}$$

The point of the chain is to convert the branch test (2.1) into a concrete placement question.

Definition 2.1 (Universal quadrupole target used in Part II)**.** *Throughout this chapter, write*

$$\boxed{N_Q^{\text{target}} := \frac{54Gc_s^5}{5a^5c^5}}. \tag{2.4}$$

Thus the isotropic Part I condition is $\widehat{m}_0^2 P_0 = N_Q^{\text{target}}$*, and the selected-mode condition later becomes* $\widehat{m}_-^2 P_{0,-} = N_Q^{\text{target}}$*.*

Theorem 2.2 (Part II conservative-response placement theorem)**.** *Inside the declared finite-throat selected-branch closure, the following chain of exact reduced statements holds.*

1. *On an $O(3)$-invariant reference branch, the real-STF angular source map is the identity and the grouped lanes satisfy*

$$D_{20,n} = D_{21,n} = D_{22,n}, \qquad N_{20,n} = N_{21,n} = N_{22,n}. \tag{2.5}$$

2. *The first weak axisymmetric quadrupole deformation has grouped signature*

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1, \qquad b = 3a. \tag{2.6}$$

3. *The first explicit finite-throat selected branch is governed by the exact D/N constants*

$$\kappa_0^2 = \frac{8}{\pi^2}, \qquad \kappa_1^2 = \frac{16}{9\pi^2}, \qquad \eta = \frac{\kappa_1^2}{\kappa_0^2} = \frac{2}{9}. \tag{2.7}$$

4. *On the natural D/N source branch, the abstract selected source map is fixed by*

$$\widehat{m}_-^2 = \frac{s_-}{\kappa_0^2}. \tag{2.8}$$

5. *The invariant selected normalization product reduces to*

$$N_-(x) = N_Q^{\text{target}}, \tag{2.9}$$

where $x = A - \lambda_-$ is the selected-mode softening depth.

6. *In dimensionless variables $\xi = x/A$ and $\delta = \Delta K_{\text{ax}}/A$, the stable branch is governed by the unique normalization locus*

$$F(\xi, \delta) = R_{\text{target}} \tag{2.10}$$

and the support-feasibility inequality

$$M_{\text{mix}} \leq G(\xi, \delta). \tag{2.11}$$

Consequently, by the end of Stage 036, the reduced theorem gap is no longer an unspecified outgoing coefficient. It is the concrete question of whether the actual moving-throat PDE places the physical defect in the admissible region of the universal (F, G) branch geometry.

Proof. Stages 024–025 establish the angular source map, isotropy collapse, and minimal one-lane normalization ratio. Stages 026–027 evaluate the first finite-throat axial overlaps and profile family. Stages 028–029 derive the selected wall profile and dynamic loading from a rank-one Schur complement. Stages 030–033 translate the selected operator into normalized-response language, prove stable-side monotonicity, fix the natural source map, and write the microscopic coupling-level equation. Stages 034–036 remove the eigenvector variables in favor of the softening depth and then non-dimensionalize the branch into the functions F and G. Each implication is exact inside the declared reduced closure. The actual placement of the physical branch remains an open realization question. □

2.2 Explicit overlap-integral normalization

2.2.1 Local goal

Stage 024 starts from the Part I grouped-bundle ratio (2.1). In Part I, the quantities $B_{A,n}$, $Z_{A,n}$, and $N_{A,n}$ were exact reduced coefficients, but the angular geometry behind them had not yet been audited. Stage 024 supplies that audit. It proves that the canonical angular source map is already correct on the isotropic branch and that the first weak-axisymmetric splitting pattern is diagnostic rather than arbitrary.

The grouped label A in this section runs over the real quadrupole ports. When the five real STF modes are paired into the project ledger's grouped convention, the three grouped lanes are $20, 21, 22$, where 21 and 22 represent cosine/sine pairs.

2.2.2 Real-STF angular basis and source-map identity

Let n^i be the unit direction on the throat sphere and let E_A^{ij} be an orthonormal real STF basis,

$$\mathrm{Tr}(E_A E_B) = \delta_{AB}. \tag{2.12}$$

Define

$$Y_A(n) = \sqrt{\frac{15}{8\pi}}\, E_A^{ij} n_i n_j. \tag{2.13}$$

Using

$$\int_{S^2} n_i n_j n_k n_l \, d\Omega = \frac{4\pi}{15} \left(\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right), \tag{2.14}$$

one obtains

$$\boxed{\int_{S^2} Y_A Y_B \, d\Omega = \delta_{AB}}. \tag{2.15}$$

If the orbital/worldtube STF quadrupole source is expanded as

$$S(n) = \sum_A S_A Y_A(n), \tag{2.16}$$

then projection onto the same port gives

$$\boxed{S_A^{\mathrm{port}} = \int_{S^2} Y_A(n) S(n) \, d\Omega = S_A}. \tag{2.17}$$

Thus the angular source-map factor is exactly one on the canonical isotropic branch:

$$\widehat{m}_0 = \widehat{m}_{\mathrm{rad}}\widehat{m}_{\mathrm{ang}}, \qquad \boxed{\widehat{m}_{\mathrm{ang}} = 1}. \tag{2.18}$$

The remaining source-map issue is radial/axial, not angular.

2.2.3 Isotropic overlap factorization

On an $O(3)$-invariant branch, take the grouped wall deformation, BdG support mode, brane-like gauge coordinate, and mixed coordinate in the separated form

$$\eta_A(s,\Omega,t) = q_A(t)\chi_\eta(s)Y_A(\Omega), \tag{2.19}$$

$$X_{\alpha,A}(s,\Omega,t) = X_{\alpha,A}(t)\phi_\alpha(s)Y_A(\Omega), \tag{2.20}$$

$$U_{r,A}(s,\Omega,t) = U_{r,A}(t)u_r(s)Y_A(\Omega), \qquad W_{r,A}(s,\Omega,t) = W_{r,A}(t)w_r(s)Y_A(\Omega). \tag{2.21}$$

The angular integral collapses by (2.15). The reduced coupling constants are therefore lane-independent:

$$\boxed{\begin{aligned}
C_\alpha &= \lambda_{B,\alpha}I_{\eta\alpha}, & I_{\eta\alpha} &= \int ds\,\mu_s(s)\chi_\eta(s)\phi_\alpha(s),\\
G_{U,r} &= \lambda_{U,r}I_{\eta u,r}, & I_{\eta u,r} &= \int ds\,\mu_s(s)\chi_\eta(s)u_r(s),\\
G_{W,r} &= \lambda_{W,r}I_{\eta w,r}, & I_{\eta w,r} &= \int ds\,\mu_s(s)\chi_\eta(s)w_r(s),\\
R_r &= \lambda_{R,r}I_{uw,r}, & I_{uw,r} &= \int ds\,\mu_s(s)u_r(s)w_r(s).
\end{aligned}} \tag{2.22}$$

The BdG support moments reduce to

$$\boxed{B_0 = \sum_\alpha \frac{C_\alpha^2}{\varpi_\alpha^2}, \qquad B_2 = \sum_\alpha \frac{C_\alpha^2}{\varpi_\alpha^4}, \qquad B_4 = \sum_\alpha \frac{C_\alpha^2}{\varpi_\alpha^6}}. \tag{2.23}$$

For each Maxwell/mixed port define

$$\Delta_r = \Omega_{U,r}^2\Omega_{W,r}^2 - R_r^2, \qquad S_r = \Omega_{U,r}^2 + \Omega_{W,r}^2, \tag{2.24}$$

$$Q_r = G_{U,r}^2\Omega_{W,r}^2 + 2G_{U,r}G_{W,r}R_r + G_{W,r}^2\Omega_{U,r}^2, \qquad H_r = G_{U,r}^2 + G_{W,r}^2, \tag{2.25}$$

$$\mathcal{P}_r = \Omega_{U,r}^2 G_{W,r} + R_r G_{U,r}. \tag{2.26}$$

Then

$$\boxed{\begin{aligned}
Z_0 &= \sum_r \frac{Q_r}{\Delta_r},\\
Z_2 &= \sum_r \frac{Q_rS_r - H_r\Delta_r}{\Delta_r^2},\\
Z_4 &= \sum_r \frac{Q_r(S_r^2-\Delta_r) - S_rH_r\Delta_r}{\Delta_r^3},\\
N_0 &= \sum_r \frac{\mathcal{P}_r^2}{\Delta_r^2},\\
N_2 &= \sum_r \frac{2\mathcal{P}_r(\mathcal{P}_rS_r - \Delta_rG_{W,r})}{\Delta_r^3},\\
N_4 &= \sum_r \frac{\Delta_r^2G_{W,r}^2 - 2\Delta_r\mathcal{P}_r^2 - 4\Delta_r\mathcal{P}_rS_rG_{W,r} + 3\mathcal{P}_r^2S_r^2}{\Delta_r^4}.
\end{aligned}} \tag{2.27}$$

The conservative operator is

$$\boxed{\begin{aligned} D(\omega) &= D_0 + D_2\omega^2 + D_4\omega^4 + O(\omega^6), \\ D_0 &= K - B_0 - Z_0, \\ D_2 &= -(M + B_2 + Z_2), \\ D_4 &= -(B_4 + Z_4). \end{aligned}} \tag{2.28}$$

Thus every truly $O(3)$-invariant reduced kernel forces

$$\boxed{\begin{aligned} D_{20,n} = D_{21,n} = D_{22,n} = D_n, \\ N_{20,n} = N_{21,n} = N_{22,n} = N_n. \end{aligned}} \tag{2.29}$$

All grouped anisotropy defects vanish on that branch.

2.2.4 First weak-axisymmetric splitting law

The leading symmetry-breaking perturbation that talks directly to the grouped real P_2 bundle is an axisymmetric quadrupole background. Write it schematically as

$$\delta K = \varepsilon\, \kappa(s) Y_{20}(\Omega). \tag{2.30}$$

The angular matrix is

$$M_{AB}^{(20)} = \int_{S^2} Y_A Y_{20} Y_B \, d\Omega. \tag{2.31}$$

The exact sixth-moment identity gives, in the five real STF lanes,

$$\boxed{M^{(20)} = \frac{\sqrt{5}}{7\sqrt{\pi}} \operatorname{diag}\!\big(1, \tfrac{1}{2}, \tfrac{1}{2}, -1, -1\big)}. \tag{2.32}$$

After regrouping cosine/sine pairs, the signature is

$$\boxed{\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1}. \tag{2.33}$$

Therefore any first-order grouped coefficient has the form

$$x_{20} = x^{(0)} + \varepsilon x^{(1)}, \qquad x_{21} = x^{(0)} + \frac{\varepsilon}{2} x^{(1)}, \qquad x_{22} = x^{(0)} - \varepsilon x^{(1)}. \tag{2.34}$$

With the grouped trace/anomaly coordinates of Part I, this implies

$$\boxed{\bar{x} = x^{(0)}, \qquad a_x = \frac{\varepsilon}{4} x^{(1)}, \qquad b_x = \frac{3\varepsilon}{4} x^{(1)}, \qquad b_x = 3a_x}. \tag{2.35}$$

For the outgoing prefactor, if

$$D_A = D_0 + \varepsilon \lambda_A D_1 + O(\varepsilon^2), \qquad N_A = N_0 + \varepsilon \lambda_A N_1 + O(\varepsilon^2), \tag{2.36}$$

then

$$P_A = \frac{N_A}{D_A} = P_0 + \varepsilon \lambda_A P_1 + O(\varepsilon^2), \qquad \boxed{P_1 = \frac{N_1 D_0 - N_0 D_1}{D_0^2}}. \tag{2.37}$$

The same diagnostic line follows:

$$a_P = \frac{\varepsilon}{4} P_1, \qquad b_P = \frac{3\varepsilon}{4} P_1. \tag{2.38}$$

Theorem 2.3 (Angular source identity and weak-axisymmetric signature)**.** *On the canonical real-STF angular basis, the isotropic source-map angular factor is exactly one. Any $O(3)$-invariant reduced kernel collapses the grouped 20/21/22 bundle to one common lane. The first pure axisymmetric quadrupole splitting of that bundle has signature $(1, 1/2, -1)$ and therefore satisfies $b = 3a$.*

Proof. The source-map statement follows from orthonormality of the real-STF harmonics. The isotropy collapse follows because all angular overlaps are diagonal and lane-independent under an $O(3)$-invariant kernel. The weak-axisymmetric signature follows from the cubic angular integral $\int Y_A Y_{20} Y_B \, d\Omega$, whose diagonal entries are proportional to $(1, 1/2, 1/2, -1, -1)$. Regrouping the cosine/sine pairs gives (2.33), and the weighted grouped inverse map gives (2.35). □

Downstream use. The identity $\widehat{m}_{\mathrm{ang}} = 1$ is used in every later selected-branch normalization equation. The line $b = 3a$ becomes the diagnostic weak-axisymmetric signature used later in the quotient/orbit and same-charge prefactor analyses.

2.3 Minimal isotropic single-mode closure

2.3.1 One-mode radial/axial model

Stage 025 strips the isotropic branch to the smallest radial/axial module that can carry the normalization product: one stable BdG support mode, one brane-like internal gauge coordinate, and one mixed $A_w/F_{\mu w}/J^w$ coordinate. Define

$$C = \lambda_B I_{\eta\phi}, \qquad G_U = \lambda_U I_{\eta u}, \qquad G_W = \lambda_W I_{\eta w}, \qquad R = \lambda_R I_{uw}. \tag{2.39}$$

Let the corresponding internal frequencies be ϖ, Ω_U, and Ω_W. The minimal Maxwell/mixed invariants are

$$\boxed{\begin{aligned} &\Delta = \Omega_U^2 \Omega_W^2 - R^2, \\ &Q = G_U^2 \Omega_W^2 + 2 G_U G_W R + G_W^2 \Omega_U^2, \\ &\mathcal{P} = \Omega_U^2 G_W + R G_U. \end{aligned}} \tag{2.40}$$

The static BdG softening is

$$X = \frac{C^2}{\varpi^2}. \tag{2.41}$$

Thus

$$D_0 = K - X - \frac{Q}{\Delta}. \tag{2.42}$$

The outgoing-transfer numerator is

$$N_0 = \frac{\mathcal{P}^2}{\Delta^2}. \tag{2.43}$$

2.3.2 Minimal explicit normalization law

The static prefactor is

$$P_0 = \frac{N_0}{D_0} = \frac{\mathcal{P}^2}{\Delta^2 \left(K - C^2/\varpi^2 - Q/\Delta\right)} = \frac{\mathcal{P}^2}{\Delta \left(K\Delta - \Delta C^2/\varpi^2 - Q\right)}. \tag{2.44}$$

Since Stage 024 fixed $\widehat{m}_{\rm ang} = 1$, the branch test is purely radial/axial:

$$\boxed{\widehat{m}_{\rm rad}^2 \frac{\mathcal{P}^2}{\Delta\,(K\Delta - \Delta C^2/\varpi^2 - Q)} = N_Q^{\rm target}}. \tag{2.45}$$

The stability conditions are

$$\boxed{\Delta > 0, \qquad D_0 = K - \frac{C^2}{\varpi^2} - \frac{Q}{\Delta} > 0}. \tag{2.46}$$

The first inequality keeps the internal (U, W) block positive; the second keeps the dressed wall lane stable.

The monotonicity in the bare stiffness and support softening is immediate:

$$\frac{\partial P_0}{\partial K} = -\frac{N_0}{D_0^2} < 0, \qquad \frac{\partial P_0}{\partial X} = +\frac{N_0}{D_0^2} > 0. \tag{2.47}$$

Thus increasing bare wall stiffness suppresses the outgoing prefactor, while increasing conservative support softening enhances it, as long as stability is maintained.

Proposition 2.4 (Minimal isotropic normalization formula)**.** *In the one-BdG-mode, one-Maxwell/mixed-pair isotropic closure, the universal outgoing quadrupole condition is exactly* (2.45)*. The only branch data still needed are C, G_U, G_W, R, ϖ, Ω_U, Ω_W, K, and $\widehat{m}_{\rm rad}$.*

Downstream use. Stage 026 evaluates the overlap factors in (2.39) on a concrete finite-throat D/N branch. Stage 029 later shows that the same mixed-sector numerator $\mathcal{P}$ controls the selected-mode dynamic loading and odd outgoing coefficient.

2.4 Concrete finite-throat axial modes

2.4.1 Finite interval and axial basis

Stage 026 chooses a concrete finite-throat interval

$$s \in [0, L] \tag{2.48}$$

with flat axial measure. The wall and brane-like zero-mode profile is the normalized Neumann/Neumann constant mode

$$u_0(s) = \frac{1}{\sqrt{L}}, \qquad -u_0'' = 0, \qquad u_0'(0) = u_0'(L) = 0. \tag{2.49}$$

The trapped support and mixed channel use the exact D/N ladder

$$f_n(s) = \sqrt{\frac{2}{L}} \sin\!\left(\frac{(n+1/2)\pi s}{L}\right), \qquad f_n(0) = 0, \qquad f_n'(L) = 0, \tag{2.50}$$

with

$$k_n = \frac{(n+1/2)\pi}{L}. \tag{2.51}$$

On the minimal branch, keep only $n = 0$:

$$f_0(s) = \sqrt{\frac{2}{L}} \sin\!\left(\frac{\pi s}{2L}\right). \tag{2.52}$$

The exact overlap of the N/N zero mode with the D/N ladder is

$$\kappa_n = \int_0^L u_0(s) f_n(s)\, ds = \frac{\sqrt{2}}{(n+1/2)\pi}. \tag{2.53}$$

Therefore

$$\boxed{\kappa := \kappa_0 = \frac{2\sqrt{2}}{\pi}}. \tag{2.54}$$

The lowest trapped half-wave is automatically the dominant axial branch because $|\kappa_n| \sim (n+1/2)^{-1}$.

2.4.2 Branch-level coefficients

Take

$$\chi_\eta = u_0, \qquad \phi = f_0, \qquad u = u_0, \qquad w = f_0. \tag{2.55}$$

Then

$$I_{\eta\phi} = \kappa, \qquad I_{\eta u} = 1, \qquad I_{\eta w} = \kappa, \qquad I_{uw} = \kappa. \tag{2.56}$$

The reduced couplings become

$$\boxed{C = \kappa\lambda_B, \qquad G_U = \lambda_U, \qquad G_W = \kappa\lambda_W, \qquad R = \kappa\lambda_R}. \tag{2.57}$$

Consequently,

$$\boxed{\begin{aligned} &\Delta = \Omega_U^2\Omega_W^2 - \kappa^2\lambda_R^2, \\ &Q = \lambda_U^2\Omega_W^2 + 2\kappa^2\lambda_U\lambda_W\lambda_R + \kappa^2\lambda_W^2\Omega_U^2, \\ &\mathcal{P} = \kappa(\Omega_U^2\lambda_W + \lambda_R\lambda_U). \end{aligned}} \tag{2.58}$$

The BdG softening is

$$X = \frac{\kappa^2\lambda_B^2}{\varpi^2}. \tag{2.59}$$

2.4.3 First finite-throat normalization test

Substitution into (2.45) gives

$$\boxed{\widehat{m}_{\mathrm{rad}}^{\,2} \frac{\kappa^2(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Delta\left(K\Delta - \Delta\kappa^2\lambda_B^2/\varpi^2 - Q\right)} = N_Q^{\mathrm{target}}}. \tag{2.60}$$

Equivalently, the required wall stiffness is

$$\boxed{K_{\mathrm{req}} = \frac{\kappa^2\lambda_B^2}{\varpi^2} + \frac{Q}{\Delta} + \frac{\widehat{m}_{\mathrm{rad}}^{\,2}\kappa^2(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{N_Q^{\mathrm{target}}\,\Delta^2}}. \tag{2.61}$$

For the constant wall profile, the quadrupole wall stiffness is

$$K = K_\eta + 6T_\Omega. \tag{2.62}$$

Thus the first concrete finite-throat branch asks whether

$$K_\eta + 6T_\Omega = K_{\mathrm{req}}. \tag{2.63}$$

Proposition 2.5 (First finite-throat normalization equation)**.** *On the constant N/N wall/brane branch coupled to the lowest D/N support/mixed half-wave, all axial overlap data collapse to $\kappa = 2\sqrt{2}/\pi$. The outgoing-normalization test is exactly* (2.60)*, equivalently the wall-stiffness condition* (2.61)*.*

Downstream use. Stage 027 tests whether this equation was an artifact of the constant wall profile. Stage 032 later reuses the same D/N mode integrals to eliminate the abstract selected source-map factor.

2.5 Nonconstant axial profile families

2.5.1 First N/N deformation family

Stage 027 turns on the first nonconstant N/N axial mode

$$u_1(s) = \sqrt{\frac{2}{L}}\cos\left(\frac{\pi s}{L}\right), \qquad -u_1'' = \frac{\pi^2}{L^2}u_1. \tag{2.64}$$

The coherent one-parameter profile family is

$$\chi_\theta(s) = \cos\theta\, u_0(s) + \sin\theta\, u_1(s), \tag{2.65}$$

and it is used for both the wall quadrupole axial profile and the brane-like internal coordinate:

$$\chi_\eta = \chi_\theta, \qquad u = \chi_\theta, \qquad \phi = w = f_0. \tag{2.66}$$

The direct wall/brane overlap remains normalized:

$$\int_0^L \chi_\theta^2\, ds = 1. \tag{2.67}$$

The D/N overlap is

$$\boxed{\kappa(\theta) = \int_0^L \chi_\theta(s) f_0(s)\, ds = \kappa_0\cos\theta + \kappa_1\sin\theta}, \tag{2.68}$$

where

$$\kappa_0 = \frac{2\sqrt{2}}{\pi}, \qquad \kappa_1 = -\frac{4}{3\pi}. \tag{2.69}$$

Equivalently,

$$\kappa(\theta) = \frac{2(-2\sin\theta + 3\sqrt{2}\cos\theta)}{3\pi}. \tag{2.70}$$

The maximum possible magnitude in this first family is

$$\boxed{|\kappa|_{\max} = \sqrt{\kappa_0^2 + \kappa_1^2} = \frac{2\sqrt{22}}{3\pi}}. \tag{2.71}$$

This exceeds the constant-branch value $2\sqrt{2}/\pi$.

2.5.2 Blind-angle no-go and max-coupling angle

The blind angle is defined by $\kappa(\theta) = 0$. From (2.68),

$$\boxed{\tan\theta_{\text{blind}} = \frac{3\sqrt{2}}{2}}. \tag{2.72}$$

At that angle, $C = G_W = R = 0$, hence $\mathcal{P} = 0$ and $N_0 = 0$. Since the right-hand side N_Q^{target} is positive, the blind-angle branch cannot realize the outgoing quadrupole normalization target.

The positive-overlap maximum occurs at

$$\boxed{\tan\theta_{\max} = \frac{\kappa_1}{\kappa_0} = -\frac{\sqrt{2}}{3}, \qquad \sin^2\theta_{\max} = \frac{2}{11}}. \tag{2.73}$$

This is a small negative admixture of the first N/N excitation.

2.5.3 Profile-dressed wall stiffness and branch condition

The quadrupole wall operator is

$$G_\eta = -T_w \frac{d^2}{ds^2} + K_\eta + 6T_\Omega. \tag{2.74}$$

On the coherent profile family,

$$\boxed{K_{\text{geo}}(\theta) = K_\eta + 6T_\Omega + \frac{\pi^2 T_w}{L^2}\sin^2\theta}. \tag{2.75}$$

The full profile-dressed normalization condition is the Stage 026 test with

$$\kappa \longrightarrow \kappa(\theta), \qquad K \longrightarrow K_{\text{geo}}(\theta). \tag{2.76}$$

Equivalently,

$$\boxed{K_{\text{geo}}(\theta) = K_{\text{req}}(\theta)}, \tag{2.77}$$

where

$$K_{\text{req}}(\theta) = \frac{\lambda_B^2 \kappa(\theta)^2}{\varpi^2} + \frac{Q(\theta)}{\Delta(\theta)} + \frac{\widehat{m}_{\text{rad}}^2 \kappa(\theta)^2 (\Omega_U^2 \lambda_W + \lambda_R \lambda_U)^2}{N_Q^{\text{target}}\,\Delta(\theta)^2}, \tag{2.78}$$

with

$$\Delta(\theta) = \Omega_U^2 \Omega_W^2 - \lambda_R^2 \kappa(\theta)^2, \tag{2.79}$$

$$Q(\theta) = \lambda_U^2 \Omega_W^2 + 2\lambda_U \lambda_W \lambda_R \kappa(\theta)^2 + \lambda_W^2 \Omega_U^2 \kappa(\theta)^2. \tag{2.80}$$

Theorem 2.6 (First profile-selection gate)**.** *The Stage 026 constant profile is the $\theta = 0$ member of an exact finite-throat profile family. The family has an exact blind-angle no-go at* (2.72)*, an exact max-coupling point at* (2.73)*, and the profile-selected branch must satisfy the dressed stiffness equation* (2.77)*.*

Downstream use. Stage 028 stops treating θ as a chosen parameter and derives it as an eigenvector of the loaded wall operator.

2.6 Loaded-profile selection and dynamic loading

2.6.1 Rank-one loaded wall eigenproblem

Work in the first two N/N wall modes, with profile vector

$$q = (q_0, q_1)^T, \qquad q_0^2 + q_1^2 = 1, \qquad q_0 = \cos\theta, \qquad q_1 = \sin\theta. \tag{2.81}$$

The bare quadrupole wall matrix is

$$K_{\text{bare}} = \begin{pmatrix} K_0 & 0 \\ 0 & K_1 \end{pmatrix}, \qquad K_0 = K_\eta + 6T_\Omega, \qquad K_1 = K_0 + \Delta K_{\text{ax}}, \tag{2.82}$$

where

$$\Delta K_{\text{ax}} = \frac{\pi^2 T_w}{L^2}. \tag{2.83}$$

The D/N overlap vector is

$$\boxed{v = (\kappa_0, \kappa_1)^T, \qquad \kappa_0 = \frac{2\sqrt{2}}{\pi}, \qquad \kappa_1 = -\frac{4}{3\pi}}. \tag{2.84}$$

The minimal attractive loading from the support/mixed branch gives

$$\boxed{K_{\text{eff}} = K_{\text{bare}} - \alpha v v^T, \qquad \alpha > 0}. \tag{2.85}$$

The exact eigenvalues are

$$\lambda_\pm = \frac{1}{2}\left[K_0 + K_1 - \alpha(\kappa_0^2 + \kappa_1^2) \pm \sqrt{\left(\Delta K_{\text{ax}} + \alpha(\kappa_0^2 - \kappa_1^2)\right)^2 + 4\alpha^2\kappa_0^2\kappa_1^2}\right]. \tag{2.86}$$

The selected lower profile angle obeys

$$\boxed{\tan(2\theta_-) = \frac{2\alpha\kappa_0\kappa_1}{\Delta K_{\text{ax}} + \alpha(\kappa_0^2 - \kappa_1^2)}}. \tag{2.87}$$

Because $\kappa_0\kappa_1 < 0$ and $\kappa_0^2 - \kappa_1^2 > 0$, positive loading drives $\theta_- < 0$. The selected profile moves toward the max-coupling branch and away from the positive blind-angle branch.

The softening threshold is

$$\boxed{\alpha_{\text{crit}} = \frac{K_0 K_1}{K_1\kappa_0^2 + K_0\kappa_1^2}}. \tag{2.88}$$

2.6.2 Dynamic origin of the loading parameter

Stage 029 replaces the phenomenological α with the Schur complement of the coupled response block. Let

$$D_\eta^{\text{bare}}(\omega) = \operatorname{diag}(K_0 - M_0\omega^2, K_1 - M_1\omega^2), \tag{2.89}$$

$$A_\phi(\omega) = \varpi^2 - \omega^2, \qquad A_U(\omega) = \Omega_U^2 - \omega^2, \qquad A_W(\omega) = \Omega_W^2 - \omega^2 - \Pi_{\text{out}}(\omega). \tag{2.90}$$

With one D/N BdG support mode, one N/N brane-like internal doublet, and one D/N mixed coordinate, the exact wall self-energy is

$$\boxed{\Sigma_{\text{wall}}(\omega) = \Xi(\omega) I_2 + \alpha(\omega) v v^T}, \tag{2.91}$$

where

$$\boxed{\Xi(\omega) = \frac{\lambda_U^2}{A_U(\omega)}, \qquad \alpha(\omega) = \frac{\lambda_B^2}{A_\phi(\omega)} + \frac{\left(A_U(\omega)\lambda_W + \lambda_R\lambda_U\right)^2}{A_U(\omega)\Delta_{UW}(\omega)}}, \tag{2.92}$$

with

$$\Delta_{UW}(\omega) = A_U(\omega)A_W(\omega) - \lambda_R^2\sigma, \qquad \sigma = v \cdot v = \kappa_0^2 + \kappa_1^2 = \frac{88}{9\pi^2}. \tag{2.93}$$

On the conservative static branch,

$$\boxed{\Xi_0 = \frac{\lambda_U^2}{\Omega_U^2}, \qquad \Delta_0 = \Omega_U^2\Omega_W^2 - \lambda_R^2\sigma, \qquad \alpha_0 = \frac{\lambda_B^2}{\varpi^2} + \frac{(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Omega_U^2\Delta_0}}. \tag{2.94}$$

The static wall matrix is

$$K_{\text{eff}}^{(0)} = \begin{pmatrix} K_0 - \Xi_0 & 0 \\ 0 & K_1 - \Xi_0 \end{pmatrix} - \alpha_0 v v^T. \tag{2.95}$$

Thus the Stage 028 angle law survives with $K_i \mapsto K_i - \Xi_0$ and $\alpha \mapsto \alpha_0$.

2.6.3 Outgoing dressing and selected odd coefficient

To first order in the passive/outgoing port,

$$\alpha(\omega) = \alpha_{\text{cons}}(\omega) + \beta(\omega)\Pi_{\text{out}}(\omega) + O(\Pi_{\text{out}}^2), \tag{2.96}$$

with

$$\boxed{\beta(\omega) = \frac{\left(A_U(\omega)\lambda_W + \lambda_R\lambda_U\right)^2}{\Delta_{\text{cons}}(\omega)^2}, \qquad \Delta_{\text{cons}}(\omega) = A_U(\omega)(\Omega_W^2 - \omega^2) - \lambda_R^2\sigma}. \tag{2.97}$$

At zero frequency,

$$\boxed{\beta_0 = \frac{(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Delta_0^2} \geq 0}. \tag{2.98}$$

For the compact outgoing $l = 2$ branch,

$$\Pi_{\text{out}}(\omega) = +i\frac{a^5}{27c_s^5}\omega^5 + O(\omega^7). \tag{2.99}$$

Projecting the odd rank-one operator onto the conservative selected eigenvector e_- gives

$$\boxed{\delta D_-^{\text{odd}}(\omega) = -i\frac{a^5}{27c_s^5}\frac{(\Omega_U^2\lambda_W + \lambda_R\lambda_U)^2}{\Delta_0^2}\,(v\cdot e_-)^2\,\omega^5 + O(\omega^7)}. \tag{2.100}$$

This is the first selected-mode version of the outgoing quadrupole bridge.

Theorem 2.7 (Loaded profile and dynamic Schur-complement theorem)**.** *The first reduced wall/BdG/Maxwell/mixed operator produces the rank-one loaded profile-selection problem exactly. Its self-energy splits as* $\Sigma_{\text{wall}} = \Xi I_2 + \alpha vv^T$*. The same directional loading coefficient inherits the passive/outgoing* $i\omega^5$ *branch, and the selected odd coefficient is* (2.100)*.*

Downstream use. Stages 030–033 translate (2.100) into normalized-response language and then into a microscopic coupling-level target.

2.7 Selected-mode normalized response

2.7.1 Selected eigenvalue and overlap

Use the compact notation

$$A = K_0 - \Xi_0, \qquad B = K_1 - \Xi_0 = A + \Delta K_{\text{ax}}. \tag{2.101}$$

Define

$$\delta_\kappa = \kappa_0^2 - \kappa_1^2, \qquad \Pi_\kappa = \kappa_0^2\kappa_1^2, \qquad \sigma = \kappa_0^2 + \kappa_1^2. \tag{2.102}$$

The selected lower eigenvalue is

$$\boxed{\lambda_- = \frac{A + B - \alpha_0\sigma - \sqrt{(\Delta K_{\text{ax}} + \alpha_0\delta_\kappa)^2 + 4\alpha_0^2\Pi_\kappa}}{2}}. \tag{2.103}$$

Let

$$R_\lambda = \sqrt{(\Delta K_{\text{ax}} + \alpha_0\delta_\kappa)^2 + 4\alpha_0^2\Pi_\kappa}. \tag{2.104}$$

The selected overlap is

$$\boxed{s_- = (v\cdot e_-)^2 = \frac{1}{2}\left[\sigma + \frac{(\Delta K_{\mathrm{ax}} + \alpha_0\delta_\kappa)\delta_\kappa + 4\alpha_0\Pi_\kappa}{R_\lambda}\right]}. \tag{2.105}$$

Equivalently, by Hellmann–Feynman,

$$\boxed{s_- = -\frac{d\lambda_-}{d\alpha_0}}. \tag{2.106}$$

2.7.2 Normalized selected response and selected prefactor

Write the selected operator as

$$D_-(\omega) = D_{-0} + D_{-2}\omega^2 + D_{-4}\omega^4 - iC_{5,-}\omega^5 + O(\omega^6), \qquad D_{-0} = \lambda_-. \tag{2.107}$$

The normalized selected response is

$$Y_-(\omega) = \frac{D_{-0}}{D_-(\omega)} = 1 + u_{2,-}\omega^2 + u_{4,-}\omega^4 + i\Gamma_{5,-}\omega^5 + O(\omega^6), \tag{2.108}$$

where

$$u_{2,-} = -\frac{D_{-2}}{D_{-0}}, \qquad u_{4,-} = \frac{D_{-2}^2 - D_{-0}D_{-4}}{D_{-0}^2}, \qquad \Gamma_{5,-} = \frac{C_{5,-}}{D_{-0}}. \tag{2.109}$$

Using the Stage 029 odd coefficient,

$$\Gamma_{5,-} = \frac{a^5}{27c_s^5}\frac{\beta_0 s_-}{\lambda_-}. \tag{2.110}$$

Therefore

$$\boxed{\Gamma_{5,-} = \frac{a^5}{27c_s^5}P_{0,-}, \qquad P_{0,-} = \frac{\beta_0 s_-}{\lambda_-} = -\beta_0\frac{d\ln\lambda_-}{d\alpha_0}}. \tag{2.111}$$

The selected-branch quadrupole target is

$$\boxed{\widehat{m}_-^2 P_{0,-} = N_Q^{\mathrm{target}}}. \tag{2.112}$$

2.7.3 Stable-side monotonicity and unique crossing

Stage 031 proves that the selected prefactor is strictly monotone on the stable branch. First,

$$\boxed{\frac{ds_-}{d\alpha_0} = \frac{2(\Delta K_{\mathrm{ax}})^2\Pi_\kappa}{R_\lambda^3} > 0}. \tag{2.113}$$

Using (2.106),

$$\boxed{\frac{dP_{0,-}}{d\alpha_0} = \beta_0\frac{(ds_-/d\alpha_0)\lambda_- + s_-^2}{\lambda_-^2} > 0} \tag{2.114}$$

whenever $\lambda_- > 0$ and $\beta_0 > 0$.

At zero loading,

$$P_{0,-}(0) = \frac{\beta_0\kappa_0^2}{K_0 - \Xi_0}. \tag{2.115}$$

The refined softening threshold is

$$\boxed{\alpha_{\rm crit} = \frac{AB}{B\kappa_0^2 + A\kappa_1^2}}. \tag{2.116}$$

As $\alpha_0 \to \alpha_{\rm crit}^-$, $\lambda_- \to 0$, while s_- remains finite and positive, so $P_{0,-} \to +\infty$. Therefore the stable branch spans

$$P_{0,-}(0) \le P_{0,-} < +\infty. \tag{2.117}$$

Theorem 2.8 (Unique stable-side selected crossing)**.** *For every target $P_{\rm target} > P_{0,-}(0)$, there exists a unique $\alpha_* \in (0, \alpha_{\rm crit})$ such that $P_{0,-}(\alpha_*) = P_{\rm target}$. Applied to the quadrupole target, $P_{\rm target} = N_Q^{\rm target}/\widehat{m}_-^2$.*

Downstream use. The monotonicity theorem justifies treating the selected branch as a one-dimensional spectral-placement problem rather than as a many-parameter matching ambiguity.

2.8 Source-map and microscopic normalization equation

2.8.1 Natural D/N source branch

Stage 032 attaches the orbital/worldtube STF quadrupole source through the same D/N axial load used by the mixed/support module:

$$L_{\rm src} = g_Q Q_{\rm STF} \int_0^L \eta(s) f_0(s)\, ds. \tag{2.118}$$

With $\eta = q_0 u_0 + q_1 u_1$, the source vector is proportional to v. Projection onto the selected eigenvector gives

$$J_- = g_Q Q_{\rm STF}(v \cdot e_-). \tag{2.119}$$

At zero loading, $J_-(0) = g_Q Q_{\rm STF}\kappa_0$. Hence

$$\boxed{\widehat{m}_- = \frac{v \cdot e_-}{\kappa_0}, \qquad \widehat{m}_-^2 = \frac{s_-}{\kappa_0^2}}. \tag{2.120}$$

Since s_- grows from κ_0^2 to $\sigma = \kappa_0^2 + \kappa_1^2$,

$$\boxed{1 \le \widehat{m}_-^2 < \frac{\sigma}{\kappa_0^2} = \frac{11}{9}}. \tag{2.121}$$

Thus the selected source map is real and modest; it does not hide the normalization burden in an arbitrary large factor.

Combining (2.112) and (2.120) gives the invariant selected product

$$\boxed{N_-(\alpha_0) := \widehat{m}_-^2 P_{0,-} = \frac{\beta_0 s_-^2}{\kappa_0^2 \lambda_-}}. \tag{2.122}$$

The target is

$$\boxed{N_-(\alpha_0) = N_Q^{\rm target}}. \tag{2.123}$$

2.8.2 Microscopic coupling variables and stability gate

For the first explicit finite-throat kernel model, Stage 033 writes

$$\boxed{A = K_0 - \frac{g_U^2}{\Omega_U^2}, \qquad \Delta_0 = \Omega_U^2\Omega_W^2 - g_R^2\sigma, \qquad \mathcal{X} = \Omega_U^2 g_W + g_R g_U}. \tag{2.124}$$

Then

$$\boxed{\beta_0 = \frac{\mathcal{X}^2}{\Delta_0^2}, \qquad \alpha_0 = \frac{g_B^2}{\varpi^2} + \frac{\mathcal{X}^2}{\Omega_U^2\Delta_0}}. \tag{2.125}$$

The exact stability gate is

$$\boxed{\Delta_0 > 0, \qquad A > 0, \qquad \alpha_0 < \alpha_{\mathrm{crit}}}, \tag{2.126}$$

where, using $\kappa_0^2 = 8/\pi^2$ and $\kappa_1^2 = 16/(9\pi^2)$,

$$\boxed{\alpha_{\mathrm{crit}} = \frac{9\pi^2 A(A + \Delta K_{\mathrm{ax}})}{8(11A + 9\Delta K_{\mathrm{ax}})}}. \tag{2.127}$$

In coupling form,

$$\frac{g_B^2}{\varpi^2} + \frac{\mathcal{X}^2}{\Omega_U^2\Delta_0} < \frac{9\pi^2 A(A + \Delta K_{\mathrm{ax}})}{8(11A + 9\Delta K_{\mathrm{ax}})}. \tag{2.128}$$

2.8.3 Onset criterion and weak-loading expansion

Because $N_-(\alpha_0)$ is strictly increasing on the stable branch, the zero-loading value is a necessary onset test. At $\alpha_0 = 0$,

$$s_-(0) = \kappa_0^2, \qquad \lambda_-(0) = A, \tag{2.129}$$

so

$$\boxed{N_-(0) = \frac{\beta_0\kappa_0^2}{A} = \frac{\mathcal{X}^2\kappa_0^2}{A\Delta_0^2}}. \tag{2.130}$$

A necessary condition for the natural positive-loading branch to hit the universal target is

$$\boxed{N_-(0) \le N_Q^{\mathrm{target}}, \qquad \text{equivalently} \qquad \mathcal{X}^2 \le \frac{N_Q^{\mathrm{target}} A\Delta_0^2}{\kappa_0^2}}. \tag{2.131}$$

If this fails, the branch starts above the target and cannot return to it because N_- only increases.

For weak loading,

$$N_-(\alpha_0) = \frac{\beta_0\kappa_0^2}{A} + \alpha_0\frac{\beta_0\kappa_0^2(4A\kappa_1^2 + \Delta K_{\mathrm{ax}}\kappa_0^2)}{A^2\Delta K_{\mathrm{ax}}} + O(\alpha_0^2). \tag{2.132}$$

With the exact D/N constants this becomes

$$N_-(\alpha_0) = \frac{8\beta_0}{\pi^2 A} + \alpha_0\frac{64\beta_0(8A + 9\Delta K_{\mathrm{ax}})}{9\pi^4 A^2\Delta K_{\mathrm{ax}}} + O(\alpha_0^2). \tag{2.133}$$

Theorem 2.9 (Microscopic selected-branch test)**.** *In the first explicit finite-throat kernel closure, the selected quadrupole normalization condition is the exact coupling-level equation*

$$\frac{\mathcal{X}^2}{\Delta_0^2}\frac{s_-(\alpha_0)^2}{\kappa_0^2\lambda_-(\alpha_0)} = N_Q^{\mathrm{target}}, \qquad \alpha_0 = \frac{g_B^2}{\varpi^2} + \frac{\mathcal{X}^2}{\Omega_U^2\Delta_0}, \tag{2.134}$$

subject to (2.126)*. The onset condition* (2.131) *is necessary for a hit on the natural positive-loading branch.*

Downstream use. Stages 034–036 remove s_-, λ_-, and the eigenvector algebra from (2.134) by passing to the softening depth and then to dimensionless functions F and G.

2.9 Softening-depth normal form

2.9.1 Softening-depth variable

Stage 034 trades the selected eigenvalue for the softening depth

$$\boxed{x := A - \lambda_-}. \tag{2.135}$$

On the stable selected branch,

$$\lambda_- = A - x, \qquad 0 \le x < A. \tag{2.136}$$

The rank-one secular equation is

$$1 = \alpha_0 \left(\frac{\kappa_0^2}{x} + \frac{\kappa_1^2}{x + \Delta K_{\rm ax}} \right). \tag{2.137}$$

Solving for total directional loading gives

$$\boxed{\alpha_0(x) = \frac{x(x + \Delta K_{\rm ax})}{\kappa_0^2(x + \Delta K_{\rm ax}) + \kappa_1^2 x}}. \tag{2.138}$$

The selected overlap follows from $s_- = dx/d\alpha_0$:

$$\boxed{s_-(x) = \frac{\left[\kappa_0^2(x + \Delta K_{\rm ax}) + \kappa_1^2 x\right]^2}{\kappa_0^2(x + \Delta K_{\rm ax})^2 + \kappa_1^2 x^2}}. \tag{2.139}$$

The invariant selected product becomes

$$\boxed{N_-(x) = \frac{\beta_0 \left[\kappa_0^2(x + \Delta K_{\rm ax}) + \kappa_1^2 x\right]^4}{\kappa_0^2(A - x)\left[\kappa_0^2(x + \Delta K_{\rm ax})^2 + \kappa_1^2 x^2\right]^2}}. \tag{2.140}$$

The selected quadrupole theorem is now the scalar equation

$$\boxed{N_-(x) = N_Q^{\rm target}, \qquad 0 \le x < A}. \tag{2.141}$$

2.9.2 Monotone loading and required support contribution

Differentiating (2.138) gives

$$\boxed{\frac{d\alpha_0}{dx} = \frac{\kappa_0^2(x + \Delta K_{\rm ax})^2 + \kappa_1^2 x^2}{\left[\kappa_0^2(x + \Delta K_{\rm ax}) + \kappa_1^2 x\right]^2} > 0}. \tag{2.142}$$

Thus x is a one-to-one branch coordinate for the stable loading.

The loading split is

$$\alpha_0 = \frac{g_B^2}{\varpi^2} + \alpha_{\rm mix}, \qquad \alpha_{\rm mix} = \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0}. \tag{2.143}$$

Therefore the required support loading at a target softening depth is

$$\boxed{\frac{g_{B,\rm req}^2}{\varpi^2} = \frac{x(x + \Delta K_{\rm ax})}{\kappa_0^2(x + \Delta K_{\rm ax}) + \kappa_1^2 x} - \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0}}. \tag{2.144}$$

The nonnegativity of this expression is the first support-feasibility test.

Theorem 2.10 (Softening-depth normal form)**.** *The first rank-one selected finite-throat branch is exactly parameterized by the scalar softening depth $x = A - \lambda_-$. The loading $\alpha_0(x)$, selected overlap $s_-(x)$, invariant normalization product $N_-(x)$, and required support contribution are given by* (2.138)*–*(2.144)*.*

Downstream use. The softening-depth form is the input to the dimensionless (F, G) functions in Stages 035–036 and to the support-completion program in Part III.

2.10 Support-feasibility frontier

2.10.1 Dimensionless D/N normalization function

Stage 035 substitutes the exact D/N constants

$$\boxed{\kappa_0^2 = \frac{8}{\pi^2}, \qquad \kappa_1^2 = \frac{16}{9\pi^2}, \qquad \eta = \frac{\kappa_1^2}{\kappa_0^2} = \frac{2}{9}}. \tag{2.145}$$

Introduce

$$\boxed{\xi = \frac{x}{A}, \qquad \delta = \frac{\Delta K_{\mathrm{ax}}}{A}, \qquad 0 \le \xi < 1}. \tag{2.146}$$

Then

$$N_-(x) = N_-(0)F(\xi, \delta), \qquad N_-(0) = \frac{\beta_0 \kappa_0^2}{A}. \tag{2.147}$$

The exact dimensionless normalization shape function is

$$\boxed{F(\xi, \delta) = \frac{(9\delta + 11\xi)^4}{81(1-\xi)(9\delta^2 + 18\delta\xi + 11\xi^2)^2}}. \tag{2.148}$$

The dimensionless target is

$$\boxed{R_{\mathrm{target}} := \frac{N_Q^{\mathrm{target}}}{N_-(0)} = \frac{N_Q^{\mathrm{target}} A}{\beta_0 \kappa_0^2}}. \tag{2.149}$$

The normalization locus is therefore

$$\boxed{F(\xi, \delta) = R_{\mathrm{target}}}. \tag{2.150}$$

The exact derivative is

$$\frac{\partial F}{\partial \xi} = \frac{(9\delta + 11\xi)^3\,(81\delta^3 + 72\delta^2 + 206\delta^2\xi + 297\delta\xi^2 + 138\xi^3)}{81(1-\xi)^2(9\delta^2 + 18\delta\xi + 11\xi^2)^3} > 0. \tag{2.151}$$

Also,

$$F(0, \delta) = 1, \qquad \lim_{\xi \to 1^-} F(\xi, \delta) = +\infty. \tag{2.152}$$

Thus:

$$\boxed{R_{\mathrm{target}} < 1 \Rightarrow \text{no stable hit}, \qquad R_{\mathrm{target}} = 1 \Rightarrow \xi_{\mathrm{req}} = 0, \qquad R_{\mathrm{target}} > 1 \Rightarrow \text{unique } \xi_{\mathrm{req}} \in (0,1)}. \tag{2.153}$$

The exact required total loading is

$$\boxed{\alpha_{\mathrm{req}}(\xi, \delta) = \frac{9\pi^2 A\xi(\xi + \delta)}{8(9\delta + 11\xi)}}. \tag{2.154}$$

2.10.2 Support-feasibility function

Stage 036 writes the support-feasibility test in the same dimensionless variable. Define

$$\boxed{M_{\rm mix} := \frac{8\alpha_{\rm mix}}{\pi^2 A} = \frac{8\mathcal{X}^2}{\pi^2 A\Omega_U^2\Delta_0}}. \tag{2.155}$$

Define the support frontier

$$\boxed{G(\xi,\delta) := \frac{8\alpha_{\rm req}}{\pi^2 A} = \frac{9\xi(\xi+\delta)}{9\delta+11\xi}}. \tag{2.156}$$

Then

$$\boxed{\frac{g_{B,\rm req}^2}{\varpi^2} = \frac{\pi^2 A}{8}\left[G(\xi_{\rm req},\delta) - M_{\rm mix}\right]}. \tag{2.157}$$

Therefore the exact support-feasibility condition is

$$\boxed{M_{\rm mix} \le G(\xi_{\rm req},\delta)}. \tag{2.158}$$

The function G is also strictly monotone:

$$\frac{\partial G}{\partial \xi} = \frac{9(9\delta^2+18\delta\xi+11\xi^2)}{(9\delta+11\xi)^2} > 0. \tag{2.159}$$

Its endpoint values are

$$G(0,\delta) = 0, \qquad G_{\max}(\delta) = \lim_{\xi\to 1^-} G(\xi,\delta) = \frac{9(1+\delta)}{9\delta+11}. \tag{2.160}$$

For fixed δ, the stable selected quadrupole branch traces the parametric frontier

$$\boxed{\xi \in [0,1) \longmapsto (F(\xi,\delta), G(\xi,\delta))}. \tag{2.161}$$

The branch is admissible exactly when the normalization target chooses a unique $\xi_{\rm req}$ and the physical mixed baseline lies below the support frontier there.

Near onset, if $R_{\rm target} = 1+\varepsilon_R$ and $\xi \ll 1$, then

$$F(\xi,\delta) = 1 + \left(1+\frac{8}{9\delta}\right)\xi + O(\xi^2), \qquad \xi_{\rm req} \simeq \frac{\varepsilon_R}{1+8/(9\delta)}, \tag{2.162}$$

while

$$G(\xi,\delta) = \xi - \frac{2\xi^2}{9\delta} + O(\xi^3). \tag{2.163}$$

Near softening, if $R_{\rm target}$ is large,

$$1-\xi_{\rm req} \simeq \frac{C(\delta)}{R_{\rm target}}, \qquad C(\delta) = \frac{(9\delta+11)^4}{81(9\delta^2+18\delta+11)^2}. \tag{2.164}$$

Theorem 2.11 (Dimensionless selected-branch admissibility frontier)**.** *For fixed $\delta > 0$, the first selected D/N quadrupole branch is governed by the monotone functions F and G in* (2.148) *and* (2.156)*. A physical point is admissible in this reduced closure if and only if*

$$\boxed{R_{\rm target} \ge 1, \qquad F(\xi_{\rm req},\delta) = R_{\rm target}, \qquad M_{\rm mix} \le G(\xi_{\rm req},\delta)}. \tag{2.165}$$

The first condition enters the unique normalization locus; the second fixes the stable softening depth; the third ensures the remaining required support loading is nonnegative.

2.11 Part II audit summary and handoff to Part III

Part II establishes the following stable outputs.

1. **Angular source map.** The real-STF angular matching matrix is the identity, so the canonical isotropic source-map factor satisfies $\widehat{m}_{\mathrm{ang}} = 1$.
2. **Isotropic grouped collapse.** An $O(3)$-invariant reduced kernel forces the grouped 20/21/22 lanes to share common D_n and N_n data.
3. **Weak-axisymmetric diagnostic.** The first pure axisymmetric quadrupole splitting has signature $(1, 1/2, -1)$, equivalently $b = 3a$.
4. **Minimal isotropic ratio.** The one-mode radial/axial module gives the explicit branch test (2.45).
5. **Concrete D/N overlaps.** The first finite-throat branch collapses to the exact overlap $\kappa = 2\sqrt{2}/\pi$.
6. **Profile family.** The first nonconstant N/N family introduces $\kappa(\theta)$, the blind-angle no-go, and the max-coupling angle.
7. **Loaded profile selection.** The selected profile is the lower eigenvector of a rank-one loaded wall matrix; positive loading drives it toward the max-coupling direction, not the blind-angle direction.
8. **Dynamic loading.** The first coupled wall/BdG/Maxwell/mixed Schur complement splits exactly as $\Xi I_2 + \alpha v v^T$, and the same loading carries the outgoing $i\omega^5$ branch.
9. **Selected normalized response.** The selected static prefactor is $P_{0,-} = \beta_0 s_-/\lambda_-$, and it is strictly monotone on the stable branch.
10. **Natural source map.** On the D/N source branch, $\widehat{m}_-^2 = s_-/\kappa_0^2$, so the selected target becomes $N_- = \beta_0 s_-^2/(\kappa_0^2 \lambda_-) = N_Q^{\mathrm{target}}$.
11. **Softening-depth normal form.** The entire first selected branch can be parameterized by $x = A - \lambda_-$.
12. **Dimensionless branch geometry.** The final reduced admissibility problem is $F(\xi, \delta) = R_{\mathrm{target}}$ plus $M_{\mathrm{mix}} \le G(\xi, \delta)$.

The chapter deliberately does not assert that the actual branch satisfies (2.165). It supplies the reduced test that the actual branch must pass. In particular, Part II does not yet compute the continuum-kernel values of A, ΔK_{ax}, β_0, M_{mix}, or the later rank-two support-completion data. Those are the job of Part III.

- ☐ The angular source-map identity $\widehat{m}_{\mathrm{ang}} = 1$ is separated from the radial/axial source-map factors.
- ☐ The grouped labels $20, 21, 22$ are used as real P_2 lane labels, not spacetime indices.
- ☐ The weak-axisymmetric signature $(1, 1/2, -1)$ and the line $b = 3a$ are preserved.
- ☐ The minimal isotropic branch obeys $\Delta > 0$ and $D_0 > 0$ before the normalization target is asked.
- ☐ The finite-throat D/N overlap value $\kappa = 2\sqrt{2}/\pi$ is used consistently.
- ☐ The blind-angle branch has $N_0 = 0$ and is therefore an exact no-go for a positive quadrupole target.

- ☐ Positive rank-one loading selects a negative profile angle and moves toward the max-coupling direction.
- ☐ The Schur complement produces $\Sigma_{\text{wall}} = \Xi I_2 + \alpha v v^T$, not a generic two-by-two load.
- ☐ The selected prefactor is $P_{0,-} = \beta_0 s_- / \lambda_-$, and the source-map-corrected product is $N_- = \beta_0 s_-^2 / (\kappa_0^2 \lambda_-)$.
- ☐ The softening-depth variable satisfies $0 \le x < A$, and the support contribution is nonnegative only when $M_{\text{mix}} \le G(\xi_{\text{req}}, \delta)$.

Part III begins where this chapter stops. It computes and organizes the continuum placement data that feed

$$R_{\text{target}} = \frac{N_Q^{\text{target}} A}{\beta_0 \kappa_0^2}, \qquad M_{\text{mix}} = \frac{8\mathcal{X}^2}{\pi^2 A \Omega_U^2 \Delta_0}, \qquad \delta = \frac{\Delta K_{\text{ax}}}{A}, \tag{2.166}$$

and then asks whether rank-two support completion, coherent D/N support, and explicit branch placement can satisfy the frontier (2.165).

The citation-safe summary is:

> Stages 024–036 derive the conservative grouped-response and selected-branch normalization language. They prove exact angular isotropy, identify the weak-axisymmetric $b = 3a$ signature, compute the first finite-throat D/N overlap structure, derive the selected wall-mode Schur complement, eliminate the abstract source map on the natural D/N branch, and reduce the selected quadrupole branch to the dimensionless admissibility test
>
> $$F(\xi_{\text{req}}, \delta) = R_{\text{target}}, \qquad M_{\text{mix}} \le G(\xi_{\text{req}}, \delta).$$
>
> They do not yet prove that the actual moving-throat branch lands in that admissible region.

Chapter 3

Support Completion, Continuum Placement, and Family–1 Branch Construction

Stage range: 037–090

Part III turns the Part II admissibility geometry into explicit support/source branch data. In that language the stable selected branch is fixed by a softening depth ξ, a bare anisotropy ratio δ, a target normalization ratio R_{target}, and a mixed loading baseline M_{mix}. The reduced admissibility test was

$$F(\xi_{\text{req}}, \delta) = R_{\text{target}}, \qquad M_{\text{mix}} \le G(\xi_{\text{req}}, \delta). \tag{3.1}$$

This chapter asks where the quantities in (3.1) come from in the explicit moving-throat branch, how the support/source sector can complete them, and whether the first concrete Family–1 wall branch survives the support side of the quadrupole test.

The endpoint is not a full branch realization. It is a reduced support/source verdict. The chapter first turns the selected branch into an exact continuum placement and support-completion problem, then converts the remaining demand into lowest-lane transport and parent-overlap thresholds, and finally evaluates the first explicit Family–1 wall-shell branch against those thresholds.

Part III at a glance

Primary question	How are the Part II placement targets completed by actual support and source data, and does the first Family–1 branch pass that reduced test?
Starting inputs	The Part II admissibility pair (F, G), the selected finite-throat branch data, and the first explicit Family–1 wall-shell closure.
Delivers	Continuum placement formulas, support-completion theorems, lowest-lane transport and parent-overlap thresholds, and the first explicit Family–1 reduced verdict.
Used by	Part IV as the point where support/source uncertainty stops being the dominant reduced obstruction, and later realization chapters as the source of Family–1 reference data and residual language.
Result anchors	**MTDC-T6** and **MTDC-T7**.

Stage packets.

Stages	Packet	Status	Carry-forward output
037–051	Continuum placement and support completion	EXACT WITHIN CLOSURE / REDUCED / CONTROLLED REDUCTION	Continuum kernel extraction, dimensionless placement laws, rank-2 support completion, coherent D/N tracking rules, and the lowest-twin sufficiency criterion.
052–069	Lowest-lane transport and parent thresholds	EXACT WITHIN CLOSURE / REDUCED / CONTROLLED REDUCTION	Non-twin asymmetry regimes, overlap and Robin reachability windows, drift-diffusion transport laws, parent-overlap gain thresholds, and the three-zone reduced support/source verdict.
070–078	Explicit Family–1 wall-shell baseline	EXACT WITHIN CLOSURE / REDUCED / CONTROLLED REDUCTION / NUMERICALLY LOCATED	GNLS wall-shell reduction, canonical tanh-wall branch, Family–1 geometry and healing-length locks, wall-depth extraction, and the first explicit Family–1 baseline verdict.
079–090	Family–1 demand window and finish line	EXACT WITHIN CLOSURE / OPEN	Demand-to-transport map, branch-product windows, master quadrupole residual, minimal-module ratio packet, and the reduced statement that support/source closure is finished once the outgoing conservative module is realized.

Claim status. Mixed. Most algebraic reductions, Schur complements, branch maps, monotonicity statements, and support-threshold identities are EXACT WITHIN CLOSURE **once the declared finite-throat, selected-mode, and Family–1 closures are adopted. The extraction of a first explicit branch from the GNLS wall shell is a** REDUCED / CONTROLLED REDUCTION **use of the parent action. Several branch values are** NUMERICALLY LOCATED **because they are exact integrals or exact roots evaluated numerically. The final realization question—whether the actual moving-throat grouped-P_2/geometry branch realizes the minimal isotropic contact-plus-pole conservative module—remains** OPEN.

What this part proves. Part III converts the abstract admissibility geometry into explicit support-completion theorems and then evaluates the first concrete Family–1 wall branch, producing the first serious reduced success and failure windows on the support/source side.

What this part does not prove. It does not finish the retarded normalization problem and it does not prove that the full PDE realizes the selected support data. The remaining obstruction is compressed to reduced residuals and branch-placement gates.

Why later parts need it. Part IV starts from the fact that the support/source side is no longer the dominant reduced uncertainty, and later realization chapters reuse the Family–1 reference data, residual language, and support-threshold packets established here.

3.1 Block contract and theorem-level output

Part III has a different contract from Part II. Part II built the selected-branch test; Part III must populate the test with actual reduced branch data. The chain is

$$\begin{aligned}&\text{continuum kernel} \longrightarrow \text{placement map} \longrightarrow \text{support completion}\\ &\qquad\longrightarrow \text{support/source operator} \longrightarrow \text{Family–1 branch verdict.}\end{aligned} \tag{3.2}$$

The chain is not a full nonlinear PDE solution. It is a verification ledger for the reduced branch machinery that any completed solution must satisfy.

Theorem 3.1 (Part III support-completion and Family–1 branch theorem)**.** *Inside the declared finite-throat selected-branch closure, the following reduced theorem chain holds.*

1. *The first continuum kernel fixes the selected-branch placement data A, ΔK_{ax}, α_{mix}, β_0, M_{mix}, and δ in closed form.*

2. *These data compress to dimensionless continuum ratios $(\epsilon_\eta, \epsilon_W, \rho, Z_W, \delta_0, \Lambda)$, with exact product law*
$$R_{\mathrm{target}} M_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon_W)}{\pi^2}. \tag{3.3}$$

3. *Turning on the first non-flat U doublet preserves scalar placement but generically breaks source/loading collinearity; the resulting selected-branch geometry is a controlled deformation governed by R_U.*

4. *With two loading directions, the selected branch remains exactly solvable because the determinant is linear in the support loading.*

5. *The coherent D/N support kernel forces the first local support completion onto a tracking branch; support enhancement is controlled by $S(\zeta;\epsilon)$, and the lowest symmetric twin lane supplies the exact doubling branch $\zeta_0 = 1$, $S_0 = 2$.*

6. *If the symmetric twin is insufficient, the non-twin lowest-lane problem is controlled by explicit overlap boost, Robin softening, drift-diffusion transport, and parent-action gain thresholds.*

7. *The first explicit Family–1 wall-shell branch fixes*
$$\Lambda_\ell = 37, \qquad \eta = 37, \qquad \chi_s = \frac{37}{2}, \qquad \kappa = \frac{12321}{5}. \tag{3.4}$$
Its natural shell-weighted wall datum is
$$\Theta_w^{(\chi)} \simeq 4.06863235008162\,\lambda_\mu^2, \tag{3.5}$$
with conservative floor
$$\Theta_w^{(J)} \simeq 0.927552032539308\,\lambda_\mu^2. \tag{3.6}$$

8. *Once the selected outgoing quadrupole normalization is imposed, the explicit support theorem depends only on the loading ratio*
$$\rho_\alpha = \frac{\alpha_{\mathrm{req}}}{\alpha_{\mathrm{mix}}}. \tag{3.7}$$
Under the natural minimal isotropic contact-plus-pole precursor, this ratio is fixed to
$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} = \frac{1}{3}, \qquad \mathrm{Pe}_{\mathrm{req}} = 0, \tag{3.8}$$
and the explicit Family–1 support/source side succeeds with large margin.

The only remaining reduced gap after this theorem chain is not support/source sufficiency. It is whether the actual moving-throat grouped-P_2/geometry branch realizes the minimal isotropic conservative quadrupole precursor in the natural contact-plus-pole way.

Proof structure. Stages 037–038 compute the continuum Schur complement and then repackage it into dimensionless placement ratios. Stages 039–044 show how non-flat internal structure and rank-2 support loading deform, but do not destroy, the selected-branch algebra. Stages 045–051 reduce coherent D/N support to a tracking branch and an exact support-ratio threshold. Stages 052–061 identify the physical resources that can exceed the lowest-twin threshold: overlap bias, Robin compliance, drift-diffusion transport, and microscopic gain. Stages 062–069 project that gain back into the parent GNLS/confinement action and close the support/source phase diagram. Stages 070–078 evaluate the first explicit Family–1 wall-shell branch. Stages 079–087 collapse its remaining support test to one loading-ratio datum. Stages 088–090 evaluate that datum for the minimal isotropic contact-plus-pole quadrupole precursor and state the final reduced status. □

3.2 Continuum-kernel extraction and placement map

3.2.1 Local goal

The Part II admissibility test (3.1) is useful only if the quantities A, ΔK_{ax}, β_0, α_{mix}, M_{mix}, and δ can be obtained from a concrete branch. Stages 037–038 provide the first such extraction. The branch is still reduced: it uses the lowest wall doublet, the first brane-like internal U block, the lowest D/N support field, and the mixed W lane. But after that reduced operator is declared, the extraction is exact.

3.2.2 Finite-throat operator and projected constants

Work on $s \in [0, L]$. The wall field η has N/N modes, the support and mixed lanes use the lowest D/N half-wave, and the flat internal U block starts with the first two N/N modes. The exact overlaps are

$$\kappa_0 = \frac{2\sqrt{2}}{\pi}, \qquad \kappa_1 = -\frac{4}{3\pi}, \qquad \sigma = \kappa_0^2 + \kappa_1^2 = \frac{88}{9\pi^2}. \tag{3.9}$$

After mass normalization, the internal static quantities are

$$\Omega_U^2 = \frac{K_U}{\mu_U}, \qquad \Omega_W^2 = \frac{K_W^{\text{eff}}}{\mu_W}, \qquad K_W^{\text{eff}} = K_W + \frac{\pi^2 T_W}{4L^2}, \tag{3.10}$$

with couplings

$$g_U = \frac{c_{\eta U}}{\sqrt{\mu_\eta \mu_U}}, \qquad g_W = \frac{c_{\eta W}}{\sqrt{\mu_\eta \mu_W}}, \qquad g_R = \frac{c_{UW}}{\sqrt{\mu_U \mu_W}}. \tag{3.11}$$

Eliminating the internal block gives the same rank-one wall loading form used abstractly in Part II,

$$\Sigma_{\text{wall}}(\omega) = \Xi(\omega) I_2 + \alpha(\omega) v v^T, \qquad v = (\kappa_0, \kappa_1)^T. \tag{3.12}$$

On the conservative static branch define

$$\Delta_0 = \Omega_U^2 \Omega_W^2 - g_R^2 \sigma, \qquad \mathcal{X} = \Omega_U^2 g_W + g_R g_U. \tag{3.13}$$

Then the selected-branch inputs are

$$\boxed{A = K_0 - \frac{g_U^2}{\Omega_U^2}, \qquad \alpha_{\text{mix}} = \frac{\mathcal{X}^2}{\Omega_U^2 \Delta_0}, \qquad \beta_0 = \frac{\mathcal{X}^2}{\Delta_0^2}, \qquad M_{\text{mix}} = \frac{8\alpha_{\text{mix}}}{\pi^2 A}, \qquad \delta = \frac{\Delta K_{\text{ax}}}{A}}. \tag{3.14}$$

In unnormalized continuum variables, with

$$K_\eta^{\mathrm{eff}} = K_\eta + 6T_\Omega, \tag{3.15}$$

these become

$$\boxed{A = \frac{K_U K_\eta^{\mathrm{eff}} - c_{\eta U}^2}{\mu_\eta K_U}, \qquad \Delta K_{\mathrm{ax}} = \frac{\pi^2 T_w}{\mu_\eta L^2}}, \tag{3.16}$$

$$\boxed{\alpha_{\mathrm{mix}} = \frac{(K_U c_{\eta W} + c_{UW} c_{\eta U})^2}{\mu_\eta K_U \left(K_U K_W^{\mathrm{eff}} - c_{UW}^2 \sigma\right)}}, \tag{3.17}$$

$$\boxed{\beta_0 = \frac{\mu_W}{\mu_\eta} \frac{(K_U c_{\eta W} + c_{UW} c_{\eta U})^2}{\left(K_U K_W^{\mathrm{eff}} - c_{UW}^2 \sigma\right)^2}}, \tag{3.18}$$

$$\boxed{M_{\mathrm{mix}} = \frac{8(K_U c_{\eta W} + c_{UW} c_{\eta U})^2}{\pi^2 (K_U K_\eta^{\mathrm{eff}} - c_{\eta U}^2)(K_U K_W^{\mathrm{eff}} - c_{UW}^2 \sigma)}, \qquad \delta = \frac{\pi^2 T_w K_U}{L^2 (K_U K_\eta^{\mathrm{eff}} - c_{\eta U}^2)}}. \tag{3.19}$$

The stability conditions are correspondingly

$$K_U K_\eta^{\mathrm{eff}} > c_{\eta U}^2, \qquad K_U K_W^{\mathrm{eff}} > c_{UW}^2 \sigma. \tag{3.20}$$

3.2.3 Dimensionless placement ledger

Stage 038 compresses the same continuum data to dimensionless ratios

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{\mathrm{eff}}}, \qquad \epsilon_W = \frac{c_{UW}^2 \sigma}{K_U K_W^{\mathrm{eff}}}, \qquad \rho = \frac{c_{UW} c_{\eta U}}{K_U c_{\eta W}}, \tag{3.21}$$

$$Z_W = \frac{c_{\eta W}^2}{K_\eta^{\mathrm{eff}} K_W^{\mathrm{eff}}}, \qquad \delta_0 = \frac{\pi^2 T_w}{L^2 K_\eta^{\mathrm{eff}}}, \qquad \Lambda = \frac{27\pi^2 G c_s^5 K_W^{\mathrm{eff}}}{20 a^5 c^5 \mu_W}. \tag{3.22}$$

The exact placement map is

$$\boxed{\delta = \frac{\delta_0}{1-\epsilon_\eta}, \qquad M_{\mathrm{mix}} = \frac{8 Z_W (1+\rho)^2}{\pi^2 (1-\epsilon_\eta)(1-\epsilon_W)}, \qquad R_{\mathrm{target}} = \frac{\Lambda (1-\epsilon_\eta)(1-\epsilon_W)^2}{Z_W (1+\rho)^2}}. \tag{3.23}$$

Multiplication gives the product law already stated in (3.3):

$$\boxed{R_{\mathrm{target}} M_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon_W)}{\pi^2} = \frac{54 G c_s^5 K_W^{\mathrm{eff}} (1-\epsilon_W)}{5 a^5 c^5 \mu_W}}. \tag{3.24}$$

This is the first useful factorization of the placement problem. The pair (ϵ_W, Λ) sets the product scale, while $(\epsilon_\eta, Z_W, \rho)$ redistributes the physical point along that product curve.

Proposition 3.2 (Continuum placement factorization)**.** *For the Stage-20 continuum kernel, the selected branch has an exact three-lane structure:*

$$\textit{geometry lane: } \delta = \frac{\delta_0}{1-\epsilon_\eta}, \qquad \textit{product lane: } R_{\mathrm{target}} M_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon_W)}{\pi^2}, \tag{3.25}$$

with $(\epsilon_\eta, Z_W, \rho)$ redistributing the physical point between R_{target} and M_{mix} at fixed product.

3.3 Non-flat U doublet and generalized source/loading mismatch

3.3.1 Split-U deformation

The flat-U doublet made the source direction, the support direction, and the mixed loading direction coincide. Stage 039 tests that simplifying assumption by adding the first axial structure to U:

$$K_{U0} = K_U, \qquad K_{U1} = K_U\left(1 + \delta_U\right), \qquad \delta_U = \frac{\pi^2 T_U}{L^2 K_U}. \tag{3.26}$$

The scalar placement formulas survive, but with

$$\boxed{\delta_{\text{split}} = \frac{\delta_0 + \epsilon_\eta \delta_U/(1+\delta_U)}{1-\epsilon_\eta}, \qquad \epsilon_{W,\text{split}} = \epsilon_W \left[1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right]}. \tag{3.27}$$

The directional effect is measured by

$$R_U = \frac{1 + \rho_0/(1+\delta_U)}{1+\rho_0}, \tag{3.28}$$

and the exact direction-splitting invariant is

$$D_{\text{dir}} = -\frac{\kappa_0 \kappa_1 g_W \rho_0 \delta_U}{1+\delta_U}. \tag{3.29}$$

Thus collinearity survives exactly iff

$$D_{\text{dir}} = 0 \quad \Longleftrightarrow \quad \delta_U = 0 \quad \text{or} \quad \rho_0 = 0. \tag{3.30}$$

On the natural constructive branch $\rho_0 > 0$, positive δ_U lowers the mixed baseline and raises the normalization demand before support completion is even considered.

3.3.2 General selected-branch deformation

Stage 040 replaces the old one-vector branch functions by a source/loading mismatch law. Let the loading vector be $z = z_0(1,q)^T$ and the source vector be $s = s_0(1,t)^T$. The selected lower eigenvalue is still written

$$\lambda_- = A_0(1-\xi), \qquad 0 \le \xi < 1. \tag{3.31}$$

For a single rank-one loading, the exact loading is

$$\alpha_{\text{req}} = \frac{A_0 \xi(\xi+\delta)}{z_0^2\left[\delta + (1+q^2)\xi\right]}, \tag{3.32}$$

and the selected eigenvector satisfies

$$\frac{e_1}{e_0} = \frac{q\xi}{\delta+\xi}. \tag{3.33}$$

Writing

$$\eta_s = \frac{s_1 z_1}{s_0 z_0}, \tag{3.34}$$

the generalized normalization function is

$$\boxed{F_{q,\eta_s}(\xi,\delta) = \frac{\left[\delta + (1+q^2)\xi\right]^2 \left[\delta + (1+\eta_s)\xi\right]^2}{(1-\xi)\left[(\delta+\xi)^2 + q^2\xi^2\right]^2}}, \tag{3.35}$$

while the required baseline loading is

$$\boxed{G_q(\xi,\delta) = \frac{\xi(\xi+\delta)}{\delta+(1+q^2)\xi}}. \tag{3.36}$$

For the split-U continuum, with $\lambda_0 = \kappa_1^2/\kappa_0^2 = 2/9$,

$$q = -\frac{\sqrt{2}}{3}R_U, \qquad \eta_s = \frac{2}{9}R_U, \tag{3.37}$$

so the deformation collapses to

$$\boxed{F_U(\xi,\delta;R_U) = \frac{\left[9\delta+(9+2R_U^2)\xi\right]^2\left[9\delta+(9+2R_U)\xi\right]^2}{81(1-\xi)\left[9\delta^2+18\delta\xi+(9+2R_U^2)\xi^2\right]^2}}, \tag{3.38}$$

$$\boxed{G_U(\xi,\delta;R_U) = \frac{9\xi(\xi+\delta)}{9\delta+(9+2R_U^2)\xi}}. \tag{3.39}$$

Setting $R_U = 1$ recovers the flat-U functions of Part II exactly.

3.4 Rank-2 support completion and continuum-selected closure

3.4.1 Two-direction selected branch

Once source/loading collinearity is broken, the selected wall branch must allow both mixed loading and support loading. Stage 041 writes

$$\frac{K_{\text{loaded}}}{A_0} = \operatorname{diag}(1, 1+\delta) - m(1,q)^T(1,q) - n(1,r)^T(1,r), \tag{3.40}$$

where m is the mixed baseline, n is the support loading, q is the mixed direction, and r is the support direction. The determinant condition for $\lambda_- = A_0(1-\xi)$ is

$$D_{\text{sel}} = \xi(\delta+\xi) - m\left[\delta+(1+q^2)\xi\right] - n\left[\delta+(1+r^2)\xi\right] + mn(q-r)^2. \tag{3.41}$$

The structural fact is that (3.41) is linear in n. Hence

$$\boxed{n_{\text{req}}(\xi,\delta;m,q,r) = \frac{\xi(\delta+\xi) - m\left[\delta+(1+q^2)\xi\right]}{\delta+(1+r^2)\xi - m(q-r)^2}}. \tag{3.42}$$

Differentiation gives the exact monotonicity law

$$\frac{\partial n_{\text{req}}}{\partial m} = -\frac{\left[\delta+(1+qr)\xi\right]^2}{\left[\delta+(1+r^2)\xi-m(q-r)^2\right]^2} < 0 \tag{3.43}$$

on every regular branch. More mixed baseline always lowers the additional support loading required to reach the same softening depth.

Two closures are then sharply distinguished. If support tracks the mixed direction, $r = q$, then

$$n_{\text{req}} = G_q(\xi,\delta) - m, \tag{3.44}$$

so no new selected-branch geometry appears. If support stays tied to the original source direction $t = \kappa_1/\kappa_0$ while the mixed direction is $q = tR_U$, then

$$n_{\text{req}}^{\text{src}} = \frac{\xi(\delta+\xi) - m\left[\delta+(1+\lambda_0 R_U^2)\xi\right]}{\delta+(1+\lambda_0)\xi - m\lambda_0(R_U-1)^2}. \tag{3.45}$$

This is the first genuinely new rank-2 support-feasibility branch.

3.4.2 Rank-2 selected-mode normalization

Stage 042 keeps the same two loading directions and introduces the source direction $s = s_0(1,t)^T$. With $n = n_{\rm req}$, the lower selected eigenvector obeys

$$\frac{e_1}{e_0} = \frac{m(q-r) + r\xi}{\delta + \xi - mq(q-r)}. \tag{3.46}$$

The two normalized overlaps are

$$\frac{(z \cdot e_-)^2}{z_0^2} = \frac{[\delta + (1+qr)\xi]^2}{D_{q,r}(\xi,\delta;m)}, \tag{3.47}$$

$$\frac{(s \cdot e_-)^2}{s_0^2} = \frac{[\delta + (1+rt)\xi - m(q-r)(q-t)]^2}{D_{q,r}(\xi,\delta;m)}, \tag{3.48}$$

where

$$D_{q,r}(\xi,\delta;m) = [\delta + \xi - mq(q-r)]^2 + [m(q-r) + r\xi]^2\,. \tag{3.49}$$

Thus the rank-2 normalization law depends on outgoing, support, and source directions. The old one-vector formula is recovered only when $r = q$.

3.4.3 Continuum-selected quadratic

Stages 043–044 insert the continuum support direction data. The physical softening depth is pinned by

$$\boxed{\xi^2 + B_{\rm cont}\xi + C_{\rm cont} = 0}, \tag{3.50}$$

with

$$B_{\rm cont} = \delta - M_{\rm mix}(1 + t^2R_U^2) - M_{\rm supp}(1 + t^2R_\phi^2), \tag{3.51}$$

$$C_{\rm cont} = -\delta(M_{\rm mix} + M_{\rm supp}) + t^2 M_{\rm mix}M_{\rm supp}(R_U - R_\phi)^2. \tag{3.52}$$

The selected-branch normalization theorem gate is then simply

$$\boxed{R_{\rm target} = F_{\rm cont}(\xi_{\rm phys})}. \tag{3.53}$$

This is the first place where the abstract selected-mode branch becomes a concrete continuum-selected branch point rather than a free ξ.

3.5 Coherent local D/N support kernel and support compensation

3.5.1 Tracking reduction

Stage 045 asks what the first coherent local D/N kernel does to the rank-2 ambiguity. The coherent kernel forces

$$g_B g_R = g_W g_S, \tag{3.54}$$

which lands exactly on the tracking surface

$$R_\phi = R_U = R_{\rm tr}. \tag{3.55}$$

The coherent tracking factor is

$$R_{\rm tr} = \frac{1 + \chi_0/(1+\delta_U)}{1+\chi_0}, \qquad \frac{1}{1+\delta_U} < R_{\rm tr} < 1 \tag{3.56}$$

on the constructive split-U branch. Hence the rank-2 branch collapses to the one-parameter tracking laws

$$M_{\mathrm{tr}} = G_{\mathrm{tr}}(\xi, \delta; R_{\mathrm{tr}}), \qquad R_{\mathrm{target}} = F_{\mathrm{tr}}(\xi, \delta; R_{\mathrm{tr}}). \tag{3.57}$$

Stage 046 then proves the monotonicity signs

$$\frac{\partial G_{\mathrm{tr}}}{\partial R} < 0, \qquad \frac{\partial F_{\mathrm{tr}}}{\partial R} > 0. \tag{3.58}$$

Since the constructive split branch has $R_{\mathrm{tr}} < 1$, it raises the required loading and lowers the normalized response relative to the flat branch. The first coherent D/N kernel therefore does not automatically rescue the target; it makes support compensation necessary.

3.5.2 Support-enhancement factor

Stage 047 introduces a coherent support lane with ratio ζ. The total tracking load is

$$M_{\mathrm{tr}} = M_{\mathrm{mix}} S(\zeta; \epsilon), \tag{3.59}$$

where

$$\boxed{S(\zeta; \epsilon) = 1 + \frac{\zeta(1-\epsilon)}{1-\epsilon\zeta}}. \tag{3.60}$$

The target ratio is independent of ζ:

$$R_{\mathrm{target}} = \frac{\Lambda(1-\epsilon_\eta)(1-\epsilon)^2}{Z_W(1+\chi_0)^2}. \tag{3.61}$$

Thus support completion changes the available loading but not the selected normalization demand.

Let M_{req} be the load required by the tracking branch at the selected ξ. Define

$$S_{\mathrm{req}} = \frac{M_{\mathrm{req}}}{M_{\mathrm{mix}}}. \tag{3.62}$$

Solving $S(\zeta; \epsilon) = S_{\mathrm{req}}$ gives

$$\boxed{\zeta_{\mathrm{req}} = \frac{S_{\mathrm{req}} - 1}{1 + \epsilon(S_{\mathrm{req}} - 2)}}. \tag{3.63}$$

Stage 048 proves that if the mixed-only branch is below the required load, there is a unique support ratio below the softening pole that reaches the target, provided the actual physical support lane can supply $\zeta_{\mathrm{phys}} \geq \zeta_{\mathrm{req}}$.

3.5.3 D/N microscopic support ratio and lowest-twin criterion

Stage 049 derives the physical support ratio of an explicit D/N support tower:

$$\boxed{\zeta_n^{\mathrm{phys}} = \frac{K_W^{\mathrm{eff}}}{K_{\phi,n}^{\mathrm{eff}}} \left(\frac{I_n}{I_0}\right)^2}. \tag{3.64}$$

For the first uniform local source,

$$\frac{I_n}{I_0} = \frac{1}{2n+1}. \tag{3.65}$$

For the same-operator twin family,

$$\boxed{\zeta_n^{\text{twin}} = \frac{1}{(2n+1)^2\,[1+xn(n+1)]}, \qquad x = \frac{\pi^2 T_X}{L^2 K_W^{\text{eff}}}}. \tag{3.66}$$

Therefore

$$\zeta_0^{\text{twin}} = 1, \qquad S_0 = S(1;\epsilon) = 2. \tag{3.67}$$

The symmetric lowest twin succeeds iff

$$\boxed{\zeta_{\text{req}} \le 1 \quad \Longleftrightarrow \quad S_{\text{req}} \le 2}. \tag{3.68}$$

For $n \ge 1$, the exact necessary condition is

$$\zeta_{\text{req}} \le \frac{1}{(2n+1)^2}, \tag{3.69}$$

and if this is satisfied the stiffness threshold is

$$x \le x_{\max}(n;\zeta_{\text{req}}) = \frac{1}{n(n+1)}\left[\frac{1}{(2n+1)^2\zeta_{\text{req}}} - 1\right]. \tag{3.70}$$

Thus the first coherent support tower is strongly biased toward the lowest twin lane.

3.5.4 Branch-product form of the lowest-twin test

Stage 051 rewrites the same result in the product variable. Define

$$C_{\text{mix}} = \frac{8\Lambda(1-\epsilon)}{\pi^2}, \tag{3.71}$$

and let $\Pi_{\text{tr}} = F_{\text{tr}}G_{\text{tr}}$ be the tracking-branch product at the selected depth. The symmetric lowest twin is sufficient iff

$$\boxed{\Pi_{\text{tr}} \le 2C_{\text{mix}} = \frac{16\Lambda(1-\epsilon)}{\pi^2}}. \tag{3.72}$$

This criterion will be reused below when the support demand is rewritten in terms of the physical outgoing branch.

3.6 Lowest-lane asymmetry, transport, and parent gain thresholds

3.6.1 Three support regimes

Stage 052 gives the exact branch-demand variable

$$\boxed{\zeta_{\text{req}}(\Pi_{\text{tr}}, C_{\text{mix}}, \epsilon) = \frac{\Pi_{\text{tr}} - C_{\text{mix}}}{C_{\text{mix}} - \epsilon(2C_{\text{mix}} - \Pi_{\text{tr}})}}. \tag{3.73}$$

The support problem splits into three exact regimes:

$$\begin{aligned}
&\Pi_{\text{tr}} \le C_{\text{mix}} && \Longrightarrow \quad \text{mixed baseline already enough,}\\
&C_{\text{mix}} < \Pi_{\text{tr}} \le 2C_{\text{mix}} && \Longrightarrow \quad \text{symmetric lowest twin enough,}\\
&\Pi_{\text{tr}} > 2C_{\text{mix}} && \Longrightarrow \quad \text{non-twin lowest-lane asymmetry required.}
\end{aligned} \tag{3.74}$$

For a general lowest support lane,

$$\zeta_0^{\text{phys}} = \frac{K_W^{\text{eff}}}{K_{\phi,0}^{\text{eff}}}\Omega_0^2. \tag{3.75}$$

Thus non-twin rescue can come from overlap boost $\Omega_0^2 > 1$, support softening $K_{\phi,0}^{\text{eff}} < K_W^{\text{eff}}$, or both.

3.6.2 Overlap boost and Robin softening

Stage 053 studies the normalized exponential source family

$$\sigma_\alpha(s) = \frac{\alpha e^{\alpha s/L}}{e^\alpha - 1}, \qquad \int_0^L \sigma_\alpha(s)\, ds = L. \tag{3.76}$$

The exact overlap boost is

$$\boxed{\Omega_{\exp}(\alpha) = \frac{\pi\alpha\,(2\alpha e^\alpha + \pi)}{(4\alpha^2 + \pi^2)(e^\alpha - 1)}}. \tag{3.77}$$

It satisfies

$$\Omega_{\exp}(0) = 1, \qquad \lim_{\alpha\to+\infty} \Omega_{\exp}(\alpha) = \frac{\pi}{2}, \qquad \Omega_0^2 \le \frac{\pi^2}{4}. \tag{3.78}$$

Pure overlap rescue is therefore possible only if $\zeta_{\mathrm{req}} \le \pi^2/4$.

Stage 054 replaces the Dirichlet mouth with Robin compliance. Let

$$y \tan y = \eta, \qquad 0 < y < \frac{\pi}{2}, \tag{3.79}$$

and define

$$x = \frac{\pi^2 T_X}{L^2 K_W^{\mathrm{eff}}}, \qquad 0 < x < 4. \tag{3.80}$$

The exact softening factor is

$$\boxed{A_K(\eta) = \frac{K_W^{\mathrm{eff}}}{K_{\phi,0}^{\mathrm{eff}}} = \frac{1}{1 - x/4 + xy^2/\pi^2}}. \tag{3.81}$$

Its endpoint window is

$$1 \le A_K \le \frac{4}{4-x}. \tag{3.82}$$

Combining overlap and compliance yields the first explicit non-twin reachability window of Stage 055.

3.6.3 Transport origin of the overlap bias

Stage 056 derives the exponential family from a stationary drift-diffusion operator:

$$\partial_t \sigma + \partial_s J = 0, \qquad J = -D_\sigma \partial_s \sigma + v_\sigma \sigma. \tag{3.83}$$

The zero-flux stationary branch has the exponential profile with Peclet number

$$\mathrm{Pe} = \frac{v_\sigma L}{D_\sigma}. \tag{3.84}$$

It is therefore natural to write the physical overlap factor as

$$\boxed{\Omega_{\mathrm{Pe}} = \frac{\pi \mathrm{Pe}\left(2\mathrm{Pe}e^{\mathrm{Pe}} + \pi\right)}{(4\mathrm{Pe}^2 + \pi^2)(e^{\mathrm{Pe}} - 1)}, \qquad \Omega_0 = 1}. \tag{3.85}$$

Stage 057 combines this with the Robin factor into the physical lowest-lane support ratio

$$\boxed{\zeta_0(\mathrm{Pe}, \eta; \kappa) = \Omega_{\mathrm{Pe}}^2 \frac{\kappa + \pi^2/4}{\kappa + y(\eta)^2}, \qquad y(\eta)\tan y(\eta) = \eta}. \tag{3.86}$$

3.6.4 Coupled support/source fixed point

Stage 058 closes the source and support operators with a self-consistent branch equation

$$\boxed{\mathrm{Pe}_* = \Xi\Delta(\mathrm{Pe}_*;\kappa,\eta)}. \tag{3.87}$$

The endpoint brackets are

$$\Xi\Delta_0(\kappa,\eta) \le \mathrm{Pe}_* \le \Xi\Delta_\infty(\kappa,\eta), \tag{3.88}$$

where, with $\alpha = \sqrt{\kappa}$,

$$\Delta_0 = \frac{\eta(\cosh\alpha - 1)}{\alpha^2(\alpha\sinh\alpha + \eta\cosh\alpha)}, \qquad \Delta_\infty = \frac{\cosh\alpha + (\eta/\alpha)\sinh\alpha - 1}{\alpha\sinh\alpha + \eta\cosh\alpha}. \tag{3.89}$$

Since ζ_0 is monotone in Pe, Stage 059 turns this into residual bounds and exact coupling thresholds. If $\mathrm{Pe}_{\mathrm{req}}$ is the Peclet bias needed to meet ζ_{req}, then

$$\boxed{\Xi_{\mathrm{fail}} = \frac{\mathrm{Pe}_{\mathrm{req}}}{\Delta_\infty(\kappa,\eta)}, \qquad \Xi_{\mathrm{suff}} = \frac{\mathrm{Pe}_{\mathrm{req}}}{\Delta_0(\kappa,\eta)}}. \tag{3.90}$$

Below Ξ_{fail} the branch fails; above Ξ_{suff} it succeeds; the intermediate strip requires the full fixed-point root.

3.6.5 Parent-action gain and overlap thresholds

Stages 060–063 derive the gain from the parent GNLS/confinement sector. Project the linearized compressional source onto $\delta\rho(s,y) = \sigma(s)\chi_\sigma(y)$, and the support onto $\phi(s)\chi_\phi(y)$, with

$$\delta V_{\mathrm{conf}}(s,y) = -g_\phi\chi_\phi(y)\phi(s). \tag{3.91}$$

For the frozen $n = 5$ EOS,

$$h'(\rho_*) = 5K\rho_*^3 = \frac{mc_{s,*}^2}{\rho_*}. \tag{3.92}$$

Define

$$N_{\sigma\sigma} = \int d^3y\,\chi_\sigma^2, \qquad N_{\phi\phi} = \int d^3y\,\chi_\phi^2, \qquad O_{\sigma\phi} = \int d^3y\,\chi_\sigma\chi_\phi. \tag{3.93}$$

Then

$$\boxed{G_{\mathrm{micro}} = \frac{\rho_* g_\phi^2 O_{\sigma\phi}^2}{mc_{s,*}^2 K_X N_{\sigma\sigma}} = \frac{\rho_* g_\phi^2 N_{\phi\phi}}{mc_{s,*}^2 K_X} C_{\sigma\phi}^2}, \tag{3.94}$$

where

$$C_{\sigma\phi}^2 = \frac{O_{\sigma\phi}^2}{N_{\sigma\sigma}N_{\phi\phi}}, \qquad 0 \le C_{\sigma\phi}^2 \le 1. \tag{3.95}$$

The fixed-point strength is

$$\Xi_{\mathrm{micro}} = \kappa G_{\mathrm{micro}}. \tag{3.96}$$

The parent amplitude thresholds are therefore exact:

$$g_{\phi,\mathrm{fail}}^2 = \frac{mc_{s,*}^2 K_X N_{\sigma\sigma} G_{\mathrm{fail}}}{\rho_* O_{\sigma\phi}^2}, \qquad g_{\phi,\mathrm{suff}}^2 = \frac{mc_{s,*}^2 K_X N_{\sigma\sigma} G_{\mathrm{suff}}}{\rho_* O_{\sigma\phi}^2}. \tag{3.97}$$

Equivalently, if g_ϕ is fixed, the required coherence floors are

$$C_{\mathrm{fail}}^2 = \frac{mc_{s,*}^2 K_X G_{\mathrm{fail}}}{\rho_* g_\phi^2 N_{\phi\phi}}, \qquad C_{\mathrm{suff}}^2 = \frac{mc_{s,*}^2 K_X G_{\mathrm{suff}}}{\rho_* g_\phi^2 N_{\phi\phi}}. \tag{3.98}$$

The Cauchy bound gives the first parent-overlap no-go: if the best possible aligned gain is below G_{fail}, no source-profile engineering in that support channel can rescue the branch.

3.6.6 Equilibrium alignment and wall figure of merit

Stage 064 shows that the parent equilibrium response is not arbitrary:

$$\chi_\sigma(y) = \frac{g_\phi \chi_\phi(y)}{H(y)}, \qquad H(y) = h'(\rho_*(y)). \tag{3.99}$$

Thus

$$I_1 = \int d^3y \frac{\chi_\phi^2}{H}, \qquad I_2 = \int d^3y \frac{\chi_\phi^2}{H^2}, \qquad C_{\sigma\phi}^2 = \frac{I_1^2}{N_{\phi\phi} I_2}. \tag{3.100}$$

In a thin layer with nearly constant H, $C_{\sigma\phi}^2 = 1$ and

$$G_{\text{eq}} = \frac{g_\phi^2 I_1}{K_X}. \tag{3.101}$$

Stages 065–066 then use a concrete thin-wall confinement

$$V_{\text{conf}}(r; a) = V_0 f\left(\frac{r-a}{\ell}\right), \qquad g_\phi = \frac{V_0}{\ell}. \tag{3.102}$$

In the thin-wall matched branch, the support/source theorem collapses to the dimensionless wall figure of merit

$$\boxed{W_{\text{wall}} = \frac{4\pi a^2 L^2 J_1 V_0^2}{T_X \ell}}. \tag{3.103}$$

The universal reduced window is

$$\boxed{W_{\text{wall}} \le \frac{\text{Pe}_{\text{req}}}{\Delta_\infty} \Rightarrow \text{fail}, \qquad W_{\text{wall}} \ge \frac{\text{Pe}_{\text{req}}}{\Delta_0} \Rightarrow \text{succeed}}. \tag{3.104}$$

Only the intermediate band requires a full root solve.

Stages 067–068 benchmark independent source/support mismatch using $\chi_\sigma = \operatorname{sech}(y/w_f)$, $\chi_\phi = e^{-y^2/w_g^2}$. The exact self-dual stationary point is

$$\frac{w_g}{w_f} = \sqrt{\pi}, \tag{3.105}$$

with

$$C_{\text{res}}^2 \simeq 0.994418836451529, \qquad P_{\text{res}} = \frac{1}{C_{\text{res}}^2} \simeq 1.005612487760576. \tag{3.106}$$

Thus the first independent-profile benchmark changes the matched thresholds by only about 0.56%. Stage 069's reduced verdict is therefore the three-zone wall theorem: universal fail below $\text{Pe}_{\text{req}}/\Delta_\infty$, universal matched success above $\text{Pe}_{\text{req}}/\Delta_0$, and only a narrow profile-sensitive band in between.

3.7 Family–1 explicit branch construction

3.7.1 GNLS wall-shell support branch

Stage 070 derives the support coefficients from the parent GNLS shell energy. Near the wall background ρ_w, the quadratic support energy for $\delta\rho(s, y) = q(s)\chi_\phi(y)$ is

$$E^{(2)}[q] = \frac{1}{2}\int_0^L ds \left[T_X q'(s)^2 + K_X q(s)^2\right], \tag{3.107}$$

with

$$T_X = \frac{\hbar^2}{4m\rho_w} N_{\phi\phi}, \qquad K_X = H_w N_{\phi\phi} + \frac{\hbar^2}{4m\rho_w} G_{\phi\phi}, \qquad H_w = \frac{mc_{s,w}^2}{\rho_w}. \tag{3.108}$$

For a thin shell with profile moments

$$I_f = \int d\xi\, f'(\xi)^2, \qquad I_g = \int d\xi\, f''(\xi)^2, \tag{3.109}$$

this gives

$$T_X = \frac{\pi a^2 \ell I_f \hbar^2}{m\rho_w}, \qquad K_X = 4\pi a^2 \ell I_f H_w + \frac{\pi a^2 I_g \hbar^2}{m\rho_w \ell}. \tag{3.110}$$

Thus

$$\kappa = \frac{K_X L^2}{T_X} = 4\left(\frac{mc_{s,w}L}{\hbar}\right)^2 + \frac{I_g}{I_f}\left(\frac{L}{\ell}\right)^2. \tag{3.111}$$

The matched thin-wall branch also has

$$W_{\rm wall} = \Xi = \frac{4\rho_w^2 V_0^2 L^2}{\hbar^2 c_{s,w}^2 \ell^2}. \tag{3.112}$$

3.7.2 Canonical tanh wall and reference geometry

Stage 071 chooses

$$f(\xi) = \frac{1+\tanh\xi}{2}, \qquad f'(\xi) = \frac{1}{2}\,\mathrm{sech}^2\,\xi. \tag{3.113}$$

The moments are exact:

$$I_f = \frac{1}{3}, \qquad I_g = \frac{4}{15}, \qquad \frac{I_g}{I_f} = \frac{4}{5}. \tag{3.114}$$

The natural local mouth closure is

$$K_m = \frac{T_X}{\ell}, \qquad \eta = \frac{K_m L}{T_X} = \frac{L}{\ell} = \Lambda_\ell. \tag{3.115}$$

Define

$$\chi_s = \frac{mc_{s,w}L}{\hbar}, \qquad \Lambda_\ell = \frac{L}{\ell}, \qquad \Upsilon_w = \frac{4\rho_w^2 V_0^2}{\hbar^2 c_{s,w}^2}. \tag{3.116}$$

Then

$$\boxed{\kappa = 4\chi_s^2 + \frac{4}{5}\Lambda_\ell^2, \qquad \eta = \Lambda_\ell, \qquad W_{\rm wall} = \Upsilon_w \Lambda_\ell^2}. \tag{3.117}$$

Stages 073–074 fix the Family–1 reference values. The carried geometry is

$$\frac{L}{a} = \frac{37}{20}, \qquad \frac{\ell}{a} = \frac{1}{20}, \qquad \Lambda_\ell = 37, \qquad \eta = 37. \tag{3.118}$$

The healing-width closure

$$\ell = \frac{\hbar}{2mc_{s,w}} \tag{3.119}$$

gives

$$\boxed{\chi_s = \frac{37}{2}, \qquad \kappa = \frac{9}{5}\Lambda_\ell^2 = \frac{12321}{5}, \qquad \alpha = \sqrt{\kappa} = \frac{128}{\sqrt{5}}}. \tag{3.120}$$

3.7.3 Family–1 threshold window and wall-depth datum

Stage 075 evaluates the exact support/source thresholds at $(\kappa, \eta) = (12321/5, 37)$:

$$\Delta_0 \left(\frac{12321}{5}, 37 \right) \simeq 1.73302079021525 \times 10^{-4}, \tag{3.121}$$

$$\Delta_\infty \left(\frac{12321}{5}, 37 \right) \simeq 2.01447565540522 \times 10^{-2}. \tag{3.122}$$

With $V_0 = \alpha_r \mu_*$, $\alpha_r = 10$, write

$$\Upsilon_w = 117\Theta_w. \tag{3.123}$$

The explicit branch theorem is

$$\boxed{\Theta_w \leq 3.62605617972939 \times 10^{-4} \, \mathrm{Pe}_{\mathrm{req}} \Rightarrow \text{fail}}, \tag{3.124}$$

$$\boxed{\Theta_w \geq 4.21495341569977 \times 10^{-2} \, \mathrm{Pe}_{\mathrm{req}} \Rightarrow \text{succeed}}. \tag{3.125}$$

Stage 076 uses the exact $n = 5$ identity $h(\rho) = mc_s^2/4$ and the healing lock to obtain

$$\Theta_w = 25\lambda_\mu^2 \rho_w^2 \tag{3.126}$$

in normalized Family–1 variables. Stage 077 extracts ρ_w from the explicit radial wall profile with the canonical support weight. The result is

$$\langle \rho_r \rangle_\chi \simeq 0.192619005556493, \qquad \langle \rho_r^2 \rangle_\chi \simeq 0.162745294003265, \tag{3.127}$$

so

$$\boxed{\Theta_w^{(\chi)} = 25\lambda_\mu^2 \langle \rho_r^2 \rangle_\chi \simeq 4.06863235008162\lambda_\mu^2}, \tag{3.128}$$

with conservative lower envelope

$$\boxed{\Theta_w^{(J)} = 25\lambda_\mu^2 \langle \rho_r \rangle_\chi^2 \simeq 0.927552032539308\lambda_\mu^2}. \tag{3.129}$$

Stage 078 compares these to the threshold window. For the natural shell-weighted datum,

$$\mathrm{Pe}_{\mathrm{suff}}^{(\chi)} \simeq 96.5285247264386\lambda_\mu^2, \qquad \mathrm{Pe}_{\mathrm{fail}}^{(\chi)} \simeq 11220.5441626259\lambda_\mu^2. \tag{3.130}$$

For the conservative floor,

$$\mathrm{Pe}_{\mathrm{suff}}^{(J)} \simeq 22.0062226330754\lambda_\mu^2, \qquad \mathrm{Pe}_{\mathrm{fail}}^{(J)} \simeq 2558.01892349205\lambda_\mu^2. \tag{3.131}$$

The branch-level verdict is that Family–1 is not wall-depth starved except for unusually large quadrupole-side demand.

3.8 First reduced verdict and residual localization

3.8.1 Demand map and Family–1 ceiling

Stages 079–081 rewrite the remaining support/source condition in the demand variables used by the outgoing quadrupole branch. Let y_{F1} be the Robin root

$$y_{\mathrm{F1}} \tan y_{\mathrm{F1}} = 37, \qquad 0 < y_{\mathrm{F1}} < \frac{\pi}{2}, \qquad y_{\mathrm{F1}} \simeq 1.52948248371470. \tag{3.132}$$

Define

$$A_{\mathrm{F1}} = \frac{\kappa_{\mathrm{F1}} + \pi^2/4}{\kappa_{\mathrm{F1}} + y_{\mathrm{F1}}^2}, \qquad \kappa_{\mathrm{F1}} = \frac{12321}{5}. \tag{3.133}$$

Numerically,

$$A_{\mathrm{F1}} \simeq 1.00005192880220. \tag{3.134}$$

The Family–1 support-ratio map is

$$\boxed{\zeta_{\mathrm{F1}}(\mathrm{Pe}) = A_{\mathrm{F1}} \Omega_{\mathrm{Pe}}^2}. \tag{3.135}$$

The hard constructive ceiling is

$$\boxed{\zeta_{\max}^{\mathrm{F1}} = A_{\mathrm{F1}} \frac{\pi^2}{4} \simeq 2.46752922945601}. \tag{3.136}$$

Thus a demanded ζ_{req} is already met at zero bias if $\zeta_{\mathrm{req}} < A_{\mathrm{F1}}$, has a unique constructive $\mathrm{Pe}_{\mathrm{req}}$ if $A_{\mathrm{F1}} \le \zeta_{\mathrm{req}} \le \zeta_{\max}^{\mathrm{F1}}$, and fails the explicit Family–1 branch if it exceeds $\zeta_{\max}^{\mathrm{F1}}$.

At $\lambda_\mu = 1$, the natural shell-weighted demand thresholds are

$$\zeta_{\mathrm{suff}}^{(\chi)} \simeq 2.46622291347846, \qquad \zeta_{\mathrm{fail}}^{(\chi)} \simeq 2.46752913273870, \tag{3.137}$$

while the conservative floor gives

$$\zeta_{\mathrm{suff}}^{(J)} \simeq 2.44257571477179, \qquad \zeta_{\mathrm{fail}}^{(J)} \simeq 2.46752736855058. \tag{3.138}$$

The natural branch nearly reaches the hard ceiling.

3.8.2 Master residual

Stage 082 packages the remaining reduced PDE into one scalar residual. Let

$$\zeta_{\mathrm{req}}(\Pi_{\mathrm{tr}}, C_{\mathrm{mix}}, \epsilon_{\mathrm{blk}}) = \frac{\Pi_{\mathrm{tr}} - C_{\mathrm{mix}}}{C_{\mathrm{mix}} - \epsilon_{\mathrm{blk}}(2C_{\mathrm{mix}} - \Pi_{\mathrm{tr}})}, \tag{3.139}$$

and

$$\zeta_{\mathrm{phys}}(\mathrm{Pe}, \eta; \kappa) = \Omega_{\mathrm{Pe}}^2 \frac{\kappa + \pi^2/4}{\kappa + y(\eta)^2}. \tag{3.140}$$

The selected support/source branch chooses Pe_* from (3.87). Therefore

$$\boxed{R_{\mathrm{quad}}(\Pi_{\mathrm{tr}}, C_{\mathrm{mix}}, \epsilon_{\mathrm{blk}}; \Xi, \eta, \kappa) = \zeta_{\mathrm{req}}(\Pi_{\mathrm{tr}}, C_{\mathrm{mix}}, \epsilon_{\mathrm{blk}}) - \zeta_{\mathrm{phys}}(\mathrm{Pe}_*(\Xi, \eta, \kappa), \eta; \kappa)}. \tag{3.141}$$

The sign convention is

$$\begin{aligned} R_{\mathrm{quad}} &< 0 \Rightarrow \text{support supply exceeds demand,} \\ R_{\mathrm{quad}} &= 0 \Rightarrow \text{exact saturation,} \\ R_{\mathrm{quad}} &> 0 \Rightarrow \text{explicit lowest-lane support fails.} \end{aligned} \tag{3.142}$$

The Family–1 specialization is

$$R^{\mathrm{F1}}_{\mathrm{quad}} = \zeta_{\mathrm{req}}(\Pi_{\mathrm{tr}}, C_{\mathrm{mix}}, \epsilon_{\mathrm{blk}}) - \zeta_{\mathrm{phys}}(\mathrm{Pe}_*(\Xi_{\mathrm{F1}}, 37, 12321/5), 37; 12321/5), \tag{3.143}$$

with

$$\Xi_{\mathrm{F1}} = W_{\mathrm{wall}} = 1369\Upsilon_w = 136900\Theta_w. \tag{3.144}$$

3.8.3 Cancellation of outgoing normalization factors

Stages 085–087 show that once the selected outgoing branch is normalized, the support theorem no longer depends separately on N_Q^{target}, β_0, s_-, λ_-, or $\widehat{m}_-$. The exact identities are

$$\Pi_{\mathrm{tr}} = \frac{N_Q^{\mathrm{target}}}{\beta_0}\alpha_{\mathrm{req}}, \qquad C_{\mathrm{mix}} = \frac{N_Q^{\mathrm{target}}}{\beta_0}\alpha_{\mathrm{mix}}, \tag{3.145}$$

so

$$\boxed{\frac{\Pi_{\mathrm{tr}}}{C_{\mathrm{mix}}} = \frac{\alpha_{\mathrm{req}}}{\alpha_{\mathrm{mix}}} =: \rho_\alpha}. \tag{3.146}$$

Consequently

$$\boxed{\zeta_{\mathrm{req}}(\rho_\alpha, \epsilon_{\mathrm{blk}}) = \frac{\rho_\alpha - 1}{1 - \epsilon_{\mathrm{blk}}(2 - \rho_\alpha)}}. \tag{3.147}$$

In the unblocked limit this is simply $\zeta_{\mathrm{req}} = \rho_\alpha - 1$.

At $\lambda_\mu = 1$ and $\epsilon_{\mathrm{blk}} = 0$, the Family–1 ratio window is

$$\rho_\alpha \le 3.46622291347846 \Rightarrow \text{guaranteed success}, \tag{3.148}$$

$$\begin{aligned} &\rho_\alpha \ge 3.46752913273870 \Rightarrow \text{guaranteed failure}, \\ &\rho_\alpha < 3.46752922945601 \Rightarrow \text{absolute constructive ceiling}. \end{aligned} \tag{3.149}$$

Thus the support/source side reduces to one number from the outgoing branch: ρ_α.

3.8.4 Minimal isotropic contact-plus-pole module

Stage 088 extracts ρ_α from the minimal isotropic conservative precursor. The precursor is

$$Y_Q^{\mathrm{cons}}(\omega) = c_0 + \frac{c_1}{1 - \omega^2/\Omega_Q^2}, \qquad c_0 = \frac{3}{4}, \qquad c_1 = \frac{1}{4}, \qquad \Omega_Q = \frac{3c_s}{2a}. \tag{3.150}$$

The natural contact-plus-pole reading is

$$Y_Q^{\mathrm{cons}}(\omega) = \frac{1}{\rho_\alpha} + \frac{\rho_\alpha - 1}{\rho_\alpha}\frac{1}{1 - \omega^2/\Omega_Q^2}. \tag{3.151}$$

Therefore

$$\boxed{\rho_\alpha = \frac{1}{c_0} = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} = \frac{c_1}{c_0} = \frac{1}{3}, \qquad \Pi_{\mathrm{tr}} = \frac{4}{3}C_{\mathrm{mix}}}. \tag{3.152}$$

This lies in the symmetric-lowest-twin regime,

$$C_{\mathrm{mix}} < \Pi_{\mathrm{tr}} < 2C_{\mathrm{mix}}, \qquad 0 < \zeta_{\mathrm{req}} = \frac{1}{3} < 1. \tag{3.153}$$

Stage 089 compares it to the explicit Family–1 support map. Since

$$\zeta_{\rm req}^{\rm min} = \frac{1}{3} < A_{\rm F1} \simeq 1.00005192880220, \tag{3.154}$$

the required transport bias is

$$\boxed{\mathrm{Pe}_{\rm req} = 0}. \tag{3.155}$$

So the explicit Family–1 support/source branch supports the minimal isotropic passive/outgoing quadrupole demand before any transport-driven overlap boost is turned on.

3.9 Part III audit summary and handoff to Part IV

Part III establishes the following stable outputs.

1. **Continuum placement is explicit.** The first finite-throat continuum operator gives closed formulas for the selected-branch data and compresses them to $(\epsilon_\eta, \epsilon_W, \rho, Z_W, \delta_0, \Lambda)$.
2. **The product law is exact.** The placement variables obey $R_{\rm target}M_{\rm mix} = 8\Lambda(1-\epsilon_W)/\pi^2$, so only part of the microscopic data sets the product scale.
3. **Non-flat internal structure matters directionally.** The split-U correction preserves scalar placement but generically breaks source/loading collinearity through $D_{\rm dir}$.
4. **Rank-2 support completion is analytic.** The two-direction determinant is linear in the support load, giving the exact $n_{\rm req}$ theorem and monotonicity in mixed baseline.
5. **The first coherent D/N kernel tracks.** The coherent local D/N condition forces $R_\phi = R_U = R_{\rm tr}$; the first split-$U$ tracking branch worsens the target relative to the flat branch.
6. **Support compensation is governed by one ratio.** The support enhancement is $S(\zeta;\epsilon)$, and the physical D/N tower gives $\zeta_n^{\rm phys}$. The lowest twin branch has $\zeta = 1$ and exactly doubles the mixed baseline.
7. **Non-twin rescue is bounded.** Overlap boost is capped by $\pi^2/4$; Robin compliance has its own exact softening ceiling; the combined physical branch is described by $\zeta_0(\mathrm{Pe}, \eta; \kappa)$.
8. **Transport bias has an operator origin.** The source profile is the stationary zero-flux solution of a drift-diffusion equation, and Pe_* is selected by $\mathrm{Pe}_* = \Xi\Delta(\mathrm{Pe}_*; \kappa, \eta)$.
9. **Parent GNLS projection fixes the microscopic gain.** The gain is the local compressional susceptibility times a squared confinement/support overlap, divided by support stiffness.
10. **Matched support/source alignment is natural.** The parent equilibrium source profile is $\chi_\sigma = g_\phi \chi_\phi / H$, with exact coherence near one on a thin active layer.
11. **The explicit Family–1 wall branch is concrete.** The tanh-wall and healing-lock closures give $\Lambda_\ell = \eta = 37$, $\chi_s = 37/2$, and $\kappa = 12321/5$.
12. **Family–1 is not wall-depth starved.** The natural shell-weighted wall datum $\Theta_w^{(\chi)}$ sits far above the success threshold for modest demand.
13. **The final support test is one loading ratio.** After outgoing normalization factors cancel, the explicit support/source theorem depends only on $\rho_\alpha = \alpha_{\rm req}/\alpha_{\rm mix}$.

14. **The minimal isotropic precursor passes.** Under the contact-plus-pole reading, $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$, and $\text{Pe}_{\text{req}} = 0$, safely inside the Family–1 success region.

The chapter deliberately does not claim that the actual moving-throat grouped-P_2/geometry branch has already been derived. It shows that if the final quadrupole branch realizes the minimal isotropic contact-plus-pole conservative precursor, then the explicit Family–1 support/source side is already sufficient. Part IV begins at precisely the remaining issue: the geometry lane, retarded closure, outgoing normalization, outlet/core/mouth compensation, and the branch mechanisms that can actually realize the passive/outgoing quadrupole module.

- ☐ The continuum variables A, ΔK_{ax}, α_{mix}, β_0, M_{mix}, and δ are tied to the Stage-20 finite-throat operator, not treated as free branch knobs.
- ☐ The dimensionless placement map uses the exact stability restrictions $0 < \epsilon_\eta < 1$, $0 < \epsilon_W < 1$, and $1 + \rho \neq 0$.
- ☐ The split-U deformation is separated into scalar placement effects and directional source/loading effects.
- ☐ The rank-2 support theorem keeps the directions q, r, and t distinct until a tracking closure is explicitly imposed.
- ☐ The coherent D/N kernel is recorded as a tracking result, not as a generic rank-2 solution.
- ☐ The lowest twin branch is used only when $\zeta_{\text{req}} \leq 1$ or equivalently $\Pi_{\text{tr}} \leq 2C_{\text{mix}}$.
- ☐ Non-twin overlap and Robin resources are bounded by their exact finite-throat ceilings.
- ☐ The drift-diffusion source family is used through the physical Peclet number, not as an arbitrary exponential ansatz.
- ☐ Parent-action gain formulas keep the source/support coherence factor and Cauchy no-go visible.
- ☐ The sech–Gaussian resonance is treated as a benchmark family, not as a replacement for the matched parent-equilibrium theorem.
- ☐ The Family–1 values $\Lambda_\ell = 37$, $\eta = 37$, $\chi_s = 37/2$, and $\kappa = 12321/5$ are marked as reference-branch closures, not universal parent-action theorems.
- ☐ The extracted values $\Theta_w^{(\chi)}$ and $\Theta_w^{(J)}$ are distinguished as natural quadratic weighting and conservative lower envelope.
- ☐ The cancellation of outgoing normalization factors is applied only after the selected outgoing branch is required to hit the quadrupole normalization target.
- ☐ The conclusion $\rho_\alpha = 4/3$, $\zeta_{\text{req}} = 1/3$, $\text{Pe}_{\text{req}} = 0$ is conditional on the minimal isotropic contact-plus-pole precursor.

The citation-safe summary is:

> Stages 037–090 derive the support-completion and explicit Family–1 branch side of the moving-throat response program. They extract the selected-branch continuum data, compress the placement problem to dimensionless ratios and a product law, treat source/loading mismatch

through rank-2 support completion, derive coherent D/N support thresholds, reduce non-twin rescue to overlap, Robin, transport, and parent-gain resources, evaluate the first explicit GNLS wall-shell Family–1 branch, and finally show that the minimal isotropic contact-plus-pole quadrupole precursor selects $\rho_\alpha = 4/3$, $\zeta_{\mathrm{req}} = 1/3$, and $\mathrm{Pe}_{\mathrm{req}} = 0$. The support/source side is therefore not the remaining bottleneck under that precursor; the remaining open task is to derive the precursor itself from the actual grouped real P_2 and geometry branch.

Chapter 4

Geometry Lane, Retarded Closure, Outlet/Core/Mouth Compensation

Stage range: 091–163

Part IV starts after the support/source side is no longer the dominant reduced uncertainty. Parts I–III built the parent wall/BdG/Maxwell/mixed response compiler, the grouped real P_2 normalization language, the selected finite-throat support front end, and the first explicit Family–1 support verdict. The question here is what still prevents the reduced passive/outgoing quadrupole branch from becoming a usable retarded branch.

The answer is progressively narrowed across Stages 091–163. First, the geometry lane is shown not to contaminate the isotropic grouped P_2 carrier at linear order, so the conservative quadrupole module collapses to the forced contact-plus-pole split

$$\widehat{Y}_Q^{\mathrm{cons}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}. \tag{4.1}$$

The endpoint is again a reduced obstruction packet, not a finished nonlinear branch. The remaining 2.5PN ambiguity collapses to the outgoing normalization factor χ_Q; outlet and core models isolate the viable compensated branch; mouth-source and co-evolving corrections then compress the last first-order obstruction to one microscopic normal coordinate.

Part IV at a glance

Primary question	Once support/source closure is no longer the bottleneck, what reduced datum still blocks a usable retarded quadrupole branch?
Starting inputs	The Part III Family–1 support verdict, the grouped passive/outgoing quadrupole packet, and the reduced outlet/core/mouth closures.
Delivers	The geometry-lane firewall, the factorization to χ_Q, the compensated outlet and core conditions, the canonical mouth branch, and the final off-family defect scalar.
Used by	Part V as the source of the surviving off-family normal coordinate and by all later realization work as the origin of the outgoing-normalization and branch-defect language.
Result anchors	**MTDC-T8**.

Stage packets.

Stages	Packet	Status	Carry-forward output
091–106	Geometry firewall and reduced retarded finish line	Exact Within Closure / Open	Isotropic geometry decoupling, the forced $3/4 + 1/4$ conservative module, factorization of the 2.5PN condition, exact outgoing $l = 2$ DtN fingerprint, and the canonical statement $\chi_Q = 1$ within reduced closure.
107–120	Outlet deformation and compensated core outlet	Exact Within Closure / Reduced / Controlled Reduction	General χ_Q deformation algebra, Robin and mixed outlet audit, compensated Robin–mixed branch, two-channel core realization, and the surviving parent-level core-balance target.
121–145	Mouth-source selection and canonical mouth branch	Exact Within Closure / Reduced / Controlled Reduction / Numerically Located	Geometric mouth selection, positive mouth-source families, boundary-layer and gain laws, self-consistent mouth closure, and the unique finite-bias canonical mouth branch.
146–163	Co-evolving corrections and final defect scalar	Exact Within Closure / Numerically Located / Open	Mouth rigidity corrections, full-profile Family–1 mouth correction, renormalized co-evolving canonical point, defect transport onto the compensated outlet, D/N similarity law, and the off-family normal coordinate $\delta_\perp$.

Claim status. Mixed. The algebraic split, obstruction formulas, harmonic decoupling, outgoing DtN expansion, deformation laws, Schur complements, compensation surfaces, fixed-point laws, and first-order transport maps are Exact Within Closure **once their declared reduced models are adopted. Stable normal-mode reductions, finite D/N tube identifications, Family–1 geometry choices, positive mouth-layer ansatz, self-matched susceptibility closure, and co-evolving fixed-point branch are** Reduced / Controlled Reduction **or** Effective Closure **where explicitly stated. The numerically located co-evolving canonical point and representative correction values are** Numerically Located**. Actual realization by the completed nonlinear moving-throat PDE remains** Open**.**

What this part proves. Part IV shows that the isotropic geometry lane is not the linear obstruction, collapses the 2.5PN retarded issue to the outgoing normalization factor, and then drives the outlet/core/mouth program to a first-order off-family defect scalar.

What this part does not prove. It does not prove that the actual branch lands on the canonical or renormalized compensated point everywhere needed for full realization. The retarded compiler is narrowed sharply, but the physical branch packet is still not closed here.

Why later parts need it. Part V starts from the surviving off-family normal coordinate, and all later realization work inherits the outgoing-normalization, mouth/core, and branch-defect language established in this chapter.

4.1 Block contract and citation boundary

The contract of Part IV is to prevent the retarded quadrupole branch from being overclaimed. It does not assert that the full nonlinear PDE has been solved. It proves that, inside the declared reduced closure sequence, the active uncertainty can be compressed from many apparent degrees of freedom to one scalar branch-realization question at a time.

The chain of compression is

$$\text{geometry lane} \longrightarrow (\epsilon_2, \epsilon_4) \longrightarrow N_Q \longrightarrow \chi_Q \longrightarrow \Delta_Q \longrightarrow \delta\gamma_W \longrightarrow \Xi_{\text{slip}} \longrightarrow \delta_\perp. \tag{4.2}$$

Each arrow is a theorem inside a reduced block, not a statement that the final scalar is already zero on the actual branch.

Theorem 4.1 (Part IV reduced retarded-compression theorem)**.** *Inside the declared grouped-P_2, static-geometry, passive/outgoing, compensated outlet, and co-evolving Family–1 mouth/core closures, the retarded quadrupole verification problem admits the following nested reductions.*

1. *The isotropic geometry lane contributes no dynamic $O(\omega^2)$ or $O(\omega^4)$ contamination to the grouped real P_2 carrier at linear order.*
2. *The conservative quadrupole module is therefore* (4.1)*.*
3. *The point-particle* 2.5*PN normalization condition factorizes as*

$$\widehat{m}_0^2 \chi_Q N_Q = 1. \tag{4.3}$$

   On the natural source-map branch, the remaining retarded obstruction is $\Delta_Q = \chi_Q - 1$.

4. *The canonical outgoing $l = 2$ DtN branch has $\chi_Q = 1$.*
5. *In the explicit compensated Robin–mixed outlet class, first-order even preservation forces $\delta\mathfrak{g} = 0$ and $\delta\kappa_W = 0$, so the remaining first-order defect is purely odd.*
6. *In the concrete D/N mixed-tube realization, that odd defect is zero whenever the tangential deformation preserves the D/N similarity law.*
7. *The first off-family deviation from the exact parent compensation family is measured by a single normal coordinate $\delta_\perp$.*

Thus the end of Part IV is not a broad family of remaining outlet defects. It is the concrete microscopic question whether the true branch has $\delta_\perp = 0$, or if not, what value that scalar takes.

Proof. Stages 091–099 prove the geometry and conservative-normalization reductions. Stages 100–106 prove the retarded factorization and canonical DtN coefficient. Stages 107–124 classify admissible low-frequency outlet/core deformations and identify the compensated Robin–mixed class. Stages 125–145 derive parent and mouth-layer parameters for that class and select the regular mouth branch. Stages 146–163 linearize the remaining defect and show that the outlet, D/N similarity, and parent compensation defects are all images of the same off-family normal coordinate. □

4.2 Geometry-lane firewall

4.2.1 Why the geometry lane had to be checked

The conservative side enters Part IV with two facts already frozen. First, the higher conservative quadrupole payload lives in the grouped real P_2 bundle. Second, the conservative PN assembly also contains a geometry completion. If the geometry completion were dynamically active through $O(\omega^4)$, then the clean one-pole grouped branch used in the retarded bridge could be contaminated before the outgoing branch was even tested.

The check therefore begins with the most general reduced isotropic ansatz needed at this order:

$$K_Q^{\text{cons}}(\omega) = K_g(\omega) + K_{P2}(\omega), \qquad K_{P2}(\omega) = \frac{K_{\text{pole}}}{1 - \omega^2/\Omega_Q^2}, \tag{4.4}$$

where

$$K_g(\omega) = K_{g,0} + K_{g,2}\omega^2 + K_{g,4}\omega^4 + O(\omega^6). \tag{4.5}$$

The minimal isotropic branch identity is

$$K_0 K_4 = 4K_2^2. \tag{4.6}$$

If $K_{g,2} = K_{g,4} = 0$, inserting (4.4) into (4.6) gives

$$K_{g,0} = 3K_{\text{pole}}, \qquad K_{\text{pole}} = \frac{K_0}{4}, \qquad K_{g,0} = \frac{3K_0}{4}, \tag{4.7}$$

which is exactly the conservative split in (4.1).

4.2.2 Exact obstruction if geometry is dynamic

If the geometry lane carries dynamic even moments, the simple split is replaced by an exact obstruction formula. Define

$$\epsilon_2 := \frac{\Omega_Q^2 K_{g,2}}{K_{\text{pole}}}, \qquad \epsilon_4 := \frac{\Omega_Q^4 K_{g,4}}{K_{\text{pole}}}. \tag{4.8}$$

Then the static pole fraction becomes

$$c_{\text{pole}} = \frac{K_{\text{pole}}}{K_0} = \frac{1+\epsilon_4}{4(1+\epsilon_2)^2}. \tag{4.9}$$

For small contamination,

$$c_{\text{pole}} = \frac{1}{4}\left(1 + \epsilon_4 - 2\epsilon_2 + O(\epsilon^2)\right). \tag{4.10}$$

Thus the static-geometry hypothesis is not a cosmetic simplification. It is the exact condition under which the pole fraction is 1/4.

4.2.3 Isotropic decoupling

The distributed wall action on an isotropic reference throat is block diagonal in angular momentum. With

$$\eta(\Omega, w, t) = g(w,t)Y_{00}(\Omega) + \sum_A q_A(w,t)Y_{2A}(\Omega) + \cdots, \tag{4.11}$$

where A ranges over the real quadrupole harmonics, every possible quadratic cross term between g and q_A is proportional to one of

$$\int_{S^2} Y_{00} Y_{2A}\, d\Omega, \qquad \int_{S^2} \nabla_{S^2} Y_{00} \cdot \nabla_{S^2} Y_{2A}\, d\Omega, \qquad \int_{S^2} Y_{00}(-\Delta_{S^2}) Y_{2A}\, d\Omega. \tag{4.12}$$

All three vanish. Therefore

$$M_{0\leftrightarrow 2} = 0 \tag{4.13}$$

inside the isotropic quadratic wall theory, and the geometry contamination parameters vanish:

$$K_{g,2} = K_{g,4} = 0, \qquad \epsilon_2 = \epsilon_4 = 0. \tag{4.14}$$

Proposition 4.2 (Second-order onset of geometry contamination)**.** *If weak anisotropy or operator noncommutation generates a bilinear mixing term $\chi M_0 qg$ between one grouped quadrupole coordinate and one scalar geometry coordinate, then after eliminating g, the first dynamic geometry contamination is quadratic in χ.*

Proof. Let

$$D_q(\omega) = K_{\mathrm{stat}} + \frac{K_{\mathrm{pole}}}{1-\omega^2/\Omega_Q^2}, \qquad D_g(\omega) = G_0 + G_2\omega^2 + G_4\omega^4 + O(\omega^6). \tag{4.15}$$

The Schur complement is

$$D_{\mathrm{eff}}(\omega) = D_q(\omega) - \frac{\chi^2 M_0^2}{D_g(\omega)}. \tag{4.16}$$

Expanding D_g^{-1} gives

$$K_{g,2}^{\mathrm{eff}} = \frac{\chi^2 M_0^2 G_2}{G_0^2}, \qquad K_{g,4}^{\mathrm{eff}} = \frac{\chi^2 M_0^2 (G_0 G_4 - G_2^2)}{G_0^3}. \tag{4.17}$$

Therefore $\epsilon_2, \epsilon_4 = O(\chi^2)$. □

The geometry-lane firewall is therefore an exact reduced selection rule: on the isotropic branch the geometry lane is dynamically inert through the relevant even orders, while weak contamination is second order in the explicit mixing source.

4.3 Retarded 2.5PN collapse

4.3.1 From conservative module to one scalar defect

With the geometry-lane check complete, the actual isotropic conservative quadrupole response can be written as

$$\overline{K}_Q^{\mathrm{cons}}(\omega) = \overline{K}_0 \left[\frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}\right]. \tag{4.18}$$

Its even moments are

$$\overline{K}_2 = \frac{\overline{K}_0}{4\Omega_Q^2}, \qquad \overline{K}_4 = \frac{\overline{K}_0}{4\Omega_Q^4}. \tag{4.19}$$

The target static normalization is

$$\overline{K}_0^{\mathrm{target}} = \frac{64 G\Omega_Q^5}{45 c^5} = \frac{54 G c_s^5}{5 a^5 c^5}, \qquad \Omega_Q = \frac{3c_s}{2a}. \tag{4.20}$$

Thus the conservative defect is the scalar

$$N_Q := \frac{\overline{K}_0}{K_0^{\mathrm{target}}}. \tag{4.21}$$

All even ratios scale by the same N_Q. The Family–1 support side is automatic on this actual isotropic branch because the forced branch value $\rho_\alpha = 4/3$ gives

$$\zeta_{\mathrm{req}}^{\mathrm{act}}(\epsilon_{\mathrm{blk}}) = \frac{1}{3 - 2\epsilon_{\mathrm{blk}}}, \tag{4.22}$$

which is below the Family–1 constructive support ceiling throughout the admissible blocked regime.

4.3.2 Retarded factorization

The retarded one-pole grouped branch is written

$$\widehat{Y}_Q^{\mathrm{ret}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1 - \omega^2/\Omega_Q^2 - i\chi_Q\sigma_Q^{\mathrm{can}}\omega^5} + O(\omega^6), \tag{4.23}$$

where

$$\sigma_Q^{\mathrm{can}} = \frac{9}{8\Omega_Q^5} = \frac{4a^5}{27c_s^5}. \tag{4.24}$$

The odd coefficient is

$$\overline{\Gamma}_5 = \chi_Q \frac{9\overline{K}_0}{32\Omega_Q^5}. \tag{4.25}$$

Consequently

$$\frac{\overline{\Gamma}_5}{2G/(5c^5)} = \chi_Q N_Q, \qquad \widehat{m}_0^2 \chi_Q N_Q = 1. \tag{4.26}$$

On the natural point-particle source-map branch, $\widehat{m}_0 = 1 + O(a^2/r^2)$, so in the strict point-particle limit the remaining obstruction is

$$\Delta_Q := \chi_Q - 1. \tag{4.27}$$

Higher odd denominator terms beginning at $O(\omega^7)$ are invisible to the point-particle 2.5PN coefficient.

4.3.3 Canonical outgoing DtN branch

Let

$$z := \frac{a\omega}{c_s}. \tag{4.28}$$

The exact outgoing spherical $l = 2$ DtN operator is

$$\Lambda_2^{\mathrm{out}}(z) = z\frac{d}{dz}\ln h_2^{(1)}(z) = -3 + \frac{z^2}{3} + \frac{z^4}{9} + i\frac{z^5}{9} + O(z^6). \tag{4.29}$$

Normalizing by the static slot gives

$$\widehat{Y}_2^{\mathrm{out}}(z) = \frac{-3}{\Lambda_2^{\mathrm{out}}(z)} = 1 + \frac{z^2}{9} + \frac{4z^4}{81} + i\frac{z^5}{27} + O(z^6). \tag{4.30}$$

Comparison with (4.23) fixes

$$\chi_Q = 1 \tag{4.31}$$

for the canonical compact passive/outgoing DtN branch. The canonical invariant coefficients are then

$$\overline{K}_0 = \frac{54Gc_s^5}{5a^5c^5}, \qquad \overline{K}_2 = \frac{6Gc_s^3}{5a^3c^5}, \qquad \overline{K}_4 = \frac{8Gc_s}{15ac^5}, \qquad \overline{\Gamma}_5 = \frac{2G}{5c^5}. \tag{4.32}$$

Theorem 4.3 (Conditional reduced 2.5PN closure)**.** *Within the reduced isotropic grouped-P_2 hierarchy, the nonspinning point-particle* 2.5*PN branch closes if the actual passive/outgoing moving-throat branch realizes the canonical outgoing DtN coefficient $\chi_Q = 1$ and the natural source-map limit $\widehat{m}_0 \to 1$. If the actual branch has $\chi_Q \neq 1$, the entire reduced failure is measured by $\Delta_Q = \chi_Q - 1$ at leading order.*

4.4 Outlet DtN and Robin outlet tests

4.4.1 General isotropic deformation algebra

The remaining branch-selection issue is whether the true moving-throat outlet is the canonical branch or a deformed one. The first explicit deformation family is

$$\Lambda_2^{\mathrm{def}}(z) = S\Lambda_2^{\mathrm{out}}(\beta z) + \Sigma_0 + \Sigma_2 z^2 + \Sigma_4 z^4 + i\Sigma_5 z^5 + O(z^6). \tag{4.33}$$

Writing the expansion coefficients as L_0, L_2, L_4, L_5, the normalized response is

$$\widehat{Y}_2^{\mathrm{def}}(z) = 1 - \frac{L_2}{L_0} z^2 + \left(\frac{L_2^2}{L_0^2} - \frac{L_4}{L_0}\right) z^4 - i\frac{L_5}{L_0} z^5 + O(z^6). \tag{4.34}$$

After imposing the canonical even fingerprint, the exact odd normalization is

$$\chi_Q = \frac{3(S\beta^5 + 9\Sigma_5)}{3S - \Sigma_0}, \tag{4.35}$$

so

$$\chi_Q - 1 = \frac{3S(\beta^5 - 1) + \Sigma_0 + 27\Sigma_5}{3S - \Sigma_0}. \tag{4.36}$$

Linearizing with

$$S = 1 + \varepsilon s, \qquad \beta = 1 + \varepsilon b, \qquad \Sigma_0 = \varepsilon a_0, \qquad \Sigma_5 = \varepsilon a_5, \tag{4.37}$$

while keeping the even slots canonical gives

$$\Delta_Q = 5b + \frac{a_0}{3} + 9a_5 + O(\varepsilon^2). \tag{4.38}$$

Pure overall scaling drops out. Pure argument rescaling cannot move χ_Q without also moving the even fingerprint unless $\beta = 1$. Additive throat-core slots are therefore the first genuinely dangerous outlet data.

4.4.2 Robin and mixed-pole tests

A pure isotropic Robin outlet is

$$\Lambda_2^{\mathrm{R}}(z) = \Lambda_2^{\mathrm{out}}(z) + \rho_R. \tag{4.39}$$

It gives

$$\chi_Q^{\mathrm{R}} = \frac{3}{3 - \rho_R}, \tag{4.40}$$

so it generically shifts the canonical branch.

A standalone hidden mixed side-channel pole is

$$\Lambda_2^{\mathrm{mix}}(z) = \Lambda_2^{\mathrm{out}}(z) - \frac{\sigma_W}{1 - \kappa_W z^2 - i\gamma_W z^5} + O(z^6). \tag{4.41}$$

The canonical even conditions force $\kappa_W = -1/9$, and then $\sigma_W = 0$. Thus a naive standalone passive hidden pole cannot be the actual branch if the conservative even fingerprint is already fixed.

4.4.3 Compensated Robin–mixed outlet

The combined outlet is

$$\Lambda_2^{\mathrm{hyb}}(z) = \Lambda_2^{\mathrm{out}}(z) + \rho_R - \frac{\sigma_W}{1-\kappa_W z^2 - i\gamma_W z^5} + O(z^6). \tag{4.42}$$

The exact canonical-even conditions have two solutions:

$$(\rho_R, \kappa_W) = (\sigma_W, 0), \qquad (\rho_R, \kappa_W) = (4\sigma_W, 1/3). \tag{4.43}$$

The first is a trivial static cancellation. The second is the nontrivial compensated branch. On it,

$$\chi_Q^{\mathrm{hyb}} = \frac{1-9\sigma_W\gamma_W}{1-\sigma_W}, \tag{4.44}$$

so the canonical outgoing value is preserved iff

$$\gamma_W = \frac{1}{9}. \tag{4.45}$$

With (4.45), the hybrid outlet collapses to a harmless pure scale deformation through the retained order.

4.5 Mixed side-channel pole realization

4.5.1 Concrete two-channel core model

The reduced hybrid outlet is realized by a concrete shell/mixed core. Let $u(\omega)$ be the isotropic $l=2$ mouth amplitude, and introduce a static shell coordinate s and a mixed side-channel coordinate q. The core equations are

$$\begin{pmatrix} K_s & \lambda \\ \lambda & -K_q D_W^{\mathrm{bare}}(z) \end{pmatrix}\begin{pmatrix} s \\ q \end{pmatrix} = u\begin{pmatrix} g_s \\ g_q \end{pmatrix}, \qquad D_W^{\mathrm{bare}}(z) = 1-\kappa_0 z^2 - i\gamma_0 z^5 + O(z^6). \tag{4.46}$$

Eliminating (s,q) gives

$$\delta\Lambda_{\mathrm{core}}(z) = \frac{g_s^2}{K_s} - \frac{(K_s g_q - \lambda g_s)^2}{K_s\left(K_s K_q D_W^{\mathrm{bare}}(z) + \lambda^2\right)}. \tag{4.47}$$

With

$$r_c := \frac{\lambda^2}{K_s K_q}, \tag{4.48}$$

this has the reduced form

$$\delta\Lambda_{\mathrm{core}}(z) = \rho_c - \frac{\sigma_c}{1-\kappa_c z^2 - i\gamma_c z^5} + O(z^6), \tag{4.49}$$

where

$$\rho_c = \frac{g_s^2}{K_s}, \qquad \sigma_c = \frac{(K_s g_q - \lambda g_s)^2}{K_s^2 K_q (1+r_c)}, \qquad \kappa_c = \frac{\kappa_0}{1+r_c}, \qquad \gamma_c = \frac{\gamma_0}{1+r_c}. \tag{4.50}$$

4.5.2 Core-balance theorem and D/N tube

The nontrivial compensated branch requires

$$\rho_c = 4\sigma_c, \qquad \kappa_c = \frac{1}{3}, \qquad \gamma_c = \frac{1}{9}. \tag{4.51}$$

The static balance condition is exactly

$$g_s^2(K_sK_q + \lambda^2) = 4(K_s g_q - \lambda g_s)^2. \tag{4.52}$$

Equivalently, defining

$$\mathfrak{r} := \frac{\lambda}{\sqrt{K_sK_q}}, \qquad \mathfrak{g} := \frac{g_q\sqrt{K_s}}{g_s\sqrt{K_q}}, \tag{4.53}$$

one obtains the parent compensation family

$$1 + \mathfrak{r}^2 = 4(\mathfrak{g} - \mathfrak{r})^2, \qquad \mathfrak{g}_\pm(\mathfrak{r}) = \mathfrak{r} \pm \frac{1}{2}\sqrt{1+\mathfrak{r}^2}. \tag{4.54}$$

The finite D/N mixed-tube realization gives

$$L_W = \frac{\pi a}{2}\sqrt{\frac{1+r_c}{3}} = \frac{\pi a}{2}\sqrt{\frac{1+\mathfrak{r}^2}{3}}. \tag{4.55}$$

Identifying this tube with the actual throat span L fixes the branch value

$$\mathfrak{r}_{\text{geom}}\left(\frac{L}{a}\right) = \sqrt{\frac{12}{\pi^2}\left(\frac{L}{a}\right)^2 - 1}. \tag{4.56}$$

On the Family–1 preferred aspect ratio $L/a = 37/20$,

$$\mathfrak{r}_{F1} = \frac{\sqrt{4107 - 100\pi^2}}{10\pi} \approx 1.77799353547498. \tag{4.57}$$

The lower compensated mouth-coupling branch is then

$$\mathfrak{g}_-^{F1} = \mathfrak{r}_{F1} - \frac{1}{2}\sqrt{1+\mathfrak{r}_{F1}^2} \approx 0.758035078944663. \tag{4.58}$$

The upper branch is $\mathfrak{g}_+^{F1} \approx 2.79795199200529$.

4.5.3 Parent-action overlap extraction

The abstract core parameters are tied to parent-action overlaps. In the first GNLS plus localized-Maxwell throat-core closure,

$$K_s = 4\pi a^2\left(\frac{H_w\ell}{3} + \frac{\hbar^2}{15m_\psi\rho_w\ell}\right), \qquad K_s = \frac{3\pi a^2\hbar^2}{5m_\psi\rho_w\ell} \quad \text{on the healing-locked branch,} \tag{4.59}$$

$$K_q = \frac{\mathcal{Z}_q}{\mu_0}\frac{\pi^2c_s^2}{4L_W^2}, \tag{4.60}$$

$$\lambda = -q_* v_{w0}\mathcal{I}_{sq}, \tag{4.61}$$

and

$$g_s = \mathcal{T}_m\frac{4\pi a^2\ell}{3}, \qquad g_q = \frac{\mathcal{Z}_q}{\mu_0}\frac{\pi}{\sqrt{2}\,L_W^{3/2}}. \tag{4.62}$$

These formulas are the parent-action path by which a future full branch calculation must decide whether (4.54) is actually realized.

4.6 Core-to-mouth compensation law

4.6.1 Positive mouth-source selection

The core compensation family selects a mouth-coupling ratio. The first physical filter is positivity of the localized mouth source. For a positive normalized axial source $\sigma(z)$ on $[0, L]$, the normalized mouth-bias factor is

$$\mathfrak{g}[\sigma] = \int_0^L \sigma(z) \cos\left(\frac{\pi z}{2L}\right) dz. \tag{4.63}$$

Since the cosine lies between 0 and 1 on the interval,

$$0 \leq \mathfrak{g}[\sigma] \leq 1. \tag{4.64}$$

Therefore the upper Family–1 branch is impossible under positive localized mouth sourcing, while the lower branch (4.58) is admissible.

The self-matched derivative source $\sigma_{\mathrm{match}}(z) = k\cos(kz)$, with $k = \pi/(2L)$, gives

$$\mathfrak{g}_{\mathrm{match}} = \frac{\pi}{4} \approx 0.785398163397448, \tag{4.65}$$

only about 3.61% in traction away from the lower compensated value. A convex positive family interpolating between this derivative profile and the uniform profile reaches the exact lower branch at a modest broadening fraction.

4.6.2 Electrochemical boundary layer

The positive exponential source family is not an arbitrary fit. It follows from a local mouth free energy and zero-flux Onsager law. Linearizing the parent mouth potential as

$$V_m(z) \approx V_1 z, \qquad V_1 = \partial_z \delta V_{\mathrm{conf}}\big|_{\mathrm{m}} - q_* \, \partial_z A_0\big|_{\mathrm{m}}, \tag{4.66}$$

the electrochemical potential is

$$\mu_\sigma^{\mathrm{chem}}(z) = \Theta_\sigma \ln \frac{\sigma}{\sigma_*} + V_1 z. \tag{4.67}$$

The zero-flux stationary solution is

$$\sigma_\Pi(z) = \frac{\Pi e^{-\Pi z/L}}{L(1-e^{-\Pi})}, \qquad \Pi := \frac{V_1 L}{\Theta_\sigma} > 0. \tag{4.68}$$

The exact bias map is

$$\mathfrak{g}_\Pi = \frac{2\Pi(2\Pi e^{\Pi} + \pi)}{(4\Pi^2 + \pi^2)(e^{\Pi} - 1)}. \tag{4.69}$$

It is strictly increasing from $2/\pi$ to 1, and the Family–1 lower branch is reached at

$$\Pi_* \approx 1.50882951349316. \tag{4.70}$$

Equivalently, the parent threshold is

$$\partial_z \delta V_{\mathrm{conf}}\big|_{\mathrm{m}} - q_* \, \partial_z A_0\big|_{\mathrm{m}} = \Pi_* \frac{\Theta_\sigma}{L}. \tag{4.71}$$

4.6.3 Coupled mouth-layer fixed point

The effective bias Π is then replaced by a coupled boundary-layer solve. For a two-component field $\mathbf{U} = (U_s, U_q)^T$,

$$\left[-\partial_x^2 \mathbf{I} + \mathbf{K}\right] \mathbf{U}(x) = \Sigma_\Pi(x)\mathbf{G}, \qquad \mathbf{U}(0) = 0, \qquad \mathbf{U}'(1) = 0. \tag{4.72}$$

After diagonalizing $\mathbf{K}$, the scalar D/N response kernel is

$$\mathcal{S}(\Pi, \kappa) = \frac{\Pi\left[\kappa \tanh \kappa + \Pi\left(e^{-\Pi} \operatorname{sech} \kappa - 1\right)\right]}{(1 - e^{-\Pi})(\kappa^2 - \Pi^2)}. \tag{4.73}$$

The full fixed-point law is

$$\Pi = \sum_{\alpha=\pm} M_\alpha \mathcal{S}(\Pi, \kappa_\alpha). \tag{4.74}$$

On the first Family–1 shell plus mixed D/N branch,

$$\Pi = M_s + M_q \mathcal{S}_q(\Pi), \qquad \mathcal{S}_q(\Pi) := \mathcal{S}\left(\Pi, \frac{\pi}{2}\right). \tag{4.75}$$

Inheriting the compensated outlet ratio gives

$$M_s = 4\Sigma_m, \qquad M_q = -\Sigma_m, \tag{4.76}$$

so

$$\Pi = \Sigma_m \left[4 - \mathcal{S}_q(\Pi)\right]. \tag{4.77}$$

At the canonical point,

$$\Sigma_m^* = \frac{\Pi_*}{4 - \mathcal{S}_q(\Pi_*)} \approx 0.451485277739090. \tag{4.78}$$

4.7 Mouth boundary-layer source theorem

4.7.1 From gains to parent core quantities

Embedding the same core Schur-complement weights into the mouth electrochemical free energy gives explicit gains

$$M_s = \frac{L g_s^2}{K_s \Theta_\sigma}, \qquad M_q = -\frac{L(K_s g_q - \lambda g_s)^2}{K_s(K_s K_q + \lambda^2)\Theta_\sigma}. \tag{4.79}$$

In normalized variables,

$$R_q := -\frac{M_q}{M_s} = \frac{(\mathfrak{g}_c - \mathfrak{r})^2}{1 + \mathfrak{r}^2}, \qquad M_s = \Sigma_0. \tag{4.80}$$

On the exact compensation family, $R_q = 1/4$. Under the self-matched mouth susceptibility closure,

$$\Sigma_0 = M_s = \frac{20}{9}\widehat{T}_m^2. \tag{4.81}$$

The self-consistent Family–1 mouth branch identifies the core mouth ratio with the source bias:

$$\mathfrak{g}_c = \mathfrak{g}_\Pi, \qquad R_q(\Pi) = \frac{(\mathfrak{g}_\Pi - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}_{F1}^2}. \tag{4.82}$$

The scalar branch law becomes

$$\Pi = \Sigma_0 \left[1 - R_q(\Pi)\mathcal{S}_q(\Pi)\right], \qquad \Sigma_0(\Pi) = \frac{\Pi}{1 - R_q(\Pi)\mathcal{S}_q(\Pi)}. \tag{4.83}$$

At Π_*,

$$R_q(\Pi_*) = \frac{1}{4}, \qquad \Sigma_0(\Pi_*) \approx 1.80594111095636, \qquad \widehat{T}_m(\Pi_*) \approx 0.901484054174205. \tag{4.84}$$

4.7.2 Regularity theorem

The explicit exponential mouth family has

$$0 < \mathfrak{g}_\Pi < 1 \qquad (\Pi > 0), \tag{4.85}$$

and $\mathfrak{g}_\Pi \to 1$ only as $\Pi \to \infty$. The equal-normalized branch $\mathfrak{g}_c = 1$ is therefore a singular point-source limit. Since $\widehat{T}_m \sim C\Pi^{1/2}$ in that limit, it is not a regular finite-traction mouth state.

Theorem 4.4 (Unique regular Family–1 canonical mouth branch)**.** *Inside the explicit Family–1 positive exponential mouth-layer closure, the upper compensated branch is excluded by positivity, the equal-normalized branch is singular, and the lower compensated branch is the unique regular finite-bias, finite-traction branch preserving the canonical outgoing quadrupole fingerprint.*

4.8 Family-1 mouth correction

4.8.1 First-order positive deformations

After branch selection, the remaining mouth-side uncertainty is finite correction around the unique regular branch. Let

$$\Sigma_*(x) = \frac{\Pi_* e^{-\Pi_* x}}{1 - e^{-\Pi_*}} \tag{4.86}$$

and consider a positive normalized deformation

$$\Sigma_\epsilon(x) = (1-\epsilon)\Sigma_*(x) + \epsilon\varsigma(x), \qquad \varsigma \geq 0, \qquad \int_0^1 \varsigma(x)\,dx = 1. \tag{4.87}$$

Only two moments matter at first order:

$$\bar{g}[\sigma] = \int_0^1 \sigma(x)c(x)\,dx, \qquad \bar{S}[\sigma] = \int_0^1 \sigma(x)K_q(x)\,dx, \tag{4.88}$$

with

$$c(x) = \cos\frac{\pi x}{2}, \qquad K_q(x) = \frac{\cosh\left(\frac{\pi}{2}(1-x)\right)}{\cosh(\pi/2)}. \tag{4.89}$$

Retuning Π to stay on the canonical lower branch gives

$$\delta\Pi = -\epsilon\frac{\bar{g}_\varsigma - \mathfrak{g}_*}{\mathfrak{g}'_*}, \qquad \mathfrak{g}'_* \approx 0.0714453558083195. \tag{4.90}$$

The traction shift is

$$\delta\widehat{T}_m = \epsilon\left[A_T(\bar{g}_\varsigma - \mathfrak{g}_*) + B_T(\bar{S}_\varsigma - \mathcal{S}_*)\right], \tag{4.91}$$

where

$$A_T \approx -4.27263956256927, \qquad B_T \approx 0.134875005736706. \tag{4.92}$$

Equivalently, all first-order source-shape sensitivity is the projection against one centered rigidity kernel.

4.8.2 Full-profile residual

The actual full mouth potential generated by the shell plus mixed D/N channels is $\Phi_*(x)$. Define the tangent-subtracted residual

$$R_*(x) := \Phi_*(x) - \Pi_* x. \tag{4.93}$$

It satisfies

$$R_*(0) = R_*'(0) = 0, \qquad R_*''(0) < 0, \tag{4.94}$$

so the full source is broader than the tangent exponential near the mouth. Expanding

$$\Sigma_{\text{act}}(x) = \Sigma_*(x)\left[1 - \widetilde{R}_*(x)\right] + O(R_*^2), \qquad \widetilde{R}_* = R_* - \langle R_* \rangle_*, \tag{4.95}$$

gives the selected moment shifts

$$\delta\mathfrak{g}_{\text{act}} = -\operatorname{Cov}_*(c, R_*), \qquad \delta\mathcal{S}_{\text{act}} = -\operatorname{Cov}_*(K_q, R_*). \tag{4.96}$$

On the explicit Family–1 branch,

$$\operatorname{Cov}_*(c, R_*) \approx 0.0648069687666328, \qquad \operatorname{Cov}_*(K_q, R_*) \approx 0.0388718368650403. \tag{4.97}$$

Thus the first-order correction broadens the source and shifts the mouth-only canonical point upward:

$$\Pi_{\text{corr}} \approx 2.41591392833607, \qquad \widehat{T}_{m,\text{corr}} \approx 1.17313803363654. \tag{4.98}$$

A one-step nonlinear iterate gives a slightly stronger but directionally consistent shift.

4.9 Co-evolving core-mouth fixed point

4.9.1 Exact co-evolving map

The fixed-core mouth correction is then replaced by a co-evolving map. For any positive normalized Σ, define

$$\mathfrak{g}[\Sigma] = \int_0^1 \Sigma(x)c(x)\,dx, \qquad \mathcal{S}[\Sigma] = \int_0^1 \Sigma(x)K_q(x)\,dx. \tag{4.99}$$

The core loading ratio is

$$\mathcal{R}[\Sigma] = \frac{(\mathfrak{g}[\Sigma] - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}_{F1}^2}. \tag{4.100}$$

With the shell and mixed Green operators $\mathcal{T}_s$ and $\mathcal{T}_q$, the full potential is

$$\Phi_{\Sigma_0}[\Sigma](x) = \Sigma_0\left[\mathcal{T}_s[\Sigma](x) - \mathcal{R}[\Sigma]\mathcal{T}_q[\Sigma](x)\right]. \tag{4.101}$$

The nonlinear fixed-point equation is

$$\Sigma(x) = \frac{e^{-\Phi_{\Sigma_0}[\Sigma](x)}}{\int_0^1 e^{-\Phi_{\Sigma_0}[\Sigma](y)}\,dy}. \tag{4.102}$$

The canonical compensation condition is

$$\mathfrak{g}[\Sigma] = \mathfrak{g}_*, \qquad \mathcal{R}[\Sigma] = \frac{1}{4}. \tag{4.103}$$

4.9.2 Renormalized canonical point

At the old canonical traction, the co-evolving fixed point remains positive but moves off exact compensation:

$$\mathfrak{g}_{\mathrm{fp}} \approx 0.693352419668063, \qquad \mathcal{S}_{\mathrm{fp}} \approx 0.621601316751401, \qquad \mathcal{R}_{\mathrm{fp}} \approx 0.282713904908238. \tag{4.104}$$

Restoring exact lower compensation requires a numerically located renormalized gain

$$\Sigma_0^{\mathrm{can}} \approx 4.651033550168876, \qquad \widehat{T}_{m,\mathrm{can}} \approx 1.446708366456762, \tag{4.105}$$

with

$$\mathfrak{g}_{\mathrm{can}} = \mathfrak{g}_*, \qquad \mathcal{R}_{\mathrm{can}} = \frac{1}{4}, \qquad \mathcal{S}_{\mathrm{can}} \approx 0.670362115673462, \qquad \Pi_{\mathrm{can}} \approx 3.871564377479009. \tag{4.106}$$

This is a Numerically Located placement inside the exact co-evolving map, not a closed-form theorem of the full PDE.

4.10 Linear defect transport

4.10.1 Mouth/core transport around the renormalized point

Linearizing the exact Family–1 loading ratio gives

$$\delta\mathcal{R} = -\frac{\delta\mathfrak{g}}{\sqrt{1+\mathfrak{r}_{F1}^2}} + O(\delta\mathfrak{g}^2) \approx -0.490216044387626\,\delta\mathfrak{g}. \tag{4.107}$$

The gains transport as

$$\delta M_s = \delta\Sigma_0, \qquad \delta M_q = -\frac{1}{4}\delta\Sigma_0 + \frac{\Sigma_0^{\mathrm{can}}}{\sqrt{1+\mathfrak{r}_{F1}^2}}\delta\mathfrak{g}. \tag{4.108}$$

The mouth-bias transport is

$$\delta\Pi = \left(1 - \frac{1}{4}\mathcal{S}_{\mathrm{can}}\right)\delta\Sigma_0 - \frac{\Sigma_0^{\mathrm{can}}}{4}\delta\mathcal{S} + \frac{\Sigma_0^{\mathrm{can}}\mathcal{S}_{\mathrm{can}}}{\sqrt{1+\mathfrak{r}_{F1}^2}}\delta\mathfrak{g}, \tag{4.109}$$

or numerically

$$\delta\Pi \approx 0.832409471081635\,\delta\Sigma_0 - 1.16275838754222\,\delta\mathcal{S} + 1.52843317823248\,\delta\mathfrak{g}. \tag{4.110}$$

4.10.2 Projection into the compensated outlet

The compensated outlet loads are

$$\rho_R = \Xi M_s, \qquad \sigma_W = -\Xi M_q, \qquad \Xi = \frac{\Theta_\sigma}{L}. \tag{4.111}$$

Define the off-compensation scalar

$$\delta\mathcal{C} := \delta\rho_R - 4\delta\sigma_W. \tag{4.112}$$

Then

$$\delta\mathcal{C} = \frac{16\sigma_*}{\sqrt{1+\mathfrak{r}_{F1}^2}}\delta\mathfrak{g}. \tag{4.113}$$

Linearizing the hybrid outlet gives

$$\delta E_2 = \frac{\delta\mathcal{C} - 9\sigma_*\delta\kappa_W}{27(1-\sigma_*)}, \qquad \delta E_4 = \frac{5\delta\mathcal{C} - 72\sigma_*\delta\kappa_W}{243(1-\sigma_*)}, \tag{4.114}$$

$$\Delta_Q = \frac{\delta\mathcal{C} - 27\sigma_*\delta\gamma_W}{3(1-\sigma_*)}. \tag{4.115}$$

Canonical even preservation $\delta E_2 = \delta E_4 = 0$ forces

$$\delta\mathcal{C} = 0, \qquad \delta\kappa_W = 0, \qquad \delta\mathfrak{g} = 0. \tag{4.116}$$

The remaining first-order defect is therefore

$$\Delta_Q = -\frac{9\sigma_*}{1-\sigma_*}\delta\gamma_W, \qquad N_Q - 1 = \frac{9\sigma_*}{1-\sigma_*}\delta\gamma_W. \tag{4.117}$$

4.10.3 Bare mixed-port slippage and D/N similarity

The concrete core identities give

$$\kappa_W = \frac{\kappa_0}{1+r_c}, \qquad \gamma_W = \frac{\gamma_0}{1+r_c}. \tag{4.118}$$

On the compensated branch,

$$\kappa_0 = \frac{1+r_c}{3}, \qquad \gamma_0 = \frac{1+r_c}{9}. \tag{4.119}$$

Under the canonical-even gate $\delta\kappa_W = 0$, the odd defect is

$$\delta\gamma_W = \frac{1}{1+\mathfrak{r}_{F1}^2}\left(\delta\gamma_0 - \frac{1}{3}\delta\kappa_0\right). \tag{4.120}$$

Thus a pure-scale bare mixed-port deformation is harmless.

The D/N tube gives

$$\kappa_0 = \frac{4L_W^2}{\pi^2 a^2}. \tag{4.121}$$

The first-order bare slippage is equivalently the D/N similarity mismatch

$$\Xi_{\mathrm{slip}} := \Xi_\gamma - 2\Xi_L, \qquad \Xi_\gamma = \frac{\delta\ln\gamma_0}{\delta\Pi_{\tan}}, \qquad \Xi_L = \frac{\delta\ln(L_W/a)}{\delta\Pi_{\tan}}. \tag{4.122}$$

Then

$$\Delta_Q = -\frac{\sigma_*}{1-\sigma_*}\Xi_{\mathrm{slip}}\,\delta\Pi_{\tan}, \qquad N_Q - 1 = \frac{\sigma_*}{1-\sigma_*}\Xi_{\mathrm{slip}}\,\delta\Pi_{\tan}. \tag{4.123}$$

Here

$$\delta\Pi_{\tan} = 0.832409471081635\,\delta\Sigma_0 - 1.16275838754222\,\delta\mathcal{S} \tag{4.124}$$

when $\delta\mathfrak{g} = 0$.

4.10.4 Parent compensation-surface rigidity and off-family normal coordinate

On the exact parent compensation family, both the D/N tube ratio and the bare odd normalization are functions of $\mathfrak{r}$:

$$\frac{L_W}{a} = \frac{\pi}{2}\sqrt{\frac{1+\mathfrak{r}^2}{3}}, \qquad \gamma_0 = \frac{1+\mathfrak{r}^2}{9}. \tag{4.125}$$

Therefore

$$\delta \ln \gamma_0 - 2\delta \ln\left(\frac{L_W}{a}\right) = 0, \qquad \Xi_{\mathrm{slip}} = 0. \tag{4.126}$$

Furthermore, on the lower branch, the canonical-even gate $\delta\mathfrak{g} = 0$ forces $\delta\mathfrak{r} = 0$ because $d\mathfrak{g}_-/d\mathfrak{r} > 0$. Thus $\Delta_Q = 0$ at first order for tangential motion inside the exact parent compensation family.

The normal coordinate measuring departure from that family is

$$\delta_\perp := \delta\mathfrak{g} - \mathfrak{g}_-'(\mathfrak{r}_*)\delta\mathfrak{r}. \tag{4.127}$$

It is equivalently the normalized first-order parent compensation defect, since

$$\mathcal{F}(\mathfrak{g},\mathfrak{r}) = 1 + \mathfrak{r}^2 - 4(\mathfrak{g}-\mathfrak{r})^2, \qquad \delta\mathcal{F} = 4\sqrt{1+\mathfrak{r}_*^2}\,\delta_\perp. \tag{4.128}$$

In microscopic parent variables,

$$\boxed{\delta_\perp = \mathfrak{g}_*\delta \ln\left(\frac{g_q K_s}{g_s \lambda}\right) + \frac{1}{4\sqrt{1+\mathfrak{r}_*^2}}\delta \ln\left(\frac{K_s K_q}{\lambda^2}\right)}. \tag{4.129}$$

This is the final output of Part IV. The last first-order reduced defect is not a diffuse outlet problem. It is the question of whether the true moving-throat branch has $\delta_\perp = 0$, or what value (4.129) takes if it leaves the compensation family.

4.11 Verification outputs and downstream use

The usable downstream citation packet from this chapter is as follows.

1. **MTDC-T8.1**: cite for the geometry-lane firewall, the obstruction variables (ϵ_2, ϵ_4), and the forced $3/4 + 1/4$ module.
2. **MTDC-T8.2**: cite for the reduction of the retarded 2.5PN problem to $\widehat{m}_0^2\chi_Q N_Q = 1$, the outgoing DtN fingerprint, and $\chi_Q = 1$ on the canonical branch.
3. **MTDC-T8.3**: cite for the low-frequency outlet deformation algebra, the Robin and hidden-pole no-go results, and the compensated Robin–mixed branch.
4. **MTDC-T8.4**: cite for the positive mouth-source theorem, the exponential boundary-layer law, the Family–1 bias point Π_*, and the coupled mouth fixed-point equations.
5. **MTDC-T8.5**: cite for the co-evolving core–mouth canonical point, linear defect transport, D/N similarity preservation, and the final off-family scalar $\delta_\perp$.

☐ The grouped P_2 carrier is kept distinct from the axisymmetric geometry lane until the isotropic decoupling theorem is invoked.

- ☐ The $3/4 + 1/4$ conservative module is cited only under the static-geometry or geometry-decoupled hypothesis; dynamic geometry contamination is tracked by (ϵ_2, ϵ_4).
- ☐ The Family–1 support success statement is conditioned on the minimal isotropic precursor $\rho_\alpha = 4/3$, $\zeta_{\rm req} = 1/3$, not on an arbitrary support branch.
- ☐ The retarded 2.5PN gate is stated as the factorized product $\widehat{m}_0^2 \chi_Q N_Q = 1$, with source-map, conservative-normalization, and outgoing-normalization factors separated.
- ☐ The exact outgoing $l = 2$ DtN fingerprint is used only after the passive/outgoing compact branch has been selected.
- ☐ Pure Robin and standalone mixed-pole outlet deformations are recorded as no-go or noncanonical branches unless the compensated Robin–mixed constraints are imposed.
- ☐ The shell/mixed core Schur complement is treated as a reduced core model; parent-action realization still requires the corresponding overlap data.
- ☐ The finite D/N tube normalization is tied to the geometric throat ratio before it is used in the compensation surface.
- ☐ Positive mouth-source selection is kept separate from arbitrary signed source fitting; the lower compensated branch is selected by the positivity theorem.
- ☐ The exponential mouth-layer law is read as the zero-flux GNLS/localized-Maxwell boundary-layer solution, not as a free curve fit.
- ☐ Numerically located mouth and co-evolving fixed points are marked as exact equations solved numerically, not as closed-form constants.
- ☐ Linear defect transport is applied only after expanding about the renormalized co-evolving canonical point.
- ☐ The final first-order obstruction is reported as $\delta_\perp$, so future uses of $\Delta_Q = 0$ must either assume or verify tangent motion on the parent compensation family.
- ☐ Branch-realization status remains explicit whenever Part IV is cited for downstream PN, anomaly, or same-charge work.

The claim firewall is especially important for this chapter. Equations such as (4.31), (4.123), and (4.129) are exact inside their declared reduced closures. They are not, by themselves, proof that the full moving-throat PDE has realized the canonical branch. Future papers should cite Part IV for the reduction and should separately state the branch-realization status whenever they need $\Delta_Q = 0$, $N_Q = 1$, $\Xi_{\rm slip} = 0$, or $\delta_\perp = 0$.

Chapter 5

Weak-Axisymmetric Transport, Quotient Coordinates, and Orbit Closure

Stage range: 164–200

Part IV ended with the co-evolving core–mouth branch reduced to a first-order off-family normal coordinate. Part V starts from that coordinate and asks a sharper question: after the Family–1 mouth/core compensation machinery is already in place, which microscopic grouped weak-axisymmetric motions still survive as genuine reduced defects, and which motions are only similarity-orbit relabelings of the same coherent branch?

The answer is that the apparent many-variable weak-axisymmetric problem collapses to two levels. On the branch side, the grouped real P_2 carrier reduces to a single transported prefactor slope,

$$\Xi_1 = \frac{P_1}{P_0}, \tag{5.1}$$

with the weak-axisymmetric grouped signature

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1. \tag{5.2}$$

Equivalently, the grouped anisotropy lies on the fixed line

$$b_{P_0} = 3a_{P_0}. \tag{5.3}$$

On the orbit side, the coherent local branch reduces to three exact quotient coordinates,

$$\mathfrak{C}_{\mathrm{tr},*}, \qquad \mathfrak{C}_{\mathrm{nt},*}, \qquad \epsilon_\eta, \tag{5.4}$$

whose finite level sets are exactly the five-parameter microscopic similarity orbits. The final reduced verdict after this part is therefore not an eight-slot branch residual plus an ambiguous orbit condition. It is the exact four-scalar packet

$$\Delta_{\mathrm{full}} = (\Delta_Q, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta), \qquad \Delta_Q := \chi_Q - 1. \tag{5.5}$$

Part V at a glance

Primary question	After Part IV reduces the retarded obstruction to one transported branch scalar, which weak-axisymmetric motions are genuine defects and which are only similarity-orbit relabelings of the same coherent branch?

Starting inputs	The Part IV off-family normal coordinate $\delta_\perp$, the compensated Family–1 mouth/core branch, and the outgoing-normalization packet.
Delivers	The scalar transfer-shape slope $\Xi_1 = P_1/P_0$, the coherent normal form and monomial quotient coordinates, the Packet-A finish line $\Delta_Q = \chi_Q - 1$, and the final verdict packet Δ_{full}.
Used by	Part VI as the realization-compiler input and Parts VII–VIII as the strict and relaxed branch packet language.
Result anchors	**MTDC-T9**.

Stage packets.

Stages	Packet	Status	Carry-forward output
164–169	Off-family drift reduction and scalar-slippage firewall	Exact Within Closure	Rewrites $\delta_\perp$ into microscopic drifts, isolates the true scalar slippage channels, and proves pure grouped real P_2 anisotropy does not linearly feed them.
170–180	Grouped outlet/load collapse to the transfer-shape scalar	Exact Within Closure	Maps grouped outlet deformations to physical slopes, derives Ξ_{load} and the static self-similarity laws, and collapses weak-axisymmetric outgoing transport to $\Xi_1 = P_1/P_0$.
181–190	Coherent normal form, monomial coordinates, and direct compilers	Exact Within Closure	Compresses the coherent branch to $(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta)$, identifies direct monomial quotient coordinates, and records observable compilers and no-go packets.
191–197	Packet-A reduction and retarded normalization finish line	Exact Within Closure / Reduced / Controlled Reduction	Defines Packet A and Packet B, fixes the outgoing $l = 2$ fingerprint, imports the natural point-particle source map, and proves $\Delta_Q = \chi_Q - 1$ is the sole Packet-A scalar.
198–200	Orbit transport and final reference-free verdict packet	Exact Within Closure	Removes representative privilege, proves exact orbit transport, and closes the home stretch with $\Delta_{\mathrm{full}} = (\Delta_Q, q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)$.

Claim status. Mixed but predominantly Exact Within Closure**. Stages 164–200 are exact algebra inside the declared co-evolving Family–1, grouped real P_2, coherent local D/N, outgoing-prefactor, Packet-A, and Packet-B reduced hierarchies. The natural point-particle source-map use in Stage 195 is a** Reduced / Controlled Reduction **import. The actual nonlinear moving-throat PDE still has to return branch data that land on the final packet; realization by that branch remains** Open **even though the compilers and quotient/orbit theorems are exact within closure.**

What this part proves. Part V reduces the apparent many-variable weak-axisymmetric defect problem to one transported branch scalar, three exact quotient coordinates, and the final four-scalar verdict packet Δ_{full}.

What this part does not prove. It does not show that the actual branch makes that packet vanish. It only proves that this is the exact reduced packet that later realization work must evaluate.

Why later parts need it. Part VI turns $\Delta_{\rm full}$ into the realization compiler, while Parts VII and VIII use the same packet language to state strict actual-branch tests and relaxed-branch extensions.

5.1 Block contract and theorem-level output

The contract of Part V is to turn the remaining first-order weak-axisymmetric and retarded branch checks into explicit finite packets. It is not trying to prove that the completed nonlinear PDE realizes those packets. Instead it proves that, once a candidate moving-throat solution gives the required reduced branch data, the verdict can be computed by exact formulas with no remaining interpretation freedom.

The block has two sides. The first side is the weak-axisymmetric branch transport:

$$\delta_\perp \longrightarrow (\mathcal{K}_A, \mathcal{G}_A) \longrightarrow (u_2^{(1)}, P_1/P_0) \longrightarrow \Xi_{\rm load} \longrightarrow \Xi_1. \tag{5.6}$$

The second side is the coherent orbit quotient:

$$(\Sigma_{\rm tr}, \Sigma_{\rm nt}, \Sigma_\eta) \longrightarrow (\mathfrak{C}_{{\rm tr},*}, \mathfrak{C}_{{\rm nt},*}, \epsilon_\eta) \longrightarrow \mathcal{M}_+/\mathcal{G}_* \longrightarrow (q_{\rm tr}, q_{\rm nt}, q_\eta). \tag{5.7}$$

The final theorem joins these with the retarded Packet-A scalar $\Delta_Q = \chi_Q - 1$.

Theorem 5.1 (Part V weak-axisymmetric quotient and orbit-closure theorem)**.** *Inside the carried moving-throat reduced hierarchy for Stages 164–200, the following theorem chain holds.*

1. *The first off-family Family–1 normal coordinate is an explicit logarithmic transport law in the moving-throat wall/BdG/Maxwell/mixed branch variables. Arbitrary first-order isotropic bundle drift is tangent to the compensated parent family; true off-bundle motion first enters through three scalar slippages.*

2. *A pure grouped real P_2 anisotropy cannot linearly feed the scalar slippage channel. Its linear effect enters the direct outlet coefficients only through*

$$\mathcal{K}_A := \delta D_{A,2} + \frac{1}{9}\delta D_{A,0}, \qquad \mathcal{G}_A := \delta N_{A,0} - P_0 \delta D_{A,0}. \tag{5.8}$$

3. *On the canonical compensated branch, these outlet coefficients are exactly the physical grouped slopes,*

$$\mathfrak{K}_1 = -D_0 u_2^{(1)}, \qquad \mathfrak{G}_1 = D_0 P_1. \tag{5.9}$$

4. *On the even-preserving weak-axisymmetric branch, all conservative grouped transport is controlled by one static operator slope, and the sole remaining linear grouped normalization defect is*

$$\Xi_{\rm load} := \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{5.10}$$

5. *The same scalar is the weighted failure of static self-similarity and, after wall-normalized outgoing-port factorization, the logarithmic slope of the effective transfer shape:*

$$\Xi_1 = \frac{P_1}{P_0} = \frac{\delta \ln \mathcal{T}^2_{{\rm eff},A}}{\epsilon \lambda_A}. \tag{5.11}$$

6. *On the coherent local D/N branch, the weak-axisymmetric observable drifts obey the triangular normal form*

$$\Theta_1 = -C_{\rm tr}\Sigma_{\rm tr}, \qquad \Xi_1 = A_{\rm tr}\Sigma_{\rm tr} + \Sigma_{\rm nt}, \qquad \mathcal{R}_1 + \Xi_1 = -\frac{\epsilon_\eta}{1-\epsilon_\eta}\Sigma_\eta. \tag{5.12}$$

7. *The three normal-form coordinates are exactly the logarithmic drifts of three direct microscopic monomials,*

$$\delta \ln \mathfrak{C}_{{\rm tr},*} = \Sigma_{\rm tr}, \qquad \delta \ln \mathfrak{C}_{{\rm nt},*} = \Sigma_{\rm nt}, \qquad \delta \ln \epsilon_\eta = \Sigma_\eta. \tag{5.13}$$

8. *The finite level sets of these three monomials are exactly the five-parameter similarity orbits $\mathcal{G}_*$, so*

$$\mathcal{M}_+/\mathcal{G}_* \cong (\mathbb{R}_{>0})^3, \qquad (\mathfrak{C}_{{\rm tr},*}, \mathfrak{C}_{{\rm nt},*}, \epsilon_\eta) \text{ are quotient coordinates.} \tag{5.14}$$

9. *The branch-side Packet A collapses to one scalar finish line,*

$$\Delta_{\rm branch} = 0 \iff \chi_Q = 1 \iff \Delta_Q := \chi_Q - 1 = 0. \tag{5.15}$$

10. *The full reference-free reduced verdict is therefore the four-scalar packet*

$$\Delta_{\rm full}^{(x|\mathcal{O}_*)} := \big(\Delta_Q(x), q_{\rm tr}^{(x\leftarrow\mathcal{O}_*)}, q_{\rm nt}^{(x\leftarrow\mathcal{O}_*)}, q_\eta^{(x\leftarrow\mathcal{O}_*)}\big), \tag{5.16}$$

with

$$\Delta_{\rm full}^{(x|\mathcal{O}_*)} = 0 \iff \Delta_{\rm branch}(x) = 0 \text{ and } x \in \mathcal{O}_*. \tag{5.17}$$

Proof structure. Stages 164–168 translate the first off-family normal coordinate into explicit microscopic transport variables and then separate on-bundle tangent motion from off-bundle slippage. Stages 169–173 prove that grouped real P_2 anisotropy does not linearly feed scalar slippage, derive the direct outlet map, and collapse the even-preserving grouped defect to $\Xi_{\rm load}$. Stages 174–180 factor $\Xi_{\rm load}$ through wall-referenced self-similarity, outgoing-load factors, port co-loading, and the effective transfer shape, giving $\Xi_1 = P_1/P_0$. Stages 181–185 pass from physical coherent-kernel variables to the triangular normal form and then to direct microscopic monomials. Stages 186–187 prove the finite orbit–fibre theorem. Stages 188–192 build the observable, transfer-shape, finite-packet, and projector compilers. Stages 193–197 close the branch-side Packet-A finish line to $\Delta_Q = \chi_Q - 1$. Stages 198–200 remove the orbit representative and combine Packet A and Packet B into (5.16). □

5.2 Weak-axisymmetric grouped signature

5.2.1 Local goal

The first job in Part V is to connect the co-evolving Family–1 off-family scalar from Part IV to the grouped real P_2 transport language. Stage 163 had left the first true normal coordinate in the form

$$\delta_\perp = \mathfrak{g}_*\delta\ln\left(\frac{g_qK_s}{g_s\lambda}\right) + \frac{1}{4\sqrt{1+\mathfrak{r}_*^2}}\delta\ln\left(\frac{K_sK_q}{\lambda^2}\right). \tag{5.18}$$

Stage 164 shows that this is not an abstract parent-coordinate expression. With

$$\mathfrak{r} = \frac{\lambda}{\sqrt{K_sK_q}}, \qquad \mathfrak{g} = \frac{g_q\sqrt{K_s}}{g_s\sqrt{K_q}}, \qquad r_c = \frac{\lambda^2}{K_sK_q} = \mathfrak{r}^2, \tag{5.19}$$

the two channels are exactly

$$\delta\ln\left(\frac{g_qK_s}{g_s\lambda}\right) = \delta\ln\left(\frac{\mathfrak{g}}{\mathfrak{r}}\right), \qquad \delta\ln\left(\frac{K_sK_q}{\lambda^2}\right) = -\delta\ln r_c. \tag{5.20}$$

Thus $\delta_\perp$ measures relative mouth-coupling drift and hybridization drift.

5.2.2 Explicit branch-variable form

On the healing-locked wall/BdG/Maxwell/mixed branch, the same scalar becomes

$$\begin{aligned}\delta_\perp = A_*\,\delta\ln\!\left(\frac{\mathcal{Z}_q}{\rho_w}\right) + 3A_*\,\delta\ln c_{s,w} + B_*\,\delta\ln c_s - \mathfrak{g}_*\,\delta\ln\mathcal{T}_m \\ -(\mathfrak{g}_*+B_*)\delta\ln v_{w0} - 2A_*\,\delta\ln a - C_*\,\delta\ln L_W,\end{aligned} \tag{5.21}$$

where

$$A_* := \mathfrak{g}_* + \frac{1}{4\sqrt{1+\mathfrak{r}_*^2}}, \qquad B_* := \frac{1}{2\sqrt{1+\mathfrak{r}_*^2}}, \qquad C_* := 2\mathfrak{g}_* + \frac{3}{4\sqrt{1+\mathfrak{r}_*^2}}. \tag{5.22}$$

The exact tangency condition $\delta_\perp = 0$ is therefore a co-transport law for $\mathcal{T}_m$ relative to $(\mathcal{Z}_q, \rho_w, c_{s,w}, c_s, v_{w0}, a, L_W)$, not a free adjustment.

Stages 165–167 then show that the explicit lower compensated branch is even more constrained. The D/N tube length co-transports with the mouth radius,

$$\delta\ln L_W = \delta\ln a, \tag{5.23}$$

and the exact branch laws reduce the apparent branch drift problem to four irreducible variables. Stage 166 chooses the four bundle observables

$$\Theta_w, \qquad K_s, \qquad K_q, \qquad P_0 = \frac{N_0}{D_0} \tag{5.24}$$

as the right variables to invert. Stage 167 proves that arbitrary first-order drift of these four bundle observables leaves the compensation ratios tangent to the parent family:

$$\delta\ln r_c = 0, \qquad \delta\ln\mathfrak{r} = 0, \qquad \delta\ln\mathfrak{g} = 0 \quad \text{for pure bundle drift.} \tag{5.25}$$

Consequently, any first-order failure after Stage 167 must be genuinely off-bundle.

5.2.3 Off-bundle slippages and grouped signature

Stage 168 introduces the three scalar slippages

$$\varepsilon_L, \qquad \varepsilon_v, \qquad \varepsilon_T, \tag{5.26}$$

which measure failure of the exact lower-branch laws for axial length, mixed background speed, and mouth traction. Stage 169 then proves a key firewall: a pure weak grouped real P_2 anisotropy cannot linearly source these scalar slippages. The reason is angular. The weak grouped perturbation has zero weighted trace at first order; scalar feed-down can only be built from quadratic grouped invariants such as

$$\mathcal{I}[x,y] = \frac{1}{5}\delta x^T G_{\mathrm{grp}}\delta y = 4a_x a_y + \frac{4}{5}b_x b_y, \qquad G_{\mathrm{grp}} = \mathrm{diag}(1,2,2). \tag{5.27}$$

Thus linear grouped real P_2 transport enters the direct outlet coefficients rather than the scalar off-bundle slippage channel.

The grouped weak-axisymmetric signature is inherited from the first axisymmetric quadrupole perturbation:

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1. \tag{5.28}$$

For any first-order grouped quantity

$$x_A = x_0 + \epsilon\lambda_A x_1 + O(\epsilon^2), \tag{5.29}$$

the weighted grouped anomalies obey

$$a_x = \frac{\epsilon}{4} x_1, \qquad b_x = \frac{3\epsilon}{4} x_1, \qquad b_x = 3a_x. \tag{5.30}$$

This fixed line is the weak-axisymmetric branch signature used throughout the rest of Part V.

Claim status. Stages 164–169 are Exact Within Closure **inside the explicit co-evolving Family–1 throat-core closure and the grouped real P_2 weak-axisymmetric expansion. They do not prove that the actual nonlinear PDE produces a particular slippage; they identify the exact variables through which such slippage would appear.**

5.3 Even-preserving and hidden-even gates

5.3.1 Direct outlet map

Stages 170–171 move the linear grouped problem from scalar slippage to direct outlet data. For each grouped lane A, define

$$\mathcal{K}_A := \delta D_{A,2} + \frac{1}{9}\delta D_{A,0}, \qquad \mathcal{G}_A := \delta N_{A,0} - P_0 \delta D_{A,0}. \tag{5.31}$$

The hidden even deformation and hidden odd normalization deformation are then

$$\delta\kappa_W^{(A)} = \frac{3(1-\sigma_*)}{\sigma_* D_0}\mathcal{K}_A, \qquad \delta\gamma_W^{(A)} = -\frac{1-\sigma_*}{9\sigma_* N_0}\mathcal{G}_A. \tag{5.32}$$

Stage 171 decomposes these into microscopic sectors:

$$\mathcal{K}_A = \mathcal{W}_A - \mathcal{B}_A - \mathcal{Z}_A, \tag{5.33}$$

with

$$\mathcal{W}_A := \frac{1}{9}\delta K_A - \delta M_A, \qquad \mathcal{B}_A := \delta B_{A,2} + \frac{1}{9}\delta B_{A,0}, \qquad \mathcal{Z}_A := \delta Z_{A,2} + \frac{1}{9}\delta Z_{A,0}, \tag{5.34}$$

and

$$\mathcal{G}_A = -P_0 \delta K_A + P_0 \delta B_{A,0} + \mathcal{N}_A, \qquad \mathcal{N}_A := \delta N_{A,0} + P_0 \delta Z_{A,0}. \tag{5.35}$$

This identifies the only four microscopic bundles that can generate a linear grouped outlet obstruction: wall baseline, BdG support, conservative Maxwell/mixed, and outgoing-transfer loading.

5.3.2 Physical slope collapse

Stage 172 rewrites the same obstruction in physical grouped-response variables. For each grouped lane,

$$u_2^{(A)} = -\frac{D_{A,2}}{D_{A,0}}, \qquad u_4^{(A)} = \frac{D_{A,2}^2 - D_{A,0}D_{A,4}}{D_{A,0}^2}, \qquad P_0^{(A)} = \frac{N_{A,0}}{D_{A,0}}. \tag{5.36}$$

On the canonical compensated branch, the two microscopic amplitudes are simply

$$\mathfrak{K}_1 = -D_0 u_2^{(1)}, \qquad \mathfrak{G}_1 = D_0 P_1. \tag{5.37}$$

Equivalently,

$$\delta\kappa_W^{(A)} = -\frac{3(1-\sigma_*)}{\sigma_*}\delta u_2^{(A)}, \qquad \delta\gamma_W^{(A)} = -\frac{1-\sigma_*}{9\sigma_*}\frac{\delta P_0^{(A)}}{P_0}. \tag{5.38}$$

So the full linear grouped outlet problem is now measurable by $u_2^{(1)}$ and P_1/P_0.

5.3.3 Even-preserving branch

Stage 173 then computes those physical slopes directly from the weak-axisymmetric grouped operator expansion

$$D_{A,0} = D_0 + \epsilon\lambda_A D_{01} + O(\epsilon^2), \quad D_{A,2} = D_2 + \epsilon\lambda_A D_{21} + O(\epsilon^2), \quad D_{A,4} = D_4 + \epsilon\lambda_A D_{41} + O(\epsilon^2), \tag{5.39}$$

$$N_{A,0} = N_0 + \epsilon\lambda_A N_{01} + O(\epsilon^2). \tag{5.40}$$

The exact slope formulas are

$$u_2^{(1)} = -\frac{D_{21} + u_2 D_{01}}{D_0}, \tag{5.41}$$

$$u_4^{(1)} = -\frac{5D_{01} + 18D_{21} + 81D_{41}}{81D_0} \quad \text{on } (u_2, u_4) = \left(\frac{1}{9}, \frac{4}{81}\right), \tag{5.42}$$

and

$$\frac{P_1}{P_0} = \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{5.43}$$

If the canonical conservative even fingerprint is preserved at first order, then

$$D_{21} = -\frac{D_{01}}{9}. \tag{5.44}$$

The hidden-even consistency relation $u_4^{(1)} = (8/9)u_2^{(1)}$ is exactly

$$D_{41} = \frac{2}{3}D_{21} + \frac{1}{27}D_{01}, \tag{5.45}$$

and therefore on the even-preserving branch

$$D_{41} = -\frac{D_{01}}{27}. \tag{5.46}$$

Thus the conservative grouped response is frozen to canonical order, and the only surviving linear grouped defect is the static loading mismatch

$$\Xi_{\text{load}} := \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0}. \tag{5.47}$$

Its lane form is

$$\Delta_Q^{(20)} = \epsilon\Xi_{\text{load}}, \qquad \Delta_Q^{(21)} = \frac{\epsilon}{2}\Xi_{\text{load}}, \qquad \Delta_Q^{(22)} = -\epsilon\Xi_{\text{load}}. \tag{5.48}$$

Claim status. Stages 170–173 are Exact Within Closure **in the grouped low-frequency bundle. The even-preserving condition is a branch restriction, not an unconditional statement that the completed PDE must satisfy it.**

5.4 Outgoing-load scalar ($\Xi_1 = P_1/P_0$)

5.4.1 Self-similarity form of the loading mismatch

Stage 174 rewrites the mismatch (5.47) as failure of static self-similarity. Let

$$\delta_K := \frac{K_1}{K}, \qquad \delta_B := \frac{B_0^{(1)}}{B_0}, \qquad \delta_Z := \frac{Z_0^{(1)}}{Z_0}, \qquad \delta_N := \frac{N_1}{N_0}, \tag{5.49}$$

and

$$\omega_B := \frac{B_0}{D_0}, \qquad \omega_Z := \frac{Z_0}{D_0}, \qquad D_0 = K - B_0 - Z_0. \tag{5.50}$$

Then

$$\Xi_{\mathrm{load}} = (\delta_N - \delta_K) + \omega_B(\delta_B - \delta_K) + \omega_Z(\delta_Z - \delta_K). \tag{5.51}$$

Define the wall-referenced self-similarity defect fields

$$\Sigma_\alpha^{(B)} := 2\delta \ln\left(\frac{c_\alpha}{\varpi_\alpha}\right) - \delta_K, \quad \Sigma_r^{(Z)} := \delta \ln\left(\frac{Q_r}{\Delta_r}\right) - \delta_K, \quad \Sigma_r^{(N)} := 2\delta \ln\left(\frac{P_r}{\Delta_r}\right) - \delta_K. \tag{5.52}$$

Then

$$\Xi_{\mathrm{load}} = \sum_r \rho_r^{(N)} \Sigma_r^{(N)} + \omega_B \sum_\alpha \rho_\alpha^{(B)} \Sigma_\alpha^{(B)} + \omega_Z \sum_r \rho_r^{(Z)} \Sigma_r^{(Z)}. \tag{5.53}$$

Static self-similarity is the sufficient condition that all three defect families vanish. On the even-preserving branch it implies $\Xi_{\mathrm{load}} = 0$ and hence no first-order grouped normalization defect.

5.4.2 Outgoing-load theorem and mixed-leg law

Stages 175–176 factor the port objects relative to the wall baseline. The remaining defect is controlled by the outgoing load factor

$$\Lambda_r := \frac{P_r}{\Delta_r}. \tag{5.54}$$

On conservative-shape-preserving branches the full remaining defect is carried only by the outgoing load slope relative to the wall. Stage 176 factors that load into a mixed leg, an interference ratio, and a hybridization ratio. In the notation of the reduced one-port block,

$$\frac{\Lambda_r^2}{K} = \mathcal{M}_r^2 \frac{(1+\mathcal{I}_r)^2}{(1-\mathcal{H}_r)^2}. \tag{5.55}$$

Consequently

$$\Sigma_r^{(N)} = 2\delta \ln \mathcal{M}_r + \frac{2\mathcal{I}_r}{1+\mathcal{I}_r} \delta \ln \mathcal{I}_r + \frac{2\mathcal{H}_r}{1-\mathcal{H}_r} \delta \ln \mathcal{H}_r. \tag{5.56}$$

If the interference and hybridization ratios are rigid, the whole outgoing defect is the raw mixed-leg drift. The zero-defect condition becomes the square-root mixed-leg law

$$\frac{G_W}{\Omega_W^2} \propto \sqrt{K}. \tag{5.57}$$

This is useful because it identifies a concrete microscopic path to zero defect rather than just a cancellation among compiled coefficients.

5.4.3 Collapse to one scalar amplitude

Stage 177 applies the weak-axisymmetric signature (5.28). Every microscopic outgoing slippage inherits the same lane pattern, so the full remaining grouped defect is one scalar amplitude:

$$\Xi_{\mathrm{load}}^{(A)} = \epsilon \lambda_A \Xi_1. \tag{5.58}$$

The same scalar is the physical outgoing-prefactor slope:

$$\Xi_1 = \frac{P_1}{P_0}. \tag{5.59}$$

Stage 178 rewrites it portwise. For

$$N^{(r)}_{A,0} = \frac{P^2_{A,r}}{\Delta^2_{A,r}}, \tag{5.60}$$

define the port slope ν_r by

$$\delta \ln N^{(r)}_{A,0} = \epsilon\lambda_A \nu_r. \tag{5.61}$$

Then

$$\Xi_1 = \bar{\nu}_N - \kappa_1, \tag{5.62}$$

where κ_1 is the wall-baseline static slope and $\bar{\nu}_N$ is the outgoing-weighted average. Thus zero defect is exact co-loading of outgoing transfer with the wall baseline.

Stages 179–180 put the same statement into transfer-shape language. Each port factors as

$$N^{(r)}_{A,0} = K_A \mathcal{T}^2_{A,r}, \qquad \mathcal{T}_{A,r} = \frac{\widehat{G}_{W,A,r} + \widehat{R}_{A,r}\widehat{G}_{U,A,r}}{1 - \widehat{R}^2_{A,r}}. \tag{5.63}$$

Therefore

$$\Xi_1 = 2\sum_r \rho^{(N)}_r \tau_r, \qquad \tau_r := \frac{\delta \ln \mathcal{T}_{A,r}}{\epsilon\lambda_A}. \tag{5.64}$$

Equivalently, all ports collapse to one effective transfer shape,

$$\mathcal{T}^2_{\mathrm{eff},A} := \sum_r \mathcal{T}^2_{A,r} = \frac{N_{A,0}}{K_A}, \tag{5.65}$$

and

$$\Xi_1 = \frac{\delta \ln \mathcal{T}^2_{\mathrm{eff},A}}{\epsilon\lambda_A}. \tag{5.66}$$

On the actual minimal one-port continuum branch,

$$\mathcal{T}^2_A = \frac{Z_{W,A}(1+\rho_A)^2}{\Omega^2_{W,A}(1-\epsilon_{W,A})^2} = \frac{27\pi^2 G c_s^5}{20a^5c^5}\frac{1-\epsilon_{\eta,A}}{R_{\mathrm{target},A}}. \tag{5.67}$$

So the remaining theorem gate is to determine whether the actual weak-axisymmetric branch keeps $\mathcal{T}_{\mathrm{eff}}$ rigid.

Claim status. Stages 174–180 are Exact Within Closure **inside the weak-axisymmetric grouped bundle and the declared outgoing-port factorization. The square-root mixed-leg law is a sufficient zero-defect branch condition under interference/hybridization rigidity; actual realization is not assumed.**

5.5 Coherent-kernel slippages

5.5.1 Actual coherent local D/N branch

Stage 181 inserts the coherent local D/N tracking branch into (5.67). On that branch,

$$\mathcal{T}^2 = \frac{Z_W(1+\chi_0)^2}{\Omega^2_W(1-\epsilon)^2} = \frac{27\pi^2 G c_s^5}{20a^5c^5}\frac{1-\epsilon_\eta}{R_{\mathrm{target}}}, \tag{5.68}$$

where

$$\epsilon = \epsilon_W \left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right). \tag{5.69}$$

The weak-axisymmetric defect is

$$\Xi_1 = \zeta_Z - \omega_W + \frac{2\chi_1}{1+\chi_0} + \frac{2\epsilon_1}{1-\epsilon}, \tag{5.70}$$

with

$$\epsilon_1 = \left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right)\varepsilon_W - \frac{2\epsilon_W}{11(1+\delta_U)^2}\delta_{U,1}. \tag{5.71}$$

The support enhancement ratio ζ drops out identically:

$$\partial_\zeta \ln \mathcal{T}^2 = 0, \qquad \partial_\zeta \ln R_{\text{target}} = 0, \qquad \partial_\zeta \Xi_1 = 0. \tag{5.72}$$

Thus coherent support can help the steady normalization load, but it cannot repair the weak-axisymmetric prefactor slope at first order.

5.5.2 Microscopic slippage decomposition

Stage 182 pushes (5.70) down to microscopic coherent-kernel slippages. The direct grouped defect depends on four slippages,

$$\Sigma_Z, \qquad \Sigma_\chi, \qquad \Sigma_\epsilon, \qquad \Sigma_\delta, \tag{5.73}$$

and selected-branch dressing introduces one more,

$$\Sigma_\eta. \tag{5.74}$$

The direct defect law is

$$\Xi_1 = \Sigma_Z + \frac{2\chi_0}{1+\chi_0}\Sigma_\chi + \frac{2\epsilon_W}{1-\epsilon}\left[\frac{11+9\delta_U}{11(1+\delta_U)}\Sigma_\epsilon - \frac{2\delta_U}{11(1+\delta_U)^2}\Sigma_\delta\right]. \tag{5.75}$$

The tracking drift is the special combination

$$\Sigma_{\text{tr}} := (1+\chi_0)\Sigma_\delta + (1+\delta_U)\Sigma_\chi, \tag{5.76}$$

which controls the tracking observable

$$\Theta_1 = -\frac{\chi_0\delta_U}{(1+\chi_0)(1+\delta_U)(1+\chi_0+\delta_U)}\Sigma_{\text{tr}}. \tag{5.77}$$

So exact tracking rigidity means $\Sigma_{\text{tr}} = 0$, but that alone does not generally force $\Xi_1 = 0$; a genuine nontracking transfer-shape slippage can remain.

Claim status. Stages 181–182 are Exact Within Closure **for the coherent local D/N tracking branch. The support-blindness statement is exact inside that branch; it does not assert that support is irrelevant to all other branch tests.**

5.6 Triangular normal form

5.6.1 Normal-form coordinates

Stage 183 packages the five slippages into three branch-adapted coordinates. The nontracking transfer-shape slippage is defined by

$$\begin{aligned}\Sigma_{\mathrm{nt}} := \Sigma_Z &+ \frac{2\epsilon_W}{1-\epsilon}\frac{11+9\delta_U}{11(1+\delta_U)}\Sigma_\epsilon \\ &- \left[\frac{2\chi_0}{1+\delta_U} + \frac{4\epsilon_W\delta_U}{11(1-\epsilon)(1+\delta_U)^2}\right]\Sigma_\delta.\end{aligned} \tag{5.78}$$

The three branch-adapted coordinates are therefore

$$(\Sigma_{\mathrm{tr}}, \Sigma_{\mathrm{nt}}, \Sigma_\eta). \tag{5.79}$$

In these coordinates the observable drifts have the exact triangular form

$$\Theta_1 = -C_{\mathrm{tr}}\Sigma_{\mathrm{tr}}, \qquad \Xi_1 = A_{\mathrm{tr}}\Sigma_{\mathrm{tr}} + \Sigma_{\mathrm{nt}}, \qquad \mathcal{R}_1 + \Xi_1 = -\frac{\epsilon_\eta}{1-\epsilon_\eta}\Sigma_\eta, \tag{5.80}$$

where

$$C_{\mathrm{tr}} = \frac{\chi_0\delta_U}{(1+\chi_0)(1+\delta_U)(1+\chi_0+\delta_U)}, \qquad A_{\mathrm{tr}} = \frac{2\chi_0}{(1+\chi_0)(1+\delta_U)}. \tag{5.81}$$

This is the first exact three-scalar normal form of the weak-axisymmetric coherent branch.

5.6.2 Inversion and interpretation

The triangular structure makes the three physical failure modes transparent.

1. Σ_{tr} is tracking failure. It is measured directly by Θ_1.
2. Σ_{nt} is nontracking transfer-shape failure. It is the residual part of Ξ_1 after subtracting tracking feed-through.
3. Σ_η is dressing failure. It measures the mismatch between direct transfer-shape drift and selected-branch demand through $\mathcal{R}_1 + \Xi_1$.

Zero defect in this normal form is

$$\Theta_1 = \Xi_1 = \mathcal{R}_1 = 0 \iff \Sigma_{\mathrm{tr}} = \Sigma_{\mathrm{nt}} = \Sigma_\eta = 0. \tag{5.82}$$

5.6.3 Branch-observable coordinates

Stage 184 identifies exact branch composites whose logarithmic drifts are the same three coordinates. The tracking factor is

$$R_{\mathrm{tr}} = \frac{1+\chi_0/(1+\delta_U)}{1+\chi_0} = \frac{1+\chi_0+\delta_U}{(1+\chi_0)(1+\delta_U)}. \tag{5.83}$$

The corrected nontracking composite is

$$\mathfrak{N}_* := \mathcal{T}^2 R_{\mathrm{tr}}^{B_*}, \qquad B_* := \frac{2(1+\chi_{0,*}+\delta_{U,*})}{\delta_{U,*}}. \tag{5.84}$$

After a harmless positive normalization of the tracking factor,

$$\mathfrak{T}_* := R_{\mathrm{tr}}^{-C_*}, \qquad C_* := \frac{(1+\chi_{0,*})(1+\delta_{U,*})(1+\chi_{0,*}+\delta_{U,*})}{\chi_{0,*}\delta_{U,*}}, \tag{5.85}$$

the branch-coordinate laws are

$$\delta \ln \mathfrak{T}_* = \Sigma_{\mathrm{tr}}, \qquad \delta \ln \mathfrak{N}_* = \Sigma_{\mathrm{nt}}, \qquad \delta \ln \epsilon_\eta = \Sigma_\eta. \tag{5.86}$$

So the next continuation point is no longer a list of microscopic slippages; it is drift of three exact branch composites.

Claim status. Stages 183–184 are Exact Within Closure **inside the coherent local D/N branch. The normal form is a compiler from reduced coherent variables to observables; it does not assert that the completed PDE has zero coordinates.**

5.7 Microscopic monomial invariants

5.7.1 Direct monomial coordinates

Stage 185 pushes the three branch composites to direct microscopic monomials. At reference-branch order,

$$\delta \ln \mathfrak{C}_{\mathrm{tr},*} = \Sigma_{\mathrm{tr}}, \qquad \delta \ln \mathfrak{C}_{\mathrm{nt},*} = \Sigma_{\mathrm{nt}}, \qquad \delta \ln \epsilon_\eta = \Sigma_\eta. \tag{5.87}$$

The tracking monomial is

$$\mathfrak{C}_{\mathrm{tr},*} := \chi_0^{1+\delta_{U,*}} \delta_U^{1+\chi_{0,*}}, \tag{5.88}$$

and the nontracking monomial is

$$\mathfrak{C}_{\mathrm{nt},*} := \frac{Z_W}{\Omega_W^2} \epsilon_W^{E_*} \delta_U^{-F_*}, \tag{5.89}$$

with

$$E_* = \frac{2\epsilon_{W,*}}{1-\epsilon_*} \frac{11+9\delta_{U,*}}{11(1+\delta_{U,*})}, \qquad F_* = \frac{2\chi_{0,*}}{1+\delta_{U,*}} + \frac{4\epsilon_{W,*}\delta_{U,*}}{11(1-\epsilon_*)(1+\delta_{U,*})^2}. \tag{5.90}$$

Using the coherent definitions, these become direct microscopic monomials in the positive variables

$$\mathbf{x} = (\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})}, K_W^{(\mathrm{eff})}, \mu_W, T_U). \tag{5.91}$$

Explicitly,

$$\mathfrak{C}_{\mathrm{tr},*} = \left(\frac{\gamma c_{\eta U}}{K_U}\right)^{1+\delta_{U,*}} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{1+\chi_{0,*}}, \tag{5.92}$$

$$\mathfrak{C}_{\mathrm{nt},*} = \left(\frac{\lambda_W^2 \mu_W}{K_\eta^{(\mathrm{eff})}(K_W^{(\mathrm{eff})})^2}\right) \left(\frac{\gamma^2 \lambda_W^2 \sigma}{K_U K_W^{(\mathrm{eff})}}\right)^{E_*} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{-F_*}, \tag{5.93}$$

and

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{(\mathrm{eff})}}. \tag{5.94}$$

5.7.2 Linear monomial-drift map

Introduce the grouped weak-axisymmetric logarithmic drift vector

$$\delta\mathbf{x} = \begin{pmatrix} \lambda_1 \\ c_1 \\ \gamma_1 \\ \kappa_U \\ \kappa_\eta \\ \kappa_W \\ \mu_1 \\ \tau_1 \end{pmatrix} = \begin{pmatrix} \delta \ln \lambda_W \\ \delta \ln c_{\eta U} \\ \delta \ln \gamma \\ \delta \ln K_U \\ \delta \ln K_\eta^{(\mathrm{eff})} \\ \delta \ln K_W^{(\mathrm{eff})} \\ \delta \ln \mu_W \\ \delta \ln T_U \end{pmatrix}_{\mathrm{grp}}. \tag{5.95}$$

Then

$$\begin{pmatrix} \delta \ln \mathfrak{C}_{\mathrm{tr},*} \\ \delta \ln \mathfrak{C}_{\mathrm{nt},*} \\ \delta \ln \epsilon_\eta \end{pmatrix} = M_* \delta\mathbf{x}, \tag{5.96}$$

with

$$M_* = \begin{pmatrix} 0 & 1+\delta_{U,*} & 1+\delta_{U,*} & -(2+\chi_{0,*}+\delta_{U,*}) & 0 & 0 & 0 & 1+\chi_{0,*} \\ 2(1+E_*) & 0 & 2E_* & F_*-E_* & -1 & -(2+E_*) & 1 & -F_* \\ 0 & 2 & 0 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}. \tag{5.97}$$

A convenient dependent minor has determinant

$$\det M_*^{(\tau,\kappa_\eta,\mu_1)} = 1+\chi_{0,*} > 0. \tag{5.98}$$

Therefore M_* has rank three and a five-dimensional kernel. The zero-defect condition is

$$\Theta_1 = \Xi_1 = \mathcal{R}_1 = 0 \iff M_*\delta\mathbf{x} = 0. \tag{5.99}$$

Claim status. Stage 185 is Exact Within Closure **in the reference coherent branch. The monomial exponents are fixed by the branch normal form, and M_* is an exact exponent matrix.**

5.8 Similarity-orbit theorem

5.8.1 Finite similarity family

Stage 186 integrates the tangent kernel of M_* to a finite five-parameter similarity family. Let

$$(\Lambda, C, \Gamma, U, W) \tag{5.100}$$

be free logarithmic co-scalings of $(\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{(\mathrm{eff})})$. Monomial preservation fixes the dependent scalings:

$$K_\eta^{(\mathrm{eff})} \mapsto e^{2C-U} K_\eta^{(\mathrm{eff})}, \tag{5.101}$$

$$T_U \mapsto \exp\left[U - \frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\Gamma + C - U)\right] T_U, \tag{5.102}$$

$$\mu_W \mapsto e^{M(\Lambda,C,\Gamma,U,W)}\mu_W, \tag{5.103}$$

where

$$\begin{aligned} M(\Lambda, C, \Gamma, U, W) &= 2C - U + 2W - 2\Lambda - E_*(2\Gamma + 2\Lambda - U - W) \\ &\quad - F_* \frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\Gamma + C - U). \end{aligned} \tag{5.104}$$

Call this family $\mathcal{G}_*$. It preserves $(\mathfrak{C}_{\mathrm{tr},*}, \mathfrak{C}_{\mathrm{nt},*}, \epsilon_\eta)$ exactly.

5.8.2 Orbit–quotient closure

Stage 187 proves the converse. On the positive microscopic state space

$$\mathcal{M}_+ = \{(\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})}, K_W^{(\mathrm{eff})}, \mu_W, T_U) > 0\}, \tag{5.105}$$

define

$$\mathcal{I}(x) = (\mathfrak{C}_{\mathrm{tr},*}(x), \mathfrak{C}_{\mathrm{nt},*}(x), \epsilon_\eta(x)). \tag{5.106}$$

For two positive states $x, \widetilde{x}$, equality of the three invariants is exactly

$$M_*\Delta\mathbf{x} = 0, \qquad \Delta\mathbf{x} := \ln(\widetilde{x}/x). \tag{5.107}$$

Solving these finite equations gives precisely the dependent transformations (5.101)–(5.103). Hence

$$\mathcal{I}(\widetilde{x}) = \mathcal{I}(x) \iff \widetilde{x} \in \mathcal{G}_* \cdot x. \tag{5.108}$$

Therefore

$$\mathcal{M}_+/\mathcal{G}_* \cong (\mathbb{R}_{>0})^3, \tag{5.109}$$

with quotient coordinates (5.4). The first grouped weak-axisymmetric observables are the infinitesimal motion in these quotient coordinates:

$$\Theta_1 = -C_{\mathrm{tr}}\delta \ln \mathfrak{C}_{\mathrm{tr},*}, \tag{5.110}$$

$$\Xi_1 = A_{\mathrm{tr}}\delta \ln \mathfrak{C}_{\mathrm{tr},*} + \delta \ln \mathfrak{C}_{\mathrm{nt},*}, \tag{5.111}$$

$$\mathcal{R}_1 + \Xi_1 = -\frac{\epsilon_{\eta,*}}{1-\epsilon_{\eta,*}}\delta \ln \epsilon_\eta. \tag{5.112}$$

5.8.3 Projector calculus

Stages 191–192 then turn the quotient coordinates into an exact finite packet and an exact microscopic projector split. With the finite log-ratio vector ordered as

$$\Delta\mathbf{x} = (\Delta_\lambda, \Delta_c, \Delta_\gamma, \Delta_U, \Delta_{K_\eta}, \Delta_W, \Delta_\mu, \Delta_T)^T, \tag{5.113}$$

the quotient packet is

$$\mathbf{q} = (q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)^T = M_*\Delta\mathbf{x}. \tag{5.114}$$

Choosing the dependent triple $(T_U, K_\eta^{(\mathrm{eff})}, \mu_W)$, the exact pivot block is

$$P_{(T,K_\eta,\mu)} = \begin{pmatrix} 1+\chi_{0,*} & 0 & 0 \\ -F_* & -1 & 1 \\ 0 & -1 & 0 \end{pmatrix}, \qquad \det P_{(T,K_\eta,\mu)} = 1+\chi_{0,*} > 0. \tag{5.115}$$

The canonical section $S_{(T,K_\eta,\mu)}$ is the right inverse satisfying $M_* S_{(T,K_\eta,\mu)} = I_3$. The exact projectors are

$$Q_{\text{quot}} := S_{(T,K_\eta,\mu)} M_*, \qquad O_{\text{orb}} := I_8 - Q_{\text{quot}}. \tag{5.116}$$

They obey

$$Q_{\text{quot}}^2 = Q_{\text{quot}}, \qquad O_{\text{orb}}^2 = O_{\text{orb}}, \qquad M_* O_{\text{orb}} = 0, \qquad M_* Q_{\text{quot}} = M_*. \tag{5.117}$$

The quotient-failure piece is supported only on the dependent triple:

$$(\Delta_T)_{\text{fail}} = \frac{q_{\text{tr}}}{1+\chi_{0,*}}, \qquad (\Delta_{K_\eta})_{\text{fail}} = -q_\eta, \tag{5.118}$$

$$(\Delta_\mu)_{\text{fail}} = \frac{F_*}{1+\chi_{0,*}} q_{\text{tr}} + q_{\text{nt}} - q_\eta. \tag{5.119}$$

Thus

$$\mathbf{q} = 0 \iff M_* \Delta\mathbf{x} = 0 \iff Q_{\text{quot}} \Delta\mathbf{x} = 0 \iff \Delta\mathbf{x} \in \ker M_*. \tag{5.120}$$

Claim status. Stages 186–192 are Exact Within Closure **for the positive coherent reduced state space. The word "orbit" here means a finite reduced monomial-preserving similarity class, not a fundamental gauge symmetry of the unreduced parent PDE.**

5.9 Four-scalar packet (Δ_{full})

5.9.1 Packet A: conservative target and outgoing finish line

Stages 193–197 sharpen the branch-side Packet A. Start from the Stage 191 branch residual

$$\Delta_{\text{branch}} = (a_2, b_2, a_4, b_4, a_{P_0}, b_{P_0}, \Delta_{\text{pole}}, \Delta_{\text{norm}}). \tag{5.121}$$

Stage 193 freezes the conservative target surface

$$a_2 = b_2 = a_4 = b_4 = 0, \qquad \Delta_{\text{pole}} := \bar\nu_4 - 4\bar\nu_2^2 = 0. \tag{5.122}$$

On the isotropic branch this is equivalent to

$$D_0 D_4 + 3D_2^2 = 0, \tag{5.123}$$

or, in bundle variables,

$$D_0(B_4 + Z_4) = 3(M + B_2 + Z_2)^2. \tag{5.124}$$

The resulting conservative carrier is

$$\widehat{Y}_Q^{\text{cons}}(\omega) = \frac{3}{4} + \frac{1}{4}\frac{1}{1-\omega^2/\Omega_Q^2}. \tag{5.125}$$

Stage 193 also proves the scalar/geometry firewall: if weak anisotropy induces the first $l = 0 \leftrightarrow l = 2$ mixing, then Schur completion gives

$$\mathcal{D}_{2,\text{eff}}(\omega,\chi) = \mathcal{D}_2(\omega) I_3 - \chi^2 C(\omega)\mathcal{D}_0(\omega)^{-1} C(\omega)^T, \tag{5.126}$$

so

$$\partial_\chi \mathcal{D}_{2,\text{eff}}(\omega,\chi)|_{\chi=0} = 0. \tag{5.127}$$

The scalar/geometry lane cannot linearly contaminate the grouped $l = 2$ conservative carrier.

Stage 194 attaches the outgoing $l = 2$ DtN fingerprint. With

$$z := \frac{a\omega}{c_s}, \tag{5.128}$$

the exact outgoing operator is

$$\Lambda_2^{\text{out}}(z) = z\frac{d}{dz}\ln h_2^{(1)}(z), \tag{5.129}$$

with normalized response

$$\widehat{Y}_2^{\text{out}}(z) = 1 + \frac{z^2}{9} + \frac{4z^4}{81} + i\frac{z^5}{27} + O(z^6). \tag{5.130}$$

Matching this to the retarded grouped module fixes

$$\chi_Q = 1 \tag{5.131}$$

on the canonical compact passive/outgoing branch. More generally, for the isotropic deformation

$$\Lambda_2^{\text{def}}(z) = S\Lambda_2^{\text{out}}(\beta z) + \Sigma_0 + \Sigma_2 z^2 + \Sigma_4 z^4 + i\Sigma_5 z^5 + O(z^6), \tag{5.132}$$

canonical even matching forces Σ_2 and Σ_4, while the outgoing scalar is

$$\chi_Q = \frac{3(S\beta^5 + 9\Sigma_5)}{3S - \Sigma_0}, \qquad \chi_Q - 1 = \frac{3S(\beta^5 - 1) + \Sigma_0 + 27\Sigma_5}{3S - \Sigma_0}. \tag{5.133}$$

Thus the only isotropic DtN data that can move χ_Q, while preserving the canonical even moments, are β, Σ_0, and Σ_5.

Stage 195 factorizes the observable odd closure:

$$\widehat{m}_0^2 \chi_Q N_Q = 1. \tag{5.134}$$

On the natural point-particle source map,

$$\widehat{m}_0 \to 1, \qquad N_Q = \chi_Q^{-1}. \tag{5.135}$$

Stage 196 proves that additional isotropic retarded tails beginning at $O(\omega^7)$ do not alter any coefficient through the point-particle 2.5PN order. Stage 197 therefore gives the exact Packet-A finish line

$$\Delta_{\text{branch}} = 0 \iff \chi_Q = 1 \iff \Delta_Q := \chi_Q - 1 = 0. \tag{5.136}$$

5.9.2 Packet B: finite and pairwise orbit lock

Stages 198–199 complete the orbit side. With the same positive microscopic state vector (5.91), keep the free coordinates

$$(\lambda_W, c_{\eta U}, \gamma, K_U, K_W^{(\text{eff})}) \tag{5.137}$$

and the dependent triple

$$(T_U, K_\eta^{(\text{eff})}, \mu_W). \tag{5.138}$$

For a fixed invariant triple $(\mathfrak{C}_{\text{tr},*}, \mathfrak{C}_{\text{nt},*}, \epsilon_{\eta,*})$, Stage 198 solves the exact dependent orbit law and defines mismatch ratios

$$(m_T, m_K, m_\mu). \tag{5.139}$$

The logarithmic quotient chart is

$$q_{\mathrm{tr}} = (1+\chi_{0,*})\ln m_T, \qquad q_\eta = -\ln m_K, \qquad q_{\mathrm{nt}} = \ln m_\mu - \ln m_K - F_* \ln m_T. \tag{5.140}$$

Orbit lock is

$$\Delta_{\mathrm{orbit}} = 0 \iff m_T = m_K = m_\mu = 1 \iff q_{\mathrm{tr}} = q_{\mathrm{nt}} = q_\eta = 0. \tag{5.141}$$

Stage 199 removes reference-point privilege. For two states $x^{(1)}, x^{(2)}$, define

$$\Delta_{\mathrm{orbit}}^{(2\leftarrow 1)} = (q_{\mathrm{tr}}^{(2\leftarrow 1)}, q_{\mathrm{nt}}^{(2\leftarrow 1)}, q_\eta^{(2\leftarrow 1)}). \tag{5.142}$$

Then

$$x^{(2)} \in \mathcal{G}_* \cdot x^{(1)} \iff q_{\mathrm{tr}}^{(2\leftarrow 1)} = q_{\mathrm{nt}}^{(2\leftarrow 1)} = q_\eta^{(2\leftarrow 1)} = 0. \tag{5.143}$$

The packet is an exact additive two-point cocycle, while the multiplicative mismatch and invariant-ratio packets obey the corresponding multiplicative cocycle laws. Thus Packet B can be attached to an orbit rather than to a selected representative.

5.9.3 Reference-free full verdict packet

Let

$$\mathcal{O}_* := \mathcal{G}_* \cdot x_* \tag{5.144}$$

be the target similarity orbit. Stage 200 defines the additive full packet

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} := \left(\Delta_Q(x), q_{\mathrm{tr}}^{(x\leftarrow\mathcal{O}_*)}, q_{\mathrm{nt}}^{(x\leftarrow\mathcal{O}_*)}, q_\eta^{(x\leftarrow\mathcal{O}_*)}\right), \qquad \Delta_Q(x) := \chi_Q(x) - 1. \tag{5.145}$$

The multiplicative form is

$$\mathcal{V}_{\mathrm{full}}^{(x|\mathcal{O}_*)} := \left(\chi_Q(x), \mathfrak{R}_{\mathrm{tr}}^{(x\leftarrow\mathcal{O}_*)}, \mathfrak{R}_{\mathrm{nt}}^{(x\leftarrow\mathcal{O}_*)}, \mathfrak{R}_\eta^{(x\leftarrow\mathcal{O}_*)}\right), \tag{5.146}$$

and the mismatch form is

$$\mathcal{M}_{\mathrm{full}}^{(x|\mathcal{O}_*)} := \left(\chi_Q(x), m_T^{(x\leftarrow\mathcal{O}_*)}, m_K^{(x\leftarrow\mathcal{O}_*)}, m_\mu^{(x\leftarrow\mathcal{O}_*)}\right). \tag{5.147}$$

The exact chart conversions are

$$\mathfrak{R}_{\mathrm{tr}} = e^{q_{\mathrm{tr}}} = (m_T)^{1+\chi_{0,*}}, \qquad \mathfrak{R}_\eta = e^{q_\eta} = \frac{1}{m_K}, \tag{5.148}$$

$$\mathfrak{R}_{\mathrm{nt}} = e^{q_{\mathrm{nt}}} = \frac{m_\mu}{m_K m_T^{F_*}}. \tag{5.149}$$

Therefore the additive, multiplicative, and mismatch packets are exact reparameterizations of the same verdict.

The final reference-free theorem is

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} = 0 \iff \Delta_{\mathrm{branch}}(x) = 0 \text{ and } x \in \mathcal{O}_*. \tag{5.150}$$

Using the sharpened Packet-A and Packet-B forms, this is equivalently

$$\Delta_{\mathrm{full}}^{(x|\mathcal{O}_*)} = 0 \iff \chi_Q(x) = 1, \qquad q_{\mathrm{tr}}^{(x\leftarrow\mathcal{O}_*)} = q_{\mathrm{nt}}^{(x\leftarrow\mathcal{O}_*)} = q_\eta^{(x\leftarrow\mathcal{O}_*)} = 0. \tag{5.151}$$

This is the endpoint of Part V. The reduced compiler algebra is finished; what remains is branch realization of the four scalar conditions.

Claim status. Stages 193–200 are Exact Within Closure **once the Packet-A isotropic grouped one-pole front end, the natural point-particle source-map reduction, and the coherent orbit quotient hierarchy are adopted. The actual PDE may still fail any of the four final scalar conditions. That failure would be a realization failure, not an algebraic incompleteness of the ledger.**

5.10 Verification outputs and downstream use

The usable downstream citation packet from this chapter is as follows.

1. **MTDC-T9.1**: cite for the lower-branch drift laws, tangent-compensation theorem, weak-axisymmetric grouped signature, direct outlet map, hidden-even relation, and the loading mismatch $\Xi_{\rm load} = N_{01}/N_0 - D_{01}/D_0$.
2. **MTDC-T9.2**: cite for the static self-similarity theorem, wall-normalized outgoing-load theorem, square-root mixed-leg law, port co-loading theorem, effective transfer-shape collapse, and coherent support-blindness.
3. **MTDC-T9.3**: cite for the microscopic slippage decomposition, triangular normal form, branch-invariant monomials, rank-three drift map, similarity orbit $\mathcal{G}_*$, and finite orbit–quotient closure.
4. **MTDC-T9.4**: cite for the branch-observable compiler, transfer-shape/outgoing-prefactor compiler, scalar no-go filters, minimal Packet A/Packet B data, and two-packet home-stretch theorem.
5. **MTDC-T9.5**: cite for the exact orbit/quotient projectors, isotropic grouped target surface, outgoing $l = 2$ DtN fingerprint, source-map reduction, higher-odd irrelevance, and Packet-A closure $\Delta_{\rm branch} = 0 \iff \chi_Q = 1$.
6. **MTDC-T9.6**: cite for the finite orbit law, pairwise orbit transport, orbit-representative independence, and final four-scalar verdict packet $\Delta_{\rm full} = (\Delta_Q, q_{\rm tr}, q_{\rm nt}, q_\eta)$.

- ☐ The off-family scalar $\delta_\perp$ is used only as the first post-Part-IV normal coordinate; after tangent compensation, first-order isotropic bundle drift is not counted as an independent failure mode.
- ☐ The lower compensated Family–1 drift laws are applied only under the D/N tube selection, healing-lock shell closure, and frozen $n = 5$ wall-EOS reduction.
- ☐ The four-bundle inversion through $(\Theta_w, K_s, K_q, P_0)$ is kept distinct from a direct microscopic PDE solve; it is an algebraic compiler once those observables are known.
- ☐ The weak-axisymmetric signature $(1, 1/2, -1)$ is not confused with arbitrary grouped anisotropy. It fixes the line $b = 3a$.
- ☐ The no-feed-down result is first order only: grouped real P_2 anisotropy can enter scalar ledgers through quadratic grouped invariants.
- ☐ The even-preserving branch is invoked before reducing the linear grouped problem to $\Xi_{\rm load}$.
- ☐ Static self-similarity is stated in wall-referenced variables. Frozen port shapes do not by themselves imply zero outgoing defect when the wall baseline moves.
- ☐ The square-root mixed-leg law is a reduced sufficient condition under interference/hybridization rigidity, not an unconditional PDE theorem.

- ☐ The scalar $\Xi_1 = P_1/P_0$ is the physical outgoing-prefactor slope and effective transfer-shape slope; it is not introduced as a fit parameter.
- ☐ Coherent support-blindness applies to the direct transfer-shape defect, not to all support-dependent selected-branch quantities.
- ☐ The triangular normal form is used with branch-adapted variables $(\Sigma_{\rm tr}, \Sigma_{\rm nt}, \Sigma_\eta)$; raw microscopic slopes require the stated compiler before interpretation.
- ☐ The monomial coordinates $(\mathfrak{C}_{{\rm tr},*}, \mathfrak{C}_{{\rm nt},*}, \epsilon_\eta)$ are exact quotient coordinates of the reduced coherent hierarchy, not fundamental gauge invariants of the unsolved parent PDE.
- ☐ The similarity orbit $\mathcal{G}_*$ identifies reduced co-scaling fibres in the positive microscopic state space $\mathcal{M}_+$; branch realization still has to be checked against the actual PDE output.
- ☐ Packet A and Packet B are kept separate until the Stage 200 theorem: Packet A is the branch-side outgoing scalar; Packet B is the orbit-side quotient triple.
- ☐ The equivalence $N_Q = \chi_Q^{-1}$ is tied to the natural source-map reduction. It should not be used when the source-map factor $m_{\hat{0}}$ is still nontrivial.
- ☐ Higher odd retarded terms are irrelevant only to the reduced point-particle 2.5PN finish line; later higher-order retarded diagnostics may need them.
- ☐ The final verdict $\Delta_{\rm full} = 0$ is an exact reduced compiler condition. It is not an assertion that the completed moving-throat PDE has already realized the condition.

Future papers should cite Part V when replacing a long weak-axisymmetric or retarded endgame by the compact packet $(\Delta_Q, q_{\rm tr}, q_{\rm nt}, q_\eta)$. They should separately state whether the actual branch has returned $\Delta_Q = 0$ and $(q_{\rm tr}, q_{\rm nt}, q_\eta) = (0, 0, 0)$, or whether those realization conditions remain open.

Chapter 6

Realization Compiler and Finite Mixed-Ray Search

Stage range: 201–218

Part V ended with the reduced home-stretch theorem in its most compact form:

$$\Delta_{\text{full}} = 0 \quad \Longleftrightarrow \quad \chi_Q = 1, \qquad q_{\text{tr}} = q_{\text{nt}} = q_\eta = 0. \tag{6.1}$$

That theorem identifies the final reduced packet, but it does not by itself say how a completed moving-throat branch should be audited in practice. Part VI is the operational answer. It turns the final quotient packet into an explicit realization compiler, rewrites the target orbit as a graph over five free microscopic coordinates, reduces graph-aligned branch searches to a scalar closure function, and then proves that the local mixed-ray search over those five free coordinates is finite through support-cardinality five.

Part VI at a glance

Primary question	Once the final reduced packet is known, how should an actual moving-throat branch be audited in practice, and how large is the resulting local search problem?
Starting inputs	The Part V verdict packet Δ_{full}, its target-orbit description, and the coherent quotient/orbit machinery already fixed upstream.
Delivers	The intrinsic realization compiler, the canonical same-free-quintuple target graph, the scalar graph-slice closure law, and the finite mixed-ray search through support-cardinality five.
Used by	Part VII as the operational back end for actual same-charge branch data; Part VIII inherits the strict front-end packet extracted downstream from that audit chain.
Result anchors	**MTDC-T10**.

Stage packets.

Stages	Packet	Status	Carry-forward output
201–203	Realization compiler, canonical projection, and target graph	Exact Within Closure	Rewrites the verdict packet intrinsically, builds the canonical repair vector and same-free-quintuple projection, and reduces graph-aligned realization to the scalar closure law $\widehat{\chi}_Q(\mathbf{y}) = 1$.
204–205	Explicit log-ray sweep and local predictor machinery	Exact Within Closure	Introduces finite graph lifts along log rays, directional Hessians, quadratic predictors, and tangency/turning-point diagnostics for the scalar closure function.
206–212	Pairwise and triple search certification	Exact Within Closure / Numerically Located	Turns local curvature envelopes into certified brackets, optimizes pairwise and triple interiors, and closes the support-≤ 3 search ledger.
213–218	Higher-support simplex closure and final local search theorem	Exact Within Closure / Numerically Located / Open	Extends the sieve to support cardinalities four and five, proves the support ceiling, and closes the local mixed-ray search with a finite preferred budget once the required branch and envelope data are supplied.

Claim status. Part VI is predominantly Exact Within Closure **inside the declared realization/orbit/free-quintuple graph and local Hessian-envelope search hierarchy. The algebraic compilers, graph maps, stationary systems, candidate-set reductions, and support-cardinality closure theorems are exact once the reduced packet and local envelope data are supplied. Any claim that the completed nonlinear moving-throat PDE actually supplies the required free-quintuple branch, Hessian envelopes, validity intervals, or winning candidate remains** Open**. Any row that depends on numerical insertion of local envelope values or candidate evaluation is marked** Numerically Located **at the realization stage, not at the algebraic compiler stage.**

What this part proves. Part VI converts the final reduced packet into an operational realization compiler, solves the target orbit as a graph over a free quintuple, and proves that the local mixed-ray search is finite through support-cardinality five.

What this part does not prove. It does not prove that the completed PDE supplies the required branch data, local envelopes, or winning realization candidate. The compiler and finite search are exact; the physical winner remains an open branch question.

Why later parts need it. Part VII is the first direct application of this compiler to actual same-charge branch data, and Part VIII inherits the strict front-end packet that Part VII extracts from the completed realization machinery.

6.1 Block contract and theorem-level output

The contract of Part VI is different from the contract of the previous grouped-response chapters. Parts III–V constructed the reduced branch and quotient packets. Part VI assumes those packets exist and asks how they are to be realized by actual branch data. The operational chain is

$$\Delta_{\text{full}} \longrightarrow \Delta^{\text{int}}_{\text{real}} \longrightarrow \Pi^{\text{can}}_{\mathcal{O}_*} \longrightarrow \mathbf{x}^{\text{graph}}_*(\mathbf{y}) \longrightarrow \widehat{\chi}_Q(\mathbf{y}) - 1 \longrightarrow \text{finite mixed-ray search.} \tag{6.2}$$

The first arrow turns the theorem packet into an intrinsic audit packet. The second and third arrows replace the abstract target orbit by an explicit graph. The fourth arrow reduces graph-aligned closure to one scalar. The last arrow proves that the local free-quintuple search is finite through the full support-cardinality ceiling.

The positive microscopic state carried throughout this part is

$$\mathbf{x} = \left(\lambda_W, c_{\eta U}, \gamma, K_U, K^{(\text{eff})}_\eta, K^{(\text{eff})}_W, \mu_W, T_U\right), \tag{6.3}$$

and the free quintuple is

$$\mathbf{y} = \left(\lambda_W, c_{\eta U}, \gamma, K_U, K^{(\text{eff})}_W\right). \tag{6.4}$$

The dependent triple is

$$(T_U, K^{(\text{eff})}_\eta, \mu_W). \tag{6.5}$$

This split is not a new physical assumption. It is the canonical section of the Stage 192 quotient map used to make the same-free-quintuple projection explicit.

Theorem 6.1 (Part VI realization compiler and finite mixed-ray closure theorem)**.** *Inside the carried realization hierarchy for Stages 201–218, the following theorem chain holds.*

1. *The final four-scalar verdict packet can be written intrinsically as*

$$\Delta^{\text{int}}_{\text{real}}(\mathbf{x} \mid \mathcal{Z}_*) = \left(\chi_Q(\mathbf{x}) - 1, q_{\text{tr}}(\mathbf{x}), q_{\text{nt}}(\mathbf{x}), q_\eta(\mathbf{x})\right), \tag{6.6}$$

where the three quotient coordinates are logarithmic ratios of the three target monomials.

2. *The canonical repair vector is supported only on the dependent triple,*

$$\Delta\mathbf{x}_{\text{rep}} = \begin{pmatrix} 0 & 0 & 0 & 0 & q_\eta & 0 & -q_{\text{nt}} + q_\eta - \dfrac{F_*}{1+\chi_{0,*}} q_{\text{tr}} & -\dfrac{q_{\text{tr}}}{1+\chi_{0,*}} \end{pmatrix}^T, \tag{6.7}$$

and exponentiating it gives the canonical target-orbit projection $\Pi^{\text{can}}_{\mathcal{O}_*}$.

3. *The target orbit $\mathcal{O}_*$ is exactly the graph of three dependent target functions over the free quintuple $\mathbf{y}$:*

$$\mathcal{O}_* = \{\mathbf{x}^{\text{graph}}_*(\mathbf{y}) : \mathbf{y} \in (\mathbb{R}_{>0})^5\}. \tag{6.8}$$

4. *The graph errors*

$$(E_T, E_K, E_\mu) = \left(\ln \frac{T_U}{T^{\text{graph}}_U}, \ln \frac{K^{(\text{eff})}_\eta}{K^{\text{graph}}_{\eta,*}}, \ln \frac{\mu_W}{\mu^{\text{graph}}_W}\right) \tag{6.9}$$

are exactly equivalent to the quotient packet:

$$E_T = \frac{q_{\text{tr}}}{1+\chi_{0,*}}, \qquad E_K = -q_\eta, \qquad E_\mu = q_{\text{nt}} - q_\eta + \frac{F_*}{1+\chi_{0,*}} q_{\text{tr}}. \tag{6.10}$$

5. *On the target graph, Packet B vanishes identically and the full closure set is the scalar graph slice*

$$\mathcal{Z}_* = \{\mathbf{x}^{\text{graph}}_*(\mathbf{y}) : \widehat{\chi}_Q(\mathbf{y}) = 1\}, \qquad \widehat{\chi}_Q(\mathbf{y}) := \chi_Q(\mathbf{x}^{\text{graph}}_*(\mathbf{y})). \tag{6.11}$$

6. *Every smooth graph-aligned branch is locally represented by a free-quintuple log ray*

$$\mathbf{y}_{\mathbf{s}}(\tau) = \mathbf{y}_{\circ} \odot e^{\tau \mathbf{s}}, \tag{6.12}$$

and closure along that ray is the scalar root problem

$$\Phi_{\mathbf{s}}(\tau) := \widehat{\chi}_Q(\mathbf{y}_{\mathbf{s}}(\tau)) = 1. \tag{6.13}$$

7. *The first- and second-order local data of the ray are*

$$\Phi_0, \quad \Phi_1 = (\mathcal{D}_{\mathbf{s}}\widehat{\chi}_Q)(\mathbf{y}_{\circ}), \quad \Phi_2 = (\mathcal{D}_{\mathbf{s}}^2\widehat{\chi}_Q)(\mathbf{y}_{\circ}), \tag{6.14}$$

or, equivalently, logarithmic data $(H_0, K_0, \underline{K}_1, \overline{K}_1)$ *after orientation. These data determine certified local brackets whenever the Stage 206 admissibility hypotheses hold.*

8. *The primitive, pairwise, triple, quadruple, and five-coordinate searches are finite certified algebraic ledgers. The support-cardinality ceiling is five because the free space has exactly five primitive axes.*

9. *The final local mixed-ray search on the positive free-quintuple simplex is reduced to the splice*

$$\tau^{\mathrm{lo}}_{\le 5,\min} = \min(\tau^{\mathrm{lo}}_{\le 4,\min}, \tau^{\mathrm{lo,int}}_{5,\min}), \qquad \tau^{\mathrm{hi}}_{\le 5,\min} = \min(\tau^{\mathrm{hi}}_{\le 4,\min}, \tau^{\mathrm{hi,int}}_{5,\min}), \tag{6.15}$$

with

$$\tau^{\mathrm{lo}}_{\le 5,\min} \le \tau^{\mathrm{best}}_{\le 5,*} \le \tau^{\mathrm{hi}}_{\le 5,\min}. \tag{6.16}$$

Proof structure. Stages 201–202 convert the reference-free four-scalar packet into an intrinsic monomial-ratio audit, a canonical same-free-quintuple projection, and an explicit target graph. Stage 203 proves that graph-aligned families lie in the kernel of the quotient map and hence reduce to the scalar closure function $\widehat{\chi}_Q - 1$. Stages 204–207 build the log-ray, Hessian, curvature-envelope, and primitive certified-ray compilers. Stages 208–209 add pairwise mixed cones and reduce the one-parameter ratio optimization to finite quartic candidate sets. Stages 210–212 do the same for genuine three-coordinate simplex interiors and then splice them into the pairwise boundary ledger. Stages 213–215 repeat the construction for four-coordinate interiors and splice the result into the support-≤ 3 ledger. Stages 216–218 identify the unique five-coordinate simplex, reduce its interior optimizer to a finite lifted algebraic candidate set, and finally prove that its boundary is exactly the already-solved support-≤ 4 search set. ☐

6.2 Exact realization compiler and target graph

6.2.1 Intrinsic realization packet

Stage 201 begins from the three target monomials already isolated by the orbit/quotient theorem,

$$\mathfrak{C}_{\mathrm{tr},*}(\mathbf{x}), \qquad \mathfrak{C}_{\mathrm{nt},*}(\mathbf{x}), \qquad \epsilon_\eta(\mathbf{x}). \tag{6.17}$$

Fix target values

$$\mathfrak{C}^{\mathrm{target}}_{\mathrm{tr},*}, \qquad \mathfrak{C}^{\mathrm{target}}_{\mathrm{nt},*}, \qquad \epsilon^{\mathrm{target}}_{\eta,*}. \tag{6.18}$$

The target orbit is then

$$\mathcal{O}_* = \left\{ \mathbf{x} > 0 : \mathfrak{C}_{\mathrm{tr},*} = \mathfrak{C}^{\mathrm{target}}_{\mathrm{tr},*}, \quad \mathfrak{C}_{\mathrm{nt},*} = \mathfrak{C}^{\mathrm{target}}_{\mathrm{nt},*}, \quad \epsilon_\eta = \epsilon^{\mathrm{target}}_{\eta,*} \right\}. \tag{6.19}$$

The intrinsic ratios are

$$\mathfrak{R}_{\mathrm{tr}} := \frac{\mathfrak{C}_{\mathrm{tr},*}}{\mathfrak{C}_{\mathrm{tr},*}^{\mathrm{target}}}, \qquad \mathfrak{R}_{\mathrm{nt}} := \frac{\mathfrak{C}_{\mathrm{nt},*}}{\mathfrak{C}_{\mathrm{nt},*}^{\mathrm{target}}}, \qquad \mathfrak{R}_{\eta} := \frac{\epsilon_\eta}{\epsilon_{\eta,*}^{\mathrm{target}}}, \tag{6.20}$$

and the logarithmic quotient coordinates are

$$q_{\mathrm{tr}} := \ln \mathfrak{R}_{\mathrm{tr}}, \qquad q_{\mathrm{nt}} := \ln \mathfrak{R}_{\mathrm{nt}}, \qquad q_\eta := \ln \mathfrak{R}_\eta. \tag{6.21}$$

Thus

$$\mathbf{x} \in \mathcal{O}_* \quad \Longleftrightarrow \quad q_{\mathrm{tr}} = q_{\mathrm{nt}} = q_\eta = 0. \tag{6.22}$$

The fully closed target set is

$$\mathcal{Z}_* := \{\mathbf{x} \in \mathcal{O}_* : \chi_Q(\mathbf{x}) = 1\}. \tag{6.23}$$

Consequently the intrinsic realization packet is exactly

$$\Delta^{\mathrm{int}}_{\mathrm{real}}(\mathbf{x} \mid \mathcal{Z}_*) = \left(\chi_Q(\mathbf{x}) - 1, q_{\mathrm{tr}}(\mathbf{x}), q_{\mathrm{nt}}(\mathbf{x}), q_\eta(\mathbf{x})\right). \tag{6.24}$$

This is the operational version of Part V's reference-free packet.

6.2.2 Canonical same-free-quintuple projection

The Stage 192 quotient map is carried in the linearized form

$$\mathbf{q} = M_* \Delta \mathbf{x}, \qquad \mathbf{q} = (q_{\mathrm{tr}}, q_{\mathrm{nt}}, q_\eta)^T. \tag{6.25}$$

The canonical section chooses the dependent triple $(T_U, K_\eta^{\mathrm{eff}}, \mu_W)$. The exact repair vector is

$$\Delta \mathbf{x}_{\mathrm{rep}} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ q_\eta \\ 0 \\ -q_{\mathrm{nt}} + q_\eta - \dfrac{F_*}{1+\chi_{0,*}} q_{\mathrm{tr}} \\ -\dfrac{q_{\mathrm{tr}}}{1+\chi_{0,*}} \end{pmatrix}. \tag{6.26}$$

It satisfies

$$M_* \Delta \mathbf{x}_{\mathrm{rep}} = -\mathbf{q}. \tag{6.27}$$

Exponentiating (6.26) gives the canonical projection

$$\begin{aligned} \Pi^{\mathrm{can}}_{\mathcal{O}_*}(\mathbf{x}) = \big(&\lambda_W, c_{\eta U}, \gamma, K_U, K_\eta^{(\mathrm{eff})} \mathfrak{R}_\eta, K_W^{(\mathrm{eff})}, \\ &\mu_W \mathfrak{R}_{\mathrm{nt}}^{-1} \mathfrak{R}_\eta \mathfrak{R}_{\mathrm{tr}}^{-F_*/(1+\chi_{0,*})}, T_U \mathfrak{R}_{\mathrm{tr}}^{-1/(1+\chi_{0,*})}\big). \end{aligned} \tag{6.28}$$

The projection lands on the target orbit, preserves the free quintuple (6.4), and is the unique point of $\mathcal{O}_*$ with that free quintuple.

Proposition 6.2 (Canonical fixed-point audit)**.** *For any positive microscopic state* $\mathbf{x}$*,*

$$\mathbf{x} \in \mathcal{Z}_* \quad \Longleftrightarrow \quad \chi_Q(\mathbf{x}) = 1 \quad \textit{and} \quad \Pi^{\mathrm{can}}_{\mathcal{O}_*}(\mathbf{x}) = \mathbf{x}. \tag{6.29}$$

6.2.3 Direct microscopic target graph

Stage 202 removes the remaining abstractness by solving the orbit constraints directly. The carried monomials are

$$\mathfrak{C}_{\mathrm{tr},*} = \left(\frac{\gamma c_{\eta U}}{K_U}\right)^{1+\delta_{U,*}} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{1+\chi_{0,*}}, \tag{6.30}$$

$$\mathfrak{C}_{\mathrm{nt},*} = \frac{\lambda_W^2 \mu_W}{K_\eta^{(\mathrm{eff})}(K_W^{(\mathrm{eff})})^2} \left(\frac{\gamma^2 \lambda_W^2 \sigma}{K_U K_W^{(\mathrm{eff})}}\right)^{E_*} \left(\frac{\pi^2 T_U}{L^2 K_U}\right)^{-F_*}, \tag{6.31}$$

$$\epsilon_\eta = \frac{c_{\eta U}^2}{K_U K_\eta^{(\mathrm{eff})}}. \tag{6.32}$$

Solving (6.30)–(6.32) at fixed $\mathbf{y}$ gives

$$\delta_{U,*}^{\mathrm{graph}}(\mathbf{y}) := \left[\frac{\mathfrak{C}_{\mathrm{tr},*}^{\mathrm{target}}}{\left(\frac{\gamma c_{\eta U}}{K_U}\right)^{1+\delta_{U,*}}}\right]^{1/(1+\chi_{0,*})}, \tag{6.33}$$

$$T_U^{\mathrm{graph}}(\mathbf{y}) = \frac{L^2 K_U}{\pi^2}\, \delta_{U,*}^{\mathrm{graph}}(\mathbf{y}), \tag{6.34}$$

$$K_{\eta,*}^{\mathrm{graph}}(\mathbf{y}) = \frac{c_{\eta U}^2}{K_U \epsilon_{\eta,*}^{\mathrm{target}}}, \tag{6.35}$$

$$\mu_W^{\mathrm{graph}}(\mathbf{y}) = \frac{\mathfrak{C}_{\mathrm{nt},*}^{\mathrm{target}}\, c_{\eta U}^2 (K_W^{(\mathrm{eff})})^2}{\epsilon_{\eta,*}^{\mathrm{target}} K_U \lambda_W^2} \left(\frac{\gamma^2 \lambda_W^2 \sigma}{K_U K_W^{(\mathrm{eff})}}\right)^{-E_*} \left(\delta_{U,*}^{\mathrm{graph}}(\mathbf{y})\right)^{F_*}. \tag{6.36}$$

The graph point is

$$\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) = \left(\lambda_W, c_{\eta U}, \gamma, K_U, K_{\eta,*}^{\mathrm{graph}}(\mathbf{y}), K_W^{(\mathrm{eff})}, \mu_W^{\mathrm{graph}}(\mathbf{y}), T_U^{\mathrm{graph}}(\mathbf{y})\right). \tag{6.37}$$

Substitution shows

$$\mathcal{O}_* = \left\{\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) : \mathbf{y} \in (\mathbb{R}_{>0})^5\right\}. \tag{6.38}$$

Thus the canonical projection and the graph projection coincide:

$$\Pi_{\mathcal{O}_*}^{\mathrm{can}}(\mathbf{x}) = \mathbf{x}_*^{\mathrm{graph}}(\pi_{\mathrm{free}}\mathbf{x}). \tag{6.39}$$

6.2.4 Graph-error realization coordinates

For a same-free-quintuple candidate state, define

$$E_T := \ln \frac{T_U}{T_U^{\mathrm{graph}}}, \qquad E_K := \ln \frac{K_\eta^{(\mathrm{eff})}}{K_{\eta,*}^{\mathrm{graph}}}, \qquad E_\mu := \ln \frac{\mu_W}{\mu_W^{\mathrm{graph}}}. \tag{6.40}$$

The exact relations to the quotient packet are (6.10). The reduced-family audit packet becomes

$$\Delta_{\mathrm{fam}}^{\mathrm{graph}}(\mathbf{y}) = \left(\chi_Q(\mathbf{x}_{\mathrm{cand}}(\mathbf{y})) - 1, E_T, E_K, E_\mu\right), \tag{6.41}$$

and the closure criterion is

$$\Delta_{\mathrm{fam}}^{\mathrm{graph}}(\mathbf{y}) = 0 \quad \Longleftrightarrow \quad \mathbf{x}_{\mathrm{cand}}(\mathbf{y}) \in \mathcal{Z}_*. \tag{6.42}$$

Claim status. Stages 201–202 are Exact Within Closure. **They are exact consequences of the already-declared monomial quotient map, target-orbit definition, and canonical dependent-triple section. They do not prove that the actual PDE branch is graph-aligned; they provide the exact coordinates in which that claim can be tested.**

6.3 Scalar graph-slice and log-ray compiler

6.3.1 Graph closure scalar

Stage 203 defines

$$\widehat{\chi}_Q(\mathbf{y}) := \chi_Q\big(\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y})\big), \qquad \widehat{\Delta}_Q(\mathbf{y}) := \widehat{\chi}_Q(\mathbf{y}) - 1. \tag{6.43}$$

Since every graph point already lies on $\mathcal{O}_*$, the closed set is exactly

$$\mathcal{Z}_* = \left\{\mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}) : \widehat{\Delta}_Q(\mathbf{y}) = 0\right\}. \tag{6.44}$$

This is the scalar graph-slice theorem. Its importance is that the realization problem drops from four scalar equations to one scalar equation as soon as the candidate family is graph-aligned.

6.3.2 Graph tangent identities

Let $\mathbf{y}(\tau)$ be a smooth free-quintuple family and define free log-tangent components

$$(\lambda_1, c_1, \gamma_1, \kappa_U, \kappa_W) = \left(\frac{d\ln\lambda_W}{d\tau}, \frac{d\ln c_{\eta U}}{d\tau}, \frac{d\ln\gamma}{d\tau}, \frac{d\ln K_U}{d\tau}, \frac{d\ln K_W^{(\mathrm{eff})}}{d\tau}\right). \tag{6.45}$$

Then the dependent graph log tangents are

$$\frac{d\ln\delta_{U,*}^{\mathrm{graph}}}{d\tau} = -\frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\gamma_1 + c_1 - \kappa_U), \tag{6.46}$$

$$\tau_1^{\mathrm{graph}} = \kappa_U - \frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\gamma_1 + c_1 - \kappa_U), \tag{6.47}$$

$$\kappa_\eta^{\mathrm{graph}} = 2c_1 - \kappa_U, \tag{6.48}$$

$$\begin{aligned}\mu_1^{\mathrm{graph}} &= 2c_1 - \kappa_U + 2\kappa_W - 2\lambda_1 - E_*(2\gamma_1 + 2\lambda_1 - \kappa_U - \kappa_W) \\ &\quad - F_*\frac{1+\delta_{U,*}}{1+\chi_{0,*}}(\gamma_1 + c_1 - \kappa_U).\end{aligned} \tag{6.49}$$

Substitution into the monomial-drift matrix gives

$$M_*\dot{\Delta\mathbf{x}}_{\mathrm{graph}} = 0. \tag{6.50}$$

So graph-lifted families are tangent to the similarity orbit at every point, and Packet B vanishes identically on the graph.

6.3.3 One-parameter crossing theorem

Let $\mathbf{y} : [\tau_-, \tau_+] \to (\mathbb{R}_{>0})^5$ be continuous and graph-lift it. If

$$\widehat{\Delta}_Q(\tau_-)\,\widehat{\Delta}_Q(\tau_+) < 0, \tag{6.51}$$

then there exists $\tau_* \in (\tau_-, \tau_+)$ with

$$\widehat{\Delta}_Q(\tau_*) = 0, \qquad \mathbf{x}_*^{\mathrm{graph}}(\mathbf{y}(\tau_*)) \in \mathcal{Z}_*. \tag{6.52}$$

This is not a numerical method by itself. It is the exact existence theorem that justifies the later scalar log-ray search.

6.3.4 Free-quintuple log rays

Stage 204 chooses the canonical local family

$$\mathbf{y_s}(\tau) = \mathbf{y}_\circ \odot e^{\tau \mathbf{s}}, \qquad \mathbf{s} = (s_\lambda, s_c, s_\gamma, s_U, s_W). \tag{6.53}$$

Every smooth positive free-quintuple path has this form to first order at any point, with $\mathbf{s} = d \ln \mathbf{y}/d\tau$. Along the log ray define

$$\mathfrak{a}_* := \frac{1 + \delta_{U,*}}{1 + \chi_{0,*}}, \tag{6.54}$$

$$\sigma_\delta = -\mathfrak{a}_*(s_\gamma + s_c - s_U), \tag{6.55}$$

$$\sigma_T = s_U + \sigma_\delta, \qquad \sigma_{K_\eta} = 2s_c - s_U, \tag{6.56}$$

$$\sigma_\mu = 2s_c - s_U + 2s_W - 2s_\lambda - E_*(2s_\gamma + 2s_\lambda - s_U - s_W) + F_*\sigma_\delta. \tag{6.57}$$

The graph-lifted ray is therefore

$$\begin{aligned}\mathbf{x}_{\mathbf{s}}^{\mathrm{graph}}(\tau) = (&\lambda_{W,\circ} e^{s_\lambda \tau}, c_{\eta U,\circ} e^{s_c \tau}, \gamma_\circ e^{s_\gamma \tau}, K_{U,\circ} e^{s_U \tau}, K_{\eta,\circ}^{\mathrm{graph}} e^{\sigma_{K_\eta} \tau}, \\ &K_{W,\circ}^{(\mathrm{eff})} e^{s_W \tau}, \mu_{W,\circ}^{\mathrm{graph}} e^{\sigma_\mu \tau}, T_{U,\circ}^{\mathrm{graph}} e^{\sigma_T \tau}).\end{aligned} \tag{6.58}$$

The three target monomials are invariant for all τ on this lift. Hence the scalarized closure function

$$\Phi_{\mathbf{s}}(\tau) := \widehat{\chi}_Q(\mathbf{y_s}(\tau)) \tag{6.59}$$

contains the entire graph-aligned realization question:

$$\mathbf{x}_{\mathbf{s}}^{\mathrm{graph}}(\tau) \in \mathcal{Z}_* \quad \Longleftrightarrow \quad \Phi_{\mathbf{s}}(\tau) = 1. \tag{6.60}$$

6.3.5 First-order and second-order root predictors

Let

$$\Phi_0 := \Phi_{\mathbf{s}}(0), \qquad \Phi_1 := \Phi_{\mathbf{s}}'(0), \qquad \Phi_2 := \Phi_{\mathbf{s}}''(0). \tag{6.61}$$

The first-order predictors are

$$\tau_{\mathrm{aff}} = \frac{1 - \Phi_0}{\Phi_1}, \qquad \tau_{\log} = -\frac{\ln \Phi_0}{L_0}, \qquad L_0 := \frac{\Phi_1}{\Phi_0}. \tag{6.62}$$

Stage 205 introduces free log coordinates

$$\boldsymbol{\ell} = (\ln \lambda_W, \ln c_{\eta U}, \ln \gamma, \ln K_U, \ln K_W^{(\mathrm{eff})}), \tag{6.63}$$

and the directional operators

$$\mathcal{D}_{\mathbf{s}} = \sum_i s_i \partial_{\ell_i}, \qquad \mathcal{H}_{\mathbf{s}} = \mathcal{D}_{\mathbf{s}}^2. \tag{6.64}$$

Assuming $\Phi_0 > 0$, define

$$L_1 := \frac{d^2}{d\tau^2} \ln \Phi_{\mathbf{s}}(\tau)\bigg|_0 = \frac{\Phi_2}{\Phi_0} - \left(\frac{\Phi_1}{\Phi_0}\right)^2. \tag{6.65}$$

Thus

$$\Phi_2 = \Phi_0(L_1 + L_0^2). \tag{6.66}$$

The quadratic affine and logarithmic discriminants are

$$\Delta_{\text{aff}} = \Phi_1^2 - 2\Phi_2(\Phi_0 - 1), \qquad \Delta_{\log} = L_0^2 - 2L_1 \ln \Phi_0. \tag{6.67}$$

When the appropriate discriminant is nonnegative, the continuous-branch predictors are

$$\tau_{\text{quad}} = \frac{2(1-\Phi_0)}{\Phi_1 + \operatorname{sgn}(\Phi_1)\sqrt{\Delta_{\text{aff}}}}, \tag{6.68}$$

$$\tau_{\log,2} = -\frac{2\ln \Phi_0}{L_0 + \operatorname{sgn}(L_0)\sqrt{\Delta_{\log}}}. \tag{6.69}$$

If $\Phi_1 = 0$ but $\Phi_0 \neq 1$, the quadratic turning-point criterion is

$$(1 - \Phi_0)\Phi_2 > 0, \tag{6.70}$$

with local predictors

$$\tau_{\pm} = \pm\sqrt{\frac{2(1-\Phi_0)}{\Phi_2}}. \tag{6.71}$$

These are the local predictor tools used by the certified search sieve.

Claim status. Stages 203–205 are Exact Within Closure **once $\widehat{\chi}_Q$ is a differentiable scalar on the target graph. The actual values of the derivatives and Hessians are PDE-returned or candidate-family data; the predictor formulas themselves are exact.**

6.4 Certified ray sieve: primitive and pairwise searches

6.4.1 Curvature-envelope brackets

Stage 206 orients the log residual. Let

$$H_{\mathbf{s}}(\tau) := \operatorname{sgn}(\ln \Phi_{\mathbf{s}}(0)) \ln \Phi_{\mathbf{s}}(\tau), \tag{6.72}$$

with

$$H_0 := H_{\mathbf{s}}(0) > 0, \qquad K_0 := H'_{\mathbf{s}}(0) < 0. \tag{6.73}$$

Assume a curvature envelope on $[0, T]$:

$$\underline{K}_1 \le H''_{\mathbf{s}}(\tau) \le \overline{K}_1. \tag{6.74}$$

Then

$$H_-(\tau) = H_0 + K_0\tau + \frac{1}{2}\underline{K}_1\tau^2, \qquad H_+(\tau) = H_0 + K_0\tau + \frac{1}{2}\overline{K}_1\tau^2 \tag{6.75}$$

sandwich the true residual. Define

$$\mathcal{T}(H_0, K_0; c) := -\frac{2H_0}{K_0 + \operatorname{sgn}(K_0)\sqrt{K_0^2 - 2cH_0}}. \tag{6.76}$$

For $K_0 < 0$, the certified bracket is

$$\tau_{\mathrm{lo}} = \mathcal{T}(H_0, K_0; \underline{K}_1), \qquad \tau_{\mathrm{hi}} = \mathcal{T}(H_0, K_0; \overline{K}_1), \tag{6.77}$$

provided both discriminants are real and $\tau_{\mathrm{hi}} \leq T$. The true local root is then unique in $[\tau_{\mathrm{lo}}, \tau_{\mathrm{hi}}]$. For a turning ray $K_0 = 0$, strict negative curvature gives

$$\tau_{\mathrm{lo}}^{(\mathrm{tp})} = \sqrt{-\frac{2H_0}{\underline{K}_1}}, \qquad \tau_{\mathrm{hi}}^{(\mathrm{tp})} = \sqrt{-\frac{2H_0}{\overline{K}_1}}. \tag{6.78}$$

6.4.2 Primitive certified table

The primitive free axes are

$$\mathfrak{I}_5 = \{\lambda, c, \gamma, U, W\}. \tag{6.79}$$

Let

$$h(\boldsymbol{\ell}) := \ln \widehat{\chi}_Q(\boldsymbol{\ell}), \qquad \sigma_0 := \operatorname{sgn}(h(\boldsymbol{\ell}_\circ)). \tag{6.80}$$

The oriented primitive gradients are

$$\Gamma_i := \sigma_0 \partial_{\ell_i} h(\boldsymbol{\ell}_\circ). \tag{6.81}$$

If $\Gamma_i \neq 0$, the canonical forward primitive direction is

$$\widehat{\mathbf{e}}_i := \varepsilon_i \mathbf{e}_i, \qquad \varepsilon_i := -\operatorname{sgn}(\Gamma_i), \qquad k_i := |\Gamma_i|. \tag{6.82}$$

The primitive curvature uses only the diagonal Hessian restriction:

$$H_1^{(i)}(\tau) = \sigma_0 \partial_{\ell_i \ell_i} h(\boldsymbol{\ell}_\circ + \tau \widehat{\mathbf{e}}_i). \tag{6.83}$$

Thus primitive certification needs no off-diagonal Hessian data. Each primitive row is decided by

$$\mathcal{R}_i^{\mathrm{prim}} := (H_0, \Gamma_i, \underline{\kappa}_i, \overline{\kappa}_i, T_i). \tag{6.84}$$

For monotone rows, the certified interval is

$$\tau_{i,\mathrm{lo}} = \mathcal{T}(H_0, -k_i; \underline{\kappa}_i), \qquad \tau_{i,\mathrm{hi}} = \mathcal{T}(H_0, -k_i; \overline{\kappa}_i), \tag{6.85}$$

with the same discriminant and validity checks as Stage 206.

6.4.3 Pairwise mixed rays

For a monotone pair (i, j), Stage 208 introduces

$$\widehat{\mathbf{s}}_{ij}(r) = \frac{\widehat{\mathbf{e}}_i + r\widehat{\mathbf{e}}_j}{\sqrt{1 + r^2}}, \qquad r \geq 0. \tag{6.86}$$

The slope magnitude is

$$k_{ij}(r) = \frac{k_i + r k_j}{\sqrt{1 + r^2}}, \tag{6.87}$$

with unique gradient-optimal ratio

$$r_{ij}^{\mathrm{grad}} = \frac{k_j}{k_i}, \qquad k_{ij}^{\mathrm{grad}} = \sqrt{k_i^2 + k_j^2}. \tag{6.88}$$

The pairwise curvature is

$$H_{1,ij}(r;\tau) = \frac{h_{ii} + 2rh_{ij} + r^2 h_{jj}}{1 + r^2}, \tag{6.89}$$

so the off-diagonal Hessian enters with cross weight

$$w_\times(r) = \frac{2r}{1 + r^2}, \tag{6.90}$$

maximized at the equal-mix ray $r = 1$. Thus the two canonical pair screens are

$$\widehat{\mathbf{s}}_{ij}^{\mathrm{grad}} = \frac{k_i \widehat{\mathbf{e}}_i + k_j \widehat{\mathbf{e}}_j}{\sqrt{k_i^2 + k_j^2}}, \qquad \widehat{\mathbf{s}}_{ij}^{\mathrm{eq}} = \frac{\widehat{\mathbf{e}}_i + \widehat{\mathbf{e}}_j}{\sqrt{2}}. \tag{6.91}$$

6.4.4 Pairwise ratio optimizer

Stage 209 closes the full one-parameter pair search. On a compact ratio window $[0, R_{ij}]$, for envelope label $\star \in \{\mathrm{lo}, \mathrm{hi}\}$, define

$$A_\star = k_i^2 - 2H_0 u_\star, \qquad B_\star = 2k_i k_j - 4H_0 v_\star, \qquad C_\star = k_j^2 - 2H_0 w_\star. \tag{6.92}$$

The discriminant numerator is

$$\Delta^\sharp_{ij,\star}(r) = A_\star + B_\star r + C_\star r^2. \tag{6.93}$$

The certified root function is

$$\tau_{ij,\star}(r) = \frac{2H_0\sqrt{1 + r^2}}{k_i + rk_j + \sqrt{A_\star + B_\star r + C_\star r^2}}. \tag{6.94}$$

Interior optimizers satisfy the stationary numerator equation

$$\mathcal{N}_{ij,\star}(r) = 2(k_j - k_i r)\sqrt{A_\star + B_\star r + C_\star r^2} + B_\star + 2(C_\star - A_\star)r - B_\star r^2 = 0. \tag{6.95}$$

Eliminating the square root gives the quartic condition

$$\mathcal{Q}_{ij,\star}(r) = \Big[B_\star + 2(C_\star - A_\star)r - B_\star r^2\Big]^2 - 4(k_j - k_i r)^2 (A_\star + B_\star r + C_\star r^2) = 0. \tag{6.96}$$

After filtering extraneous roots, each envelope optimizer is found in a finite set containing admissible endpoints and at most four admissible quartic roots. Therefore one pair requires at most twelve exact candidate evaluations across the lower and upper envelopes.

Claim status. Stages 206–209 are Exact Within Closure **once the local gradient, Hessian-envelope, and validity data are supplied. If those inputs come from numerical PDE evaluation, the final ray rankings are** Numerically Located **insertions into exact certified formulas.**

6.5 Three- and four-coordinate simplex sieves

6.5.1 Three-coordinate simplex and boundary reduction

For a monotone primitive triple (i, j, k), the positive spherical simplex is

$$\Delta_{ijk}^{+} = \left\{(a_i, a_j, a_k) \in \mathbb{R}_{\geq 0}^{3} : a_i^2 + a_j^2 + a_k^2 = 1\right\}. \tag{6.97}$$

The ray is

$$\widehat{\mathbf{s}}_{ijk} = a_i \widehat{\mathbf{e}}_i + a_j \widehat{\mathbf{e}}_j + a_k \widehat{\mathbf{e}}_k. \tag{6.98}$$

Each boundary edge is exactly one Stage 209 pairwise cone. Hence only the interior of Δ_{ijk}^{+} is new at support-cardinality three.

The gradient-optimal point is

$$\mathbf{a}_{ijk}^{\mathrm{grad}} = \frac{(k_i, k_j, k_k)}{\sqrt{k_i^2 + k_j^2 + k_k^2}}, \tag{6.99}$$

and the equal-mix barycenter is

$$\mathbf{a}_{ijk}^{\mathrm{eq}} = \frac{(1, 1, 1)}{\sqrt{3}}. \tag{6.100}$$

The total off-diagonal leverage weight is

$$w_{\Sigma}(\mathbf{a}) = \mathbf{2}(\mathbf{a_i a_j} + \mathbf{a_i a_k} + \mathbf{a_j a_k}) = (\mathbf{a_i} + \mathbf{a_j} + \mathbf{a_k})^{\mathbf{2}} - \mathbf{1} \leq \mathbf{2}, \tag{6.101}$$

with equality at (6.100). Thus the gradient screen maximizes first-order descent, while the equal-mix screen maximizes three-way cross-Hessian leverage.

6.5.2 Triple interior optimizer

On the interior patch $a_i > 0$, write

$$(a_i, a_j, a_k) = \frac{(1, r, s)}{\sqrt{1 + r^2 + s^2}}. \tag{6.102}$$

For either envelope label $\star$, define the quadratic discriminant numerator

$$\Delta_{ijk,\star}^{\sharp}(r, s) = A_\star + B_\star r + C_\star s + D_\star r^2 + E_\star rs + F_\star s^2. \tag{6.103}$$

The certified root is

$$\tau_{ijk,\star}(r, s) = \frac{2H_0\sqrt{1 + r^2 + s^2}}{k_i + rk_j + sk_k + \sqrt{\Delta_{ijk,\star}^{\sharp}(r, s)}}. \tag{6.104}$$

The stationary equations can be written as two square-root numerator equations. Eliminating the square root gives a quartic cross-consistency polynomial and sextic square conditions. The finite pre-candidate set has at most

$$4 \cdot 6 = 24 \tag{6.105}$$

complex roots counting multiplicity when the intersection is zero-dimensional. After filtering for admissibility and unsquared stationarity, the interior optimizer is finite.

6.5.3 Up-to-three-coordinate splice

There are $\binom{5}{2} = 10$ primitive pairs and $\binom{5}{3} = 10$ primitive triples. Stage 212 imports every optimized pairwise interval

$$\tau^{\mathrm{lo}}_{ij,\min} \le \tau^{\mathrm{best}}_{ij,*} \le \tau^{\mathrm{hi}}_{ij,\min} \tag{6.106}$$

and every triple interior interval

$$\tau^{\mathrm{lo,int}}_{ijk,\min} \le \tau^{\mathrm{best,int}}_{ijk,*} \le \tau^{\mathrm{hi,int}}_{ijk,\min}. \tag{6.107}$$

For a triple $T = (i, j, k)$, the boundary minima are

$$\beta^{\mathrm{lo}}_T := \min(\tau^{\mathrm{lo}}_{ij,\min}, \tau^{\mathrm{lo}}_{ik,\min}, \tau^{\mathrm{lo}}_{jk,\min}), \quad \beta^{\mathrm{hi}}_T := \min(\tau^{\mathrm{hi}}_{ij,\min}, \tau^{\mathrm{hi}}_{ik,\min}, \tau^{\mathrm{hi}}_{jk,\min}). \tag{6.108}$$

The closed-triple interval is

$$\tau^{\mathrm{lo},\triangle}_{T,\min} = \min(\beta^{\mathrm{lo}}_T, \tau^{\mathrm{lo,int}}_{ijk,\min}), \qquad \tau^{\mathrm{hi},\triangle}_{T,\min} = \min(\beta^{\mathrm{hi}}_T, \tau^{\mathrm{hi,int}}_{ijk,\min}). \tag{6.109}$$

The full support-≤ 3 search is therefore bounded by

$$\tau^{\mathrm{lo}}_{\le 3,\min} := \min(\tau^{\mathrm{lo}}_{2,\min}, \tau^{\mathrm{lo,int}}_{3,\min}), \qquad \tau^{\mathrm{hi}}_{\le 3,\min} := \min(\tau^{\mathrm{hi}}_{2,\min}, \tau^{\mathrm{hi,int}}_{3,\min}). \tag{6.110}$$

The exact evaluation budget through three active coordinates is at most

$$10 \times 12 + 10 \times 48 = 600 \tag{6.111}$$

candidate evaluations.

6.5.4 Four-coordinate simplex and boundary reduction

For a primitive quadruple $Q = (i, j, k, l)$,

$$\Delta^{+}_{ijkl} = \left\{ (a_i, a_j, a_k, a_l) \in \mathbb{R}^4_{\ge 0} : a_i^2 + a_j^2 + a_k^2 + a_l^2 = 1 \right\}. \tag{6.112}$$

Every codimension-one face is one of the Stage 212 closed triple simplices. The gradient-optimal interior point is

$$\mathbf{a}^{\mathrm{grad}}_{ijkl} = \frac{(k_i, k_j, k_k, k_l)}{\sqrt{k_i^2 + k_j^2 + k_k^2 + k_l^2}}, \tag{6.113}$$

and the equal-mix point is

$$\mathbf{a}^{\mathrm{eq}}_{ijkl} = \frac{(1,1,1,1)}{2}. \tag{6.114}$$

The total off-diagonal leverage is

$$w_\Sigma(\mathbf{a}) = 2 \sum_{p<q} a_p a_q = \left(\sum_p a_p \right)^2 - 1 \le 3, \tag{6.115}$$

with equality at (6.114). The support-cardinality-four gate asks whether either canonical interior screen, or later the full interior optimizer, beats the imported triple-face boundary ledger.

6.5.5 Four-coordinate interior optimizer and up-to-four splice

On an interior chart, Stage 214 introduces three ratio coordinates (r, s, t), one auxiliary square-root variable y, and a lifted stationary system with degree pattern

$$(3, 3, 3, 2). \tag{6.116}$$

Thus the preferred lifted candidate bound is

$$3 \cdot 3 \cdot 3 \cdot 2 = 54 \tag{6.117}$$

per envelope. Stage 215 then splices the four-coordinate interior intervals to the imported support-≤ 3 ledger. There are $\binom{5}{4} = 5$ primitive quadruples, so the four-coordinate interior budget is

$$5 \times 2 \times 54 = 540. \tag{6.118}$$

Together with (6.111), the support-≤ 4 search budget is

$$600 + 540 = 1140. \tag{6.119}$$

The certified support-≤ 4 interval is

$$\tau^{\mathrm{lo}}_{\leq 4,\min} := \min(\tau^{\mathrm{lo}}_{\leq 3,\min}, \tau^{\mathrm{lo,int}}_{4,\min}), \qquad \tau^{\mathrm{hi}}_{\leq 4,\min} := \min(\tau^{\mathrm{hi}}_{\leq 3,\min}, \tau^{\mathrm{hi,int}}_{4,\min}). \tag{6.120}$$

Claim status. Stages 210–215 are Exact Within Closure **for the simplex algebra and** Numerically Located **only when actual local packet values are inserted. Boundary reduction is exact because each simplex face is one of the previously closed lower-support searches.**

6.6 Unique five-coordinate simplex and final local closure

6.6.1 The unique five-coordinate positive simplex

Stage 216 observes that, because the free space is the five-coordinate log space

$$\mathfrak{I}_5 = \{\lambda, c, \gamma, U, W\}, \tag{6.121}$$

there is only one support-cardinality-five simplex:

$$\Delta_5^+ = \left\{ \mathbf{a} = (a_\lambda, a_c, a_\gamma, a_U, a_W) \in \mathbb{R}^5_{\geq 0} : \sum_{p \in \mathfrak{I}_5} a_p^2 = 1 \right\}. \tag{6.122}$$

Its boundary faces are precisely the five Stage 215 primitive quadruple closed simplices. If the imported face intervals are denoted by $\mathcal{I}^{\square}_{\widehat{p}}$, then

$$\beta_5^{\mathrm{lo}} := \min_{p \in \mathfrak{I}_5} \tau^{\mathrm{lo},\square}_{\widehat{p},\min}, \qquad \beta_5^{\mathrm{hi}} := \min_{p \in \mathfrak{I}_5} \tau^{\mathrm{hi},\square}_{\widehat{p},\min}. \tag{6.123}$$

Equivalently,

$$\beta_5^{\mathrm{lo}} = \tau^{\mathrm{lo}}_{\leq 4,\min}, \qquad \beta_5^{\mathrm{hi}} = \tau^{\mathrm{hi}}_{\leq 4,\min}. \tag{6.124}$$

6.6.2 Five-way canonical screens

The five-coordinate slope magnitude is

$$k_5(\mathbf{a}) = \sum_{p \in \mathfrak{I}_5} a_p k_p. \tag{6.125}$$

The unique gradient-optimal interior point is

$$\mathbf{a}_5^{\mathrm{grad}} = \frac{(k_\lambda, k_c, k_\gamma, k_U, k_W)}{\sqrt{k_\lambda^2 + k_c^2 + k_\gamma^2 + k_U^2 + k_W^2}}, \tag{6.126}$$

and the equal-mix barycenter is

$$\mathbf{a}_5^{\mathrm{eq}} = \frac{(1,1,1,1,1)}{\sqrt{5}}. \tag{6.127}$$

The total ten-way cross leverage is

$$w_\Sigma(\mathbf{a}) = 2\sum_{p<q} a_p a_q = \left(\sum_p a_p\right)^2 - 1 \le 4, \tag{6.128}$$

with equality at (6.127). These two screens are the exact five-way analogues of the lower-cardinality gradient and equal-mix screens. They are not the full search; they are the final pre-optimizer gate.

6.6.3 Five-coordinate interior optimizer

Stage 217 solves the last continuum problem. On the chart $a_\lambda > 0$, write

$$(a_\lambda, a_c, a_\gamma, a_U, a_W) = \frac{(1, r, s, t, u)}{\sqrt{1 + r^2 + s^2 + t^2 + u^2}}. \tag{6.129}$$

The certified root function has the same form as before:

$$\tau_\star(r,s,t,u) = \frac{2H_0\sqrt{1 + r^2 + s^2 + t^2 + u^2}}{k_\lambda + rk_c + sk_\gamma + tk_U + uk_W + \sqrt{\Delta_\star^\sharp(r,s,t,u)}}. \tag{6.130}$$

The stationary conditions are four square-root numerator equations. Introducing

$$y := \sqrt{\Delta_\star^\sharp(r,s,t,u)} \tag{6.131}$$

gives the lifted polynomial system

$$\mathcal{F}_{r,\star} = 0, \quad \mathcal{F}_{s,\star} = 0, \quad \mathcal{F}_{t,\star} = 0, \quad \mathcal{F}_{u,\star} = 0, \quad \mathcal{F}_{\Delta,\star} = 0, \tag{6.132}$$

with degree pattern

$$(3,3,3,3,2). \tag{6.133}$$

Therefore the preferred lifted Bézout candidate bound is

$$3^4 \cdot 2 = 179 \tag{6.134}$$

per envelope. Across lower and upper envelopes the support-five interior contributes at most

$$2 \times 162 = 324 \tag{6.135}$$

preferred candidate evaluations. A fallback square-root-free projected chart gives bound 750 per envelope, hence fallback support-five budget 1500.

6.6.4 Boundary identification and support ceiling

Stage 218 proves that the boundary of Δ_5^+ is exactly the already-solved support-≤ 4 search set. The proper nonempty support strata have counts

$$5 \text{ facets} + 10 \text{ ridges} + 10 \text{ edges} + 5 \text{ vertices} = 30 = 2^5 - 2. \tag{6.136}$$

Every proper support subset is contained in at least one quadruple face, and the faces are carried as closed simplices. Thus

$$\partial\Delta_5^+ = \mathcal{S}_{\leq 4}^{\mathrm{loc}}. \tag{6.137}$$

There is also no higher-support local search, because the free log space has only five primitive axes:

$$1 \leq \#\operatorname{supp}(\mathbf{a}) \leq 5. \tag{6.138}$$

The true best local closure time is therefore

$$\tau_{\leq 5,*}^{\mathrm{best}} = \min(\tau_{\leq 4,*}^{\mathrm{best}}, \tau_{5,*}^{\mathrm{best,int}}). \tag{6.139}$$

6.6.5 Final splice and evaluation budget

Combining the imported support-≤ 4 interval with the support-five interior interval gives the final certified interval

$$\tau_{\leq 5,\min}^{\mathrm{lo}} = \min(\tau_{\leq 4,\min}^{\mathrm{lo}}, \tau_{5,\min}^{\mathrm{lo,int}}), \qquad \tau_{\leq 5,\min}^{\mathrm{hi}} = \min(\tau_{\leq 4,\min}^{\mathrm{hi}}, \tau_{5,\min}^{\mathrm{hi,int}}). \tag{6.140}$$

Then

$$\tau_{\leq 5,\min}^{\mathrm{lo}} \leq \tau_{\leq 5,*}^{\mathrm{best}} \leq \tau_{\leq 5,\min}^{\mathrm{hi}}. \tag{6.141}$$

The preferred exact total evaluation budget is

$$1140 + 324 = 1464. \tag{6.142}$$

The fallback projected-chart budget is

$$1140 + 1500 = 2640. \tag{6.143}$$

Thus even the fallback full local search is finite.

Theorem 6.3 (Final local mixed-ray closure)**.** *Within the declared local free-quintuple mixed-ray class, the full positive-support search is exhausted by the support-≤ 4 boundary ledger and the unique support-five interior packet. There are no support-cardinality families beyond five. Any remaining ambiguity after Stage 218 is finite-candidate ambiguity inside the already-enumerated algebraic candidate sets, not a new continuum search.*

Claim status. Stages 216–218 are Exact Within Closure **for the boundary identification, support ceiling, lifted finite candidate compiler, splice theorem, and evaluation budget. The realized winning ray, if any, remains** Numerically Located **or** Open **until actual PDE-derived envelope and validity data are inserted and evaluated.**

6.7 What Part VI does and does not prove

Part VI proves that the local realization search has a finite, auditable structure. It does not prove that the actual nonlinear moving-throat PDE returns a branch on $\mathcal{Z}_*$. It also does not prove that the Hessian envelopes, validity maps, or candidate brackets take any particular numerical values. Those are branch data.

The safe downstream citation is therefore:

> Inside the declared free-quintuple graph and local mixed-ray search class, the realization problem reduces to the exact graph-error packet $(\chi_Q - 1, E_T, E_K, E_\mu)$ and the finite support-cardinality-five mixed-ray sieve. Actual branch realization remains a separate calculation.

For later use, the practical audit recipe is:

1. compute the PDE-returned free quintuple and dependent triple;
2. evaluate the graph target $(T_U^{\mathrm{graph}}, K_{\eta,*}^{\mathrm{graph}}, \mu_W^{\mathrm{graph}})$;
3. form $(\chi_Q - 1, E_T, E_K, E_\mu)$;
4. if the branch is graph-aligned, evaluate $\widehat{\chi}_Q$ and its local log-gradient/Hessian envelopes;
5. insert those data into the completed support-≤ 5 finite ledger.

This is exactly the point at which Part VII can start inserting same-charge and rigid-mouth actual-branch data without reopening the realization compiler.

Chapter 7

One-Port Same-Charge Sieve, Actual-Branch Tests, and Rigid-Mouth Orbit Lock

Stage range: 219–242

Part VI closed the local realization compiler: once a branch is expressed in the free-quintuple graph coordinates, the remaining local mixed-ray search is finite through support-cardinality five. Part VII is the first use of that completed back end on the same-charge, same-sign, one-port mixed-sector problem. Its job is not to add another algebraic search layer. Its job is to insert actual one-port wall/BdG/Maxwell/mixed branch data into the compiler, decide what the same-charge mixed sector can and cannot do, and then translate the surviving branch tests into the rigid-mouth physical normal form.

The conceptual outcome is simple but important. The mixed sector remains real and necessary, but it does not supply a new long-range same-charge attraction. The static one-port bundle produces only short-range product kernels. The linear dynamic bundle produces no new spatial kernel class and its passive/outgoing correction is phase lag at first order. What survives is a sharply constrained short-range/resonant corridor and, on the actual rigid-mouth branch, a two-coordinate orbit-lock packet. The rigid-mouth same-charge problem is diagonal in the physical logarithmic chart

$$U := \ln \frac{\mathcal{T}^2}{\mathcal{T}_{\rm ref}^2}, \qquad V := \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}, \tag{7.1}$$

and the strict orbit-lock point is

$$U = 0, \qquad V = 0. \tag{7.2}$$

The selected passive/outgoing support side is also no longer a three-regime ambiguity: the minimal isotropic quadrupole precursor fixes

$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\rm req} = \frac{1}{3}, \qquad \Pi_{\rm tr} = \frac{4}{3}C_{\rm mix}, \tag{7.3}$$

placing the selected branch on the symmetric lowest-twin support slice.

Part VII at a glance

Primary question What survives when the finished realization compiler is applied to actual one-port same-charge branch data, and how should the surviving branch tests be read in physical variables?

Starting inputs	The Part VI realization compiler, the one-port wall/BdG/Maxwell/mixed closure, and the selected passive/outgoing branch data carried forward from earlier parts.
Delivers	The same-charge kernel firewall, the transported $(\Delta_{\text{norm}}, \Xi_1)$ ceilings, the rigid-mouth chart (U, V), the orbit-lock theorem, and the selected twin-support placement packet.
Used by	Part VIII as the strict front end for the relaxed/open-system companion; later papers should cite this chapter for the strict same-charge verdict and its surviving branch packet.
Result anchors	**MTDC-T11**.

Stage packets.

Stages	Packet	Status	Carry-forward output
219–223	One-port kernel firewall and primitive compatibility	EXACT WITHIN CLOSURE / NUMERICALLY LOCATED	Fixes the short-range static and dynamic kernel classes, proves the phase-lag and resonance/linewidth constraints, and identifies the finite-throat compatibility surface where conservative and outgoing conditions can coexist.
224–232	Transported ceilings, Ξ_1 mechanism sieve, and finite branch windows	EXACT WITHIN CLOSURE / REDUCED / CONTROLLED REDUCTION / NUMERICALLY LOCATED / OPEN	Compiles the actual-branch prefactor packet, imports the strict 5PN even gate, filters the surviving mixed-sector mechanisms, and narrows the unresolved gate to static placement and coherent orbit lock.
233–239	Rigid-mouth orbit-lock compiler and physical normal form	EXACT WITHIN CLOSURE / OPEN	Pulls the surviving branch packet into direct observables, builds the packet projectors, computes the dressing compiler, and diagonalizes the strict branch test in the physical chart (U, V) with orbit lock at $U = V = 0$.
240–242	Selected twin-support slice and final front-end packet	EXACT WITHIN CLOSURE / OPEN	Fixes the symmetric lowest-twin support slice, ranks the primitive carriers on that branch, and defines the final strict front-end packet that later relaxed-branch work must inherit rather than overwrite.

Claim status. Part VII is mixed in status. The static and dynamic one-port kernels, the projectors, the physical chart, and the selected support-ratio compilers are EXACT WITHIN CLOSURE **inside the declared one-port, weak-axisymmetric, selected-branch, and rigid-mouth closures. Numerical statements in Stages 222–223 and Stage 232 are** NUMERICALLY LOCATED **insertions into exact formulas. Any claim that the completed nonlinear moving-throat PDE actually realizes the winning static placement, dynamic window, rigid-mouth orbit-lock point, or selected twin-support point remains** OPEN**. No result in this part should be cited as a proof of a new same-charge long-range attraction.**

What this part proves. Part VII applies the realization machinery to the one-port same-charge problem, rules out a new long-range same-charge attraction inside the linear one-port closure, and compiles the surviving strict branch conditions into the rigid-mouth (U, V) chart and selected twin-support packet.

What this part does not prove. It does not prove that the completed PDE actually lands on the surviving static placement, dynamic window, rigid-mouth orbit-lock point, or selected twin-support point. Those remain actual-branch realization questions.

Why later parts need it. Part VIII starts from the strict front end extracted here and builds the relaxed/open-system extension around it without erasing the strict same-charge verdict.

7.1 Block contract and theorem-level output

The contract of Part VII is to separate three questions that are easy to conflate. First, there is the *kernel-class question*: does the mixed sector create a new long-range same-charge attractive law? Stages 219–221 answer no within the one-port linear closure. Second, there is the *window question*: can a short-range static or resonant dynamic corridor survive the exact transported ceilings? Stages 222–232 reduce that question to finite branch inequalities and show that the support/source side is not the active bottleneck in the currently injected data. Third, there is the *actual branch question*: does the PDE-selected rigid-mouth branch land on the orbit-lock and selected-support packets? Stages 233–242 compile that question into direct observables.

The operational chain is

$$\begin{aligned} \mathcal{K}_{\mathrm{red}} &\longrightarrow \mathfrak{V}_{\mathrm{mix}}(x,\omega) \longrightarrow (\Delta_{\mathrm{norm}}, \Xi_1) \\ &\longrightarrow (R_{\mathrm{tr}}, \mathcal{T}^2, \epsilon_\eta) \longrightarrow (U, V) \\ &\longrightarrow \text{selected twin-support placement packet.} \end{aligned} \tag{7.4}$$

Theorem 7.1 (Part VII same-charge and rigid-mouth selected-branch theorem)**.** *Inside the declared one-port wall/BdG/Maxwell/mixed closure, weak-axisymmetric grouped response closure, rigid-mouth actual-branch closure, and selected twin-support closure, the following statements hold.*

1. *The static same-charge one-port bundle is a positive quadratic response on the admissible side $\Omega_U^2 > 0$, $\Delta > 0$, $D_0 > 0$. It produces only product kernels built from the primitive source profiles. For the carried source pair x^{-3} and $\mathrm{e}^{-2\kappa x}/x$, the only static product families are*

$$x^{-6}, \qquad e^{-2\kappa x}x^{-4}, \qquad e^{-4\kappa x}x^{-2}. \tag{7.5}$$

2. *The linear dynamic mixed bundle preserves the same spatial families. A passive/outgoing correction at first order is pure phase lag and carries no real conservative barrier lowering off pole.*

3. *Near a simple conservative pole, every relevant susceptibility has the local form*

$$\chi_*(\omega) = \frac{A_*}{\delta - i\gamma_*}, \qquad \delta = \omega - \omega_*, \tag{7.6}$$

   so

$$\frac{|\Re\chi_*|}{|\Im\chi_*|} = \frac{|\delta|}{\gamma_*}. \tag{7.7}$$

   The maximum conservative gain occurs exactly when conservative and absorptive magnitudes are equal.

4. *On the explicit finite-throat primitive branch, the conservative one-pole condition and outgoing normalization condition are simultaneously compatible only on the exact surface*

$$\frac{N_0}{P_{0,\text{target}}} = \frac{3(M + B_2 + Z_2)^2}{B_4 + Z_4}. \tag{7.8}$$

The transported dynamic window is finite on that surface.

5. *The actual weak-axisymmetric branch test is the transported prefactor packet*

$$(\Delta_{\text{norm}}, a_{P_0}, b_{P_0}), \qquad b_{P_0} = 3a_{P_0}, \qquad \Xi_1 = \frac{P_1}{P_0}. \tag{7.9}$$

The robust one-scalar ceiling is

$$\frac{\Delta_{\text{norm}} + T_{\text{quad}}}{\hat{m}_0^2}(1 + |\varepsilon\Xi_1|) \le P_{\text{crit}}, \qquad T_{\text{quad}} := \frac{54Gc_s^5}{5a^5c^5}. \tag{7.10}$$

6. *On the rigid-mouth actual branch, the physical packet is diagonal in the variables* (U, V). *The microscopic dependent correction is*

$$(\Delta_T, \Delta_{K_\eta}, \Delta_\mu) = (0, -V, U - V). \tag{7.11}$$

Therefore the static-only correction changes only μ_W*, while the post-static dressing correction is the equal* $K_\eta^{(\text{eff})}$–μ_W *ray.*

7. *Rigid-mouth orbit lock is the codimension-two Cartesian condition*

$$U = 0, \qquad V = 0, \tag{7.12}$$

equivalently

$$\mathcal{T}^2 = \mathcal{T}_{\text{ref}}^2, \qquad \epsilon_\eta = \epsilon_{\eta,\text{ref}}. \tag{7.13}$$

8. *The selected passive/outgoing support side lands exactly on the symmetric lowest-twin slice*

$$\rho_\alpha = \frac{4}{3}, \qquad \zeta_{\text{req}} = \frac{1}{3}, \qquad C_{\text{mix}} < \Pi_{\text{tr}} < 2C_{\text{mix}}. \tag{7.14}$$

9. *The actual moving-throat front-end packet to evaluate is*

$$(\epsilon, \varrho_{\text{phys}}, \sigma_{\text{phys}}, \textit{ranking region}, R_{\text{tr}}, R_{\text{target}}, \epsilon_\eta, d\ln R_{\text{tr}}, d\ln R_{\text{target}}, d\ln\epsilon_\eta, N_Q - 1)\,. \tag{7.15}$$

Any failure of the actual branch to realize equations (7.12), (7.14) *and* (7.15) *is a branch-realization failure, not a failure of the algebraic compiler.*

7.2 One-port same-charge static kernel

Stage 219 begins with the same isotropic one-port bundle already used by the grouped normalization chain. After the stable BdG support mode is integrated out, the effective static wall stiffness is

$$K_* := K - \frac{C^2}{\varpi^2}. \tag{7.16}$$

The one-port Maxwell/mixed block is controlled by

$$\Delta := \Omega_U^2\Omega_W^2 - R^2, \qquad Q := G_U^2\Omega_W^2 + 2G_UG_WR + G_W^2\Omega_U^2, \qquad P := \Omega_U^2G_W + RG_U, \tag{7.17}$$

with static wall operator

$$D_0 := K_* - \frac{Q}{\Delta}. \tag{7.18}$$

The same quantities control the outgoing prefactor:

$$N_0 = \frac{P^2}{\Delta^2}, \qquad P_0 = \frac{N_0}{D_0}. \tag{7.19}$$

Thus the static same-charge audit is not independent of the quadrupole normalization program; it probes the same reduced bundle.

The reduced static energy is

$$V_{\rm stat}(q,U,W;r) = \frac{1}{2}K_*q^2 + \frac{1}{2}\Omega_U^2U^2 + \frac{1}{2}\Omega_W^2W^2 - RUW - G_UqU - G_WqW - J_q(r)q - J_U(r)U - J_W(r)W. \tag{7.20}$$

In matrix form,

$$V_{\rm stat} = \frac{1}{2}X^T\mathcal{K}_{\rm red}X - J^TX, \qquad X = (q,U,W)^T, \tag{7.21}$$

where

$$\mathcal{K}_{\rm red} = \begin{pmatrix} K_* & -G_U & -G_W \\ -G_U & \Omega_U^2 & -R \\ -G_W & -R & \Omega_W^2 \end{pmatrix}. \tag{7.22}$$

The determinant identity is

$$\det\mathcal{K}_{\rm red} = \Delta D_0. \tag{7.23}$$

The admissible static side is therefore

$$\Omega_U^2 > 0, \qquad \Delta > 0, \qquad D_0 > 0. \tag{7.24}$$

On this side, minimization gives

$$X_*(r) = \mathcal{K}_{\rm red}^{-1}J(r), \qquad \delta V_{\rm mix}(r) = -\frac{1}{2}J(r)^T\mathcal{K}_{\rm red}^{-1}J(r) \le 0. \tag{7.25}$$

So the one-port static response is attractive or neutral at quadratic order. But its spatial family is inherited from the source products, not newly generated.

The wall–mixed cross susceptibility is especially important:

$$\chi_{qW} := (\mathcal{K}_{\rm red}^{-1})_{qW} = \frac{P}{\Delta D_0}. \tag{7.26}$$

With $\Lambda := P/\Delta$, one has

$$N_0 = \Lambda^2, \qquad P_0 = \frac{\Lambda^2}{D_0}, \qquad \chi_{qW} = \frac{\Lambda}{D_0}, \qquad \chi_{qW}^2 = \frac{P_0}{D_0}. \tag{7.27}$$

So static mixed softening and outgoing quadrupole normalization are linked by the same load factor and denominator.

For the primitive source profiles

$$\mathcal{S}_Q(x) = \frac{1}{x^3}, \qquad \mathcal{S}_Y(x) = \frac{e^{-2\kappa x}}{x}, \tag{7.28}$$

and the source vector

$$J(x) = \begin{pmatrix} \beta_Q \mathcal{S}_Q(x) \\ \beta_U \mathcal{S}_Y(x) \\ \beta_W \mathcal{S}_Y(x) \end{pmatrix}, \tag{7.29}$$

the static correction is

$$\widetilde{\delta V}_{\text{mix}}(x) = -\frac{1}{2}\left[\frac{\mathcal{C}_6}{x^6} + 2\mathcal{C}_4 \frac{e^{-2\kappa x}}{x^4} + \mathcal{C}_2 \frac{e^{-4\kappa x}}{x^2}\right]. \tag{7.30}$$

The coefficients are susceptibility-weighted source products. The key theorem is the family statement, not the coefficient details: no $-1/x$, no $-e^{-2\kappa x}/x$, and no longer-range static attraction appears inside this one-port static closure.

Theorem 7.2 (Static one-port square-law suppression). *Inside the admissible one-port static closure, the mixed bundle creates no new long-range same-charge attractive law. For the primitive source pair x^{-3} and $e^{-2\kappa x}/x$, it produces only the three product families in equation* (7.30)*. Static same-charge survival is therefore a coefficient-engineering or near-softening question on already-short-range families, not a new spatial-kernel theorem.*

Claim status. Stage 219 is Exact Within Closure **within the isotropic one-port static closure. The absence of a new long-range family is exact for the stated source families and bundle assumptions. The magnitude of the coefficients and their realized branch values remain branch data.**

7.3 Linear dynamic mixed-bundle kernel

Stages 220–221 ask whether time dependence rescues what the static kernel does not provide. The dynamic lift keeps the same three coordinates but replaces the static coefficients by

$$K_B(\omega) := K - M\omega^2 - \frac{C^2}{\varpi^2 - \omega^2}, \qquad A(\omega) := \Omega_U^2 - \omega^2, \qquad W(\omega) := \Omega_W^2 - \omega^2 - \Pi(\omega). \tag{7.31}$$

Then

$$\Delta_\Pi(\omega) := A(\omega)W(\omega) - R^2, \qquad Q_\Pi(\omega) := G_U^2 W(\omega) + 2G_U G_W R + G_W^2 A(\omega), \tag{7.32}$$

and

$$D_\Pi(\omega) := K_B(\omega) - \frac{Q_\Pi(\omega)}{\Delta_\Pi(\omega)}. \tag{7.33}$$

The dynamic stiffness matrix is

$$\mathcal{K}_{\text{dyn}}(\omega) = \begin{pmatrix} K_B(\omega) & -G_U & -G_W \\ -G_U & A(\omega) & -R \\ -G_W & -R & W(\omega) \end{pmatrix}, \tag{7.34}$$

with determinant

$$\det \mathcal{K}_{\text{dyn}}(\omega) = \Delta_\Pi(\omega) D_\Pi(\omega). \tag{7.35}$$

The complex response functional is

$$\mathfrak{V}_{\rm mix}(x,\omega) = -\frac{1}{2}J(x,\omega)^T\mathcal{K}_{\rm dyn}(\omega)^{-1}J(x,\omega). \tag{7.36}$$

Its real part is the in-phase barrier reshaping; its imaginary part is the quadrature pumping/leakage channel.

For the same primitive source families as Stage 219, one gets the exact dynamic product-family theorem

$$\mathfrak{V}_{\rm mix}(x,\omega) = -\frac{1}{2}\left[\frac{\mathcal{C}_6(\omega)}{x^6} + 2\mathcal{C}_4(\omega)\frac{e^{-2\kappa x}}{x^4} + \mathcal{C}_2(\omega)\frac{e^{-4\kappa x}}{x^2}\right]. \tag{7.37}$$

Thus linear monochromatic driving does not create a new radial family. It promotes the same coefficients to complex susceptibilities.

The outgoing port enters only through the WW slot. With $e_W = (0,0,1)^T$,

$$\partial_\Pi \mathcal{K}_{\rm dyn}^{-1} = \mathcal{K}_{\rm dyn}^{-1} e_W e_W^T \mathcal{K}_{\rm dyn}^{-1}, \tag{7.38}$$

so

$$\partial_\Pi \mathfrak{V}_{\rm mix} = -\frac{1}{2}\left[e_W^T\mathcal{K}_{\rm dyn}^{-1}J\right]^2. \tag{7.39}$$

On the conservative off-pole branch the bracket is real. Therefore, for a passive/outgoing port

$$\Pi(\omega) = i\Gamma(\omega) + O(\Gamma^2), \qquad \Gamma(\omega) \geq 0, \tag{7.40}$$

the first correction is

$$\delta\mathfrak{V}^{(1)}_{\rm mix}(x,\omega) = -\frac{i}{2}\Gamma(\omega)\mathcal{T}_J(\omega)^2, \tag{7.41}$$

where $\mathcal{T}_J = e_W^T\mathcal{K}_{\rm cons}^{-1}J$. Consequently

$$\Re\delta\mathfrak{V}^{(1)}_{\rm mix} = 0, \qquad \overline{P}^{(1)}_{\rm abs}(x,\omega) = \frac{\omega\Gamma(\omega)}{2}\mathcal{T}_J(\omega)^2 \geq 0. \tag{7.42}$$

This is the phase-lag no-go: passive/outgoing dressing gives absorption at first order, not conservative lowering.

Stage 221 then studies the only remaining dynamic escape hatch: resonant dispersive enhancement of the already-short-range families. Near a simple conservative pole, write

$$F_\Pi(\omega) = F_0'(\omega_*)\delta - \Pi(\omega_*)Z_* + O(\delta^2, \Pi\delta, \Pi^2), \qquad \delta = \omega - \omega_*. \tag{7.43}$$

For $\Pi(\omega_*) = i\Gamma_*$, the reduced susceptibility has the universal form

$$\chi_s(\omega) \approx \frac{A_*}{\delta - i\gamma_*}, \qquad A_* := \frac{\mathcal{N}_s(\omega_*)}{F_0'(\omega_*)}, \qquad \gamma_* := \frac{\Gamma_* Z_*}{|F_0'(\omega_*)|}. \tag{7.44}$$

This gives

$$|\Re\chi_*| = \frac{|A_*|}{\gamma_*}\frac{r}{1+r^2}, \qquad |\Im\chi_*| = \frac{|A_*|}{\gamma_*}\frac{1}{1+r^2}, \qquad r := \frac{|\delta|}{\gamma_*}. \tag{7.45}$$

The exact maximum of the conservative part occurs at $r = 1$, where $|\Re\chi_*| = |\Im\chi_*|$. If one demands a low-loss window

$$|\Im\chi_*| \leq \eta|\Re\chi_*|, \qquad 0 < \eta \leq 1, \tag{7.46}$$

then the best possible conservative magnitude is

$$\sup_{|\Im\chi_*|\le\eta|\Re\chi_*|} |\Re\chi_*| = \frac{|A_*|}{\gamma_*}\frac{\eta}{1+\eta^2}. \tag{7.47}$$

For a spatial family $S_j(x)$, a necessary low-loss survival condition is

$$\frac{|A_j|}{\gamma_*}\frac{\eta}{1+\eta^2}S_j(x)^2 \ge 2\Delta V_{\mathrm{req}}(x). \tag{7.48}$$

Theorem 7.3 (Linear dynamic no-free-lunch theorem)**.** *Inside the one-port linear dynamic closure, time dependence does not create a new same-charge spatial kernel. Passive/outgoing dressing is pure phase lag at first order off pole. Near a pole, useful conservative gain and absorptive loading obey the exact line-shape tradeoff equation* (7.45)*. Therefore any surviving linear dynamic corridor must be a high-quality, residue-to-linewidth enhancement of one of the already-known short-range families.*

Claim status. Stages 220–221 are Exact Within Closure **for the dynamic bundle algebra, phase-lag identity, simple-pole normal form, and low-loss survival inequality. Whether a completed branch supplies a useful pole with large enough residue-to-linewidth ratio remains** Open**.**

7.4 Actual-branch prefactor ceiling

Stages 222–224 turn the general dynamic survival inequality into a concrete transported branch test. Stage 222 chooses the finite interval $s \in [0, L]$, the lowest N/N zero mode

$$u_0(s) = \frac{1}{\sqrt{L}}, \tag{7.49}$$

and the lowest D/N half-wave

$$f_0(s) = \sqrt{\frac{2}{L}}\sin\frac{\pi s}{2L}. \tag{7.50}$$

The overlap is

$$\kappa = \int_0^L u_0(s)f_0(s)\,ds = \frac{2\sqrt{2}}{\pi}. \tag{7.51}$$

Thus

$$C = \kappa\lambda_B, \qquad G_U = \lambda_U, \qquad G_W = \kappa\lambda_W, \qquad R = \kappa\lambda_R. \tag{7.52}$$

The conservative pole condition is a quartic in $y = \omega^2$:

$$\begin{aligned} F(y) = &\Big((K - My)(\varpi^2 - y) - C^2\Big)\Big((\Omega_U^2 - y)(\Omega_W^2 - y) - R^2\Big) \\ &- (\varpi^2 - y)\Big(G_U^2(\Omega_W^2 - y) + 2G_UG_WR + G_W^2(\Omega_U^2 - y)\Big). \end{aligned} \tag{7.53}$$

So the primitive finite-throat branch has four conservative poles in the generic admissible case.

For a simple wall-like pole with the outgoing quadrupole port

$$\Pi_{\mathrm{out}}(\omega) = i\Gamma_5\omega^5, \qquad \Gamma_5 = \frac{a^5}{27c_s^5}, \tag{7.54}$$

the residue/linewidth figure for the pure quadrupolar same-charge family simplifies to

$$\mathcal{R}_{Q,*} = \frac{|A_{Q,*}|}{\gamma_*} = \frac{27c_s^5}{a^5\omega_*^5 N(\omega_*)}. \tag{7.55}$$

The derivative of the pole cancels; the figure is controlled by the pole frequency and outgoing transfer factor.

Stage 223 then imposes the isotropic one-pole target surface. On the isotropic branch,

$$D_0 = K - B_0 - Z_0, \qquad D_2 = -(M + B_2 + Z_2), \qquad D_4 = -(B_4 + Z_4). \tag{7.56}$$

The one-pole identity and outgoing-normalization condition solve for the same wall stiffness. Their equality is exactly

$$\frac{N_0}{P_{0,\text{target}}} = \frac{3(M + B_2 + Z_2)^2}{B_4 + Z_4}. \tag{7.57}$$

Equivalently, the primitive branch induces the compatible target

$$P_{0,\text{target,compat}} = \frac{N_0(B_4 + Z_4)}{3(M + B_2 + Z_2)^2}. \tag{7.58}$$

This is the exact compatibility surface from the conservative one-pole side to the outgoing-normalization side.

Stage 224 transports the finite dynamic survival windows from the primitive compatibility family into the actual branch packet. The final 5PN branch packet contains

$$\Delta_{\text{branch}} = (a_2, b_2, a_4, b_4, a_{P_0}, b_{P_0}, \Delta_{\text{pole}}, \Delta_{\text{norm}}), \tag{7.59}$$

where

$$\Delta_{\text{norm}} = \hat{m}_0^2 \bar{P}_0 - T_{\text{quad}}, \qquad T_{\text{quad}} := \frac{54Gc_s^5}{5a^5c^5}. \tag{7.60}$$

Hence

$$\bar{P}_0 = \frac{\Delta_{\text{norm}} + T_{\text{quad}}}{\hat{m}_0^2}. \tag{7.61}$$

The grouped inverse map gives

$$P_{20} = \bar{P}_0 + 4a_{P_0}, \qquad P_{21} = \bar{P}_0 - a_{P_0} + b_{P_0}, \qquad P_{22} = \bar{P}_0 - a_{P_0} - b_{P_0}. \tag{7.62}$$

For a ceiling P_{crit}, the all-lane transported condition is

$$\max\{P_{20}, P_{21}, P_{22}\} \le P_{\text{crit}}. \tag{7.63}$$

On the weak-axisymmetric branch,

$$\lambda_{20} = 1, \qquad \lambda_{21} = \frac{1}{2}, \qquad \lambda_{22} = -1, \tag{7.64}$$

and

$$P_A = \bar{P}_0(1 + \varepsilon\lambda_A\Xi_1), \qquad \Xi_1 = \frac{P_1}{P_0}. \tag{7.65}$$

Therefore

$$a_{P_0} = \frac{\varepsilon\bar{P}_0\Xi_1}{4}, \qquad b_{P_0} = \frac{3\varepsilon\bar{P}_0\Xi_1}{4}, \tag{7.66}$$

and the all-lane ceiling collapses to

$$\bar{P}_0(1 + |\varepsilon\Xi_1|) \le P_{\text{crit}}. \tag{7.67}$$

Substituting equation (7.61) gives the actual-branch form already stated in equation (7.10).

Theorem 7.4 (Transported actual-branch prefactor ceiling)**.** *Inside the transported primitive-family window closure, an actual weak-axisymmetric branch clears the robust all-lane ceiling only if*

$$\frac{\Delta_{\mathrm{norm}} + T_{\mathrm{quad}}}{\hat{m}_0^2}(1 + |\varepsilon \Xi_1|) \le P_{\mathrm{crit}}. \tag{7.68}$$

If the branch is exactly calibrated, $\Delta_{\mathrm{norm}} = 0$, the same condition becomes a lower bound on the source-map factor,

$$\hat{m}_0^2 \ge \frac{T_{\mathrm{quad}}(1 + |\varepsilon \Xi_1|)}{P_{\mathrm{crit}}}. \tag{7.69}$$

Claim status. Stages 222–224 are exact for the primitive branch formulas, compatibility surface, and packet compiler. Pole censuses, compatibility-point values, and numerical headroom budgets are Numerically Located**. The use of primitive-family ceilings as actual-branch sufficient conditions is a transported reduced test, not an unconditional theorem of the full PDE.**

7.5 Strict 5PN even-gate package

Stages 225–226 insert the weak-axisymmetric prefactor into the stricter first-order 5PN package. At first order write

$$D_A = D_0 + \varepsilon\lambda_A D_{01} + O(\varepsilon^2), \qquad N_A = N_0 + \varepsilon\lambda_A N_{01} + O(\varepsilon^2). \tag{7.70}$$

Then

$$\Xi_{\mathrm{load}} = \frac{N_{01}}{N_0} - \frac{D_{01}}{D_0} = \frac{P_1}{P_0} = \Xi_1. \tag{7.71}$$

The stricter imported first-order even gates are

$$K_1 := D_{21} + \frac{D_{01}}{9}, \qquad H_{\mathrm{even}} := D_{41} - \frac{2}{3}D_{21} - \frac{D_{01}}{27}. \tag{7.72}$$

Thus the same scalar Ξ_1 that controls the same-charge prefactor ceiling is only one member of the strict first-order package. A branch must also pass the conservative even gates if it is to remain on the imported 5PN one-pole/hidden-even structure.

Stage 225 first imposes a weaker compensation surface that preserves the conservative grouped response on the explicit branch. It then builds a primitive logarithmic-slope compiler from

$$(x_K, x_M, x_{\lambda_B}, x_{\varpi}, x_{\lambda_U}, x_{\lambda_W}, x_{\lambda_R}, x_{\Omega_U}, x_{\Omega_W}) \tag{7.73}$$

to

$$D_{01}, \qquad D_{21}, \qquad D_{41}, \qquad N_{01}, \qquad \Xi_1. \tag{7.74}$$

The first sieve shows that wall-only and pure-BdG-only mechanisms do not supply the desired nontrivial same-charge load once the conservative compensation is enforced. A mixed-sector family survives.

Stage 226 imports equation (7.72) and tightens the survivor. The sharpest surviving subcorridor has

$$D_{01} = D_{21} = D_{41} = 0, \qquad \Xi_1 = \Xi_{\mathrm{load}} = \frac{N_{01}}{N_0}. \tag{7.75}$$

In words: after the strict even gates and conservative-shape preservation are imposed on the noncanonical compatibility branch, the remaining same-charge effect is not generic mixed anisotropy. It is pure outgoing-transfer enhancement with the conservative one-pole bundle frozen at first order.

Theorem 7.5 (Strict first-order package and pure-transfer survivor)**.** *Inside the imported first-order 5PN package, the same-charge load scalar is exactly* $\Xi_{\rm load} = P_1/P_0$*. Passing the strict even gates requires controlling* K_1 *and* $H_{\rm even}$ *in addition to* $\Xi_{\rm load}$*. On the explicit compatibility branch with Stage 225 conservative-shape preservation, the sharp surviving subcorridor is the pure-transfer corridor equation* (7.75)*.*

Claim status. Stages 225–226 are Exact Within Closure **for the first-order compilers and even-gate identities. The mechanism sieve is exact within the primitive slope space, while the actual PDE realization of the pure-transfer corridor is** Open**.**

7.6 Pure-transfer corridor and co-loading no-go

Stages 227–228 analyze the pure-transfer survivor. Define the outgoing-load factor

$$\Lambda := \frac{P}{\Delta}. \tag{7.76}$$

In the pure-transfer corridor,

$$\Xi_1 = \frac{N_{01}}{N_0} = 2\frac{P_{01}}{P} - 2\frac{\Delta_{01}}{\Delta} = 2\delta \ln \Lambda. \tag{7.77}$$

The one-port factorization is

$$\Lambda = \frac{G_W}{\Omega_W^2}\frac{1+I}{1-H}, \qquad I := \frac{RG_U}{\Omega_U^2 G_W}, \qquad H := \frac{R^2}{\Omega_U^2\Omega_W^2}. \tag{7.78}$$

The three pieces are: the raw mixed leg G_W/Ω_W^2, the interference ratio I, and the hybridization/blocking ratio H.

The outgoing-rigidity sieve asks which of those pieces can remain fixed while Ξ_1 survives. If both interference and hybridization are rigid,

$$\delta \ln I = 0, \qquad \delta \ln H = 0, \tag{7.79}$$

then the simultaneous pure-transfer and conservative constraints leave only the trivial pure-transfer drift on the concrete branch. This is the first co-loading no-go: the same-charge effect cannot be reduced to a naive mixed-leg slogan under both rigidities.

Stage 228 then splits the pure-transfer slope into numerator and denominator pieces:

$$\Xi_1 = 2(\pi_1 - \delta_1), \qquad \pi_1 := \frac{P_{01}}{P}, \qquad \delta_1 := \frac{\Delta_{01}}{\Delta}. \tag{7.80}$$

The exact rigid-subcorridor theorem is:

1. pure-transfer plus $\pi_1 = 0$ leaves a one-dimensional numerator-rigid survivor;
2. pure-transfer plus $\delta_1 = 0$ leaves a one-dimensional denominator-rigid survivor;
3. imposing both $\pi_1 = 0$ and $\delta_1 = 0$ kills the corridor.

Both one-dimensional survivors pass the first wall-like dynamic-window audit on the explicit compatibility point. The first active ceiling remains the transported static $|\varepsilon\Xi_1|$ budget, not the dynamic wall-like window.

Theorem 7.6 (Pure-transfer and rigidity sieve)**.** *Inside the strict pure-transfer corridor, the same-charge scalar is the logarithmic outgoing-load slope* $\Xi_1 = 2\delta \ln \Lambda$. *The factorization equation* (7.78) *separates raw mixed-leg, interference, and hybridization contributions. Simultaneous interference and hybridization rigidity kills the nontrivial pure-transfer drift on the concrete compatibility branch, while either numerator-rigid or denominator-rigid splitting alone leaves a one-dimensional survivor.*

Claim status. Stages 227–228 are Exact Within Closure **for the pure-transfer identities and rigidity splits. The branch-specific dynamic-window survival statements are** Numerically Located **insertions into exact ceiling formulas.**

7.7 Selected-branch dynamic-window compiler

Stages 229–232 replace the free numerator/denominator split by the selected-branch softening-depth language and then pull the result back to physical continuum coordinates.

The selected normalization product is

$$N_-(x) = \frac{\beta_0 s_-(x)^2}{\kappa_0^2(A-x)}, \qquad 0 \le x < A, \tag{7.81}$$

with selected overlap

$$s_-(x) = \frac{\left[\kappa_0^2(x+\Delta K_{\mathrm{ax}}) + \kappa_1^2 x\right]^2}{\kappa_0^2(x+\Delta K_{\mathrm{ax}})^2 + \kappa_1^2 x^2}. \tag{7.82}$$

With dimensionless variables

$$\xi := \frac{x}{A}, \qquad \delta := \frac{\Delta K_{\mathrm{ax}}}{A}, \tag{7.83}$$

Stage 229 defines a log-slope classifier $\mathcal{R}_{ND}$ that compares the numerator-like and denominator-like content of the selected branch. The exact threshold

$$\delta = \frac{8}{9} \tag{7.84}$$

separates always-denominator selected branches from branches that begin numerator-dominant. The selected branch is never literally either rigid Stage 228 subcorridor; it is an exact co-loading product. However it is always denominator-like near softening, and it is denominator-like on the entire stable branch for $\delta \ge 8/9$.

Stage 230 uses the classifier to mix the carried numerator-rigid and denominator-rigid dynamic data. The share weights are

$$w_{\mathrm{num}} = \frac{\mathcal{R}_{ND}}{1+\mathcal{R}_{ND}}, \qquad w_{\mathrm{den}} = \frac{1}{1+\mathcal{R}_{ND}}. \tag{7.85}$$

The dynamic sign theorem says the upper wall-like pole always worsens, while the lower wall-like pole flips sign only at one classifier threshold. Most importantly, the selected-branch dynamic ceilings are everywhere weaker than the universal transported static ceilings. Therefore the first kill condition remains the static $|\varepsilon \Xi_1|$ budget.

Stage 231 pulls $\mathcal{R}_{ND}$ back through the continuum placement map. The physical classifier is

$$\mathcal{R}_{\mathrm{phys}}(\delta, R_{\mathrm{target}}) := \mathcal{R}_{ND}\big(\xi_{\mathrm{req}}(\delta, R_{\mathrm{target}}), \delta\big), \tag{7.86}$$

where ξ_{req} is the stable solution of

$$F(\xi, \delta) = R_{\mathrm{target}}. \tag{7.87}$$

The monotonicity theorem is

$$\partial_{R_{\text{target}}}\mathcal{R}_{\text{phys}} < 0. \tag{7.88}$$

So larger normalization demand pushes the physical selected branch in the denominator-like and dynamically safer direction. The static-first theorem survives the pullback.

Stage 232 injects the known 5PN Family-1 support/source data. The selected passive/outgoing branch requires

$$\zeta_{\text{req}} = \frac{1}{3}, \qquad \rho_{\alpha}^{\text{req}} = \frac{4}{3}. \tag{7.89}$$

The numerically located Family-1 support/source branch returns $\zeta_{\text{phys}} \approx 2.4675$ for the carried wall-depth extractions, so the support/source side overshoots the canonical demand by a large margin. This does not prove same-charge success; it says the first active unresolved bottleneck stays at the actual PDE-selected static orbit-lock/coherent placement point.

Theorem 7.7 (Selected-branch static-first theorem)**.** *Inside the selected-branch dynamic-window compiler, the wall-like dynamic window does not become the first same-charge kill condition. After continuum pullback and known support/source injection, the support/source side is non-bottlenecked in the carried data. The live unresolved test remains the actual static orbit-lock/coherent placement packet.*

Claim status. Stages 229–231 are Exact Within Closure **for the selected-branch classifier, share compiler, monotonic pullback, and static-first theorem. Stage 232 is** Numerically Located **for the injected Family-1 support/source roots and margins. The actual PDE-selected static placement remains** Open**.**

7.8 Rigid-mouth orbit-lock compiler

Stages 233–236 translate the surviving static placement problem into rigid-mouth packet language. The first firewall is that mouth rigidity and operator rigidity are not the same statement. Rigid-mouth or tracking lock is represented by

$$\delta \ln R_{\text{tr}} = 0, \tag{7.90}$$

whereas $D_{01} = 0$ is the stronger algebraic statement that the effective grouped operator does not drift at first order.

The direct finite branch observables are

$$(R_{\text{tr}}, R_{\text{target}}, \epsilon_{\eta}). \tag{7.91}$$

Relative to a reference branch, the finite quotient coordinates include

$$q_{\text{tr}} = -C_{*} \ln \frac{R_{\text{tr}}}{R_{\text{tr,ref}}}, \tag{7.92}$$

$$q_{\text{nt}} = B_{*} \ln \frac{R_{\text{tr}}}{R_{\text{tr,ref}}} + \ln \frac{1-\epsilon_{\eta}}{1-\epsilon_{\eta,\text{ref}}} - \ln \frac{R_{\text{target}}}{R_{\text{target,ref}}}, \tag{7.93}$$

$$q_{\eta} = \ln \frac{\epsilon_{\eta}}{\epsilon_{\eta,\text{ref}}}. \tag{7.94}$$

At first order, Stage 234 gives the direct compiler

$$\Theta_1 = \delta \ln R_{\text{tr}}, \qquad \Xi_1 = -\delta \ln R_{\text{target}} - c_{\eta}\delta \ln \epsilon_{\eta}, \qquad \mathcal{R}_1 = \delta \ln R_{\text{target}}, \tag{7.95}$$

with

$$c_\eta := \frac{\epsilon_{\eta,*}}{1-\epsilon_{\eta,*}}. \tag{7.96}$$

On the rigid-mouth slice, $q_{\rm tr} = 0$, so the static scalar is

$$\Xi_1 = -R_1 - c_\eta E_1, \qquad R_1 := \delta \ln R_{\rm target}, \qquad E_1 := \delta \ln \epsilon_\eta. \tag{7.97}$$

The two-observable static gate is therefore a condition on (R_1, E_1), not on support/source amplitude.

Stage 235 makes the packet structure exact. Define

$$\mathbf{x}_{\rm rm} := \begin{pmatrix} R_1 \\ E_1 \end{pmatrix}, \qquad \mathbf{q}_{\rm rm} := \begin{pmatrix} q_{\rm nt} \\ q_\eta \end{pmatrix} = \begin{pmatrix} \Xi_1 \\ E_1 \end{pmatrix}. \tag{7.98}$$

Then

$$\mathbf{q}_{\rm rm} = M_{\rm rm}\mathbf{x}_{\rm rm}, \qquad M_{\rm rm} := \begin{pmatrix} -1 & -c_\eta \\ 0 & 1 \end{pmatrix}, \qquad M_{\rm rm}^2 = I_2. \tag{7.99}$$

The direct-space projectors are

$$P_{\rm nt} = \begin{pmatrix} 1 & c_\eta \\ 0 & 0 \end{pmatrix}, \qquad P_\eta = \begin{pmatrix} 0 & -c_\eta \\ 0 & 1 \end{pmatrix}. \tag{7.100}$$

The static strip $\Xi_1 = 0$ is codimension one. Full rigid-mouth orbit lock is codimension two:

$$q_{\rm nt} = 0, \qquad q_\eta = 0 \quad \Longleftrightarrow \quad R_1 = 0, \qquad E_1 = 0. \tag{7.101}$$

The static-blind dressing line is

$$\Xi_1 = 0 \quad \Longleftrightarrow \quad \mathbf{x}_{\rm rm} = \begin{pmatrix} -c_\eta q_\eta \\ q_\eta \end{pmatrix}. \tag{7.102}$$

Thus clearing the first static same-charge ceiling need not make the true orbit-failure amplitude small.

Stage 236 identifies the microscopic carrier. On the rigid-mouth slice the dependent correction is

$$\begin{pmatrix} \Delta_T^{(q)} \\ \Delta_{K_\eta}^{(q)} \\ \Delta_\mu^{(q)} \end{pmatrix} = \begin{pmatrix} 0 \\ -q_\eta \\ q_{\rm nt} - q_\eta \end{pmatrix}. \tag{7.103}$$

The static-blind line $q_{\rm nt} = 0$ maps to the equal-drift ray

$$\begin{pmatrix} \Delta_T \\ \Delta_{K_\eta} \\ \Delta_\mu \end{pmatrix} = -q_\eta \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}. \tag{7.104}$$

So after the static strip is cleared, the remaining orbit defect is not generic throat motion. It is one scalar equal-drift $K_\eta^{(\rm eff)}$–μ_W dressing amplitude.

Theorem 7.8 (Rigid-mouth packet and dependent-plane theorem)**.** *On the rigid-mouth slice, the first static same-charge gate tests only $q_{\rm nt} = \Xi_1$. The dressing coordinate q_η is invisible to that gate and maps microscopically to the equal-drift ray $-q_\eta(0,1,1)^T$ in the dependent triple. Therefore the static gate is necessary but not sufficient for orbit lock.*

Claim status. Stages 233–236 are Exact Within Closure **for the rigid-mouth packet compilers and projectors. The physical assumption that the actual branch is rigid-mouth/track-locked, and the realized value of q_η, remain** Open **branch data.**

7.9 Physical logarithmic chart (U, V)

Stages 237–239 diagonalize the rigid-mouth packet in physical branch variables. Stage 237 begins from the finite rigid-mouth packet

$$q_{\rm nt} = \ln\frac{1-\epsilon_\eta}{1-\epsilon_{\eta,\rm ref}} - \ln\frac{R_{\rm target}}{R_{\rm target,ref}}, \qquad q_\eta = \ln\frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}. \tag{7.105}$$

The first static gate $q_{\rm nt} = 0$ is the finite curve

$$\frac{R_{\rm target}}{R_{\rm target,ref}} = \frac{1-\epsilon_\eta}{1-\epsilon_{\eta,\rm ref}}. \tag{7.106}$$

Parameterizing by q_η,

$$\epsilon_\eta = \epsilon_{\eta,\rm ref}e^{q_\eta}, \qquad \frac{R_{\rm target}}{R_{\rm target,ref}} = \frac{1-\epsilon_{\eta,\rm ref}e^{q_\eta}}{1-\epsilon_{\eta,\rm ref}}. \tag{7.107}$$

Conversely, once the static gate is cleared,

$$q_\eta = \ln\left(\frac{1-(1-\epsilon_{\eta,\rm ref})R_{\rm target}/R_{\rm target,ref}}{\epsilon_{\eta,\rm ref}}\right). \tag{7.108}$$

The microscopic extractor is

$$q_\eta = 2\ln\frac{c_{\eta U}}{c_{\eta U,\rm ref}} - \ln\frac{K_U}{K_{U,\rm ref}} - \ln\frac{K_\eta^{(\rm eff)}}{K_{\eta,\rm ref}^{(\rm eff)}}. \tag{7.109}$$

At first order,

$$q_\eta = 2c_1 - \kappa_U - \kappa_\eta. \tag{7.110}$$

The dressing coordinate is support-blind:

$$\partial_\varsigma q_\eta = 0, \qquad \partial_{M_{\rm tr}} q_\eta = 0. \tag{7.111}$$

Stage 238 then uses the coherent transfer-shape identity

$$R_{\rm target}\mathcal{T}^2 = \Lambda_0(1-\epsilon_\eta). \tag{7.112}$$

This converts the same-charge packet into physical variables. Relative to a reference branch,

$$q_{\rm nt} + \frac{B_*}{C_*}q_{\rm tr} = \ln\frac{\mathcal{T}^2}{\mathcal{T}^2_{\rm ref}}. \tag{7.113}$$

On the rigid-mouth slice, $q_{\rm tr} = 0$, so

$$q_{\rm nt} = \ln\frac{\mathcal{T}^2}{\mathcal{T}^2_{\rm ref}}, \qquad q_\eta = \ln\frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}. \tag{7.114}$$

Thus the first static same-charge gate is exactly transfer-shape rigidity:

$$q_{\rm nt} = 0 \quad \Longleftrightarrow \quad \mathcal{T}^2 = \mathcal{T}^2_{\rm ref}. \tag{7.115}$$

The post-static gate is dressing rigidity:

$$q_\eta = 0 \quad \Longleftrightarrow \quad \epsilon_\eta = \epsilon_{\eta,\rm ref}. \tag{7.116}$$

Stage 239 packages this in the Cartesian variables

$$U := \ln \frac{\mathcal{T}^2}{\mathcal{T}^2_{\rm ref}}, \qquad V := \ln \frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}. \tag{7.117}$$

Then

$$q_{\rm nt} = U, \qquad q_\eta = V. \tag{7.118}$$

The target ratio factorizes as

$$\frac{R_{\rm target}}{R_{\rm target,ref}} = \frac{1-\epsilon_{\eta,\rm ref}e^V}{1-\epsilon_{\eta,\rm ref}} e^{-U}. \tag{7.119}$$

The dependent microscopic correction is

$$\begin{pmatrix} \Delta_T \\ \Delta_{K_\eta} \\ \Delta_\mu \end{pmatrix} = \begin{pmatrix} 0 \\ -V \\ U-V \end{pmatrix}. \tag{7.120}$$

A left inverse reads

$$U = \Delta_\mu - \Delta_{K_\eta}, \qquad V = -\Delta_{K_\eta}. \tag{7.121}$$

The pure transfer-shape axis has image

$$(U,0) \longmapsto (0,0,U), \tag{7.122}$$

so clearing the static defect changes only μ_W. The pure dressing axis has image

$$(0,V) \longmapsto -V(0,1,1), \tag{7.123}$$

the equal $K_\eta^{(\rm eff)}$–μ_W ray. Therefore full orbit lock is

$$U = 0, \qquad V = 0. \tag{7.124}$$

At first order,

$$\begin{pmatrix} \Delta_T \\ \Delta_{K_\eta} \\ \Delta_\mu \end{pmatrix}^{(1)} = \begin{pmatrix} 0 \\ -\delta \ln \epsilon_\eta \\ \delta \ln \mathcal{T}^2 - \delta \ln \epsilon_\eta \end{pmatrix}. \tag{7.125}$$

Theorem 7.9 (Rigid-mouth Cartesian orbit-lock theorem)**.** *On the rigid-mouth actual branch, the physical logarithmic coordinates (U,V) are the finite quotient packet. The first static gate is $U=0$. The post-static dressing gate is $V=0$. The full orbit-lock point is the Cartesian codimension-two condition $(U,V)=(0,0)$, equivalently $\mathcal{T}^2 = \mathcal{T}^2_{\rm ref}$ and $\epsilon_\eta = \epsilon_{\eta,\rm ref}$.*

Claim status. Stages 237–239 are Exact Within Closure **for the finite static-blind curve, physical transfer-shape factorization, support-blindness, Cartesian chart, dependent correction, and orbit-lock theorem. Whether the actual branch has $U = V = 0$ remains** Open**.**

7.10 Selected twin-support branch and placement

Stages 240–242 close the selected support side and attach it to the actual branch packet. Stage 240 begins with the selected product identities

$$\Pi_{\mathrm{tr}} = \frac{N_Q^{(\mathrm{target})}}{\beta_0}\alpha_{\mathrm{req}}, \qquad C_{\mathrm{mix}} = \frac{N_Q^{(\mathrm{target})}}{\beta_0}\alpha_{\mathrm{mix}}. \tag{7.126}$$

Therefore

$$\frac{\Pi_{\mathrm{tr}}}{C_{\mathrm{mix}}} = \frac{\alpha_{\mathrm{req}}}{\alpha_{\mathrm{mix}}} =: \rho_\alpha. \tag{7.127}$$

For the minimal isotropic conservative quadrupole precursor

$$Y_Q^{\mathrm{cons}}(\omega) = c_0 + \frac{c_1}{1-\omega^2/\Omega_Q^2}, \qquad c_0 = \frac{3}{4}, \qquad c_1 = \frac{1}{4}, \qquad \Omega_Q = \frac{3c_s}{2a}, \tag{7.128}$$

the contact/pole inverse compiler gives

$$\rho_\alpha = \frac{1}{c_0} = \frac{4}{3}, \qquad \zeta_{\mathrm{req}} := \rho_\alpha - 1 = \frac{1}{3}. \tag{7.129}$$

Therefore

$$\Pi_{\mathrm{tr}} = \frac{4}{3}C_{\mathrm{mix}}. \tag{7.130}$$

With

$$\varrho := \frac{\pi^2\Pi_{\mathrm{tr}}}{16\Lambda}, \qquad C_{\mathrm{mix}} = \frac{8\Lambda(1-\epsilon_*)}{\pi^2}, \tag{7.131}$$

this becomes

$$\varrho = \frac{2(1-\epsilon_*)}{3}, \qquad S_{\mathrm{req}} = \frac{4}{3}. \tag{7.132}$$

Thus the selected branch lies strictly in the symmetric lowest-twin window:

$$C_{\mathrm{mix}} < \Pi_{\mathrm{tr}} < 2C_{\mathrm{mix}}. \tag{7.133}$$

Stage 241 turns this into a primitive ranking phase diagram. The selected curve is

$$\epsilon_* = 1 - \frac{3\varrho}{2}, \qquad \sigma = \frac{4}{3\varrho} - 2, \qquad 0 < \varrho < \frac{2}{3}. \tag{7.134}$$

For the constructive coherent parameter $0 < \beta < 2/11$, only two crossover thresholds survive:

$$\varrho_{W\Lambda} = \frac{2(1+\beta^2)}{3(2+\beta^2)}, \qquad \varrho_{U\Lambda} = \frac{2(1+\beta^2)}{3(1+\beta+\beta^2)}. \tag{7.135}$$

They obey

$$0 < \varrho_{W\Lambda} < \varrho_{U\Lambda} < \frac{2}{3}. \tag{7.136}$$

The ranking theorem is

$$0 < \varrho < \varrho_{W\Lambda} \implies q_\chi > q_Z > q_\Lambda > q_W > |q_U|, \tag{7.137}$$

$$\varrho_{W\Lambda} < \varrho < \varrho_{U\Lambda} \implies q_\chi > q_Z > q_W > q_\Lambda > |q_U|, \tag{7.138}$$

$$\varrho_{U\Lambda} < \varrho < \frac{2}{3} \quad \Longrightarrow \quad q_\chi > q_Z > q_W > |q_U| > q_\Lambda. \tag{7.139}$$

So the selected branch is dominated by interference/overlap/wall-blocking/outgoing-scale corrections, not by a large split-U correction.

Stage 242 finally places the actual coherent branch on this curve. The selected physical support coordinate is

$$\epsilon = \epsilon_W \left(1 - \frac{2}{11}\frac{\delta_U}{1+\delta_U}\right). \tag{7.140}$$

Therefore

$$\varrho_{\rm phys} = \frac{2}{3}(1-\epsilon), \qquad \sigma_{\rm phys} = \frac{2\epsilon}{1-\epsilon} = \frac{4}{3\varrho_{\rm phys}} - 2. \tag{7.141}$$

The coherent placement observables are

$$R_{\rm tr} = \frac{1+\chi_0/(1+\delta_U)}{1+\chi_0}, \tag{7.142}$$

$$R_{\rm target} = \frac{\Lambda(1-\epsilon_\eta)(1-\epsilon)^2}{Z_W(1+\chi_0)^2}, \tag{7.143}$$

together with ϵ_η. The finite coherent orbit packet in physical placement variables is

$$q_{\rm tr} = (1+\delta_{U,*})\ln\frac{\chi_0}{\chi_{0,\rm ref}} + (1+\chi_{0,*})\ln\frac{\delta_U}{\delta_{U,\rm ref}}, \tag{7.144}$$

$$q_{\rm nt} = \ln\frac{Z_W}{Z_{W,\rm ref}} - \ln\frac{\Lambda}{\Lambda_{\rm ref}} + E_*\ln\frac{\epsilon_W}{\epsilon_{W,\rm ref}} - F_*\ln\frac{\delta_U}{\delta_{U,\rm ref}}, \tag{7.145}$$

$$q_\eta = \ln\frac{\epsilon_\eta}{\epsilon_{\eta,\rm ref}}. \tag{7.146}$$

The infinitesimal orbit-lock conditions are equivalently

$$d\ln R_{\rm tr} = 0, \qquad d\ln R_{\rm target} = 0, \qquad d\ln\epsilon_\eta = 0. \tag{7.147}$$

The outgoing finish line is separate:

$$N_Q = 1, \qquad \text{or equivalently } \chi_Q = 1 \text{ on the natural source-map branch.} \tag{7.148}$$

The practical front-end packet to be returned by the completed PDE is equation (7.15); if the support lane is tracked separately, the support packet is

$$(\zeta, M_{\rm mix}, S(\zeta;\epsilon), M_{\rm tr}), \qquad S(\zeta;\epsilon) = 1 + \frac{\zeta(1-\epsilon)}{1-\zeta\epsilon}. \tag{7.149}$$

The support variable ζ affects the support baseline but not the orbit packet:

$$\partial_\zeta R_{\rm tr} = \partial_\zeta R_{\rm target} = \partial_\zeta \epsilon_\eta = \partial_\zeta q_{\rm tr} = \partial_\zeta q_{\rm nt} = \partial_\zeta q_\eta = 0. \tag{7.150}$$

Theorem 7.10 (Selected twin-support placement and orbit-packet split)**.** *Inside the selected passive/outgoing branch, the support loading ratio is fixed to $\rho_\alpha = 4/3$, so the branch lies on the symmetric lowest-twin support slice. The realized coordinate on that slice is $\varrho_{\rm phys} = 2(1-\epsilon)/3$. The orbit-lock packet is determined by $(\chi_0, \delta_U, Z_W, \epsilon_W, \epsilon_\eta, \Lambda)$ and is exactly blind to the support ratio ζ. Therefore support compensation and orbit lock are separate tests that must both be passed by the same completed moving-throat branch.*

Claim status. Stages 240–242 are Exact Within Closure **for the support-ratio extraction, selected curve, primitive ranking thresholds, placement formulas, and support/orbit split. Actual placement of the nonlinear branch and the final value of $N_Q - 1$ remain** Open **unless supplied by a completed PDE calculation.**

7.11 What Part VII does and does not prove

Part VII proves that the same-charge mixed sector has a much sharper reduced structure than the older placeholder language suggested. It proves, within the stated closures, that the one-port static mixed bundle cannot be cited as a source of a new long-range attraction; that the linear dynamic mixed bundle cannot be cited as a new radial kernel class; that resonant help is constrained by an exact dispersive/absorptive tradeoff; that the actual weak-axisymmetric branch test is the transported scalar packet $(\Delta_{\mathrm{norm}}, \Xi_1)$; that the rigid-mouth actual branch diagonalizes in the physical chart (U, V); and that selected support lies on the symmetric lowest-twin slice.

It does not prove that the completed nonlinear moving-throat PDE realizes the favorable point. In particular, it does not prove any of the following:

1. that the true branch has a new same-charge long-range attractive force;
2. that the true branch supplies a high-quality pole satisfying the dynamic survival window;
3. that the true branch lands within the transported static $|\varepsilon\Xi_1|$ ceiling;
4. that the rigid-mouth assumptions are realized by the full PDE;
5. that $U = V = 0$ holds on the actual branch;
6. that selected twin-support placement and outgoing normalization $N_Q = 1$ both hold without additional branch data.

The safe downstream citation is therefore:

> Inside the declared one-port, weak-axisymmetric, rigid-mouth, and selected-support closures, the same-charge mixed sector reduces to short-range static product kernels, a resonance/linewidth dynamic window, a transported prefactor packet $(\Delta_{\mathrm{norm}}, \Xi_1)$, the rigid-mouth Cartesian orbit-lock condition $U = V = 0$, and the selected lowest-twin support slice $\rho_\alpha = 4/3$. Actual branch realization remains a separate PDE calculation.

For later use, the practical audit recipe is:

1. compute the actual one-port bundle data (D_0, N_0, P_0) and check the static product-kernel assumptions;
2. evaluate the branch-compatible target surface equation (7.57);
3. form $(\Delta_{\mathrm{norm}}, \Xi_1)$ and test the transported ceiling equation (7.68);
4. if the branch is rigid-mouth, compute (U, V) from equation (7.117);
5. test orbit lock by $U = V = 0$;
6. compute ϵ, ϱ_{phys}, the selected ranking region, the support packet, and $N_Q - 1$ using equations (7.15) and (7.149).

This is exactly the point at which Part VIII can either preserve the strict front-end packet or declare a relaxed/open-system companion with an explicit recovery map back to the Stage 242 front end.

Chapter 8

Relaxed Open-System Branch, Barrier Compiler, Export Kernel, and Material Thresholds

Stage range: 243–253

Part VII ended at the strict selected-support and rigid-mouth front end. The exact one-port same-charge audit had already fixed the admissible short-range kernel families, the actual-branch packet had already been compressed to orbit-lock variables, and the selected passive/outgoing support side had already landed on the lowest-twin support slice. Part VIII is deliberately different. It is not another attempt to prove the strict conservative branch. It is the relaxed/open-system companion that asks what happens when three standard-recovery suppressions are lifted in a controlled way around the Stage 242 front end.

The conceptual firewall is the most important part of this chapter. The relaxed branch does *not* undo the same-charge verdict from Part VII. It does not produce a new long-range same-charge attraction, and it does not introduce a new linear dynamic kernel class. The relaxed branch is a codimension-three lift of the strict selected-support/orbit packet. Its standard-recovery slice is

$$\ell_w = 0, \qquad f_U = 0, \qquad a = b = 0, \tag{8.1}$$

which implies

$$S_{\mathrm{leak}} = 0, \qquad \mathcal{W}_w = 0, \qquad U = 0, \qquad V = 0, \qquad \mathcal{D}_{UV} = 0, \qquad \varsigma \equiv 1. \tag{8.2}$$

Away from this slice, the relaxed branch can reshape a near-contact barrier by leakage, scalar-photon work, non-rigid dressing drain, and compensated source response. Those are branch-local open-system or support/dressing/source effects. They do not alter the long-range kernel firewall.

The central stationary object of this part is the lowered front end

$$V_{\mathrm{eff}}^{(247)}(r) = V_{\mathrm{short}}^{(1p)}(r) - \lambda_L S_{\mathrm{leak}}(r) - \lambda_W \mathcal{W}_w^{\mathrm{sess}}(r) - \Delta E_{UV}(r) - \mathcal{M}_\sigma(r). \tag{8.3}$$

The central dynamic object is the event-chain energy law

$$E = \frac{m_s}{2}\dot{r}^{\,2} + V_{\mathrm{eff}}^{(247)}(r). \tag{8.4}$$

The central microscopic export object is the super-Ohmic active-leg kernel

$$K_{\mathrm{exp}}(s) = \Gamma_3 s^3 + \Gamma_5 s^5. \tag{8.5}$$

The central materials companion is the four-ratio screen

$$\Pi_{\mathrm{ep}} \geq 1, \qquad \Pi_\chi \geq 1, \qquad \Pi_k \geq 1, \qquad \Pi_T(T) \geq 1. \tag{8.6}$$

Part VIII at a glance

Primary question	If the strict same-charge branch is kept fixed, what controlled open-system extensions can be built around it without undoing the strict kernel verdict?
Starting inputs	The Part VII strict front-end packet, including the rigid-mouth orbit-lock variables, the selected twin-support slice, and the short-range kernel firewall.
Delivers	The relaxed-constraint branch declaration, leakage/work and non-rigid (U, V) compilers, the relaxed barrier and event-chain diagnostics, the microscopic export kernel, and the material-threshold companion.
Used by	Future relaxed-branch, export-kernel, and material-threshold papers; no later part in the current ledger supersedes this handoff block.
Result anchors	**MTDC-T12**.

Stage packets.

Stages	Packet	Status	Carry-forward output
243–246	Relaxed branch declaration and lane compilers	Effective Closure / Exact Within Closure	Declares the codimension-three lift of the strict Stage 242 front end, fixes the exact recovery slice, and compiles the leakage/work, non-rigid mouth, and compensated multimode source lanes.
247–250	Lowered barrier, event-chain, and Goldilocks diagnostics	Exact Within Closure / Effective Closure / Reduced / Controlled Reduction / Numerically Located	Builds the relaxed stationary barrier, the one-dimensional event-chain and tunneling diagnostics, the helicity-export readback, and the crossing-versus-collapse window used to identify viable relaxed events.
251–253	Export kernel, heat partition, and material screening	Exact Within Closure / Reduced / Controlled Reduction / Effective Closure / Numerically Located / Open	Derives the cubic/quintic active-leg export kernel, partitions exported energy between vacuum and lattice channels, and compiles the physical calibration and material-threshold screen without claiming that the completed PDE realizes those open-system coefficients.

Claim status. Part VIII is a relaxed/open-system companion. Its algebraic compilers are Exact Within Closure **inside the declared branch, one-mode, characteristic-timescale, single-growth, and physical-calibration closures. The stationary and dynamic readbacks that**

use Session-I through Session-V benchmark numbers are Numerically Located **insertions into exact formulas. The branch declaration itself is an** Effective Closure **extension around the strict Stage 242 front end, and any claim that the completed nonlinear moving-throat PDE actually realizes the relaxed packet, the microscopic export coefficients, or the physical material thresholds remains** Open**. No result in this part should be cited as a proof of a new same-charge long-range attraction or as a proof that the relaxed branch validates the strict conservative branch.**

What this part proves. Part VIII constructs the relaxed/open-system companion around the strict Stage 242 packet, including leakage/work, non-rigid mouth, compensated source, barrier, export, heat-partition, and material-screening compilers.

What this part does not prove. It does not rescue the strict same-charge conservative branch, does not create a new long-range same-charge attraction, and does not prove that the completed PDE realizes the relaxed packet or its physical screening thresholds.

Why later parts need it. This is the handoff block for future relaxed-branch, export-kernel, and material-threshold papers; no later part inside the current ledger supersedes it.

8.1 Block contract and theorem-level output

The contract of Part VIII is to keep three layers separate. The first layer is the strict front end from Part VII: selected twin-support placement, rigid-mouth orbit lock, and the outgoing normalization packet. The second layer is the relaxed branch around that front end: leakage, non-rigid mouth response, and compensated source shape. The third layer is the physical companion: event-chain survival, export, heat partition, and material screening. The chapter is useful only if those layers remain visibly separated.

The operational chain is

$$\begin{aligned}\mathcal{P}_{242} &\longrightarrow \mathfrak{B}_{243}^{\mathrm{relax}} \longrightarrow (S_{\mathrm{leak}}, \mathcal{W}_w, U, V, \mathcal{M}_\sigma)\\ &\longrightarrow V_{\mathrm{eff}}^{(247)} \longrightarrow (r_\pm, I_{\mathrm{new}}, t_{\mathrm{cross}})\\ &\longrightarrow K_{\exp}(s) \longrightarrow (\Pi_{\mathrm{ep}}, \Pi_\chi, \Pi_k, \Pi_T).\end{aligned} \tag{8.7}$$

The strict branch can be recovered from this chain by imposing equation (8.1). The relaxed branch cannot be used to infer that the strict branch succeeds; it can only be used to analyze the open-system bypass companion.

Theorem 8.1 (Part VIII relaxed-branch and export-companion theorem)**.** *Inside the declared relaxed-constraint branch, the one-mode leakage/work closure, the non-rigid mouth/dressing closure, the compensated two-mode source closure, the reduced one-dimensional event-chain closure, the characteristic crossing/collapse closure, the microscopic cubic/quintic export closure, and the physical-calibration companion, the following statements hold.*

1. *The relaxed branch is a codimension-three lift of the Stage 242 front end with exact recovery map*

$$\ell_w = 0, \qquad f_U = 0, \qquad a = b = 0. \tag{8.8}$$

On this slice all lifted leakage, work, non-rigid, and compensated-source observables reduce to the strict standard-recovery values.

2. *The same-charge static and linear dynamic kernel span remains short-ranged:*

$$\mathcal{K}_{\mathrm{SR}} = \mathrm{span}\left\{ r^{-6}, \frac{e^{-2\kappa r}}{r^4}, \frac{e^{-4\kappa r}}{r^2} \right\}. \tag{8.9}$$

In particular,

$$\lim_{r\to\infty} r\,\delta V_{\mathrm{stat}}(r) = 0, \qquad \lim_{r\to\infty} r\,\Re\mathfrak{V}_{\mathrm{dyn}}(r,\omega) = 0. \tag{8.10}$$

3. *On the selected-support branch, the leakage/work lane is explicit and support-side:*

$$S_{\mathrm{leak}}(r) = \frac{8\sqrt{2}\,\eta_{\mathrm{leak}}\mu_w\rho_0}{\pi^{5/2}\lambda^3}\Lambda(r)\varrho(r), \tag{8.11}$$

$$\mathcal{W}_w^{\mathrm{sess}}(r) = \frac{512\,\eta_{\mathrm{leak}}^2\mu_w q\rho_0}{\pi^4\lambda^2}\Lambda(r)^2\varrho(r)^2. \tag{8.12}$$

These quantities are independent of the orbit-lock variables $(R_{\mathrm{tr}}, R_{\mathrm{target}}, \epsilon_\eta)$ *within the declared pullback.*

4. *The non-rigid mouth packet is exact in the physical logarithmic chart*

$$U = \ln\frac{\mathcal{T}^2}{\mathcal{T}_{\mathrm{ref}}^2}, \qquad V = \ln\frac{\epsilon_\eta}{\epsilon_{\eta,\mathrm{ref}}}. \tag{8.13}$$

With

$$\Delta_{UV} = a_U a_V - \chi_{UV}^2, \tag{8.14}$$

the stationary point is

$$U = \frac{a_V f_U}{\Delta_{UV}}, \qquad V = \frac{\chi_{UV} f_U}{\Delta_{UV}}, \qquad \Delta_{UV} > 0. \tag{8.15}$$

The corresponding drain is nonnegative:

$$\mathcal{D}_{UV} = \frac{\chi_{UV}^2 a_V f_U^2}{\Delta_{UV}^2} \geq 0. \tag{8.16}$$

5. *The first compensated two-mode source family is an exact two-moment compiler:*

$$\sigma_{a,b}(x) = 1 + a\cos(\pi x) + b\cos(2\pi x), \qquad \int_0^1 \sigma_{a,b}(x)\,dx = 1, \tag{8.17}$$

with

$$\mathfrak{g}(a,b) = \frac{2}{\pi}\left(1 + \frac{a}{3} - \frac{b}{15}\right), \qquad \mathcal{S}(a,b) = \frac{2\tanh(\pi/2)}{\pi}\left(1 + \frac{a}{5} + \frac{b}{17}\right). \tag{8.18}$$

Its normalized two-moment map has determinant $14/425 > 0$*, so the first compensated source family spans the two carried source moments exactly.*

6. *The relaxed stationary barrier is exactly equation* (8.3)*, and the lowering identity is*

$$V_{\mathrm{short}}^{(1p)}(r) - V_{\mathrm{eff}}^{(247)}(r) = \lambda_L S_{\mathrm{leak}}(r) + \lambda_W \mathcal{W}_w^{\mathrm{sess}}(r) + \Delta E_{UV}(r) + \mathcal{M}_\sigma(r). \tag{8.19}$$

Hence, when the imported packet scalars and weights are nonnegative, $V_{\mathrm{eff}}^{(247)} \leq V_{\mathrm{short}}^{(1p)}$ *pointwise.*

7. *The dynamic event chain is determined by the conserved energy equation* (8.4)*. The lowered-branch classical threshold speed is*

$$v_{\mathrm{crit,new}} = \sqrt{\frac{2}{m_s}\big(V_{\mathrm{peak}} - V(r_0)\big)}, \tag{8.20}$$

while the pure-Coulomb contact threshold is

$$v_{\mathrm{contact,Coul}} = \sqrt{\frac{2}{m_s}\left(\frac{1}{r_{\mathrm{contact}}} - \frac{1}{r_0}\right)}. \tag{8.21}$$

If $v_{\mathrm{crit,new}} < v_0 < v_{\mathrm{contact,Coul}}$*, the lowered branch reaches contact while the pure Coulomb branch still turns back.*

8. *The subbarrier WKB action is*

$$I_{\mathrm{new}}(E) = \frac{1}{\hbar_{\mathrm{eff}}} \int_{r_-(E)}^{r_+(E)} \sqrt{2m_s\big(V(r) - E\big)}\, dr, \tag{8.22}$$

and the transmission ratio relative to Coulomb is

$$\frac{T_{\mathrm{new}}(E)}{T_{\mathrm{Coul}}(E)} = \exp\!\big[-2\big(I_{\mathrm{new}}(E) - I_{\mathrm{Coul}}(E)\big)\big]. \tag{8.23}$$

9. *The helicity-export comparison is diagnostic. With orientation label* $\sigma = \pm 1$*, the reduced closure gives*

$$\dot{H}_{\mathrm{sub},\sigma} = \Gamma_0(1 + \sigma\alpha_h), \qquad |\alpha_h| < 1 \tag{8.24}$$

for positive export in both branches. The aligned branch dominates whenever $\alpha_h > 0$*, but this does not by itself prove a microscopic spin sector.*

10. *The crossing-versus-collapse safe edge is*

$$E_{\mathrm{edge}} = V_{\mathrm{peak}} + \frac{m_s}{\mu_\eta}\frac{g_{UV}\chi_{\mathrm{peak}}\lambda_{\mathrm{eff}}^2}{2}, \tag{8.25}$$

equivalently

$$v_{\mathrm{safe,min}}^2 = v_{\mathrm{crit,new}}^2 + \frac{\lambda_{\mathrm{eff}}^2 g_{UV}\chi_{\mathrm{peak}}}{\mu_\eta}. \tag{8.26}$$

Under the declared characteristic closure this is a one-sided lower edge.

11. *The microscopic active-leg export kernel is super-Ohmic:*

$$K_{\exp}(s) = \Gamma_3 s^3 + \Gamma_5 s^5, \qquad \Gamma_3, \Gamma_5 \geq 0. \tag{8.27}$$

The positive growth root is reduced but not eliminated by any finite passive (Γ_3, Γ_5) *in this minimal closure. The exact event-safe condition is*

$$\widehat{\Gamma}_3 + s_c^2\widehat{\Gamma}_5 \geq \frac{s_0^2 - s_c^2}{s_c^3}, \qquad \widehat{\Gamma}_n := \Gamma_n/\mu_\eta. \tag{8.28}$$

12. The heat partition is channel-resolved. For event-shape quotient $\mathfrak{r}_V = \mathcal{I}_3/\mathcal{I}_2$*,*

$$f_{\rm lat}(\mathfrak{r}_V) = \frac{\Gamma_3^{\rm lat} + \Gamma_5^{\rm lat}\mathfrak{r}_V}{\Gamma_3 + \Gamma_5\mathfrak{r}_V}, \tag{8.29}$$

and on a single-growth cold event $\mathfrak{r}_V = s^2$*. At the safe edge, the total minimum exported energy is*

$$E^{\rm safe}_{\rm exp,min} = \frac{V_{\rm in}^2}{2}(e^2-1)\mu_\eta(s_0^2 - s_c^2). \tag{8.30}$$

13. The physical-material companion is an exact calibration image of the Stage 252 event-equivalent rate. With calibration factor $\Upsilon_{\rm lat} > 0$*,*

$$(\lambda_{\rm ep}\omega_D)^{(253)}_{\rm min} = \frac{f_{\rm lat}(s_c)\mu_\eta(s_0^2 - s_c^2)}{\Upsilon_{\rm lat}\zeta_{\rm ep}s_c t_*}. \tag{8.31}$$

Candidate hosts are screened by equation (8.6)*.*

Every item involving actual branch data, microscopic coefficients, physical units, or candidate material constants remains a realization or calibration question beyond the algebraic compiler.

8.2 Relaxed constraint branch declaration

Stage 243 declares the relaxed branch without yet assembling a barrier. Its purpose is to make the Session-I relaxations legal inside the derivation tree while preserving the same-charge short-range verdict. The inherited base object is the Stage 242 front-end packet

$$\mathcal{P}_{242} = \Big(\epsilon, \varrho_{\rm phys}, \sigma_{\rm phys}, \text{ranking region}, R_{\rm tr}, R_{\rm target}, \epsilon_\eta, d\ln R_{\rm tr}, d\ln R_{\rm target}, d\ln\epsilon_\eta, N_Q - 1\Big), \tag{8.32}$$

with selected-support packet

$$\mathcal{S}_{242} = (\zeta, M_{\rm mix}, S(\zeta;\epsilon), M_{\rm tr}). \tag{8.33}$$

The strict slice is

$$\mathfrak{B}_{\rm std} = \{J^w = 0,\ U = 0,\ V = 0,\ \varsigma(z) \equiv 1\} \subset (\mathcal{P}_{242}, \mathcal{S}_{242}). \tag{8.34}$$

This definition is useful because it names exactly what is being relaxed: transverse-current suppression, rigid-mouth orbit lock, and positive-source Family-1 closure.

8.2.1 Open-system leakage/work lane

The open-system lift begins with the exact projected-continuity leakage identity

$$S_{\rm leak} = -\big[W(w)j^w\big]_{-\infty}^{+\infty} + \int_{-\infty}^{+\infty} W'(w)j^w(w)\,dw. \tag{8.35}$$

For the declaration-level Gaussian closure,

$$W(w) = \frac{e^{-w^2}}{\sqrt{\pi}}, \qquad j^w(w) = \ell_w j_0\, we^{-w^2}, \qquad E_w(w) = E_0 we^{-w^2}, \tag{8.36}$$

so the boundary term vanishes and

$$S_{\rm leak} = -\frac{\sqrt{2}}{4}\ell_w j_0, \qquad \mathcal{W}_w = \frac{\sqrt{2\pi}}{8}\ell_w j_0 E_0. \tag{8.37}$$

The declaration packet for this lane is

$$L_w = (S_{\rm leak}, \mathcal{W}_w). \tag{8.38}$$

Both components vanish on $\ell_w = 0$.

8.2.2 Non-rigid mouth/dressing lane

The non-rigid lift uses the minimal free energy

$$\mathcal{F}_{UV}(U,V) = \frac{1}{2}k_U U^2 + \frac{1}{2}k_V V^2 - \chi_\lambda UV - f_U U. \tag{8.39}$$

Stationarity gives

$$U = \frac{k_V f_U}{k_U k_V - \chi_\lambda^2}, \qquad V = \frac{\chi_\lambda f_U}{k_U k_V - \chi_\lambda^2}, \qquad \frac{V}{U} = \frac{\chi_\lambda}{k_V}. \tag{8.40}$$

The branch is admissible when

$$k_U k_V - \chi_\lambda^2 > 0, \tag{8.41}$$

and the drain is

$$\mathcal{D}_{UV} = \chi_\lambda UV = \frac{\chi_\lambda^2 k_V f_U^2}{(k_U k_V - \chi_\lambda^2)^2} \geq 0. \tag{8.42}$$

This lane is the finite relaxed continuation of the earlier weak-axisymmetric transfer-shape scalar: at infinitesimal order, $\Xi_1 = \delta U$.

8.2.3 Compensated sign-changing source lane

The compensated source lift is

$$\varsigma(z) = 1 + a\cos(\pi z) + b\cos(2\pi z), \qquad z \in [0,1], \tag{8.43}$$

with exact unit mean. Setting $y = \cos(\pi z)$,

$$\varsigma(y) = 1 - b + ay + 2by^2, \qquad -1 \leq y \leq 1. \tag{8.44}$$

The interior stationary point and value are

$$y_* = -\frac{a}{4b}, \qquad \varsigma(y_*) = 1 - b - \frac{a^2}{8b}, \tag{8.45}$$

with the boundary candidates $1 + a + b$ and $1 - a + b$. Thus the first source relaxation is not a vague sign-changing profile; it is an exact mean-preserving quadratic problem.

The full declaration packet is

$$\mathfrak{B}_{243}^{\mathrm{relax}} = \{(\mathcal{P}_{242}, \mathcal{S}_{242});\ L_w,\ L_{UV},\ L_\varsigma\}, \tag{8.46}$$

where

$$L_w = (S_{\mathrm{leak}}, \mathcal{W}_w), \quad L_{UV} = (U, V, \mathcal{D}_{UV}), \quad L_\varsigma = (\varsigma, \varsigma_{\min}, \mathrm{signchg}). \tag{8.47}$$

8.2.4 Short-range theorem for the lifted branch

The most important Stage 243 output is the short-range firewall. With primitive profiles

$$\mathcal{S}_Q(r) = r^{-3}, \qquad \mathcal{S}_Y(r) = \frac{e^{-2\kappa r}}{r}, \tag{8.48}$$

the static one-port correction is

$$\delta V_{\mathrm{stat}}(r) = -\frac{1}{2}\left[\frac{\mathcal{C}_6}{r^6} + 2\mathcal{C}_4\frac{e^{-2\kappa r}}{r^4} + \mathcal{C}_2\frac{e^{-4\kappa r}}{r^2}\right], \tag{8.49}$$

and the linear monochromatic dynamic bundle has the same spatial span:

$$\Re\mathfrak{V}_{\mathrm{dyn}}(r,\omega) = -\frac{1}{2}\left[\frac{\mathcal{C}_6(\omega)}{r^6} + 2\mathcal{C}_4(\omega)\frac{e^{-2\kappa r}}{r^4} + \mathcal{C}_2(\omega)\frac{e^{-4\kappa r}}{r^2}\right]. \tag{8.50}$$

Thus the relaxed branch is a short-range/open-system bypass branch. It can reshape near-contact energetics, but it cannot be cited as a new long-range same-charge force.

8.3 Selected leakage and scalar-photon work compiler

Stage 244 takes the declared $J^w \neq 0$ lane and attaches it to the selected-support packet. The parent theory supplies both exact identities needed for the compiler: projected continuity gives equation (8.35), and the localized Maxwell energy ledger gives

$$\partial_t u_{\mathrm{EM}} + \partial_A S^A_{\mathrm{EM}} = -J^A E_A = -(J^a E_a + J^w E_w). \tag{8.51}$$

Thus S_{leak} and $J^w E_w$ are legitimate reduced observables once the mixed lane is opened.

8.3.1 Gaussian one-mode leakage law

The Stage 244 one-mode closure uses

$$W_\lambda(w) = \frac{e^{-w^2/\lambda^2}}{\lambda\sqrt{\pi}}, \qquad \phi_\lambda(w) = \frac{2w}{\sqrt{\pi}\lambda^3}e^{-w^2/\lambda^2}, \tag{8.52}$$

with

$$E_w(w;r) = -\mathcal{E}_0(r)\phi_\lambda(w), \qquad j^w(w;r) = \mu_w\rho_0 E_w(w;r), \qquad J^w = qj^w. \tag{8.53}$$

The exact projected leakage is

$$S_{\mathrm{leak}}(r) = \frac{\sqrt{2}\mu_w\rho_0}{2\sqrt{\pi}\lambda^3}\mathcal{E}_0(r). \tag{8.54}$$

The exact one-mode bulk work scalar is

$$\mathcal{W}_w^{\mathrm{bulk}}(r) = q\int j^w E_w\,dw = \frac{\sqrt{2}\mu_w q\rho_0}{2\sqrt{\pi}\lambda^3}\mathcal{E}_0(r)^2, \tag{8.55}$$

so

$$\mathcal{W}_w^{\mathrm{bulk}}(r) = q\mathcal{E}_0(r)S_{\mathrm{leak}}(r). \tag{8.56}$$

The Session-I scalar-photon work variable is the rescaled scalar

$$\mathcal{W}_w^{\mathrm{sess}}(r) = \frac{2\mu_w q\rho_0}{\lambda^2}\mathcal{E}_0(r)^2 = 2\sqrt{2\pi}\lambda\,\mathcal{W}_w^{\mathrm{bulk}}(r), \tag{8.57}$$

or equivalently

$$\mathcal{W}_w^{\mathrm{sess}}(r) = \frac{4\pi q\lambda^4}{\mu_w\rho_0}S_{\mathrm{leak}}(r)^2. \tag{8.58}$$

8.3.2 Selected-support pullback

The selected-support demand from Stage 242 is

$$\Pi_{\mathrm{tr}} = \frac{4}{3}C_{\mathrm{mix}} = \frac{32\Lambda(1-\epsilon)}{3\pi^2} = \frac{16}{\pi^2}\Lambda\varrho, \qquad \varrho = \frac{2}{3}(1-\epsilon). \tag{8.59}$$

Stage 244 chooses the simplest amplitude pullback

$$\mathcal{E}_0(r) = \eta_{\mathrm{leak}}\Pi_{\mathrm{tr}}(r). \tag{8.60}$$

Substitution gives the selected-branch leakage law equation (8.11) and the selected-branch work law equation (8.12). In the (Λ, ϵ) notation the same work law is

$$\mathcal{W}_w^{\mathrm{sess}}(r) = \frac{2048\eta_{\mathrm{leak}}^2\mu_w q\rho_0}{9\pi^4\lambda^2}\Lambda(r)^2(1-\epsilon(r))^2. \tag{8.61}$$

The formulas depend only on selected-support variables. Therefore,

$$\partial_{R_{\mathrm{tr}}}S_{\mathrm{leak}} = \partial_{R_{\mathrm{target}}}S_{\mathrm{leak}} = \partial_{\epsilon_\eta}S_{\mathrm{leak}} = 0, \tag{8.62}$$

and the same statement holds for both work scalars within the declared pullback.

The orientation parity is exact:

$$S_{\mathrm{leak}}(-\eta_{\mathrm{leak}}) = -S_{\mathrm{leak}}(\eta_{\mathrm{leak}}), \qquad \mathcal{W}_w(-\eta_{\mathrm{leak}}) = \mathcal{W}_w(\eta_{\mathrm{leak}}). \tag{8.63}$$

Thus the leakage sign is orientation-sensitive, but the exported-work magnitude is not.

8.4 Non-rigid mouth and (U, V) drain packet

Stage 245 compiles the abstract non-rigid lane into the actual finite physical variables used by the rigid-mouth chart. The carried chart is equation (8.13); hence

$$\mathcal{T}^2 = \mathcal{T}_{\mathrm{ref}}^2 e^U, \qquad \epsilon_\eta = \epsilon_{\eta,\mathrm{ref}}e^V. \tag{8.64}$$

The selected-branch identity

$$R_{\mathrm{target}}\mathcal{T}^2 = \Lambda_0(1-\epsilon_\eta) \tag{8.65}$$

then gives

$$\frac{R_{\mathrm{target}}}{R_{\mathrm{target,ref}}} = \frac{1-\epsilon_{\eta,\mathrm{ref}}e^V}{1-\epsilon_{\eta,\mathrm{ref}}}e^{-U}. \tag{8.66}$$

So the finite non-rigid packet has two independent legs: transfer shape and dressing.

The reduced free energy in session notation is

$$\mathcal{F}_{\mathrm{nr}}(U, V; r) = \frac{1}{2}a_U U^2 + \frac{1}{2}a_V V^2 - \chi_{UV}(r)UV - f_U(r)U. \tag{8.67}$$

The stationary solution is equation (8.15), with Δ_{UV} given by equation (8.14). In finite physical variables,

$$\mathcal{T}^2(r) = \mathcal{T}_{\mathrm{ref}}^2 \exp\left(\frac{a_V f_U(r)}{\Delta_{UV}(r)}\right), \tag{8.68}$$

$$\epsilon_\eta(r) = \epsilon_{\eta,\mathrm{ref}} \exp\left(\frac{\chi_{UV}(r) f_U(r)}{\Delta_{UV}(r)}\right), \tag{8.69}$$

$$\frac{R_{\text{target}}(r)}{R_{\text{target,ref}}} = \frac{1-\epsilon_{\eta,\text{ref}}\exp(\chi_{UV} f_U/\Delta_{UV})}{1-\epsilon_{\eta,\text{ref}}}\exp(-a_V f_U/\Delta_{UV}). \tag{8.70}$$

The dependent microscopic correction inherited from the rigid-mouth plane is

$$(\Delta_T, \Delta_{K_\eta}, \Delta_\mu) = (0, -V, U-V), \tag{8.71}$$

so the non-rigid stationary packet induces

$$\mathbf{y}^{\text{dep}}_{\text{nr}}(r) = \begin{pmatrix} 0 \\ -\chi_{UV} f_U/\Delta_{UV} \\ (a_V - \chi_{UV}) f_U/\Delta_{UV} \end{pmatrix}. \tag{8.72}$$

The drain $\mathcal{D}_{UV}$ is given by equation (8.16); it is nonnegative even when U and V have opposite signs.

At weak-axisymmetric order, write

$$U = \varepsilon u_1 + O(\varepsilon^2), \qquad V = \varepsilon v_1 + O(\varepsilon^2). \tag{8.73}$$

Then

$$\Xi_1^{\text{nr}} = u_1, \qquad \mathcal{R}_1^{\text{nr}} + \Xi_1^{\text{nr}} = -\frac{\epsilon_{\eta,\text{ref}}}{1-\epsilon_{\eta,\text{ref}}} v_1, \tag{8.74}$$

$$\mathcal{R}_1^{\text{nr}} = -u_1 - \frac{\epsilon_{\eta,\text{ref}}}{1-\epsilon_{\eta,\text{ref}}} v_1. \tag{8.75}$$

Thus the direct outgoing-prefactor scalar remains the U leg, while active dressing adds an independent selected-branch residual.

Within the declared closure, this packet is orbit-side/support-blind:

$$\partial_\Lambda U = \partial_\varrho U = \partial_\Lambda V = \partial_\varrho V = 0, \tag{8.76}$$

provided f_U and χ_{UV} are treated as orbit-side data rather than support-placement coordinates. This complements the support-side result from Stage 244.

8.5 Compensated multimode source compiler

Stage 246 compiles the source lane that Stage 243 only declared. The carried source kernels on the dimensionless mouth coordinate $x \in [0,1]$ are

$$c(x) = \cos\frac{\pi x}{2}, \qquad K_q(x) = \frac{\cosh\left(\frac{\pi}{2}(1-x)\right)}{\cosh(\pi/2)}. \tag{8.77}$$

The two source moments are

$$\mathfrak{g}[\sigma] = \int_0^1 \sigma(x)c(x)\,dx, \qquad \mathcal{S}[\sigma] = \int_0^1 \sigma(x)K_q(x)\,dx. \tag{8.78}$$

The carried Family-1 mixed-to-shell loading ratio is

$$\mathcal{R}[\sigma] = \frac{(\mathfrak{g}[\sigma]-\mathfrak{r}_{F1})^2}{1+\mathfrak{r}_{F1}^2}, \qquad \mathfrak{r}_{F1} = \sqrt{\frac{12}{\pi^2}\left(\frac{37}{20}\right)^2 - 1}. \tag{8.79}$$

Stage 246 keeps this law and extends the source family beyond the positive corridor.

The compensated family is equation (8.17). Its exact minimum is

$$\sigma_{\min}(a,b) = \begin{cases} 1+b-|a|, & b \le 0 \text{ or } |a| \ge 4b, \\ 1-b-\dfrac{a^2}{8b}, & b > 0 \text{ and } |a| \le 4b. \end{cases} \tag{8.80}$$

Thus

$$\mathrm{signchg}(a,b) \iff \sigma_{\min}(a,b) < 0. \tag{8.81}$$

The exact moment formulas are equation (8.18). In normalized variables

$$\widetilde{g} := \frac{\pi}{2}\mathfrak{g} - 1, \qquad \widetilde{S} := \frac{\pi}{2\tanh(\pi/2)}\mathcal{S} - 1, \tag{8.82}$$

the two-moment map is

$$\begin{pmatrix} \widetilde{g} \\ \widetilde{S} \end{pmatrix} = \begin{pmatrix} 1/3 & -1/15 \\ 1/5 & 1/17 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix}, \qquad \det M_{\mathrm{src}} = \frac{14}{425}. \tag{8.83}$$

The inverse is

$$a = \frac{85}{42}\widetilde{S} + \frac{25}{14}\widetilde{g}, \qquad b = \frac{425}{42}\widetilde{S} - \frac{85}{14}\widetilde{g}. \tag{8.84}$$

For the transported branch,

$$a(r) = a_0 s(r), \qquad b(r) = b_0 s(r), \qquad s(r) = \frac{r_\sigma^2}{r^2 + r_\sigma^2}. \tag{8.85}$$

The output packet is

$$\mathcal{M}_\sigma(r) = \big(\sigma(x;r), \sigma_{\min}(r), \mathrm{signchg}(r), \mathfrak{g}[\sigma](r), \mathcal{S}[\sigma](r), \mathcal{R}[\sigma](r)\big), \tag{8.86}$$

with

$$\mathfrak{g}[\sigma](r) = \frac{2}{\pi}\left(1 + \frac{a(r)}{3} - \frac{b(r)}{15}\right), \qquad \mathcal{S}[\sigma](r) = \frac{2\tanh(\pi/2)}{\pi}\left(1 + \frac{a(r)}{5} + \frac{b(r)}{17}\right). \tag{8.87}$$

On the Session-I orientation $a_0 > 0$, $b_0 < 0$, the minimum is boundary-controlled:

$$\sigma_{\min}(r) = 1 - (a_0 - b_0)s(r), \qquad \mathrm{signchg}(r) \iff r < r_\sigma\sqrt{a_0 - b_0 - 1}. \tag{8.88}$$

This is the source-side packet inserted into the stationary barrier compiler.

8.6 Relaxed stationary barrier front end

Stage 247 assembles the first stationary lowered barrier from the one-port short-range baseline and the three relaxed packets. The one-port static bundle is controlled by

$$\Delta = \Omega_U^2\Omega_W^2 - R_{\mathrm{mix}}^2, \qquad Q = G_U^2\Omega_W^2 + 2G_UG_WR_{\mathrm{mix}} + G_W^2\Omega_U^2, \tag{8.89}$$

$$P = \Omega_U^2 G_W + R_{\mathrm{mix}}G_U, \qquad D_0 = K_* - \frac{Q}{\Delta}. \tag{8.90}$$

With primitive short-range source amplitudes $\beta_Q, \beta_U, \beta_W$, the product-family coefficients are

$$\mathcal{C}_6 = \chi_{qq}\beta_Q^2, \qquad \mathcal{C}_4 = \chi_{qU}\beta_Q\beta_U + \chi_{qW}\beta_Q\beta_W, \tag{8.91}$$

$$\mathcal{C}_2 = \chi_{UU}\beta_U^2 + 2\chi_{UW}\beta_U\beta_W + \chi_{WW}\beta_W^2. \tag{8.92}$$

The collected baseline is

$$V_{\text{short}}^{(1p)}(r) = \frac{1}{r}\left(1 + \frac{1}{2}e^{-2\kappa r}\right) - \frac{A_6}{r^6} - A_4\frac{e^{-2\kappa r}}{r^4} - A_2\frac{e^{-4\kappa r}}{r^2}, \tag{8.93}$$

where

$$A_6 = 3\alpha_6 + \frac{1}{2}\mathcal{C}_6, \qquad A_4 = \mathcal{C}_4, \qquad A_2 = \alpha_2 + \frac{1}{2}\mathcal{C}_2. \tag{8.94}$$

This baseline inherits the same short-range firewall from Stages 219–243.

The imported relaxed scalars are

$$\mathfrak{L}(r) := \Lambda(r)\varrho(r), \qquad S_{\text{leak}} = \frac{8\sqrt{2}\eta_{\text{leak}}\mu_w\rho_0}{\pi^{5/2}\lambda^3}\mathfrak{L}, \tag{8.95}$$

$$\mathcal{W}_w^{\text{sess}} = \frac{512\eta_{\text{leak}}^2\mu_w q\rho_0}{\pi^4\lambda^2}\mathfrak{L}^2, \qquad \Delta E_{UV} = \eta_{UV}\mathcal{D}_{UV}, \tag{8.96}$$

$$\mathcal{M}_\sigma(r) = \xi_R\big[\mathcal{R}_\infty - \mathcal{R}(r)\big], \qquad \mathcal{R}_\infty = \frac{(2/\pi - \mathfrak{r}_{F1})^2}{1 + \mathfrak{r}_{F1}^2}. \tag{8.97}$$

The stationary compiler is equation (8.3); the lowering identity is equation (8.19).

The Session-I benchmark is useful as an audit point, not as a general theorem. At the strongest softening point, the Stage 247 decomposition reads

$$3.74163698 - 0.26971918 \times 0.31069599 - 1.51632107 - 0.21064278 - 0.18386120 = 1.74701126. \tag{8.98}$$

This shows that, once the one-port baseline is fixed, the recorded stationary lowering can be decomposed into work, U/V drain, compensated source response, and one remaining leakage-to-barrier embedding coefficient. It does not change the short-range kernel class.

8.7 Dynamic event-chain compiler

Stage 248 promotes $V_{\text{eff}}^{(247)}$ into a reduced scattering/event-chain compiler. Write

$$V(r) := V_{\text{eff}}^{(247)}(r). \tag{8.99}$$

The reduced radial equation is

$$m_s\ddot{r} = -V'(r), \tag{8.100}$$

with conserved energy equation (8.4). If the branch has a single barrier peak

$$V'(r_{\text{peak}}) = 0, \qquad V''(r_{\text{peak}}) < 0, \qquad V_{\text{peak}} = V(r_{\text{peak}}), \tag{8.101}$$

then the finite-radius threshold law is equation (8.20). The pure-Coulomb contact threshold is equation (8.21). The classical comparison window is therefore

$$v_{\text{crit,new}} < v_0 < v_{\text{contact,Coul}}. \tag{8.102}$$

For subbarrier energy $V(r_0) < E < V_{\text{peak}}$, turning points solve

$$V(r_-(E)) = E, \qquad V(r_+(E)) = E, \qquad r_-(E) < r_+(E), \tag{8.103}$$

and transport as

$$\frac{dr_\pm}{dE} = \frac{1}{V'(r_\pm(E))}. \tag{8.104}$$

The WKB action and transmission ratio are equations (8.22) and (8.23). The pure-Coulomb outer turning point is exact,

$$r_{\text{turn,Coul}}(E) = \frac{1}{E}, \tag{8.105}$$

and the Coulomb action closes as

$$I_{\text{Coul}}(E) = \frac{\sqrt{2m_s}}{\hbar_{\text{eff}}}\left[\frac{\pi}{2\sqrt{E}} - \sqrt{r_{\text{contact}}(1 - Er_{\text{contact}})} - \frac{\arcsin\sqrt{Er_{\text{contact}}}}{\sqrt{E}}\right] \tag{8.106}$$

for $Er_{\text{contact}} < 1$.

Near the top, with

$$V(r) = V_{\text{peak}} - \frac{K_{\text{peak}}}{2}(r - r_{\text{peak}})^2 + O((r - r_{\text{peak}})^3), \qquad K_{\text{peak}} = -V''(r_{\text{peak}}), \tag{8.107}$$

the leading top action is

$$I_{\text{top}}(E) = \frac{\pi(V_{\text{peak}} - E)}{\hbar_{\text{eff}}}\sqrt{\frac{m_s}{K_{\text{peak}}}} + O((V_{\text{peak}} - E)^{3/2}). \tag{8.108}$$

The dynamic diagnostics carried forward are

$$\Xi_{\text{turn}}(E) = \Xi_1(r_+(E)), \qquad \lambda_{\text{th}}(E) = \left|\frac{E}{V'(r_+(E))}\right|. \tag{8.109}$$

These tags are later used in the helicity and Goldilocks compilers; they do not alter the scalar event chain.

8.8 Helicity export diagnostic

Stage 249 attaches a hidden-channel export diagnostic to the Stage 248 event chain. It is diagnostic rather than constitutive. The parent projected helicity ledger gives the exact subscale helicity

$$h_{\text{sub}} = \overline{h} - h_{\text{res}} = \overline{\mathbf{A}' \cdot \mathbf{B}'}, \tag{8.110}$$

with transfer equation

$$\partial_t h_{\text{sub}} + \nabla_3 \cdot \mathbf{F}_{h,\text{sub}} = -2\overline{\mathbf{E}' \cdot \mathbf{B}'}. \tag{8.111}$$

On a reduced brane control volume,

$$\frac{dH_{\text{sub}}}{dt} = -\Phi_h(t;E) - 2\mathcal{C}_h(t;E), \tag{8.112}$$

where Φ_h is the unresolved-helicity flux through the control boundary and $\mathcal{C}_h$ is the unresolved covariance source.

The minimal orientation closure introduces $\sigma = +1$ for aligned and $\sigma = -1$ for anti-aligned, with

$$\Phi_{h,\sigma} = \Phi_{h,0} + \sigma\Phi_{h,1}, \qquad \mathcal{C}_{h,\sigma} = \mathcal{C}_{h,0} + \sigma\mathcal{C}_{h,1}. \tag{8.113}$$

Thus

$$\dot{H}_{\mathrm{sub},\sigma} = \Gamma_0 + \sigma\Gamma_1 = \Gamma_0(1+\sigma\alpha_h), \qquad \alpha_h = \frac{\Gamma_1}{\Gamma_0}. \tag{8.114}$$

The instantaneous ratio and inverse are

$$R_{\mathrm{inst}} = \frac{\dot{H}_{\mathrm{sub},+}}{\dot{H}_{\mathrm{sub},-}} = \frac{1+\alpha_h}{1-\alpha_h}, \qquad \alpha_h = \frac{R_{\mathrm{inst}}-1}{R_{\mathrm{inst}}+1}. \tag{8.115}$$

Over a common interval,

$$R_{\mathrm{int}} = \frac{\mathcal{H}_+}{\mathcal{H}_-} = \frac{1+\bar{\alpha}_h}{1-\bar{\alpha}_h}, \qquad \bar{\alpha}_h = \frac{R_{\mathrm{int}}-1}{R_{\mathrm{int}}+1}. \tag{8.116}$$

Any common export scale cancels from these ratios.

The benchmark packet is

$$\begin{aligned}(\Xi_{\mathrm{turn}}, \lambda_{\mathrm{th}}, R_{\mathrm{pk}}) &= (0.34437471, 0.42826825, 4.94653917),\\ (R_{\mathrm{int}}, \alpha_{\mathrm{pk}}, \bar{\alpha}_h) &= (4.10920923, 0.66366992, 0.60854999).\end{aligned} \tag{8.117}$$

The status is deliberately limited: this proves a hidden-channel unloading preference inside the declared closure, not intrinsic spin or a Pauli-sector theorem.

8.9 Crossing-vs-collapse Goldilocks window

Stage 250 turns the dynamic branch into a survival inequality. The exact path-time integral from the Stage 248 energy law is

$$T_{\mathrm{traj}}(E; r_a \to r_b) = \sqrt{\frac{m_s}{2}} \int_{r_b}^{r_a} \frac{dr}{\sqrt{E-V(r)}}. \tag{8.118}$$

The characteristic-width closure used by Session III compresses the active barrier region to

$$t_{\mathrm{cross}}(E) = \lambda_{\mathrm{eff}} \sqrt{\frac{m_s}{2(E-V_{\mathrm{peak}})}}. \tag{8.119}$$

This is a reduced timescale closure, not a replacement for the exact path integral.

The active dressing leg uses the local unstable normal form

$$\mu_\eta \ddot{V} = g_{UV}\chi_{\mathrm{peak}} V. \tag{8.120}$$

Hence

$$t_{\mathrm{collapse}} = \sqrt{\frac{\mu_\eta}{g_{UV}\chi_{\mathrm{peak}}}}. \tag{8.121}$$

The survival ratio is

$$\mathcal{S}(E) = \frac{t_{\mathrm{cross}}(E)}{t_{\mathrm{collapse}}} = \lambda_{\mathrm{eff}} \sqrt{\frac{m_s g_{UV}\chi_{\mathrm{peak}}}{2\mu_\eta(E-V_{\mathrm{peak}})}}. \tag{8.122}$$

The lower safe edge is equation (8.25), and the speed-space identity is equation (8.26).

If $\mu_\eta = \alpha m_s$, the edge becomes

$$E_{\mathrm{edge}} = V_{\mathrm{peak}} + \frac{g_{UV}\chi_{\mathrm{peak}}\lambda_{\mathrm{eff}}^2}{2\alpha}, \tag{8.123}$$

so simultaneous scaling of particle mass and wall inertia cancels out of the edge. The real control packet is

$$(\lambda_{\mathrm{eff}}, \chi_{\mathrm{peak}}, g_{UV}, \mu_\eta/m_s). \tag{8.124}$$

With the trigger-width specialization,

$$\lambda_{\mathrm{eff}} = \lambda_{\mathrm{th}}(r_{\mathrm{turn}}) = \left|\frac{V(r_{\mathrm{turn}})}{V'(r_{\mathrm{turn}})}\right|. \tag{8.125}$$

The benchmark edge and sensitivity statements are therefore direct consequences of the dependence

$$E_{\mathrm{edge}} - V_{\mathrm{peak}} \propto \lambda_{\mathrm{eff}}^2 \chi_{\mathrm{peak}}. \tag{8.126}$$

Within this closure the safe window is one-sided because t_{collapse} is energy-independent and $t_{\mathrm{cross}}(E)$ decreases monotonically. Any upper edge requires an additional high-energy failure mechanism outside Stage 250.

8.10 Microscopic export kernel

Stage 251 replaces the phenomenological $\gamma_{\mathrm{tot}}\dot{V}$ envelope law by the first microscopic odd export operator seen by the active dressing leg. Project

$$V(t) = \Pi_{V0} q_0(t) + \Pi_{V-} q_-(t) + \cdots, \tag{8.127}$$

where q_0 is the derivative-coupled scalar outlet and q_- is the selected mixed quadrupole outlet.

For the scalar lane, take

$$g_{W,0}(\omega) = \eta_0\omega, \qquad g_{A,0}(\omega) = 0, \qquad \Pi_0^{\mathrm{out}}(\omega) = i\gamma_1\omega + O(\omega^3). \tag{8.128}$$

With

$$\Delta_0 = \Omega_{U,0}^2 \Omega_{W,0}^2 - R_0^2, \tag{8.129}$$

the scalar outgoing factor gives

$$\Gamma_{3,0} = \gamma_1 \eta_0^2 \frac{\Omega_{U,0}^4}{\Delta_0^2}, \qquad \Gamma_3 = \Pi_{V0}^2 \Gamma_{3,0}. \tag{8.130}$$

For the selected quadrupole outlet,

$$\Gamma_{5,-} = \frac{a^5}{27c_s^5} P_{0,-}, \qquad P_{0,-} = \frac{\beta_0 s_-}{\lambda_-}, \qquad \Gamma_5 = \Pi_{V-}^2 \Gamma_{5,-}. \tag{8.131}$$

Thus the microscopic kernel is equation (8.27).

The active-leg equation is

$$\mu_\eta \ddot{V} - \kappa_V V + \Gamma_3 V^{(3)} - \Gamma_5 V^{(5)} = \mathcal{S}_V(t), \qquad \kappa_V = g_{UV}\chi_{\mathrm{peak}}. \tag{8.132}$$

In Laplace form,

$$\mathcal{D}_V(s) = \mu_\eta s^2 - \kappa_V + \Gamma_3 s^3 + \Gamma_5 s^5. \tag{8.133}$$

The passive power identity is

$$\dot{V}(\Gamma_3 V^{(3)} - \Gamma_5 V^{(5)}) = \frac{d}{dt}\mathcal{S}_{\mathrm{odd}} - \Gamma_3 \ddot{V}^2 - \Gamma_5 \dddot{V}^2, \tag{8.134}$$

so the exported power is

$$\mathcal{P}_{\text{exp}} = \Gamma_3 \ddot{V}^2 + \Gamma_5 \dddot{V}^2 \geq 0. \tag{8.135}$$

For homogeneous growth $V = V_0 e^{st}$, the characteristic polynomial is

$$F(s) = \Gamma_5 s^5 + \Gamma_3 s^3 + \mu_\eta s^2 - \kappa_V. \tag{8.136}$$

For positive $(\Gamma_3, \Gamma_5, \mu_\eta, \kappa_V)$, $F(0) < 0$, $F(s) \to +\infty$, and $F'(s) > 0$ for $s > 0$. Therefore there is exactly one positive growth root. The small-kernel slowdown is

$$s_+ = s_0 - \frac{\Gamma_3 s_0^2 + \Gamma_5 s_0^4}{2\mu_\eta} + O(\Gamma^2), \qquad s_0 = \sqrt{\kappa_V/\mu_\eta}. \tag{8.137}$$

Thus finite passive cubic/quintic export slows but does not remove the unstable root in the minimal closure.

For an event with crossing rate $s_c = 1/t_{\text{cross}}$, strict monotonicity of F gives the safe half-plane equation (8.28). In the benchmark cold event, this becomes

$$\widehat{\Gamma}_3 + 0.3013336471\widehat{\Gamma}_5 \geq 289.61004918. \tag{8.138}$$

This is the exact event-level replacement for the old γ_{safe} inequality.

8.11 Vacuum/lattice heat partition and material thresholds

Stages 252 and 253 form the physical companion to the export kernel. Stage 252 keeps the reduced microscopic language; Stage 253 maps it into physical threshold variables.

8.11.1 Channel-resolved heat partition

Split the microscopic coefficients as

$$\Gamma_3 = \Gamma_3^{\text{vac}} + \Gamma_3^{\text{lat}}, \qquad \Gamma_5 = \Gamma_5^{\text{vac}} + \Gamma_5^{\text{lat}}, \qquad \Gamma_n^{\text{vac}}, \Gamma_n^{\text{lat}} \geq 0. \tag{8.139}$$

For an event window $[0, T]$, define

$$\mathcal{I}_2(T) = \int_0^T \ddot{V}^2 \, dt, \qquad \mathcal{I}_3(T) = \int_0^T \dddot{V}^2 \, dt. \tag{8.140}$$

The integrated energies are

$$E_{\text{vac}} = \Gamma_3^{\text{vac}} \mathcal{I}_2 + \Gamma_5^{\text{vac}} \mathcal{I}_3, \qquad E_{\text{lat}} = \Gamma_3^{\text{lat}} \mathcal{I}_2 + \Gamma_5^{\text{lat}} \mathcal{I}_3. \tag{8.141}$$

With $\mathfrak{r}_V = \mathcal{I}_3/\mathcal{I}_2$, the fractions are

$$f_{\text{vac}}(\mathfrak{r}_V) = \frac{\Gamma_3^{\text{vac}} + \Gamma_5^{\text{vac}} \mathfrak{r}_V}{\Gamma_3 + \Gamma_5 \mathfrak{r}_V}, \qquad f_{\text{lat}}(\mathfrak{r}_V) = \frac{\Gamma_3^{\text{lat}} + \Gamma_5^{\text{lat}} \mathfrak{r}_V}{\Gamma_3 + \Gamma_5 \mathfrak{r}_V}. \tag{8.142}$$

The partition drift obeys

$$\frac{df_{\text{lat}}}{d\mathfrak{r}_V} = \frac{\Gamma_5^{\text{lat}} \Gamma_3^{\text{vac}} - \Gamma_3^{\text{lat}} \Gamma_5^{\text{vac}}}{(\Gamma_3 + \Gamma_5 \mathfrak{r}_V)^2}. \tag{8.143}$$

So slow events sample the cubic split, fast rough events sample the quintic split.

For a single-growth cold event

$$V(t) = V_{\text{in}} e^{st}, \qquad 0 \le t \le T, \tag{8.144}$$

$$\mathcal{I}_2 = \frac{V_{\text{in}}^2 s^3}{2}(e^{2sT} - 1), \qquad \mathcal{I}_3 = s^2 \mathcal{I}_2, \qquad \mathfrak{r}_V = s^2. \tag{8.145}$$

The event-equivalent rates are

$$\gamma_{\text{vac}}^{\text{eq}}(s) = \Gamma_3^{\text{vac}} s^2 + \Gamma_5^{\text{vac}} s^4, \qquad \gamma_{\text{lat}}^{\text{eq}}(s) = \Gamma_3^{\text{lat}} s^2 + \Gamma_5^{\text{lat}} s^4. \tag{8.146}$$

At the safe edge, with $T = 1/s_c$ and $\Gamma_3 s_c^3 + \Gamma_5 s_c^5 = \mu_\eta (s_0^2 - s_c^2)$, the total minimum exported energy is equation (8.30), and the channel energies are obtained by multiplying by $f_{\text{vac}}(s_c)$ or $f_{\text{lat}}(s_c)$.

The old Session-IV $3 : 1$ heat split becomes a coefficient-surface condition:

$$f_{\text{lat}}(s_c) = \frac{3}{4} \iff \Gamma_3^{\text{lat}} + \Gamma_5^{\text{lat}} s_c^2 = 3\big(\Gamma_3^{\text{vac}} + \Gamma_5^{\text{vac}} s_c^2\big). \tag{8.147}$$

It is speed-independent only on the special family where both cubic and quintic channels have the same lattice fraction.

8.11.2 Physical calibration dictionary

Stage 253 introduces the explicit physical dictionary

$$t^{\text{phys}} = t_* t, \qquad r^{\text{phys}} = \frac{\lambda_{\text{phys}}}{\lambda_{\text{ref}}} r, \qquad E^{\text{phys}} = E_* E. \tag{8.148}$$

The safe-edge lattice event-equivalent rate is

$$\gamma_{\text{lat,safe}}^{\text{eq}} = f_{\text{lat}}(s_c) \mu_\eta \frac{s_0^2 - s_c^2}{s_c}. \tag{8.149}$$

A coarse transport projection factor $\Upsilon_{\text{lat}} > 0$ defines the physical lattice-turnover rate

$$\gamma_{\text{lat,turn}}^{\text{phys}} = \frac{\gamma_{\text{lat,safe}}^{\text{eq}}}{\Upsilon_{\text{lat}} t_*} = \zeta_{\text{ep}} \lambda_{\text{ep}} \omega_D. \tag{8.150}$$

Thus the electron-phonon turnover threshold is equation (8.31), and the product form is

$$(\lambda_{\text{ep}} \omega_D t_{\text{cross}}^{\text{phys}})_{\text{min}}^{(253)} = \frac{f_{\text{lat}}(s_c) \mu_\eta (s_0^2 - s_c^2)}{\Upsilon_{\text{lat}} \zeta_{\text{ep}} s_c^2}. \tag{8.151}$$

The legacy Session-V slice is recovered by choosing

$$\Upsilon_{\text{lat}}^{\text{sess}} = \frac{\gamma_{\text{lat,safe}}^{\text{eq}}}{\gamma_{\text{lattice}}^{\text{red}}}. \tag{8.152}$$

This is a calibration slice, not a new branch law.

For the harmonic interstitial companion,

$$r_{\text{turn}}^{\text{phys}} = \frac{r_{\text{turn}}}{\lambda_{\text{ref}}} \lambda_{\text{phys}}, \tag{8.153}$$

and

$$\chi_{\lambda,\text{lattice}}(r_{\text{turn}}) = \frac{2\lambda_{\text{phys}}}{r_{\text{turn}}^{\text{phys}}} = \frac{2\lambda_{\text{ref}}}{r_{\text{turn}}}. \tag{8.154}$$

This is only a geometric trigger. The force-matched stiffness requires

$$k_{\mathrm{eff,req}} = \frac{E_* \lambda_{\mathrm{ref}}^2}{\lambda_{\mathrm{phys}}^2} \frac{|V'_{\mathrm{red}}(r_{\mathrm{turn}})|}{r_{\mathrm{turn}}} = \mathcal{K}_{\mathrm{turn}} \frac{E_*}{\lambda_{\mathrm{phys}}^2}, \tag{8.155}$$

where

$$\mathcal{K}_{\mathrm{turn}} = \frac{\lambda_{\mathrm{ref}}^2}{r_{\mathrm{turn}}} |V'_{\mathrm{red}}(r_{\mathrm{turn}})|. \tag{8.156}$$

For $\lambda_{\mathrm{phys}} = a_{\mathrm{int}}/2$,

$$k_{\mathrm{eff,req}} = 4\mathcal{K}_{\mathrm{turn}} \frac{E_*}{a_{\mathrm{int}}^2}. \tag{8.157}$$

The Korringa companion is the cleanest physical map. With

$$T_1 T = \mathcal{K}_{\mathrm{corr}}, \qquad T_1 \geq t_{\mathrm{cross}}^{\mathrm{phys}}, \tag{8.158}$$

the ceiling temperature is

$$T_{\max} = \frac{\mathcal{K}_{\mathrm{corr}}}{t_{\mathrm{cross}}^{\mathrm{phys}}} = \frac{s_c \mathcal{K}_{\mathrm{corr}}}{t_*}. \tag{8.159}$$

8.11.3 Material-screening ratios

The companion ends with four screening ratios:

$$\Pi_{\mathrm{ep}} = \frac{\Upsilon_{\mathrm{lat}} \zeta_{\mathrm{ep}} \lambda_{\mathrm{ep}} \omega_D t_*}{\gamma_{\mathrm{lat,safe}}^{\mathrm{eq}}} = \frac{\lambda_{\mathrm{ep}} \omega_D}{(\lambda_{\mathrm{ep}} \omega_D)_{\min}^{(253)}}, \tag{8.160}$$

$$\Pi_\chi = \frac{2\lambda_{\mathrm{ref}}}{r_{\mathrm{turn}}}, \tag{8.161}$$

$$\Pi_k = \frac{k_{\mathrm{eff}} \lambda_{\mathrm{phys}}^2}{\mathcal{K}_{\mathrm{turn}} E_*} = \frac{k_{\mathrm{eff}}}{k_{\mathrm{eff,req}}}, \tag{8.162}$$

$$\Pi_T(T) = \frac{\mathcal{K}_{\mathrm{corr}}}{T t_{\mathrm{cross}}^{\mathrm{phys}}} = \frac{T_{\max}}{T}. \tag{8.163}$$

The materials companion is then

$$\Pi_{\mathrm{ep}} \geq 1, \qquad \Pi_\chi \geq 1, \qquad \Pi_k \geq 1, \qquad \Pi_T(T) \geq 1. \tag{8.164}$$

This is not a claim that any particular host passes. It is the reduced threshold stack that a candidate host must satisfy after the physical unit map and material constants are supplied.

8.12 Citation and downstream-use guardrails

Part VIII is useful because it lets downstream papers talk about the relaxed branch without contaminating the strict branch. The safe citation language is:

> The relaxed/open-system companion is a codimension-three lift of the strict selected-support/orbit front end. It has an exact recovery map to the strict slice, preserves the no-new-long-range same-charge verdict, and supplies separate leakage/work, non-rigid dressing, compensated-source, barrier, dynamic-event, export-kernel, heat-partition, and material-screening compilers.

The unsafe citation language is:

> The relaxed barrier branch proves the strict selected same-charge branch, or supplies a new long-range attraction.

That statement is false under the status discipline of this ledger. The relaxed branch is a companion extension around the strict front end. It is valuable precisely because it states what can change when open-system and non-rigid effects are allowed while preserving what cannot change: the same-charge long-range kernel verdict.

The active open tasks after this part are also explicit. The completed moving-throat PDE still has to realize or reject the strict selected support/orbit/outgoing packet from Part VII. If the relaxed companion is used, the actual branch must additionally return the leakage amplitude, non-rigid (U, V) packet, compensated source packet, microscopic export coefficients (Γ_3, Γ_5) and their vacuum/lattice split, the unit map $(t_*, \lambda_{\mathrm{phys}}, E_*)$, and the coarse transport factor Υ_{lat}. Without those realized data, Part VIII remains an exact compiler rather than a completed physical-branch theorem.

Appendix A

Reader Provenance Summary

This appendix records where the reader build fits inside the larger derivation archive. It is intentionally compact; the full source registry remains in the archive build.

A.1 Canonical source layers

Layer	Repo location	Role
Archive paper build	`research/pde_ledger/paper/pde_ledger.tex`	Authoritative master ledger with full frontmatter and stage appendices.
Reader paper build	`research/pde_ledger/paper/pde_ledger_reader.tex`	Readable theorem-block path through Parts I–VIII plus compact provenance and verification notes.
Theorem-block synthesis layer	`paper/parts/` and `paper/appendices/stage_appendix_part*.tex`	Inline narrative that organizes Parts I–VIII and the archive stage appendices.
Stage-card layer	`paper/stages/stage_NNN.tex`	Canonical per-stage provenance and audit anchors; all 253 cards are assembled into the archive stage appendices by `\input`.
Per-stage notes and audit layer	`notes/stages/` and `notes/stages/review/`	Decisive provenance authority beneath both paper layers when origin, value, or audit-history questions arise.
SymPy audits	`research/pde_ledger/scripts/`	Primary symbolic verification corpus and numerical stress harnesses.
Mathematica audits	`research/pde_ledger/mathematica/`	Secondary CAS replay layer and saved audit outputs.

A.2 Current stage-source rule

The paper source is two-layered, not a single assembly tree. The theorem-block synthesis layer is inline in `paper/parts/` and in `paper/appendices/stage_appendix_part*.tex`. The stage-card layer is `paper/stages/`;

those cards are the canonical per-stage provenance and audit anchors, and the archive appendices input them in stage order. For provenance questions beneath either paper layer, the decisive audit record is the per-stage notes layer, especially `notes/stages/` and `notes/stages/review/`, read together with the relevant scripts, notebooks, and trackers.

A.3 Escalation path

When a chapter in the reader build is not enough, the escalation path is:

1. reader chapter and its stage range;
2. theorem-block synthesis in the matching main part or archive stage appendix;
3. canonical stage card plus the matching SymPy/Mathematica audit scripts;
4. `notes/stages/`, `notes/stages/review/`, and any matching per-stage notes tree for decisive provenance questions;
5. verification trackers for cross-layer audit status.

Appendix B

Reader Verification Summary

This appendix gives the reader build a compact view of the verification stack without embedding raw transcripts or a stage-by-stage file manifest.

PDE-ledger paths below are relative to `research/pde_ledger/`; `docs/...` paths are repo-root relative.

B.1 Current baseline

Metric	Current archive baseline
Total stages	253
Stages with SymPy symbolic audits	237
Stages with Mathematica symbolic audits	179
Stages with dedicated numerical-stress harnesses	14
Stages with written review records	192
Citation-support checkpoint trust status	25 strong; 0 moderate; 0 weak
Checkpoint stages with dedicated numerical-stress backing	003, 185, 248, 253

B.2 Verification layers

Layer	Repo location	Reader-build interpretation
Canonical stage files	`paper/stages/` and `paper/appendices/stage_appendix_part*.tex`	Source archive and human-facing derivation ledger.
Paper theorem blocks	`paper/parts/`	Curated derivation and citation layer.

Layer	Repo location	Reader-build interpretation
SymPy audits	`scripts/`	Stagewise symbolic checks and numerical stress harnesses.
Mathematica audits	`mathematica/`	Independent second-CAS replay layer.
Verification trackers	repo-local control layer	Repo-level tracking of path inventory, trust, coverage, and cleanup sequencing.

B.3 Falsification stack

The first two layers below establish internal consistency and value reconciliation. They cannot, by themselves, detect a fit dressed as a derivation; the adversarial fit-vs-derive layer is the layer-2 audit for that failure mode, and it is now complete with no surviving fatal flaw.

Layer	Status as of 2026-06-12
Internal-consistency red-team (SymPy + Mathematica dual-engine, script audit)	Complete: 253/253 stages, with two full independent passes.
Script-to-doc value reconciliation (second-pass augmentation)	Complete.
Adversarial fit-vs-derive audit	Layer-2 fit-vs-derive audit COMPLETE: no surviving fatal flaw. The free-choice, published-target, HIGH, and internal-consistency completeness-critic surfaces are cleared in the toy-analog audit record. Two non-fatal `verdict_logged` PARTIALs remain recorded for close-out scoping: Stage 192 card re-attribution and γ_0 badge scoping, both contingent on existing user-gated items. Archive disclosure is generated at `paper/appendices/fit_insertion_index.tex` and `paper/appendices/parameter_provenance_audit.tex`; the raw audit record remains under `redteam_adversarial/`.
Stage-1 branch realization (target-blind numerical PDE)	Gated on the adversarial layer by `docs/branch_realization_execution_plan.md`; not started.

B.4 How to locate the audits

The stage number is the reader-facing lookup key. Raw per-stage filenames are intentionally omitted from the printed stage cards so the derivation pages stay legible.

- SymPy symbolic audits follow `moving_throat_pde_stageNNN_*_sympy_audit.py`.
- Mathematica symbolic audits follow `moving_throat_pde_stageNNN_*_mathematica_audit.wl`.

- Numerical stress harnesses usually follow `stageNNN*_stress.py` and `stageNNN*_stress.wl`, with occasional shared-range names such as `stage003_021_*`.
- The exact path inventory and current coverage state are maintained in the control documents listed above rather than repeated inside every stage.

B.5 What is intentionally omitted here

The reader build does not include:

- raw script transcripts;
- the full stage-by-stage audit registry;
- the full reproducibility map;
- the exact per-stage file-path inventory;
- the full source-file index.

That omission is deliberate. The reader build is a navigation layer. The archive build and repo control documents remain the full audit register.

B.6 Citation rule

Use the reader build to find the right theorem block and the right status language. Use the archive build and repo artifacts when the claim is load-bearing enough that a future paper or referee needs the full audit trail.